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Financial Mathematics, Derivatives and Structured Products

 Springer

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Dedicated to our loved ones,

Raymond: to Katharine

Yves: to Véronique, Antoine, Céline, and Sophie

Spike: to Carie

Xun: to Yun

Preface

This book originates from different courses taught by the authors:

- “Financial Mathematics” (undergraduate course) at The Chinese University of Hong Kong by the first author
- “Financial Mathematics I” (undergraduate course) at the National University of Singapore and Hong Kong Polytechnic University by the fourth author
- “Mathematical Financial Modelling” (master’s course) at The Chinese University of Hong Kong by the second and third authors
- “Mathematical Investment Models” and “Structured Products and Solutions” (master’s courses) at Hong Kong University of Science and Technology by the second author.

Many examples and analysis from industry’s point of view are featured in this book and provided by the second author who has 20+ years banking experience in financial engineering for his work in a large French bank and two other top investment banks. At the same time, the book also contains theoretical content and bridges the gap between theory and practice. It takes a different route from the existing financial mathematics books and will attract both students and practitioners with or without science or finance background.

The primary audience would be undergraduate and postgraduate students. The book can also be used as a textbook for the following courses:

- Financial Mathematics (undergraduate level)
- Stochastic Modelling in Finance (postgraduate level)
- Financial Markets and Derivatives (undergraduate level)
- Structured Products and Solutions (undergraduate/postgraduate level).

This book aims to equip readers with the necessary knowledge of financial markets for working as product structurers, traders, sales, or risk managers. Both

financial practitioners and academic people may find interests in this book because it intends to unify the derivatives modelling and the financial engineering practice in the market.

Hong Kong
Sep 2018

Raymond H. Chan
Yves ZY. Guo
Spike T. Lee
Xun Li

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Acronyms

ABS	Asset-backed securities
ADI	Alternating direction implicit
ADTV	Average daily trading volume
AKI	American Knock-In
ALM	Asset liability management
APT	Arbitrage pricing theory
ASB	Accelerated Share Buy-Back
ASCOT	Asset Swapped Convertible Option Transaction
ATM	At-the-money
ATMF	At-the-money forward
ATMS	At-the-money spot
AUM	Assets under management
AXKI	At Expiry Knock-In
BA	Banker's acceptance
BBM	Buy below the market
BCBS	Basel Committee on Banking Supervision
BEN	Bonus Enhanced Note
BGM	Brace–Gatarek–Musiela
BPV	Basis point value
BSM	Black–Scholes–Merton
CAL	Capital allocation line
CAPM	Capital asset pricing model
CB	Convertible bond
CBBC	Callable bull/bear contract
CBOE	Chicago Board Options Exchange
CCS	Cross-currency swap
CD	Certificate of deposit
CDO	Collateralized debt obligation
CDS	Credit default swap
CET	Central European time

CEV	Constant elasticity of variance
CIR	Cox–Ingersoll–Ross
CLN	Credit-linked note
CME	Chicago Mercantile Exchange
CMS	Constant maturity swap
CP	Commercial paper
CPPI	Constant proportion portfolio insurance
CSA	Credit support annex
CTA	Commodity trading advisor
CTD	Cheapest-to-deliver
CVA	Credit valuation adjustment
DAC	Daily Range Accrual Callable Note
DCI	Dual Currency Investment
DCM	Debt capital market
DVA	Debit valuation adjustment
DVP	Delivery versus payment
EAD	Exposure at default
ECM	Equity capital market
ELN	Equity-linked note
EPE	Expected positive exposure
ERR	Expiry reference rate
ETF	Exchange-traded fund
Eonia	Euro Overnight Index Average
EPS	Earnings-per-share
Euribor	Euro Interbank Offered Rate
FCN	Fixed Coupon Callable Note
FRA	Forward rate agreement
FRN	Floating rate note
FX	Foreign exchange
GMRA	Global Master Repurchase Agreement
GMSLA	Global Master Securities Lending Agreement
HJM	Heath–Jarrow–Morton
HNW	High net worth
HWM	High water mark
IA	Independent amount
ICE	Intercontinental Exchange
ICMA	International Capital Market Association
IIC	International Index Company
IPO	Initial public offering
IRB	Internal ratings-based
IRR	Internal rate of return
IRS	Interest rate swap
ISDA	International Swaps and Derivatives Association
ITM	In-the-money

KMV	Kealhofer, McQuown, and Vašíček
KO	Knock-Out
LGD	Loss given default
LGM	Linear Gaussian model
Libor	London Interbank Offered Rate
LMM	Libor market model
LTV	Loan-to-value
MD	Modified duration
MPT	Modern Portfolio Theory
MTA	Minimum transfer amount
MtM	Mark-to-market
MVO	Mean-variance optimization
NAV	Net asset value
NDF	Non-deliverable forward
NDS	Non-deliverable swap
NPV	Net present value
OBPI	Option-based portfolio insurance
OIS	Overnight indexed swap
O/N	Overnight (as in deposit rate)
OTC	Over-the-counter
OTM	Out-of-the-money
PCA	Principal component analysis
PD	Participating dealer
PDE	Partial differential equation
PFE	Potential future exposure
PIDE	Partial integro-differential equation
P/L	Profit & Loss
PnL	Profit & Loss
PR	Participation rate
PV	Present value
Repo	Repurchase agreement
RICI	Rogers International Commodity Index
RWA	Risk-weighted asset
SAM	Sell above the market
SBL	Securities borrowing and lending
SDE	Stochastic differential equation
SPV	Special purpose vehicle
STIR	Short-term interest rate
T-bills	Treasury bills
TER	Total expense ratio
TIPS	Treasury inflation-protected securities
TIPP	Time invariant portfolio protection
T/N	Tomorrow/Next (as in deposit rate)
TRF	Target redemption forward
TRS	Total return swap

WTI	West Texas Intermediate
VaR	Value at risk
VWAP	Volume-weighted average price
ZC(B)	Zero coupon (bond)

Notation

W	week, e.g., 1W means a 1-week tenor
M	month, e.g., 3M means a 3-month tenor
Y	year, e.g., 1Y means a 1-year tenor
k	thousand, e.g., USD50k
m/mio	million, e.g., USD10m
a.s.	almost surely
bn	billion, e.g., USD1bn
bps	basis point, which is $0.01\% = 0.0001$
c.d.f.	cumulative distribution function
i.i.d.	independent and identically distributed
LHS	left-hand side of an equation
RHS	right-hand side of an equation
r.v.	random variable
s.t.	such that
p.a.	per annum
p.d.f.	probability density function
w.r.t.	with respect to
$\overset{(d)}{\Longleftrightarrow}$	equivalence in distribution
$a \wedge b$	$\min\{a, b\}$
$a \vee b$	$\max\{a, b\}$
$(X)^+$	$\max\{0, X\}$
$\cdot$	vector inner product
$e, \exp$	exponential function
$\ln, \log$	natural logarithm

Part I

Financial Markets

Chapter 1

Introduction to Financial Markets



There are a few objectives in this book. The first is to study the main properties of basic derivatives such as forward/futures, swaps, vanilla and exotic options. The second is to provide a modelling framework within which those derivatives can be priced and hedged. The third is to introduce the structured products, which are customized solutions offered to investors.

1.1 Investable Assets and Instruments

The assets for investors in the financial market are mainly liquid investable assets such as bonds, stocks, foreign currencies and commodities. The liquidity of an asset is measured by the easiness and cost to convert it into cash. Not all assets are transferable, for instance, the bank deposits. A transferable asset traded in the market is often identified by a code given by a well-recognized institution.¹

The basic types of assets are *debts* and *equities* which are directly linked to the economical activities. The debt securities take the form of bonds, bills, deposits, etc. Equity securities issued by corporations are called *stocks* or *shares* representing claims of the ownership of the corporation. One can invest directly in a certain asset and receive the return or performance of the asset. For taking benefit of the price drop of an asset, one may borrow the asset, sell it at its current price and then, buy the asset back in the future (hopefully at a lower price) and return it to the lender.

¹Taking the BNP Paribas SA stock as example, there are RIC (Reuters Instrument Code) “bnpp.pa”, ISIN (International Securities Identification Number) code “FR0000131104” and BBGID (Bloomberg Global Identifier) “BNP FP Equity”, etc. In the financial market, terminals from Bloomberg, Reuters are currently the most used ones providing real-time market data, news, trading/pricing platform and other services.

A *derivative contract* (or simply, *derivative*) is a financial instrument that derives its value from the price(s) of some specific transferable asset(s) or security(ies) (called *underlying asset(s)* or simply *underlying(s)*) such as equity securities, fixed-income securities, foreign currencies, or commodities. The most common derivatives are *forward/futures contracts*, *swaps* and *options*.

A *forward* is an agreement between two parties in which the buyer will purchase a certain asset from the seller at a pre-agreed price, called strike price or exercise price, at a future time (called expiry date or maturity date). Mathematically, the payoff for the buyer is $S_t - K$, where S_t is the asset's price at expiry time t and K is the delivery price. The level K which makes the forward contract value at zero is called the (market) *forward price* of the underlying asset.

A *futures contract* is a standardized forward contract listed at an organized market such as a stock exchange.

A *swap* is a derivative contract whereby two counterparties exchange cash flows or securities. The standard interest rate swap (IRS) is a typical example in which one party will pay the cash flows based on a pre-agreed fixed rate and receive the floating rate based cash flows applied to a specified notional amount.

Roughly speaking, an *option* is a financial instrument which offers a return (called payoff or payout) that is asymmetric to the upside and downside performance of the underlying asset(s). The basic options (nicknamed as Vanilla options) are call options and put options. A *call option* gives the owner the right but not the obligation to buy a certain asset from the seller at a pre-agreed price, called the strike price or exercise price. The payoff at expiry t of a call option is $\max\{S_t - K, 0\}$. Similarly, a *put option* gives the owner the right but not the obligation to sell the underlying asset at the strike price (or exercise price). The payoff at expiry t is $\max\{K - S_t, 0\}$. Any option other than a Vanilla option is referred to as an *exotic* option, for instance, Barrier option, Asian option, etc.

A *structured product* is a pre-packaged investment strategy based on derivatives which offers a customized risk-return objective. In the past two decades, we have witnessed a revolutionary development for structured products and derivatives in financial markets around the world.

1.2 Investment Returns and Risks

The investment return of a non-dividend paying asset for the period between time s and time t ($s < t$), is defined as:

$$R = \frac{P_t - P_s}{P_s} \quad \text{or} \quad R = \frac{P_t}{P_s} - 1, \quad \text{where } P_t \text{ is the price of the asset at time } t.$$

The return rate is an annualized concept in general. There are different types of return rates. For instance, the IRR (Internal Rate of Return) is the value of r which makes the following equality:

$$P_t = P_s(1 + r)^\tau, \text{ where } \tau \text{ represents the number of years for the period } [s, t].$$

Usual measures for the rate of return of an asset or a portfolio are

Price return: it is measured by the portfolio's value at the beginning and the end of the period. The dividend payments during the period are ignored.

Total return: it is obtained with all the dividends re-invested back into the assets of the portfolio with the same proportion. It represents the return of a fully funded portfolio.

Excess return: It is defined as the portfolio's total return minus the financing cost or a relevant interest rate reference. It represents the return of a self-financed portfolio.

Example 1.1 Consider two assets: Asset 1 and Asset 2. Asset 2 pays a \$1 dividend per unit at time t . The financing cost is 1% on the borrowed amount.

	P_s	P_t	Price return	Total return	Excess return
Asset 1	19	20	5.26%	5.26%	4.26%
Asset 2	90	99	10%	11.11%	10.11%

Note that the total return for Asset 2 is calculated from $\frac{99+1}{90} - 1$, i.e., assuming that the \$1 dividend per unit of the asset is reinvested.

In financial market, the term *long* means “in the possession of” and the term *short* means the opposite. A long position of an asset will realize a profit if the asset price goes up and suffer a loss if the asset price drops. The short position of an asset will realize a profit if the asset's price drops and lose money if the asset's price goes up. The short position can be built by borrowing the asset and then selling it to the market. For closing the short position, the asset must be bought back and returned to the asset lender. Financial analysts use the words “bull/bullish” and “bear/bearish” for describing the rising trend and falling trend of the price of the assets. The term *arbitrage* used in finance refers to the transactions leading to a risk-free profit.

Example 1.2 If a trader observes three foreign exchange prices USD/JPY = 113.65 (i.e. 113.65 Yen per US dollar), EUR/USD = 1.0922, EUR/JPY = 124.23, he can make a risk-free profit of about 0.10 JPY per EUR by the following transactions²:

²For illustration purpose, we assume zero bid/ask spread and zero transaction cost.

- sell EUR/buy JPY at 124.23, and simultaneously,
- buy EUR/sell USD at 1.0922 and buy USD/sell JPY at 113.65.

The most important risks for the financial products include

Market Risk: it is the risk on the return or mark-to-market (MtM) price of the product caused by the changes of the market price of the related financial assets and other financial factors;

Counterparty Risk: it arises when one of the parties fails to fulfill obligation towards the other party;

Liquidity Risk: the uncertainty about the terms and ability to make, unwind or hedge a transaction.

1.3 Performance Measures

There are many indicators that are used for measuring the performance of an investment strategy or an investment fund (aggregated form of investment) other than its total return R . The most popular ones include

Sharpe ratio = $(R - r)/$ “standard deviation of $(R - r)$ ”, where r is the return of a risk-free investment.³ It is the most used measure for risk-adjusted excess return.

Information ratio = $(R - R_{BM})/$ “standard deviation of $(R - R_{BM})$ ”, where R_{BM} is the return of the relevant benchmark (e.g. an equity index, if the investment portfolio is composed of stocks).

Sortino ratio = $(R - r)/$ “downside deviation”, where “downside deviation” is the standard deviation obtained with negative returns only.⁴ This measure is preferred by people who consider that the Sharpe ratio punishes the “good risk” as the positive returns are also used in its standard deviation calculation.

Maximum drawdown: the peak-to-trough decline in percentage during the considered period of an investment. For the considered period $[t_1, t_2]$, it is defined as $\frac{m(\hat{t}, t_2)}{M(t_1, t_2)} - 1$, where $M(t_1, t_2)$ is the highest price which is achieved at time $\hat{t}$, and $m(\hat{t}, t_2)$ is the lowest price after $\hat{t}$.

³Strictly speaking, there is no such thing as risk-free investment or risk-free asset. People often approximate it by the return of government debt or bank deposit. The real financing rate can also be used in certain circumstances.

⁴Downside deviation can be calculated by $\sqrt{\frac{252}{n} \sum_{i=1}^n \min\{R_i, 0\}^2}$ where R_i represents the return of the i -th business day, n the total number of business days for the considered period, and 252 the number of business days in a year.

1.4 Financial Markets

A financial market designates the aggregate of participants, organizations and facilities in which people trade financial securities (e.g. stocks and bonds), currencies and commodities at prices that reflect supply and demand. The participants include financial institutions (such as banks, insurance companies, pension funds, mutual funds, hedge funds), individual investors and corporations.

The financial market can be detailed by the type of services it offers:

Capital markets: which provide financing through the issuance of shares and debts, and enable the subsequent trading thereof. Capital markets include debt and equity markets.

- *Equity market* (also known as *Stock market*) allows the participants to buy and sell stocks of (publicly traded) companies. The value of a stock reflects the view about the expected dividend payments, future earnings, and resources that the company will control.
- *Debt market* (or *Fixed-income market*) includes *bond market* which deals in government, corporate and other bonds for long term financing, and *money market* for short term (up to 1 year) debt securities such as bank deposits, treasury bills, certificates of deposit, commercial papers, etc.

Foreign exchange market: where currencies are bought and sold.

Commodity market: where commodities such as precious metals, industrial metals, energy products and agricultural products are traded. Futures contracts are the most convenient instruments for commodities trading activities. A futures contract may be sold out before the commodity is due to be delivered.

Derivatives market: where futures, swaps, options and other derivatives are transacted.

The financial market can also be classified with other criteria, such as

Primary market: where new issues are first sold through IPOs (Initial Public Offerings). The primary market business for debts and stocks is covered respectively by DCM (Debt Capital Market) and ECM (Equity Capital Market) divisions of the Corporate Finance entity in the investment banks.

Secondary market: for all subsequent trading after IPO between market participants. It constitutes the support for the financial products for investment and risk management.

When classified by market organization, there are

Listed market: public auction market such as Stock Exchanges, e.g. Euronext, New York Exchange, Nasdaq, Hong Kong Exchange; The exchanges are usually private companies and may be listed themselves.

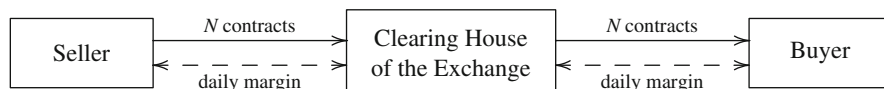
OTC (Over-The-Counter) market: where buy/sell contracts are made between the two counterparties. OTC market is often animated by brokers as intermediary agents.

1.4.1 Centralized Clearing

As an OTC derivative involves potential payments between the parties in the future, the *counterparty risk* that one party does not pay as obligated in the contract can not be neglected. To mitigate the counterparty risk, the long and short parties of a derivative contract may transact with the “Central Clearing House” which covers the risk by a collateral deposit system known as the margining system.

For illustration purpose, we take the case of futures contracts transacted at a Futures Exchange serving as the central clearing house.

- The buyer and seller should have their “margin account” in place before trading futures.
- At the contract inception, both parties will deposit an “initial margin”, fixed by the exchange according to the type and price of futures, as collateral which are typically cash or government bonds.
- At the end of each trading session, each party will have their margin account debited or credited for the daily P/L (Profit and Loss, or PnL).
- A minimum margin level called “maintenance margin” is required for every margin account. If the account value is below this level, a “margin call” will be issued for bringing back the account to the level of initial margin.



1.4.2 OTC Derivatives Documentation and Counterparty Risk Management

The financial institutions offering derivative products and services control and monitor their counterparty risk with the trading counterparty. Some products such as a deposit or a structured note imply one-way counterparty risk where the end investor takes the counterparty risk of the financial institution but the financial institution has no risk from the end investor. However, in some other products such as forward and swap, both parties are taking counterparty risk. The growth of OTC derivatives business prompted the standardization for the contract terms as well as the counterparty credit risk management which are widely used by the institutions and corporates actively involved in financial market.

ISDA (International Swaps and Derivatives Association) Master Agreement, initially developed in the 1980s to cover the IRS and currency swaps, has been progressively updated to include the derivatives such as forward, swap and option linked to equities, commodities and funds. It sets out standard terms applied to all trades between the two parties.

In general, two parties set up their trading relationship for OTC derivatives by negotiating the applicability and eventual adjustment of the standard terms. The document containing the agreed terms is the ISDA Master Agreement signed by both parties.

CSA (Credit Support Annex) is the document for credit support (i.e. collateral) for derivative transactions. It defines the acceptable collaterals with the “haircuts”. In particular, it defines the “Threshold” which is the consolidated MtM level of all the trades to trigger the margin call. The “Independent Amount (IA)” is the initial margin (collateral) required by one party (usually the dealer) to the other party (usually the end user) for mitigating the counterparty risk linked to an OTC transaction. It is returned only after the termination of the transaction. Its level depends on the volatility of the mark-to-market value of the trade as well as the credit worthiness of the counterparty. During the life of the trade, “variation margin” will be exchanged according to its mark-to-market.

The institutions and corporates actively transacting derivatives usually establish an ISDA/CSA Master Agreement with their counterparties. Normally, the credit agreement between two financial institutions is a two-way CSA in which both parties may post margins for their OTC trades. Between a financial institution and a corporate (especially the small ones), the credit agreement may be a one-way CSA, meaning that only the corporate posts margins to the financial institution. Under a master agreement (ISDA/CSA or any bespoke master agreement), the specific terms and conditions of each OTC derivative trade will take a short form called *term sheet* or *transaction supplement*. The individual investors, small corporates and other non-active entities involved in financial market normally trade with financial institutions with a bespoke agreement or a long form confirmation which contains all terms and conditions for each trade.

1.5 Securities Borrowing & Lending and Repo

A repurchase agreement is a contract for the sale of a security (e.g. stock or bond) with a commitment by the seller to buy the same security back from the buyer at a specified price at a future date. During the tenor of the trade, the seller (also called the lender) of the security surrenders the legal ownership of the security. There are two activities based on the repurchase agreement: Securities Lending and Repo.

Securities Borrowing & Lending (SBL) transaction allows the lender to lend securities to the borrower on either “Open” (i.e. anytime callable) or “Term” (a fixed tenor) basis. Upon the trade termination, the securities will be returned to the lender. The borrower posts collateral with daily adjustment and pays fees to the lender. The fee rate depends on the borrow supply/demand for the underlying security. The eligible collateral can be cash or other securities negotiated by the parties. The cash collateral level is usually $\sim 105\%$ of the latest closing price of the security. The International Securities Lending Association has developed a standard agreement called *Global Master Securities Lending Agreement (GMSLA)* which is followed by most of the institutions. The motivation for the borrower includes short position recovering, hedging of derivatives, corporate action arbitrage, etc.

In a *sale and repurchase agreement (Repo)*, one counterparty (the repo seller) is borrowing money and providing collateral (mostly fixed-income assets) for the loan. See Fig. 1.1 for reference. The seller gains access to funds at lower funding costs than are typically available elsewhere as the loan is collateralized. The collateral eligibility and haircuts are negotiable between the repo counterparties. The standard agreement for Repo is *Global Master Repurchase Agreement (GMRA)*, published by the International Capital Market Association (ICMA). A Reverse Repo is the opposite transaction seen by the other counterparty of the Repo trade. Some central banks use Repo/Reverse Repo operations to regulate the money supply in the financial system.

If the collateral is held at a third party, usually a custodian bank or an international central securities depository, the transaction is called a *Tri-Party Repo* or *Tri-Party Securities Lending*. The third party will provide services such as the valuation and adjustment of the collateral. The risk in a Tri-Party Repo transaction is the correlation of the default probability of the counterparty and the value of the collateral in custody.

Although most Repo activities take place on the OTC market, there exists *Stock Exchange Repo* (e.g. Shanghai Stock Exchange Repo) whereby the exchange determines the collateral pool and haircuts, standardizes the contract features such as size and tenor, and facilitates clearing and pledge of collateral.

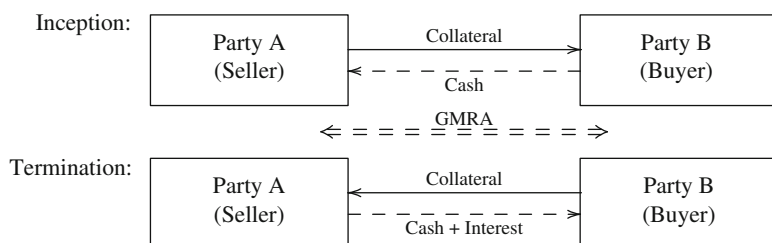


Fig. 1.1 Illustration of a Repo transaction

1.6 Derivatives Activities

The derivative instruments are used by the end-users for various purposes including but not limited to

Exposure for a specific view: derivatives can be used for gaining exposure to a specific view on future price changes. For instance, one believes that the price of an equity index will move significantly in either direction for a certain time horizon, he/she may buy a call and a put at the same time.

Hedging: which means offsetting or reducing the impact of the price change of the considered asset or product. For instance, a US dollar based company is expected to receive a cash flow in Euro in 3 months. The company, concerned by the currency exchange rate risk, may choose to short a 3 month EUR/USD forward transaction for selling the cash flow in Euro into US dollars at a fixed exchange rate. So, at any time until the expiry of the forward contract, the change of the USD value of the future cash flow in Euro will be offset by the value change of the forward contract.

Leverage: derivatives allow leveraged exposure to the underlying asset. For instance, an investor has a bullish view on a stock for a horizon of one month. But instead of investing for N stocks, the investor chooses to buy a 1-month call option. He then obtains the upside exposure for the N stocks with only the premium of, say, 3% of value of the stocks. In that sense, his investment is leveraged.

The investment banks as well as some other financial institutions offer the derivative products as a service to the clients (end-users) and earn commissions for their service. Their business model is to offer products with none or limited risk taking by *replicating* the payoffs of the products with other liquid securities so that the profit or loss of the clients will be offset by the replication position. It is a common mis-understanding that the derivative provider (underwriter, issuer) and the client are playing a zero-sum game where the client makes money when the provider loses money and vice versa. Derivative is a double-edged sword: it can help to hedge out risks but it may also allow the user to increase the risk exposure. The risk of some derivatives may be complex and difficult to understand for non sophisticated investors. The regulators for financial industry as well as the financial institutions keep implementing measures to prevent selling products that are not matching the profile of investors.

The pricing of a derivative product is based on its replication cost which has no or limited impact from the demand or offer for the product. The pricing and hedging (replication) of derivatives are based on the same techniques which constitute one of the main purposes of this book. In contrast, the price of a basic asset such as stock, currency, commodity is driven by the long term fundamentals and the short term demand/offer. There are abundant literature and textbooks on the pricing of these assets. Readers may refer to [64].

Exercises

1. An order book contains a list of buy (known as bid) orders with corresponding buy sizes and sell (known as ask) orders with corresponding sell sizes raised by market participants. The following is an order book for a security A.

Bid volume	Bid price	Ask price	Ask volume
500	100	101	500
1000	99	102	500
1500	98	103	1000
⋮	⋮	⋮	⋮

- An investor wants to buy 2000 shares of security A using a *market order*, i.e., regardless of price level. Calculate the average price of their total positions in security A.
2. Let P_t be the price of a non-dividend paying asset at time t . Find out under what condition the *log-return* defined as $\ln(P_t/P_s)$ for $s < t$ is a close approximate of the standard return $(P_t - P_s)/P_s$.
 3. Investors are becoming more aware of alternative asset classes for diversifying risk. Look up the one known as *aircraft leasing* and describe its characteristics.
 4. Consider a new private fund with initial NAV (Net Asset Value) of 100. The investment started on 2 Jan 2017 and is fully invested in Stock A. On 15 April 2017, the fund switched all the investment into Stock B. On 3 July 2017, fund paid a cash dividend of 3% of the fund NAV. The interest rate is 3% applicable with the formula $3\% \times \text{number of days}/365$ for calculating the interest. For simplifying the exercise, we make the assumptions of zero transaction cost, zero management fee and that we can trade a fraction of a stock. The stock prices are given as follows

Date	Price of Stock A	Price of Stock B
2 Jan 2017	20	180
15 Apr 2017	22	160
3 Jul 2017	21.5	190
15 Sep 2017	23	198

Calculate the below values at 15 Sep 2017:

- Fund NAV
- Total Return
- Price Return
- Excess Return

Chapter 2

Interest Rate Instruments



In this chapter, we will cover the well-known fixed-income instruments and their derivatives, and most importantly, how to construct a yield curve from liquid market products. The yield curve serves as the foundation for extracting interest rates that are used in financial calculation.

A fixed-income instrument, also referred to as a debt security, generally pays the return on a fixed schedule, though the amount of the payments can vary. The risks of debt securities are mainly interest rate risk, credit risk and liquidity risk. Fixed-income instruments can be classified by their maturities (defined at the time of issuance):

short term: money market instruments including treasury bill, commercial paper (CP), bank deposit, and etc.;

long term: bonds, notes, asset-backed securities (ABS), and etc.

2.1 Interest Rate Conventions

The maturity, a time measure with units in years, is an important parameter in fixed-income products. The financial calculation in each fixed-income instrument follows specific conventions.

2.1.1 Day Count Convention

The day count convention determines the time period in years to be used for calculating the interest for the investment in financial instruments. It is also used for calculating the present value of financial assets through discounting. Generally, the first day ($d_1/m_1/y_1$) of the considered period is included and the last day ($d_2/m_2/y_2$) is excluded.

1. **30/360 methods** are a class of calculation methods with the following general formula

$$\text{Day Count Fraction} = \frac{360 \times (y_2 - y_1) + 30 \times (m_2 - m_1) + (d_2 - d_1)}{360},$$

where y , m , and d stand for year, month, and day respectively as in a calendar system.

A non-exhaustive list includes “30/360 US” (also called “bond basis”), “30E/360” (also called “Eurobond basis”), in which different treatment of d_1 and d_2 is involved.¹

2. **ACT methods**, where ACT stands for *actual*, are widely-used and mainly utilize the number of days between two dates, denoted by $d_2 - d_1$. For instance,

ACT/360 (also called “money market basis”): $\frac{d_2 - d_1}{360}$

ACT/365 (also called “English money market basis”): $\frac{d_2 - d_1}{365}$

ACT/ACT ISDA: $\frac{\text{Days in a non-leap year}}{365} + \frac{\text{Days in a leap year}}{366}$.

2.1.2 Business Day Convention

A business day convention specifies the adjustment when a date used in calculation is not a business day. The most common ones are: *Following* – the following business day; *Preceding* – the preceding business day; *Modified Following* – the following business day unless the day is in the next calendar month, in which case the adjusted date is the preceding business day; *End-of-Month* – where the start date of a period is on the final business day of a particular calendar month, the end date is on the final business day of the end month.²

2.2 Interest Rate Types and Zero Coupon Bond

An interest rate represents the time value of money, measured as the value change per annum (e.g. 2.10% p.a.). It is the cost of borrowing money or the return for lending money. There are several types of interest rate. Some of them are used in

¹For instance, in 30/360, if d_1 is 31, then change d_1 to 30. If d_2 is 31 and d_1 is 30 or 31, then change d_2 to 30. But under 30E/360, if d_1 is 31, then change d_1 to 30. If d_2 is 31, then change d_2 to 30.

²E.g. Start date Friday, 29-Apr-2011, period 1 month: end date will be 31-May-2011 (29-Apr is the last business day of the month).

the financial contracts for the determination of cashflows; Some others, e.g. the continuous interest rate, are not directly used in real financial contracts and have been introduced for comparison and calculation purposes.

2.2.1 Simple Rate

Simple interest rate (r_m) is applied to calculate the terminal amount M_T based on the initial amount M according to the following formula:

$$M_T = M(1 + r_m T).$$

In the money market, the interest-bearing instruments (e.g. bank deposit, certificate of deposit, etc.) use this type of interest rate. Hence, it is also referred to as *money market rate*.

On the other hand, a non-interest-bearing instrument in the money market such as a treasury bill and a commercial paper is quoted in price for receiving a fixed amount at maturity. The implied rate measuring the discount is called *discount rate* (r_d) which satisfies: $M = M_T(1 - r_d T)$.

2.2.2 Compound Rate

A *compound* rate assumes the re-investment of the interests from the investment.

Suppose you deposit M in a bank at the interest rate of r with annual payments. At the end of year one, the bank account value will become $M(1 + r)$. After two years, you will have $M(1 + r) \cdot (1 + r) = M(1 + r)^2$. After T years, your account value will be $M_T = M(1 + r)^T$. If it is compounded m times per annum (i.e. the interests are paid m times a year), the terminal value of the investment after T years will be

$$M_T = M \left(1 + \frac{r}{m} \right)^{m \times T}. \quad (2.1)$$

2.2.3 Continuous Rate

The pricing of financial derivatives often involves continuous-time modelling. The interest rate used in continuous time modelling is a compound rate applied to infinitesimal time intervals.

In (2.1) for compound interest rate, if we apply compounding with very small time intervals of Δt per annum, we obtain the continuous interest rate r as the limiting case:

$$M_T = \lim_{\Delta t \rightarrow 0} M \left(1 + \frac{r}{1/\Delta t} \right)^{(1/\Delta t) \times T} = M e^{rT}.$$

Alternatively, we can assume the differential equation $dM_t = rM_t dt$ which is the limiting expression of $\Delta M_t = rM_{t-\Delta t} \Delta t$ with $\Delta t \rightarrow 0$. Its resolution shows the same result: $M_T = M e^{rT}$.

More generally, if the continuous rate is a function of time, the term *instantaneous interest rate* will be used. It is denoted as r_t . The resolution of $dM_t = r_t M_t dt$ over the time period $[t, T]$ leads to

$$M_T = M_t e^{\int_t^T r_s ds}.$$

The constant rate $r_t = r$ can be considered as a particular case:

$$M_T = M_t e^{r(T-t)}.$$

2.2.4 Relationship Between Interest Rates

One can easily transform rates of different types through the equivalence in cash-flow value. For instance, the continuous rate r is not directly observable from the financial instruments in the market. However, r can be derived from the money market rate r_m by

$$1 + r_m T_m = e^{rT}, \text{ or equivalently } r = \ln(1 + r_m T_m)/T,$$

where T_m is the day count fraction under the day count convention of the money market instrument while T usually follows ACT/365 or ACT/ACT.

Due to the interchangeability of the interest rate types, we are indifferent regarding the choice of the rate type to be used in financial calculation.

2.2.5 Zero Coupon Bond

A *zero coupon (ZC) bond* (also called *discount bond*), denoted by $P_t(T)$, represents the time- t value of an instrument paying out \$1 value at maturity T . For simplifying the notation, we will use $P(T)$ for $P_0(T)$. The zero coupon bonds constitute the building blocks for financial calculation. It can be used as a *discount factor* for calculating the present value of future cashflows.

Zero coupon bonds for different maturities can be extracted from market traded instruments (the extraction process, known as *bootstrapping*, will be explained in the last section). Conversely, the interest rate for a specific instrument in the market can be represented by zero coupon bonds. Since the interest rates embedded in different instruments follow different calculation conventions which may not be compared in a straightforward way, it is convenient to use zero coupon bonds as the base tradable assets in financial calculation.

The market price (or fair value) of a future cash-flow M_T is called the *present value (PV)* of M_T . It can be expressed as

$$PV = P(T) \times M_T.$$

Note that the term *zero coupon bond* under modelling context is an abstraction of the **zero coupon bond** issued by a real entity to be presented in the next section.

2.3 Money Market

2.3.1 Money Market Instruments

Money market instruments are short term fixed-income securities with tenor up to 1 year, typically,

- Deposit: issued by deposit taking institutions such as banks and credit unions; Deposits are not transferable.
- Treasury Bills (T-Bills): transferable short term government debt;
- Certificate of Deposit (CD): transferable short term debt issued by banks;
- Commercial Paper (CP): transferable short term debt issued by corporates;
- Banker's Acceptance (BA): involved in international trade transactions, it is a promised future payment, or time draft, which is accepted and guaranteed by a bank. After acceptance, the holder of the draft can hold it to maturity or sell it at the discounted value. It is a transferable instrument.
- Repo: a collateralized loan which is not transferable.

2.3.2 Interbank Lending Market

It is a market for the short term lending/loan between banks for various purposes including the regulatory capital requirement. The interest rates in the interbank market are published every day. Financial institutions including banks, mortgage lenders and credit card agencies set their own rates relative to them.

Financial products often use these reference rates for their returns. The most used reference rates include

Libor (London Interbank Offered Rate): it is a daily indicator for the average offered interest rates from the contributor banks in London published at around 11:55am London time by ICE (Intercontinental Exchange). So far, the major convertible currencies (currently, USD, EUR, GBP, JPY, CHF) are covered for the tenors: overnight, one week, 1M, 2M, 3M, 6M and 12M.

Euribor (Euro Interbank Offered Rate): similar to Libor, Euribor is a daily indicator for the average offered interest rates from the panel banks in Eurozone published by the European Money Markets Institute at around 11:02am CET (Central European Time).

Eonia (Euro Overnight Index Average): computed as a weighted average of all overnight unsecured lending transactions in the Euro interbank market.

Fed funds rate (Federal funds rate): the effective fed funds rate is the weighted average overnight rate at which depository banks in US lend to each other.

The overnight rate in a country is monitored by the central bank, which may influence the rate level according to its monetary policy.

2.4 Bonds

2.4.1 Bond Features and Types

Bond is a long term debt security. It has a *face* or *par value* (also called the *principal amount*) which is the payment promised by the bond issuer. Bonds may be classified according to the coupon payments:

coupon bearing bond making periodic payments known as *coupons* over the lifetime of the bond, plus the repayment of the face value (usually at maturity). Typically, the coupon payment frequency is every 6 months or every year. The *fixed rate bond* pays coupons with a fixed rate. It is the most common bond type. The *floating rate note (FRN)* pays coupons linked to a reference rate, e.g., 6M USD Libor+spread.

zero coupon bond which does not pay any coupon and only pays the bondholder the face value at maturity. Its sensitivity to interest rate is higher than a coupon bearing bond of the same maturity. Zero coupon bonds can be stripped from a coupon-bearing bond.

Bonds can also be classified according to other criteria such as the type of issuer, special features:

Issuer: *Treasury bonds (Govies)* – issued by governments, often considered as credit risk free instruments; *Corporate bonds* – usually with higher yield than Treasury bonds; *Municipal bonds* – often give tax benefits to interest income; *Foreign bonds* – issued by foreign corporates, e.g.: samurai bonds (issued in Japan, denominated in JPY), dim-sum bonds (in Hong Kong, CNH), panda bonds (in China, CNY), masala bonds (outside of India, INR), and etc.;

Seniority: *Senior bonds* have higher priority than *Subordinated bonds* for repayment and claim on the asset in case of issuer's liquidation;

Others: A *convertible bond* (CB) can be converted into a predetermined amount of the issuing company's equity at the discretion of the bondholder; *Inflation index bond* has its coupons and principal indexed on an inflation index, e.g., treasury inflation protected securities (TIPS).

Some bonds embed the *callable feature* giving the issuer the right to buy back the bond at a predefined price on the "call dates", or the *puttable feature* which allows the bondholder the right to sell back the bond at a predefined price on the "put dates".

2.4.2 Main Risks for a Bond

The main risks for a bond are

- *credit risk* which is linked to the probability of default of the issuer before the maturity of the bond; Higher the default risk, lower the price of the bond;
- *market risk* which is linked to the interest rate changes. Higher interest rate will result in lower bond price because the future cashflows will be less valuable;
- *liquidity risk* which is linked to the transaction bid/ask spread. A bond with smaller issuance size tends to have higher liquidity risk.

2.4.3 Bond Quotation and Yield to Maturity

The *bond price* is the sum of present values of all the expected future cash flows.³ In the market, bonds are quoted in either of the units below:

Dirty price (also called **full price** or **invoice price**), which is commonly used in the European bond markets, is the price that the buyer will pay to the seller. For a fixed coupon bond, it is simply

$$\text{Bond Dirty Price} = \sum_{i=1}^n CF_i \times P(t_i),$$

where CF_i is the future cash flow of the bond payable at time t_i , $i = 1, \dots, n$;

Clean price, which is mostly used in the US bond markets and is defined as

$$\text{Clean Price} = \text{Dirty Price} - \text{Accrued Coupon}.$$

³The expectation is calculated with the "risk-neutral probability measure" which will be presented in the modelling part of the book.

The accrued coupon is the interest that has been earned but not yet paid. The coupon calculation method is described in the bond prospectus which usually follows the money market convention.

We note that

- (1) Transaction settlement is always done at dirty price;
- (2) Clean price is more stable over time than dirty price which changes its value due to the accrued coupon, in addition to any economic reasons.

Yield to maturity (also called *IRR*, the Internal Rate of Return) of a fixed coupon bond is the unique yield y which satisfies the following relationship:

$$\text{Bond Full Price} = \sum_{i=1}^n \frac{CF_i}{(1 + y/q)^{t_i}},$$

where

q : number of coupon payments per year, e.g. 2 for half-yearly coupon payments;
 t_i : the number of periods from settlement date to the i -th coupon payment date. For instance, with ACT/ACT convention, $t_1 = \tau/m_1$ and $t_i = i - 1 + \tau/m_1$ for $i > 1$ where τ is the number of calendar days from the settlement date to the 1st future coupon payment date and m_1 is the total number of days from the previous coupon payment (or the bond settlement date at issuance) date to the 1st coupon payment date.

If a bond pays a constant yearly fixed coupon rate C with final redemption at par, the yield to maturity satisfies

$$\text{Bond Full Price} = \sum_{i=1}^{n-1} \frac{C}{(1 + y)^{t_i}} + \frac{1 + C}{(1 + y)^{t_n}},$$

where n is the number of coupon payments.

In the market, there are several other measures used for the return of the bond investment, e.g., current yield, simple yield (also called Japanese yield),⁴ etc.

Example 2.1 Consider a fixed coupon bond in the secondary market:

Face value: \$1,000	Maturity date: 15 Nov 2017
Coupon rate: 3% p.a.	Day count: ACT/ACT
Coupon frequency: semi-annual	

⁴Current Yield = Coupon Rate/Clean Price, which does not reflect the price impact on disposal of the bond. Simple Yield = $(C + Adj)/\text{Clean Price}$, where $Adj = (\text{Final Redemption Value} - \text{Clean Price})/\text{Life to Maturity}$.

An investor bought the bond at 99.583% of the face value (or \$995.83) with settlement date at 05 Jan 2017. The future coupon dates are on 15 May and 15 Nov. There are 130 days between the settlement date and the next coupon date (15 May 2017). The total number of days between the previous coupon date (15 Nov 2016) and the next coupon date is 181.

The *yield to maturity* is 4% because $\frac{3\%/2}{(1+4\%/2)^{\frac{130}{181}}} + \frac{1+3\%/2}{(1+4\%/2)^{1+\frac{130}{181}}} = 99.583\%$ = Price of the bond. There are 51 days from the previous coupon date to settlement date 05 Jan 2017. Hence, $AccruedCoupon = \frac{3\%}{2} \times \frac{51}{181} = 0.4227\%$ and $CleanPrice = 99.583\% - 0.4227\% = 99.160\%$.

2.4.4 Duration and Modified Duration

Duration (also called Macaulay Duration) is defined to be the averaged life of present value of all future cash flows from the bond:

$$\text{Duration} = \sum_{i=1}^n t_i \times \frac{CF_i}{(1+y)^{t_i}} / \text{Price}.$$

Modified Duration (MD) measures the percentage change of price w.r.t. the yield change which is defined to be

$$MD = -\frac{d\text{Price}}{dy} / \text{Price}.$$

The direct calculation of the derivative leads to

$$MD = \text{Duration} / (1 + y).$$

2.5 Credit Rating

Credit rating is a measure for the riskiness of the bond given by a credit rating agency. The three major agencies are Standard & Poor's, Moody's and Fitch. For instance, the decreasing credit rating levels from S&P are AAA, AA, A, *BBB*, *BB*, *B*, *CCC*, *CC*, *C*.⁵ Each category has three notches, e.g. A+, A, A−. The bonds with ratings at or above *BBB*− are called *investment grade (IG) bonds*.

⁵For AAA bonds, the averaged default probability in 5 years is ~0.10%. For *BBB* bonds, the default probability in 5 years is ~1.9%.

2.6 Forward Rate Agreement

The *forward rate agreement (FRA)* is a loan/deposit engagement with a pre-agreed fixed rate (K) which will start at a time in the future T_1 and will terminate at time T_2 . There are 3 dates in an FRA contract ($T_0 < T_1 < T_2$):

T_0 : inception date;

T_1 : expiry (or effective date) of the forward contract;

T_2 : termination date.

The notation $T_1 \times T_2$ is used to put emphasis on the effective date and the termination date for an FRA contract.

Example 2.2 The following is a 3×6 FRA trade.

Notional amount: USD10m	Fixed rate: 1%
Effective date: in 3 months	Reference rate: USD 3M Libor
Terminal date: in 6 months	

The settlement to the long position of an FRA may be either of the below amounts:

$$\left\{ \begin{array}{ll} \text{Notional Amount} \times \frac{(r - K)\delta}{1 + r\delta}, & \text{for settlement at } T_1, \text{ which is the market} \\ & \text{practice for FRA} \\ \text{Notional Amount} \times (r - K)\delta, & \text{for settlement at } T_2, \end{array} \right.$$

where

r : the reference rate (e.g. Libor) fixed at T_1 for the period $T_1 \rightarrow T_2$;

δ : day count fraction calculated with the applicable convention;

K : fixed rate in the FRA contract.

Proposition 2.1 *For the market FRA with zero PV at inception, the fixed rate K is*

$$K = \frac{1}{\delta} \left(\frac{P(T_1)}{P(T_2)} - 1 \right).$$

Proof The proof is done by replication (i.e. hedging) with zero coupon bonds of maturities T_2 and T_1 . The short position of FRA for 1 dollar notional amount can be hedged with the below self-financing transactions.

At inception T_0 : Short T_2 -ZC bond A of $\frac{P(T_1)}{P(T_2)}$ dollars nominal. Its PV is $\frac{P(T_1)}{P(T_2)}P(T_2) = P(T_1)$; Long a T_1 -ZC bond B of 1 dollar nominal. Its PV is also $P(T_1)$. Hence the total investment is flat.

At effective date T_1 : Buy back A at its market price $\frac{1}{1+r\delta} \frac{P(T_1)}{P(T_2)}$, in which $\frac{1}{1+r\delta}$ is the PV of 1 dollar at T_2 ; Receive 1 dollar as B matures at T_1 .

In the absence of arbitrage, the PnL at T_2 should be equal to the FRA settlement amount:

$$\text{PnL} = 1 - \frac{1}{1+r\delta} \frac{P(T_1)}{P(T_2)} = \frac{(r-K)\delta}{1+r\delta}.$$

We obtain hence $K = \frac{1}{\delta} \left(\frac{P(T_1)}{P(T_2)} - 1 \right)$. □

2.7 Interest Rate Futures

Interest rate futures are exchange listed forward contracts with standardized features. Their prices reflect market expectation for interest rates and constitute a building block of the interest rate market. The main advantages of futures contracts are

- mitigated counterparty risk
- leveraged exposure
- high liquidity
- price transparency
- low transaction cost

Interest rate futures are actively used for hedging interest rate risks. They are also preferred instruments for market participants to get exposure for their view on interest rate market.

2.7.1 Short-Term Interest Rate (STIR) Futures

STIR futures are standardized FRAs listed at the exchange. The time- t quotation of the futures contract with expiry at T is

$$\text{Price}(t, T) = 100 \times [1 - r(t, T)],$$

where $r(t, T)$ is the forward rate which is normally the time- t market expected value of the interest rate (e.g. USD 3M Libor) to be observed at T . The Eurodollar futures, launched by the Chicago Mercantile Exchange (CME) in 1981, is a typical example of STIR futures.

Example 2.3 The following is a Eurodollar futures contract. The PnL per contract for the investor will be determined by the purchase and sale price difference in basis points times the BPV in the table.

Contract unit	USD 1 million face value, 90-day Eurodollar time deposits
Quotation	99.250
Final settlement	Cash settlement for the price difference
Contract months	Nearest 40 months (i.e., 10 years) in the March quarterly cycle (Mar, Jun, Sep, Dec), plus the nearest 40 months not in the March quarterly cycle e.g. on Jan 30, 2016, we see the contracts: Feb-16, Mar-16, Apr-16, May-16, Jun-16,...
Last trading day	2nd London business day prior to the 3rd Wednesday of contract month Trading in expiring contract closes at 11:00 a.m.
Margin	Daily margin settlement for MtM P/L
Basis point value(BPV)	$\$1,000,000 \times (\frac{90}{360}) \times 0.01\% = \25

2.7.2 Treasury Bond Futures

The underlying of a *Treasury Bond Futures* contract is a virtual (or “notional”) security with standardized features for the coupon rate, the maturity and the size. The futures contract applies physical delivery at the expiry. There is a set of deliverable bonds associated with each contract which is determined by the exchange.

For example, the most active futures in Europe is of 10Y maturity with EUR 100,000 size and 6% coupon.⁶ The set of deliverable bonds is published by EUREX with remaining maturities of 8.5Y $\rightarrow$ 10.5Y. In US, there are 2Y, 3Y, 5Y, 10Y Treasury Note Futures, Treasury Bond Futures (deliverable bonds of 15Y $\rightarrow$ 25Y) and Ultra T-Bond Futures (deliverable bonds of >25 Y).

⁶The coupon level may be subject to change in order to be close to the market yield levels at the launch of new contracts.

Treasury bond futures are related to notional bonds but delivered with real bonds from a pool of eligible bonds and notes. For each deliverable bond, a conversion factor (CF)⁷ is applied to convert the futures contract price into the “price for the bond”:

$$\text{Price for the Bond} = CF \times \text{Futures Price} + \text{Accrued Coupon}.$$

At settlement, for each deliverable bond, the short party will

- receive the cash amount based on “price for the bond”
- deliver the bond from the pool of eligible ones.

The market price of the bond to be delivered and the “price for the bond” will be normally different. The bond showing the lowest value of “market price” minus “price for the bond” is called the *Cheapest-To-Deliver (CTD)* bond. This phenomenon is due to the fact that the market yield curve is not 6% flat in general. Moreover, the bonds have different sensitivities to the yield curve change due to the different characteristics including coupon and time to maturity.

2.8 Interest Rate Swap

An interest rate swap (IRS) is a contract that entails periodic payments between two counterparties for the term of the trade. The swap market is very large and liquid. Swaps are used to hedge long term interest rate risks. The market standard IRS (or plain vanilla IRS) is the “Fixed-Floating” trade with two legs:

Fixed leg: one party will pay the cash-flows calculated with a fixed interest rate as predetermined in the contract;

Floating leg (or Variable leg): the other party will pay the cash-flows calculated with a reference (e.g. USD 3M Libor) + spread. For each period, the Libor rate based floating rate will be fixed at the *beginning* of the period and the cashflow will be paid at the *end* of the period.

⁷For US Treasury Bond Futures, the conversion factor is calculated as the clean price of the delivered bond as if it were yielding 6%. (Note that the price of the Futures contract is at par of the face value if the market yield is 6% flat.)

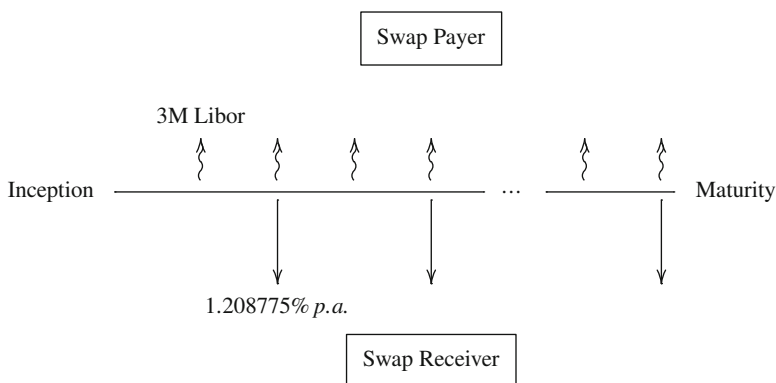
For example, the CF for a 5% bond with 15 years and 3 months remaining maturity is

$$CF = \left[\sum_{i=0}^{30} \frac{2.5\%}{(1 + 3\%)^i} + \frac{100\%}{(1 + 3\%)^{30}} \right] / (1 + 3\%)^{\frac{3}{6}} - 2.5\% \times \frac{3}{6}.$$

In the market vocabulary, the fixed rate payer (receiver) is referred to as the “swap payer” (“swap receiver”). Other combinations (“Fixed-Fixed”, “Floating-Floating”) can be composed by “Fixed-Floating” trades.

An illustration example of a swap is as follows:

Maturity:	5 years
Notional Amount:	USD 10m
Fixed Leg:	1.208775%, payable at the end of each semi-annually period
Floating Leg:	USD 3M Libor, payable at each quarter end



The valuation of an interest rate swap in general is based on the cost of replication. The time-0 PV of a receiver swap is simply the PV of the cash-flows of fixed leg minus the PV of the cash-flows of floating leg:

$$\text{Swap } PV = PV^f - PV^v = \sum_{j=1}^m P(t_j) s \delta_j - \sum_{i=1}^n P(t_i) F(t_{i-1}, t_i) \delta_i,$$

where

m, n : respective number of remaining cash-flows for the fixed leg and the floating leg;

s : rate of the fixed leg;

$F(t_{i-1}, t_i)$: expected forward rate⁸ for the i -th period (t_{i-1}, t_i) ;

δ_i : the day count fraction for the i -th period (t_{i-1}, t_i) .

⁸The expectation needs to be calculated under risk-neutral probability measure as explained in modelling part of the book.

2.8.1 Standard Swap

For standard (Vanilla) IRS, the valuation can be further simplified. From Proposition 2.1, the market quote for the floating rate is $F(t_{i-1}, t_i) = \frac{1}{\delta_i} \left(\frac{P(t_{i-1})}{P(t_i)} - 1 \right)$. Replacing $F(t_{i-1}, t_i)$ in the PV of the floating leg:

$$\begin{aligned} PV^v &= \sum_{i=1}^n P(t_i) F(t_{i-1}, t_i) \delta_i \\ &= [P(t_0) - P(t_1)] + [P(t_1) - P(t_2)] \cdots + [P(t_{n-2}) - P(t_{n-1})] \\ &\quad + [P(t_{n-1}) - P(t_n)] \\ &= P(t_0) - P(t_n). \end{aligned}$$

At the inception of the swap, $P(t_0) = 1$. Hence, we obtain $PV^v = 1 - P(t_n)$. This can be understood by the fact that the PV of an investment receiving regular market floating rate based interests and the initial investment back at the end should just be the value of the initial investment. A market swap is quoted with the fixed rate level which makes the swap valuation at zero at inception, i.e. $PV^f = PV^v$. From the above, we have

$$\sum_{j=1}^m P(t_j) s \delta_j = 1 - P(t_n).$$

Hence the rate is given as

$$s = \frac{1 - P(t_n)}{A(t_m)}.$$

where $A(t_m) = \sum_{j=1}^m P(t_j) \delta_j$ is called the *annuity factor*.

2.8.2 Overnight Indexed Swap (OIS)

An overnight indexed swap (OIS) is an interest rate swap where the cash-flow of the floating leg is based on a floating rate calculated from the overnight rates such as Eonia or effective fed funds rate for the concerned period.

Consider the example of Eonia which is quoted on ACT/360. The floating rate for OIS of the k -th period (t_{k-1}, t_k) will be calculated as the geometric average of Eonia fixings:

$$R_k = \left[\prod_{i=1}^{N_{fix}^k} \left(1 + \frac{r_i \times n_i}{360} \right) - 1 \right] \frac{360}{N_k},$$

where

R_k : rate for OIS of the k -th period

N_{fix}^k : total number of fixings of the overnight rate for the k -th period

r_i : Eonia fixing rate for day i

n_i : number of calendar days covered by r_i

N_k : total number of calendar days for the k -th period

The rate of OIS implies less risk than the corresponding interbank lending rate such as Libor and Euribor because OIS rate is based on actual trades, which makes it more difficult to be manipulated.

Because of counterparty risk, the derivatives trades are normally collateralized. The cash collaterals are in general paid with overnight interest rate. As a consequence, it is considered as a secured lending rate, i.e. a proxy of risk-free rate. It is natural to apply OIS implied rates for discounting in financial calculation. Even though Libor and Euribor are widely used as reference for swaps, OIS is gaining increasing popularity. The OIS yield curve construction is similar to that for Libor rate based instruments.

2.8.3 Other Swaps

There are other types of interest rate swap (called exotic swaps in general). Their valuation methods usually involve interest rate term structure modelling. The exotic swap valuation is often found to be the valuation of the comparable standard swap plus an extra term known as the *convexity adjustment*. We will mention only two of the most popular ones: constant maturity swap and in-arrears swap.

A *constant maturity swap* (CMS) is a swap in which the reference floating rate used for each settlement is the prevailing market rate of the standard swap of a specified fixed tenor. For example, in a 2 year maturity CMS swap linked to USD10Y swap rate, the cash flow for the floating leg is based on the market rate of the USD10Y swap rate observed at each settlement.

It is worth highlighting that the name *CMS rate* in a structured product refers to the swap rate of specified tenor (e.g. USD20Y swap rate) to be observed on the scheduled date(s) of the product.

The *in-arrears swap* is a swap in which the floating rate is determined at the end of the period, instead of observing at the beginning for a standard swap.

2.9 Yield Curve Construction

2.9.1 Yield Curve

A *yield curve* is a set of interest rate levels with equivalent credit quality across different tenors (also called pillars) in the same currency.⁹ The relationship between the interest rate levels of different tenors is called *interest rate term structure*.

The yield curve used in the financial market is also called *zero coupon curve* or simply *zero curve* because each point shows the market interest rate for a ZC bond (i.e. discount factor) of the related pillar. The zero coupon curve is built with the most liquid financial market instruments which are involved in the interest rate risk hedging. The interest rate level in any basis (e.g. continuous, money market, etc.) can be implied from the ZC bonds.

2.9.2 Interpolation Method for Yield Curve

For any date t which is not on the pillars of the curve, the zero coupon bond $P(t)$ can be calculated by interpolation from its neighbouring dates t_1 and t_2 where $t_1 < t < t_2$.

One commonly used interpolation method consists of assuming a constant forward rate r between t_1 and t_2 , such that $P(t_2) = P(t_1) \times e^{-r(t_2-t_1)/365}$, i.e. $r = \ln \frac{P(t_1)}{P(t_2)} \times \frac{365}{t_2-t_1}$. The interpolation formula for the zero coupon bond is then

$$P(t) = P(t_1) \times e^{-r(t-t_1)/365}.$$

2.9.3 Bootstrapping Method

Bootstrapping consists of calculating the ZC bonds implied by the instruments step-by-step, from the shortest tenor to the longest. The ZC bond at time t is calculated from the already known ZC bonds for the dates prior to t and the rates implied from market instruments covering the tenor until time t . The calculation depends on the type of instrument. For instance, if we know $P(t_1)$ and the money market rate r implied from t_1 to t_2 ($t_1 < t_2$), then the bootstrapping procedure gives $P(t_2) = \frac{P(t_1)}{1+rT}$, where T is the day count fraction for $t_2 - t_1$.

We note that one can also solve a system of equations instead of using bootstrapping iterations.

⁹For currencies other than the funding currency of the institution, it makes sense to include the impact from Cross Currency Swaps (CCS) of the market; see Chap. 4 for the product.

Table 2.1 Instruments for Euro rates – 27 Nov 2014

Instrument	Tenor	Expiry date	Begin	End	Rate or Price
Deposit	O/N		27 Nov 14	28 Nov 14	0.0000%
Deposit	T/N		28 Nov 14	01 Dec 14	0.1500%
Deposit	1M		01 Dec 14	02 Jan 15	0.0400%
Deposit	2M		01 Dec 14	02 Feb 15	0.0400%
Deposit	3M		01 Dec 14	02 Mar 15	0.1400%
Futures		15-Dec-14	17 Dec 14	17 Mar 15	99.910
Futures		16-Mar-15	18 Mar 15	18 Jun 15	99.915
Futures		15-Jun-15	17 Jun 15	17 Sep 15	99.930
Futures		14-Sep-15	16 Sep 15	16 Dec 15	99.925
Futures		14-Dec-15	16 Dec 15	16 Mar 16	99.910
Futures		14-Mar-16	16 Mar 16	16 Jun 16	99.890
Futures		13-Jun-16	15 Jun 16	15 Sep 16	99.870
Futures		19-Sep-16	21 Sep 16	21 Dec 16	99.840
Swap	3Y		01 Dec 14	01 Dec 17	0.2570%
Swap	4Y		01 Dec 14	03 Dec 18	0.3208%
Swap	5Y		01 Dec 14	02 Dec 19	0.3960%
Swap	6Y		01 Dec 14	01 Dec 20	0.4840%
Swap	7Y		01 Dec 14	01 Dec 21	0.5830%

2.9.4 Illustration Example for Yield Curve Construction

We illustrate the methodology through an example of Euro zero curve with the instruments in Table 2.1.

Deposit Instruments

Bootstrapping starts from the instrument with the shortest tenor, i.e., the O/N deposit rate.

Overnight (O/N) deposit rate		
Convention	ACT/360	
Rate	r	$r = 0.0000\%$
Day count	t_1 : today	$t_1 = 27 \text{ Nov } 14$
	t_2 : next business day	$t_2 = 28 \text{ Nov } 14$
	$T = (t_2 - t_1)/360$	$T = 1/360$
$P(t_1)$		$P(t_1) = 1$
$P(t_2)$	$P(t_1)/(1 + r \times T)$	$P(t_2) = 1$

Next we move on to the T/N deposit rate. Here we have to pay attention to the weekend involved (29 Nov 2014 was a Saturday) leading to 3 calendar days in the calculation.

T/N deposit rate		
Convention	ACT/360	
Rate	r	$r = 0.1500\%$
Day count	t_1 : business day after today	$t_1 = 28 \text{ Nov } 14$
	t_2 : business day after t_1	$t_2 = 01 \text{ Dec } 14$
	$T = (t_2 - t_1)/360$	$T = 3/360$
$P(t_1)$		$P(t_1) = 1$
$P(t_2)$	$P(t_1)/(1 + r \times T)$	$P(t_2) = 0.9999875$

Then, we consider other deposit rates with longer tenor. In our example, the calculation for the 1M tenor is

1-Month deposit rate		
Convention	ACT/360	
Rate	r	$r = 0.0400\%$
Day count	t_1 : settlement date	$t_1 = 01 \text{ Dec } 14$
	t_2 : one month after t_1	$t_2 = 02 \text{ Jan } 15$
	$T = (t_2 - t_1)/360$	$T = 32/360$
$P(t_1)$		$P(t_1) = 0.9999875$
$P(t_2)$	$P(t_1)/(1 + r \times T)$	$P(t_2) = 0.9999519$

Similarly, calculations for 2M and 3M tenor can be done.

Short Term Interest Rate Futures

In our example, the interest rate futures are being used which are quoted in *price* at the exchange, i.e. $\text{Price} = 1 - r$ where r is 3M Euribor observed at the expiry date of the contract. We use the term “Termination Date (TD)” for date which is 3 months after the expiry date of the contract.

For the first futures contract, we need to find the ZC bond at the beginning date of the contract, i.e., 17 Dec 14, which is not yet available. This term has to be interpolated from the nearby $P(\text{T/N})$ and $P(1\text{M})$.

From the second futures contract onward, the calculation will be $P(t_2) = \frac{P(t_1)}{1+r \times T}$, where t_2 and t_1 are the Termination Dates of Current Contract and the Preceding Contract, r being the implied rate for the current contract.

In case that the periods covered by two successive futures overlap or have a hole in between, any smooth treatment may be applied for this joining period. If the

First futures contract		
Convention	ACT/360	
Price	P	Price = 99.91%
Implied rate	$r = 1 - P$	$r = 1 - 99.91\% = 0.09\%$
Day count	t_1 : TD of the proceeding contract	$t_1 = 17 \text{ Dec } 14$
	t_2 : TD of the current contract	$t_2 = 17 \text{ Mar } 15$
	$T = (t_2 - t_1)/360$	$T = 90/360 = 0.25$
$P(t_1)$	Interpolated	$P(t_1) = 0.9999797$
$P(t_2)$	$P(t_1)/(1 + r \times T)$	$P(t_2) = 0.9997448$

impact is negligible, one may choose to simply adjust the period length of one of the contracts by keeping the implied rate unchanged.

Swaps

Features of standard EUR swaps	
Fixed leg	Variable leg
Day count: ACT/360	Day count: ACT/360
Frequency: annual	Frequency: semi-annual

The bootstrapping of swap contracts is based on the below no-arbitrage relationship:

$$P(\text{Settlement Date}) - P(t_n) = s \sum_{i=1}^n \delta_i P(t_i).$$

We take the ZC bond for the 3Y pillar as example where we have already obtained the following numbers through bootstrapping of instruments with shorter tenors:

$$P(\text{Settlement Date}) = 0.9999875, P(1Y) = 0.9991986, \delta_1 = 1, \\ P(2Y) = 0.9979836, \delta_2 = 1, \delta_3 = 1, s = 0.2570\%.$$

We can solve for $P(3Y)$ which makes the following equality hold:

$$P(\text{Settlement Date}) - P(3Y) = s \times [\delta_1 \times P(1Y) + \delta_2 \times P(2Y) + \delta_3 \times P(3Y)].$$

Applying the numerical numbers, we obtain $P(3Y) = 0.9923045$.

Exercises

1. Measure the year fraction from 2 Jan 2017 to 1 Mar 2017 with each of the following day count conventions: 30/360, ACT/365 and ACT/ACT ISDA. Modified Following is applied for business day convention.
2. For the period from 2 Jan 2014 to 31 Dec 2014, the money market interest rate for a certain currency is 2% under the day count convention ACT/360, what is the equivalent rate level under ACT/365 day count convention?
3. From the data in the yield curve example presented in the last section, calculate
 - (i) 4M deposit rate (for the period 2 Dec 2014 – 2 Apr 2015)
 - (ii) zero coupon bond values for 13 Feb 2014 and 2 May 2016 using the interpolation method presented in the last section
4. Consider the following 3×6 FRA contract issued on Tuesday, 8 May 2018:

Notional amount	USD 30 million
Expiry (i.e. Effective date)	3 months (8 August 2018, Wed)
Termination date	6 months (8 November 2018, Thu)
USD 3M Libor	2.36%
USD 6M Libor	2.52%
Day count convention	ACT/360

- (i) Calculate the market forward rate for the FRA contract (i.e. the rate at which the contract value is zero at inception);
 - (ii) At expiry, the 3M Libor on the settlement date is 2.60%, what is the P/L for the long party?
5. An institution has obtained a loan of USD10mio for 3 years at the interest rate of USD 6M Libor + 1.5% with semi-annual interest payments. With this loan, the institution decided to invest into a high yield bond paying 5% p.a. with semi-annual coupon payments.

Describe the risks for the institution.

The institution decided to enter into a 3Y standard IRS (Interest Rate Swap) transaction with a bank in which the institution receives USD 6M Libor and pays 1.70% fixed rate with zero cost. The payment dates coincide for the bond, the loan and the IRS.

- (i) Has the IRS reduced the interest rate risk for the institution?
- (ii) Who is the payer of the swap transaction? (institution or the bank)?
- (iii) Calculate the net cash flow amount the institution will receive every 6 months.
- (iv) For the floating leg of this standard IRS, when is the floating rate fixed for each 6 month period (i.e. at the beginning or the end)? when is the floating coupon paid (at the beginning or the end)?

Chapter 3

Equities and Equity Indices



3.1 Equity

Equity is the claim of the ownership of a firm. Equity securities issued by corporations are called *stocks* or *shares*.¹ The securities traded in the equity market can be *publicly traded stocks*, which are listed on the stock exchange, or *privately traded stocks*.

The two main equity securities are

Common stock (ordinary stock): a common shareholder has the voting right and is entitled to the dividend. In case of liquidation of the company, the shareholder of a common share has the lowest priority for the assets.

Preferred stock: the holder of a preferred stock has no voting right but has the priority for the dividend payment and liquidation assets over the common share. Many preferred shares pay dividends in the form of fixed coupons, like a perpetual bond. There are several types of preferred shares: a convertible preferred share gives right to its holder to convert the preferred share to a common share; a cumulative preferred share allows the dividends omitted in the past (e.g. due to profitability issue) to be paid later.

A company's *free float* refers to the number of shares that are immediately tradable in the public market. A *restricted share*, usually held by an insider (such as company's executive officer and employee), is not fully transferable until certain conditions are met. We have

$$\text{Free Float} = \text{Outstanding Shares} - \text{Restricted Shares}.$$

The market capitalization of a company is then defined as the market price multiplied by the number of outstanding shares.

¹In the context of this book, the words *stock* and *share* are used interchangeably.

3.2 Stock Dividend

A *stock dividend* is a payment (e.g. yearly or quarterly) made by a corporation to its shareholders as a distribution of profits. The dividend amount is decided by the Board of the company at each time. For a detailed discussion on the dividend policy and practice, we refer to [72].

When a dividend is declared, an *ex-dividend date* is specified. It is the date on which shares bought no longer come attached with the right to be paid the declared dividend. Other factors remaining equal, the share price will normally drop on the ex-dividend date by the amount equivalent to the after-tax dividend. Before ex-dividend date, the stock is said to trade *cum dividend*.

A stock dividend can be either a *cash dividend* in monetary form, or a *stock dividend* which involves the company issuing more shares to its existing shareholders. For example, a 5% stock dividend means that the shareholders receive one new share for every twenty already owned shares. Unlike the cash dividend, the stock dividend does not change the company's market capitalization. In modelling, stock dividends are also called proportional dividends.

Suppose that a stock pays a dividend value of D with ex-dividend date t_d . Let $S_{t_d}^-$ be the stock price just before the ex-dividend date. The quantity $d = D/S_{t_d}^-$ is called the *dividend yield*.

Obviously, $S_{t_d}^-$ cannot be equal to S_{t_d} . If it was, the strategy of buying the stock immediately before t_d , collecting the dividend, and selling straight away, would yield a risk-free profit. In fact, in the absence of other factors such as taxes, the asset price must fall by exactly the amount of the dividend payment. That is, for a stock paying both proportional dividend at rate d^{prop} and cash dividend amount D^{cash} , we have

$$S_{t_d} = S_{t_d}^- (1 - d^{prop}) - D^{cash}.$$

In continuous-time modelling, the proportional dividend rate is the value d such that $e^{-d} = 1 - d^{prop}$. Hence,

$$S_{t_d} = S_{t_d}^- e^{-d} - D^{cash}.$$

3.3 Stock Split, Reverse Stock Split, Rights Issue

A *stock split* is a corporate action in which a company divides each existing share into multiple shares which impacts the share price by the same ratio. It increases the number of the company's outstanding shares while keeping the market capitalization unchanged. If a stock's price has reached such a high level making it unaffordable to small investors, the stock split helps to increase the stock's liquidity.

A *reverse stock split* is the opposite of *stock split*. A company having its stock price dropped to a very low level such that one *tic* (i.e. the minimum quotation unit) for its price becomes significant may consider undertaking a reverse stock split for bringing back the price to a normal unit.

A company can raise money through a *rights issue* which is offered to existing shareholders for purchasing new shares in the company with a subscription price at a discount to the market price. The right can be either *renounceable*, allowing the shareholder to trade it in the market, or *non-renounceable* which is not tradable.

3.4 Equity Index

A stock market index is a measurement targeting to describe the performance of the stock market. It is computed from the price of selected stocks. An index is a mathematical construct which does not allow direct investment. People may invest through index futures, mutual funds or exchange-traded funds (ETFs) tracking the index.

An equity index may be classified according to its calculation method. Let k be the number of stocks included in the index, n_t^i the total number of outstanding (free-float) shares for stock i on day t , and S_t^i the stock price. The two main classes of index are as follows:

- *Capitalization Weighted Index:*

The majority of the equity indices belongs to this type, e.g. S&P 500, Euro Stoxx 50. The calculation for each index is a variation of the below general method:

$$Index_t = Index_{t-1} \times \frac{\sum_{i=1}^k n_t^i S_t^i}{\sum_{i=1}^k n_t^i S_{t-1}^i}.$$

- *Price Weighted Index:*

There are only a few equity indices of this type. Dow Jones Industrial Average and Nikkei Index are the most well-known ones. The calculation is as follows:

$$Index_t = \frac{\sum_{i=1}^k S_t^i}{Divisor_t},$$

where $Divisor_t$ serves for scaling the price continuity by adjusting its value in case of some corporate actions and other external factors. As the stock price alone does not contain any particular information, the methodology of relating the weight to its price level is obviously not a rational approach.

3.5 Equity Forward, Cash & Carry and Reverse Cash & Carry

An *equity forward* on a single stock is an OTC contract between two parties where the long party is engaged to buy from the short party a certain quantity N (called notional) of the stock at a pre-agreed delivery price K at the expiry of the contract. The payoff of the long party is

$$\text{Payoff} = N \times (S_T - K),$$

where S_T is the stock price at expiry T .

The market forward price (denoted as F , the value of K such that the forward contract value is zero) for equities can be derived with no-arbitrage condition. In its general form, a stock is assumed to pay both cash and proportional dividend. The stock borrow/lending fee rate is called *repo rate*² in modelling for equities.

Proposition 3.1 *Let S_0 be stock price at time 0.*

- (1) *For a non-dividend paying stock with zero repo rate and constant interest rate r , the forward price F is*

$$F = S_0 e^{rT}.$$

- (2) *For the general case, let r_t be the instantaneous interest rate, λ_t the instantaneous repo rate and d_{t_k} the rate for the proportional dividend at time t_k .³ The forward price F is*

$$F = S_0 e^{\int_0^T (r_t - \lambda_t) dt - \sum_{t_k \leq T} d_{t_k}}.$$

Proof

- (1) The short party can hedge the forward contract by the *Cash & Carry Strategy* which consists of buying one stock at time 0 financed by a loan and holding the stock until the expiry. The table shows that, by setting $K = S_0 e^{rT}$, the total P/L of the short forward position and the long stock position with a loan is zero which means perfect hedging.

²The *repo rate* may be an abused vocabulary. Actually, Repo rate in fixed-income repo trade refers to the rate at which money is borrowed/lent with collateral.

³For the case including cash dividends, we refer to [61], Chapter 1, which includes also the presence of credit risk.

Time	Value of the forward	Stock position	Monetary position	Net position
0	0	S_0	$-S_0$	0
T	$-(S_T - S_0 e^{rT})$	S_T	$-S_0 e^{rT}$	0

- (2) We choose to re-invest all the repo return⁴ into the stock. In this case, one stock at time 0 will become $e^{\int_0^T \lambda_t dt + \sum_{t_k \leq T} d_{t_k}}$ stocks at time T . For hedging the forward on one stock, the short party needs to buy $e^{-\int_0^T \lambda_t dt - \sum_{t_k \leq T} d_{t_k}}$ stocks at the inception.

Time	Value of the forward	Stock position	Monetary position	Net position
0	0	$S_0 e^{-\int_0^T \lambda_t dt - \sum_{t_k \leq T} d_{t_k}}$	$-S_0 e^{-\int_0^T \lambda_t dt - \sum_{t_k \leq T} d_{t_k}}$	0
T	$-(S_T - S_0 e^{\int_0^T (r_t - \lambda_t) dt - \sum_{t_k \leq T} d_{t_k}})$	S_T	$-S_0 e^{\int_0^T (r_t - \lambda_t) dt - \sum_{t_k \leq T} d_{t_k}}$	0

Similarly, the long position of a forward can be hedged with *Reverse Cash & Carry Strategy* which consists of selling the stocks borrowed in the SBL (stock borrowing/lending) market. The cash from the stock short-selling is invested in money market to earn interests. For taking into account the repo and dividends, the long party borrows $e^{-\int_0^T \lambda_t dt - \sum_{t_k \leq T} d_{t_k}}$ stocks at inception but returns 1 stock at T . Hence,

Time	Value of the forward	Stock position	Monetary position	Net position
0	0	$-S_0 e^{-\int_0^T \lambda_t dt - \sum_{t_k \leq T} d_{t_k}}$	$S_0 e^{-\int_0^T \lambda_t dt - \sum_{t_k \leq T} d_{t_k}}$	0
T	$S_T - S_0 e^{\int_0^T (r_t - \lambda_t) dt - \sum_{t_k \leq T} d_{t_k}}$	$-S_T$	$S_0 e^{\int_0^T (r_t - \lambda_t) dt - \sum_{t_k \leq T} d_{t_k}}$	0

□

Remark 3.1 For a forward contract with $K \neq F$, the value at the inception can be derived by

$$S_T - K = (S_T - F) + (F - K).$$

⁴The repo rate is normally positive. But under certain special market environment, it can be negative.

The first term $S_T - F$ is the payoff of the market forward contract which has zero initial value. So, for a non-dividend paying stock with zero repo rate and constant interest rate r , the initial value of the forward contract is

$$S_0 - e^{-rT} K.$$

For the general case, the contract initial value is

$$\text{Forward Initial Value} = e^{-\int_0^T r_t dt} (F - K) = S_0 e^{-\int_0^T \lambda_t dt - \sum_{t_k \leq T} d_{t_k}} - e^{-\int_0^T r_t dt} K.$$

3.6 Equity Index Futures

An *equity index futures* contract is a standardized equity forward linked to an underlying equity index listed at an exchange. The contract is usually defined by *index point value* which is the P/L per unit change of the underlying equity index. At the contract expiry, the futures price coincides with the index price and the “implied” P/L of the long party is $(\text{Index Price} - \text{Initial Futures Price}) \times \text{Index Point Value}$. Actually, on daily basis, there is a P/L adjustment through the *margin* payment between the exchange and the investor with a methodology from the exchange.

Example 3.1 E-mini S&P500 futures contracts have an index point value of \$50. An investor has bought 100 contracts at the price of 2,010.50. If S&P500 index price is 2,001.00 at the contract expiry, then, the investor’s implied P/L will be $100 \times (2,001.00 - 2,010.50) \times \$50 = -\$47,500.00$, which means a loss to the investor.

The quotation for futures is usually subject to a minimum value called the *tick* (e.g. 0.25, or \$12.5 for E-mini S&P 500 futures). The observed difference $(\text{Futures Price} - \text{current Index Price})$ is called the *basis* of the futures contract.

3.7 Equity Swap

The *equity swap* is an OTC contract between two parties where one party will receive cash flows linked to the performance of underlying equity securities and pay the funding in cash form, usually linked to a fixed or floating rate (e.g. 3M Libor + spread). It is normally a *total return swap (TRS)* where the benefits from the dividends as well as any other income/capital gain are paid to the receiver of the equity performance.

The equity swap can be used to obtain leveraged exposure to the underlying if the long party does not fully collateralize the trade. Swaps with regular settlements known as *resettable swaps* have lower counterparty risk and require lower IA (independent amount), which implies higher leverage.

Exercises

1. Consider the stock CCB listed in the Hong Kong market (ticker: 0939.HK). It had announced a cash dividend of \$0.2780 in RMB (or \$0.319504 in HKD) for the year 2017 and the ex-dividend date is 22 June 2017. The closing price on 21 June 2017 is 6.42. The market information on 22 June 2017 is

Date	Open	High	Low	Close
2017-06-22	6.11	6.21	6.1	6.12

- (i) Explain why the opening price on 22 June 2017 is not $6.42 - 0.319504$?
 - (ii) Calculate the daily total return for 22 June 2017.
 - (iii) An investor considers adjusting the historical price on 21 June 2017 by $6.42 - 0.319504$. Another investor argues that it should be done by $6.42 \times \frac{6.12}{6.12+0.319504}$. Compare the two approaches.
2. Given a non-dividend paying stock with its current price at \$100. The interest rate in the market is 2% p.a. (ACT/365).
 - (i) Calculate the 1 Year Forward price, stock lending rate is 2% p.a. (ACT/365).
 - (ii) Assume that an investor is long a 6-month forward at the forward price of \$101 for 10,000 stocks. At the expiry date, the stock price is \$115, what is the P/L of the investor?
 - (iii) Describe how to hedge a long position of an equity forward contract.
 3. Explain the sentence “Unlike the cash dividend, the stock dividend does not change the company’s market capitalization.”
 4. An investor has bought one front-month mini Hang Seng Index futures contract at the price of 30,000 and wants to unwind the position before expiry. Suppose the exchange fee is \$3.5, commission levy \$0.1 and broker commission \$5 per contract. At one moment the bid and ask prices for the futures contract are quoted as 30,020 and 30,021. What is the P/L for the investor?
 5. The P&L of Equity futures contract is determined by an index point value. It is useful to measure the performance of a position in futures with the usual method, i.e. $Notional \times (\frac{Price_T}{Price_0} - 1)$.
For E-mini S&P Futures contracts presented in the example of the section of Equity Index Futures, determine the equivalent *Notional* for 1,000 contracts.

Chapter 4

Foreign Exchange Instruments



The FX (foreign exchange) market (also called Forex or currency market) facilitates the trading of currencies. It starts from Monday morning in Wellington (New Zealand) and closes on Friday evening in New York (USA).

4.1 Quotation Conventions

A currency is represented by a three letter ISO code with the first two letters for the country and the last letter, in most of the cases, for the currency name, e.g. USD, CNY.¹

The currency exchange rate is quoted as the units of a “counter currency” (also called “domestic” currency or ccy2) per “base currency” (also called “foreign” currency or ccy1). In the market, the default order for being the base currency is EUR, GBP, AUD, NZD, USD, CAD, CHF, JPY, etc (for instance, EUR/USD, USD/JPY).

The FX rate between any two non-USD currencies is called *cross-rate* which can be implied from the quotes against USD for each currency. For example, $\text{USD/CHF} = 0.9973$ and $\text{USD/JPY} = 112.74$ implies $\text{CHF/JPY} = 112.74/0.9973 = 113.045$.

In the market, the term *pip* is used in quotations for the last meaningful digit: 1 pip for EUR/USD and most of the other pairs is 0.0001; 1 pip for USD/JPY is 0.01. 100 pips is one *big figure*.

¹Financial market practitioners sometimes use informal vocabularies for currency name: “Cable” or Sterling for GBP, “Kiwi” for New Zealand Dollar, etc. [21].

4.2 FX Spot, Forward, Swap, Non-deliverable Forward (NDF)

A *spot* transaction is an immediate purchase or sales of one currency for another currency for delivery in usually two working days which is known as T+2 (T+1 for some pairs, e.g. USD/CAD).

A *forward* (also called *outright*) is an OTC contract between two parties to exchange currencies on a future date at a pre-set rate (called forward rate or strike rate). The P/L for the long party can be calculated mathematically by an (implicit) payoff in domestic currency as

$$\text{Payoff} = \text{Notional} \times (X_T - K),$$

where

Notional: notional amount in *foreign* currency

X_T : foreign exchange rate at expiry

K : pre-agreed rate of the contract

The *forward rate* is the rate K at which, the initial valuation of the forward contract is zero. In the market, the forward is quoted by the so-called “forward points” or “swap points” which equals to (Market Forward Rate – Spot Rate).

The non-convertible currencies (e.g. CNY, KRW) can not be delivered in an offshore country outside the (onshore) country of the currency. There is a special instrument called *NDF* (Non-Deliverable Forward) used in the market for hedging FX risk. It is an outright contract quoted for a non-convertible currency against a convertible currency (typically USD). At maturity, the counterparties settle, in the convertible currency, only the difference between the contracted NDF rate and the prevailing spot rate on the agreed notional amount (typically in USD).

An *FX futures contract* is a standardized forward listed at the Exchange.² The currency futures are far less popular than forward in OTC market.

An *FX swap* is an OTC agreement and is a combination of two transactions:

- (1) a spot transaction (buy or sell) at spot rate
- (2) a forward transaction (sell or buy) at the pre-agreed FX swap rate (by arbitrage, market FX swap rate which makes the value of the FX swap at zero is also the market forward rate).

²For instance, the E-mini Euro FX Futures of CME has the contract unit of 125, 000 euros and the expiries on the first 3 immediate consecutive months and 20 months in the March quarterly cycle (Mar, Jun, Sep, Dec).

4.3 Interest Rate Parity for FX Forward

Proposition 4.1 *The market forward rate $F(T)$ is*

$$F(T) = X_0 \frac{P^f(T)}{P^d(T)},$$

where

X_0 : current foreign exchange rate (i.e. the spot rate)

$P^f(T)$: zero coupon bond in foreign currency (base currency)

$P^d(T)$: zero coupon bond in domestic currency (counter currency)

or, equivalently,

- with money market rates (R^d and R^f for domestic and foreign currencies respectively):

$$F(T) = X_0 \frac{1 + R^d T_d}{1 + R^f T_f},$$

where T_d and T_f are calculated with the day count conventions of domestic and foreign currencies respectively;

- with continuous rates (r^d and r^f for domestic and foreign currencies respectively, and T_c for the day count fraction using ACT/365):

$$F(T) = X_0 e^{(r^d - r^f)T_c}.$$

Proof There will be an arbitrage if the above relationship is not satisfied. If $F(T) >$

$X_0 \frac{P^f(T)}{P^d(T)}$, then, one can build the following arbitrage position at inception:

- sell one unit of forward at $F(T)$
- buy $P^f(T)$ foreign currency, financed by a loan of $X_0 P^f(T)$ in domestic currency

At maturity, we have

- from the forward position: short 1 unit of foreign currency, long $F(T)$ units of domestic currency
- from the hedging position: long 1 unit of foreign currency, short $X_0 \frac{P^f(T)}{P^d(T)}$ in domestic currency

$\Rightarrow$ The total P/L will be $F(T) - X_0 \frac{P^f(T)}{P^d(T)} > 0$ which is an arbitrage.

Similarly, we can prove the case for $F(T) < X_0 \frac{P^f(T)}{P^d(T)}$. □

4.4 Cross Currency Swap and Non-deliverable Swap (NDS)

The CCS (Cross Currency Swap) consists of exchanging the principals at the beginning with the spot rate, exchanging the interests during the tenor and exchanging the principals back at maturity.

Example of a USD/JPY cross currency swap assuming USD/JPY Spot rate = 110.03:

At Inception:

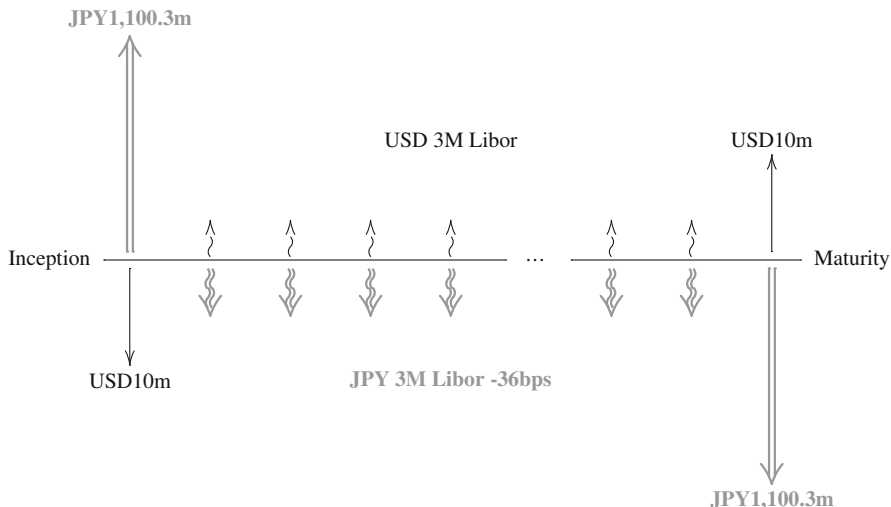
- Party A pays party B USD10m
- Party B pays party A $\text{USD10m} \times 110.03 = \text{JPY1,100.3m}$

During the Trade Life:

- Party A pays party B JPY Libor –36 bps (if it is negative, party B will pay party A) on JPY notional amount
- Party B pays party A interest of USD Libor on USD notional amount

At Maturity:

- Party B pays party A USD10m
- Party A pays party B JPY1,100.3m



CCS is a very useful instrument. As an example of its application, please refer to Chap. 24 for “Funding Management for Structured Notes”.

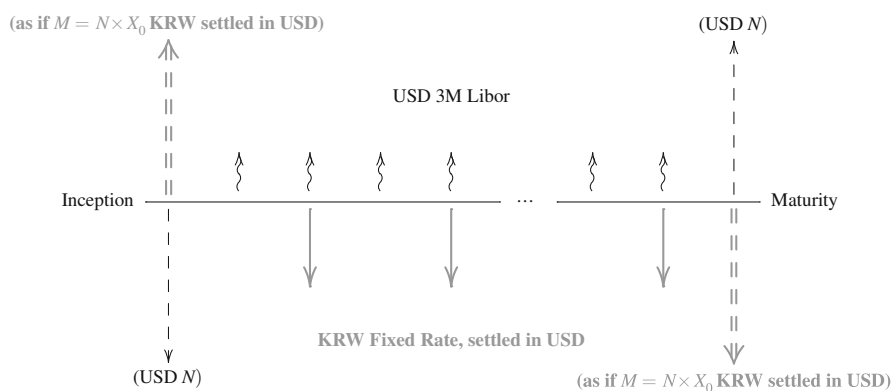
For non-convertible currencies (e.g. KRW, CNY), there is a non-deliverable instrument which is used in the market for hedging FX rate risks. It is called

NDS (Non-Deliverable Swap), a cross currency swap where one currency is non-convertible and the other one is convertible (typically USD). The particular feature is that all cash-flows are settled in the convertible currency by applying the prevailing exchange rate at the payment time. This implies that

- at inception: no initial notional exchange
- at maturity: settlement in the convertible currency for (Notional $\times$ Difference between FX Rate at Maturity and FX Rate at Inception).

The interest rate on the non-convertible currency is normally a fixed rate instead of floating rate for a cross currency swap.

Graphical Illustration of NDS on USD/KRW



Exercises

1. A student wants his HKD100,000 converted into CNY at a money exchange where the bid/ask for CNY/HKD are shown as 1.2 and 1.25 respectively. How much CNY will he get?
2. A linked exchange rate system is set up between HKD and USD. In 2015, Hong Kong Monetary Authority (HKMA, the de facto central bank in Hong Kong) announced to implement the “Convertibility Undertaking” which buys USD in the market at $\text{USD/HKD} = 7.75$ and sells USD at $\text{USD/HKD} = 7.85$. Are 7.75 and 7.85 bid or ask prices such that the necessary condition for triggering “Convertibility Undertaking” is satisfied?
3. Though USD/JPY trades for most of the time in a trading day but there are certain periods during which USD/JPY is significantly more volatile. Explain this phenomenon.

4. Assume that the spot price of USD/CNH is 6.6741 (meaning 6.6741 CNH per USD). The 3-month money market rates are 4.2% for CNH (ACT/360) and 1.17% for USD (ACT/360).
- (i) What is the base currency for the pair USD/CNH?
 - (ii) If the spot price has increased by 20 pips, what is the new spot price?
 - (iii) If the forward price for 3M tenor is 6.6751, what is the swap point?
 - (iv) What is the 3-month forward price of USD/CNH, assuming the period has 90 calendar days?
 - (v) Someone is willing to buy or sell a 3-month USD/CNH forward at the rate of 6.6800. How can I build an arbitrage position of USD20m notional to get a profit? And what is the P/L?

Chapter 5

Commodities



In this chapter we outline the basic concepts and instruments related to commodities. For a more detailed introduction, we refer to [22, 75].

5.1 Commodities Overview

Commodity is raw or processed goods that are used in the production of other goods or for direct consumption. We list the major commodity categories with examples:

Precious Metals: Gold, Silver, Platinum

Energy: Crude Oil (WTI-West Texas Intermediate and Brent), Natural Gas, Electricity

Metals: Base (copper, zinc, nickel, aluminium), Ferrous (iron, steel, iron ore)

Agricultural: Grain/Seeds (wheat, corn, soybean, rice, oats), Soft commodities (coffee, sugar, cocoa, cotton, rubber), Livestock (live cattle, lean hogs, pork bellies)

A *commodity index* tracks the weighted average price of a basket of commodities, which may be based on spot or futures prices. The below ones are well-known commodity indices, and each applies its specific methodology and diversified exposure¹: S&P GSCI Index, Dow Jones-UBS Index, Thomson Reuters/Jefferies CRB Index, Rogers International Commodity Index (RICI).

The investment exposure to commodity asset class is usually realized through the following instruments:

- Commodity futures listed at the exchange
- Mutual fund making commodity related investments, or commodity ETF

¹Actually, the diversification is limited for most of the commodity indices. Energy often represents a heavy weight.

- OTC swap, option with commodity underlying
- Structured product linked to commodity or commodity index

5.2 Commodity Forward and Futures

The shortest dated market commodity transaction is normally considered as the Spot transaction. A commodity *Forward* is an OTC contract between two parties. The Long party is engaged to buy from the Short party a certain quantity (Notional) of the underlying commodity at a pre-defined price (called Strike Price or Forward Price) at the expiry date of the contract. Mathematically, the payoff of the long party is

$$\text{Payoff} = \text{Notional} \times (S_T - K),$$

where

Notional: quantity of the commodity

S_T : commodity price at expiry

K : delivery price which is agreed at the inception of the contract

The *settlement* of the final payoff may be either of the below methods

- Physical Delivery: only real commodity users may choose physical settlement
- Cash Settlement: this is the most common for financial market participants

Commodity Futures contracts are standardized forward contracts that are actively transacted at an Exchange. Holders of the contracts need to sell their holding to the market before expiry if they do not want the delivery of the physical commodity.

*Example 5.1*² The Coffee C contract is the world benchmark for Arabica coffee. The contract prices physical delivery of exchange-grade green beans, from one of 20 countries of origin in a licensed warehouse to one of several ports in the U.S. and Europe, and etc.

Contract size	37,500 pounds
Price quotation	Cents and hundredths of a cent up to two decimal places
Contract listing	March, May, July, September, December
Settlement	Physical delivery
Min. price movement	5/100 cent/lb., equivalent to \$18.75 per contract
Grade/standards	A Notice of Certification is issued based on testing the grade of the beans and the flavor
Deliverable growths	Mexico, Salvador, Guatemala, Costa Rica, Nicaragua, Kenya, New Guinea, ...
Deliverable points	Exchange licensed warehouses in the Port of New York District and the Port of Virginia (each at par), the Port of Antwerp, ...
...	

²Source: extracted from Intercontinental Exchange (ICE) Website.

Spot-Forward Relationship

Except precious metals, it is difficult to implement Cash-and-Carry strategies for hedging forward contracts. Actually, the forward/futures prices depend on several factors including demand/offer, geopolitical and environmental changes.

In modelling, the below concepts are used for establishing the relationship between Spot and Forward:

interest rate (denoted by r): the financing rate

storage cost (denoted by c): actually $r + c$ represents the carry cost

convenience yield (denoted by y): it represents the benefit for holding the physical commodities instead of the holding of forward/futures contracts. Its value can be positive or negative.

The relationship between the forward price expiring at time T , $F(T)$, and the spot price S_0 is then

$$F(T) = S_0 e^{(r+c-y)T}.$$

Commodity Price Term Structure

In commodities market, people often observe two typical shapes for the futures price curve:

Backwardation: the downward sloping curve for forward/futures is called *Backwardation*.

Contango: The upward sloping curve for forward/futures is called *Contango*.

The futures price eventually converges to the spot price by construction. For a contango market, the futures contracts are trading at premium to the spot price. It may imply that the carry cost increases with time. The reason for backwardation may be shortages, political events or weather concerns. Due to the difficulties for implementing arbitrage strategies, the futures prices are mostly driven by supply/demand. Contango is the situation where the future price is anticipated to be higher than the spot price while the backwardation is the situation where the future price is expected to be less expensive than the spot price.

From the spot forward relationship, a large spread between c and y will give the contango structure while a negative one the backwardation.

Seasonality

Some of the commodities, e.g. natural gas, electricity, show strong patterns of seasonality due to the consumption seasonality and storage difficulties.

Roll-Over of Futures

A Futures contract has a fixed expiry date on which the price of the futures contract converges to the Spot price of the commodity. An investor may choose to “roll” his futures contract (i.e. close the contract about to expire and enter into a new contract with a later expiry date) before its expiration to the futures in a further-out month.

It is customary to roll the front month contract to the second nearest contract because the near contracts are more liquid. If the spot price of the commodity stays constant, the “roll yield” for the investor will be

- *positive*: in a backwardated futures market
- *negative*: in a contango futures market

Instead of rolling the front month futures to the second nearest futures, more sophisticated rolling strategies can be used according to the dynamic term structure of the commodity.

5.3 A Special Commodity: Gold

Gold is the most popular precious metal for jewellery and investment. It is considered as

- a “currency”: relatively independent of the interest rate policies of any government. In some of the investment banks, the trading of gold is carried out by the FX trading team.
- a safe-haven asset: throughout history gold has been a safe-haven asset providing shelter in periods of financial crisis or turmoil.
- an inflation hedging asset: the gold price is intimately linked to the real inflation rate of the economy.

The price reference largely used by the financial market is the *London Gold Fixing* by LBMA (London Bullion Market Association). It is a gold price auction taking place twice daily at 10:30am and 3:00pm London time with the price set in US dollars per fine troy ounce. The *spot* transaction is settled at T+2.

Gold is virtually an indestructible metal as it does not corrode. This helps to develop a special type of transaction called *gold lease* through which the lender of gold earns an income which can be paid in cash or in gold. For instance, a central bank may have interest to lend out the physical gold to a jewellery manufacturer.

The market forward price is $F(T) = S_0 e^{(r - r_{gl})T}$ where r_{gl} is the continuous gold lease rate. This is easily understood because $r - r_{gl}$ is the real carry cost rate. For a gold swap, it consists of buying physical gold at spot price and selling back at forward price from the borrower’s perspective. For physically settled transactions, *Loco* represents the place at which gold with a certain grade is physically held and to which a particular price applies. Examples are loco London, loco Zurich, and etc.

Exercises

1. Explain why commodities are known to bring diversification benefits for investors.
2. Name the differences between a stock market index and a commodity index.
3. The forward curve is a graph that displays futures prices against futures maturities.
 - (i) Plot the forward curves for backwardation and contango structure;
 - (ii) From a graphical point of view, explain which one benefits the futures roll-over and how the slope of the forward curve comes into play?
4. The current price of a commodity is \$50, the continuous interest rate is 1.50% flat, the storage cost is 0.32% p.a. (continuous rate), and convenience yield is 0.60% p.a. (continuous rate).
 - (i) What is the forward price for a 2-year maturity contract with zero cost? Is it a Backwardation or Contango?
 - (ii) An investor is long 50,000 commodity forward with the forward price at \$52. Assume commodity price at expiry \$52, what is the investor's payoff?

Chapter 6

Credit Derivatives



6.1 Credit Default Swap (CDS)

First introduced by JP Morgan in 1997, *CDS (Credit Default Swap)* remains the most widely used credit derivative instrument despite the negative reputation in the wake of the 2008 financial crisis.

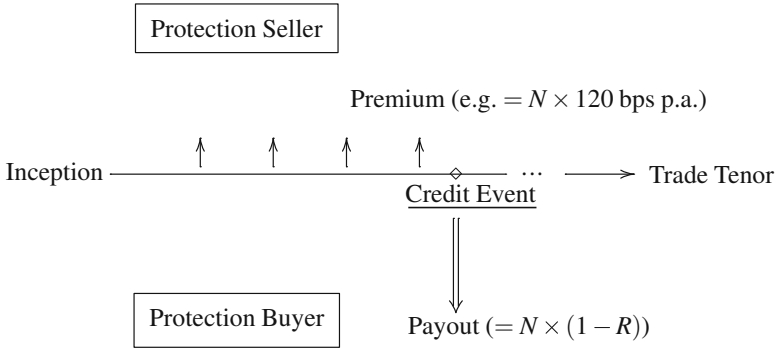
CDS is an OTC transaction where the *protection buyer* (CDS Buyer)

- makes periodic payments (called CDS spread or premium) to the *protection seller* (CDS Seller) until either the credit event or the trade tenor if no credit event occurred
- will be made whole for their exposure on the related bonds of the reference entity.

There are several standard credit events which are typically referred to in credit derivative transactions under ISDA definitions: “Bankruptcy”, “Failure to Pay”, “Restructuring” as well as “Repudiation/Moratorium”, “Obligation Acceleration”, “Obligation Default”. There are two types of settlement in case of a credit event:

Physical Settlement: the protection buyer has to deliver a bond of seniority at least equal to that of the reference obligation plus the accrued CDS interest to the protection seller for receiving the face value of the bonds (Notional Amount) in return

Cash Settlement: the protection buyer will receive “ $\text{Notional} \times (1 - \text{Recovery Rate})$ ”. The recovery rate will be determined by the Credit Event Auction.

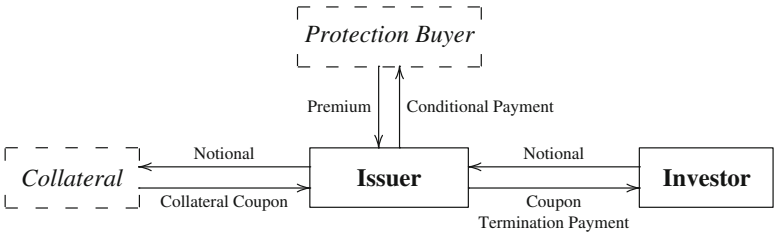


N : Notional amount
 R : Recovery Rate; For senior unsecured debt, R is assumed to be around 40% in general.

General relationship between the default probability λ and the spread s is $s \approx (1 - R)\lambda$.¹

6.2 CLN (Credit Linked Note)

A Credit Linked Note (CLN) is a debt instrument whose value is linked to credit events of one or a basket of reference entities. In case of a Credit Event, there could be a physical or cash settlement based on recovery rate from the credit auction. A CLN can be structured with a short position of the CDS on the reference entity for receiving the premium and an investment into safe assets which may serve as collateral for the CDS and may also generate income. The net income will be used to pay to the CLN investor.



¹The proof will be given in Chap. 22.

The most popular CLN linked to a basket of reference entities is the FTD (First-To-Default) Note which offers a higher yield than single reference entity linked CLN by taking the risk of the first credit event which has a higher probability to occur. An example is a 3Y USD Note paying 5% p.a. coupons until the first credit event for any of the 5 companies in the basket.

6.3 Credit Index

A credit default swap index is a standardized credit security which offers the credit exposure of a portfolio of reference entities. For example:

- CDX NA IG: 125 investment grade (IG) companies in North America (NA), administered by CDS Index Company and marketed by Markit Group Limited,
- iTraxx Europe: 125 investment grade companies in Europe, managed by the International Index Company (IIC), also owned by Markit.

These portfolios are updated (rolled over) on March 20 and September 20 every year.

Credit indices are OTC products and can be traded with licensed dealers providing liquidity. The indices trade with a fixed coupon (e.g. 1%) that is usually paid quarterly (20th of Mar, Jun, Sep, Dec). The coupon is paid by the buyer of protection on the index (who is taking a short position of the index). Upfront payments are made at the initiation and the close of the trade to reflect the change in price of the index. The indices are quoted on a clean basis.

Example 6.1 The following is an illustration example from “Markit Credit Indices – A Primer, 2013”: An index is launched with the price of 100 on Sep 20th, fixed coupon of 60 bps, 5Y tenor.

On Nov 30th, an investor buys USD10m protection on the index when the spread has moved to 90 and corresponding price is 98.67 (the price is par minus the PV of the spread differences).

Upfront payment: $\text{USD}10\text{m} \times (100 - 98.67)/100 = \text{USD}133,000$

Receiving the accrual up to trade date: $71/360 \times \text{USD}10\text{m} \times 0.0060 = \text{USD}11,833.33$

Net Outflow = $\text{USD}133,000 - \text{USD}11,833.33 = \text{USD}121,166.67$

On Dec 20th, the investor pays the fixed coupon: $0.0060 \times \text{USD}10\text{m} \times 91/360 = \text{USD}15,166.67$.

On Mar 13th, he closes the trade when the spread is 120 and the price is 97.44 and pays the accrued interest up to trade date and receives payment: $\text{Inflow} = \text{USD}10\text{m} \times [(100 - 97.44)/100 - 0.0060 \times 84/360] = \text{USD}242,000$.

Upon credit event, the protection buyer obtains the payout for the relevant reference entity according to its weight in the index:

$$\text{Notional} \times \text{Weight} \times (1 - \text{Recovery Rate}).$$

Cash settlement is normally the default settlement mechanism.

Following a credit event of a constituent in the index, a new version of the index is published which assigns a zero percent weight on the defaulted entity. The notional amount on the index trade is reduced by its weight in the index. For instance, consider a USD10m trade on an index with 100 names. After one default occurred, the new version of the index will contain 99 names and will have a revised notional of USD9.9m.

6.4 Collateralized Debt Obligation (CDO)

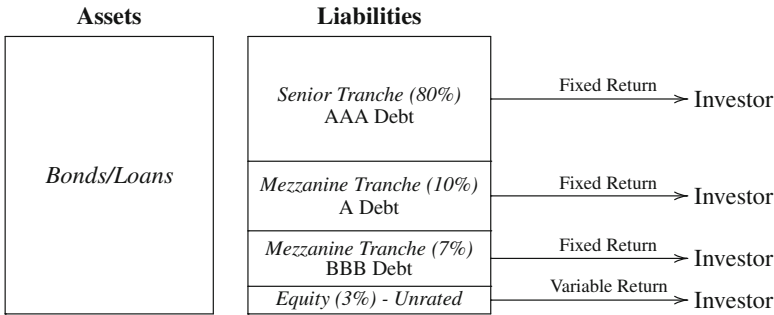
The ABS (Asset-Backed Security) with Bonds and Loans as underlying assets is known as a Cash CDO (Collateralized Debt Obligation). A CDO is commonly structured through an *SPV* (Special Purpose Vehicle, in the form of a company) which buys bonds/loans as *assets* financed by the *debts* and *equities* of the SPV sold to investors. The net revenue (after all costs and losses due to defaults) from the assets is distributed to the debt and equity investors of the SPV.

Multiple tranches corresponding to the debt and equity of the SPV are issued with different payment priorities. The tranches are categorized as Senior debt, Mezzanine debt and Equity with increasing level of risk compensated by increasing return rate. A tranche is associated with a certain return and two threshold levels for determining the principal loss:

- *attachment point*: the default ratio of the portfolio for triggering loss in this tranche
- *detachment point*: it is the attachment point of the next tranche

An illustration example of a CDO with 4 tranches is as follows:

Tranche	Credit rating	Return rate	Attachment	Detachment	Absorbing portfolio loss
Senior	AAA	1%	20%	100%	from 20% until 100%
Mezzanine	AA	1.6%	10%	20%	from 10% until 20%
Mezzanine	BBB	4%	3%	10%	from 3% until 10%
Equity	not rated	variable, e.g. 12%	0%	3%	from 0% until 3%



Synthetic CDO

If credit default swaps are used instead of bonds for the credit exposure, the CDO is called a synthetic CDO. The premiums from the component CDS’s constitute the revenue cash flow for the structure. Synthetic CDOs may be easier and less costly to create than the Cash CDO if there is the CDS supply. However, as the synthetic CDO is built with CDS instead of bonds, its size is not limited by the available bonds and may be out of control.

Single Tranche CDO on Credit Index

CDO tranches are traded on some of the credit indices. An investor can choose the exposure to a particular portion of the index loss distribution. For example, the tranches for CDX NA IG are 0–3%, 3–7%, 7–10%, 10–15%, 15–30%, 30–100%.

The protection buyer of a tranche makes regular (e.g. quarterly) coupon payments to the protection seller and receives a payment in case of a credit event. Like the credit index trade, upfront payments are made at initiation and close of a single tranche trade to reflect the change in price. Coupon payments (e.g. 500 or 100 bps per annum) are made on the remaining notional until the tranche gets fully written down due to credit events or until maturity.

Following a credit event, the equity tranche is reduced by $(100 - \text{Recovery Rate})$ while the notional of the super senior tranche is reduced by the recovery rate (because the recovered amount on the defaulted entity can no longer be lost). The principal of any other tranche except the super senior tranche is unaffected but will have a smaller cushion protecting them against further losses. Readers can refer to [52] for detailed explanation with illustration examples.

Exercises

1. What is a naked CDS and what type of investors is more likely to be interested in them?
2. Consider an investment grade company and its corresponding 5Y CDS contract. Are the equity price and the CDS spread mostly moving in tandem or inversely related? Explain your rationale.
3. Explain why an FTD Note (say, on a basket of 5 names) is often said to be a leveraged investment.
4. Assume a CDO of a total asset value of USD500m invested in high yield bonds with weighted average return of 5%. The total annual fee (management fees + other fees/charges) is 1% on the total asset value.

Tranches	Notional in M USD	Interest rate
AAA rated	N	2.5%
A rated	150	3.5%
BBB rated	100	R
Equity (unrated)	50	9.5%

Please determine the “ N ” and “ R ” in the above table after 1Y investment.

5. On Dec 1, 2017, the five-year CDS spread on corporate bond of A is 3.0%. The same corporate bond’s yield is 6% and 5-year treasury bond yield is 2.58%. Calculate the basis and explain how can you arbitrage this opportunity.

Chapter 7

Investment Funds



7.1 Funds

A fund is a collective investment scheme that pools money from investors. It is managed by investment advisors known as fund managers from a management firm which is legally and financially distinct from the fund and its assets. A fund has the objective to produce higher risk adjusted returns than its benchmark (usually an index).

There are two main types of funds: *mutual funds* which are open to all investors, and *hedge funds* for sophisticated and accredited investors.

7.1.1 Fund Price and Fees

The value of a fund unit is represented by its *net asset value* (NAV) which is defined as

$$NAV = \frac{\text{Value of the assets held by the fund} - \text{Liabilities}}{\text{Number of shares (or units)}}.$$

There are different charges linked to the investment in a fund:

- management fee: a fixed percentage applied to the NAV (e.g. 1.2% p.a.); the fee level depends on the skill/involvement/added value of the fund manager
- custodian fee: e.g. 0.05% of the fund NAV
- sales charges or loads: *front-end load* for subscription, *back-end load* for redemption
- other expenses including distribution and marketing costs.

The *Total Expense Ratio (TER)* measures the total cost to the investor. It is defined as

$$TER = \frac{\text{Management fee} + \text{All the other fund expenses}}{\text{Total asset value}}.$$

7.1.2 Fund Organization

A fund can be legally set up as

- a unit trust: the investors are trust beneficiaries
- a corporation: the investors are shareholders
- a limited partnership: the fund manager is the general partner and investors are limited partners. It is mostly used for hedge funds and private equity funds.

There are different forms for a fund, including

open-end fund: all buy/sell trades are realized in the primary market through subscription and redemption. The open-end funds constitute the majority of the funds.

closed-end fund: a fixed number of non-redeemable shares are issued at initial offering. The shares are then traded in the secondary market. The share price may deviate from NAV of the fund because of demand/supply (often at discount of the fund NAV).

exchange-traded fund (ETF): an index tracking fund listed at an exchange. Its price may be influenced by demand/offer but with limited deviation from its NAV because the ETF shares may be created or redeemed at NAV by the ETF PD (Participating Dealer) if there is an arbitrage opportunity.

fixed maturity fund: a fund having a fixed maturity. It targets to invest in the bonds with similar maturity as the fund itself. So, the fund is close to a principal protected income product (subject to no default).

7.1.3 Share Classes

One may invest in a mutual fund through a certain *share class* which, usually named by a single letter, offers a type of exposure with its fee/cost structure and NAV. For instance, based on the same investment fund, “A” may be the share class with USD denomination while “B” is the share class with EUR denomination; “I” is often used for institutional share class which has a lower management fee but higher minimum subscription amount; “Distribution” share class pays regular dividends whilst “Accumulation” share class does not pay dividends but reinvests them.

7.1.4 Fund Derivatives

Investors can get exposure through derivative structures usually provided by an investment bank who has an agreement with the fund under which, the investment bank trades at NAV and sometimes receives a rebate of part of the management fee. The most common structures are as follows:

- Leveraged exposure: total return swap, leveraged Note, etc.
- Protected exposure¹:
 - option based structure: the investor may purchase a call option or a Note with an embedded call option with the fund as underlying. The volatility target mechanism is often necessary for avoiding the option provider to hedge the fund volatility risk.
 - gap risk based structure: Constant Proportion Portfolio Insurance (CPPI) or Time Invariant Portfolio Protection (TIPP) which applies dynamic investment allocation between risky and low-risk assets for mitigating gap risk.

7.2 Mutual Funds

Mutual funds are usually accessible to all types of investors including individuals. They are highly regulated and supervised with high standard of transparency and diversification:

- in US, regulated by Investment Company Act of 1940
- in Hong Kong, by SFC (Securities and Futures Commission)
- in Europe, the UCITS qualification offers the passport for selling the fund across European countries

Mutual funds are also managed by licensed investment advisers (i.e. fund managers): e.g. in Hong Kong, the license Type 9 of SFC covers asset management. A mutual fund's assets are usually safeguarded by a custodian bank. For open-ended funds (excluding the sales charge), there is daily liquidity for subscription/redemption. A prospectus (or explanatory memorandum) is required for each fund. A mutual fund is overseen by a board of directors for funds set-up as a corporation and an annual report to fund investors is required.

Mutual funds are generally classified by the focus of investments which is described in the prospectus:

- money market fund: investing in money market instruments and bonds with short expiration date
- bond fund: e.g. emerging market bond fund, high yield bond fund, senior loan bond fund

¹cf. Chap. 8 for options and Chap. 26 for volatility target mechanism and gap risk structures.

- equity fund: such as mid-cap fund, high dividend yield fund
- hybrid fund: with its investment allocated to several asset classes

The management fees for an actively managed mutual fund may range from 0.25% to 1.50% of the NAV. For passive funds such as ETFs and index funds, the management fee level is lower (e.g. 0.15%).

7.3 Hedge Funds

A hedge fund aims to achieve an absolute return by investing in a range of markets with a wide variety of styles and financial instruments. It is restricted to sophisticated or accredited investors with a minimum investment size.² The hedge fund managers often have significant stakes in the funds, which serves to align their own interests with those of the investors.

The hedge funds and their managers have historically been less regulated compared to mutual funds. A hedge fund is normally not very transparent, offering limited liquidity for subscription/redemption (e.g. monthly, semi-annually). This makes it difficult to create structured products linked to hedge funds.

The legal structure of a hedge fund varies depending on its location and investors. In the US, they are generally structured as limited partnerships so that the investors are not subject to both entity-level and personal-level taxation. Many hedge funds are established in offshore financial centres to avoid adverse tax consequences for investors.

A hedge fund has one or more *prime brokers* providing execution, custodian, leverage and short-term financing, securities lending. Prime brokers may even help fund managers to meet potential investors.

Fees charged by hedge fund management firms are as follows:

- Management fees: typically 2% on NAV
- Early redemption fee (or early withdrawal fee)
- Performance fee: typically 20% on the out-performance above its *HWM (High Water Mark)* which is the highest NAV level (after fees) over a specified past period or *Hurdle* which is a benchmark rate (e.g. Libor + spread).

Examples of hedge fund strategies are:

- *global macro*: investment strategy in share, bond or currency markets in anticipation of global macroeconomic events;
- *event-driven*: seeking investment opportunities in corporate events;

²For instance, the minimum subscription size is USD50k for a single hedge fund in Hong Kong.

- distressed securities: traded at a deep discount due to restructuring, bankruptcy, etc.
- risk arbitrage: buying and selling the stocks of two or more merging companies.
- special situations: spin-off, share buy-back, security issuance, asset sales, etc.
- *relative value*: it is a market neutral strategy taking advantage of relative discrepancies in price between securities. The list of such strategy includes fixed-income arbitrage, equity market neutral, convertible bond arbitrage, capital structure arbitrage and regulatory arbitrage;
- *managed futures fund* (also known as CTA – Commodity Trading Advisor): trades mainly futures in commodities, stock indices, forex, etc. The fund uses quantitative techniques, such as trend-following, technical pattern recognition, to take long and/or short positions.

An efficient way to offer products linked to a hedge fund consists of setting up a *managed account*, typically through the asset management arm of an investment bank. The hedge fund manager will be mandated as investment advisor for making the same investment as the hedge fund. But the managed account is transparent with improved liquidity and smaller entry size. The investment bank may offer structured products on the managed account because the hedging (or more specifically, delta hedging, which will be introduced in the next chapter) can be done with the constituent investments in the account. To financial institutions, the regulatory risk weight for the investment in a managed account is often lower than the risk weight for the investment in the hedge fund itself.

Exercises

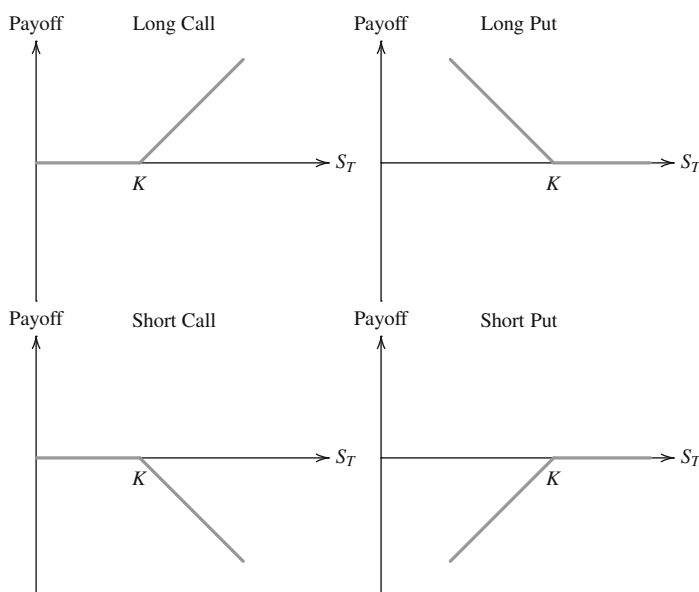
1. Find out what are *target-date funds* and list their pros and cons.
2. An open-end fund has reported a net asset value of \$1 per share for all the historical trading days. Find out the type of this fund and explain such phenomenon.
3. Investors are becoming more aware of the Environmental, Social and Governance (ESG) criteria in an investment process. Explain how such criteria make an impact on index funds.

Chapter 8 Options



An option is a financial instrument which may offer an asymmetrical exposure to the performance of the underlying. There is a large variety of options. But the call and put options are the most basic ones.

We recall that the *call option* gives the owner the right but not the obligation to buy the underlying asset from the seller at a pre-agreed price, called the strike price or exercise price. The payoff of a call option at expiry T is $\max\{S_T - K, 0\}$ where S_T , K represent respectively the final underlying's price and the strike price of the option. Similarly, a *put option* gives the owner the right but not the obligation to sell a certain asset at the strike price. The payoff at expiry T is $\max\{K - S_T, 0\}$. Graphically,



Options can be used for several purposes including

- hedging: for instance, if an investor holding XYZ share is concerned by its price drop in the short term, he may buy a put option to hedge his long position of the stock.
- leverage: for instance, instead of investing directly N dollars into an asset, one can choose to buy a *call* offering the same upside exposure by paying the option premium which is a fraction of N .
- directional exposure with protection: for speculating with a bearish view, one may build a short position with a forward or by short-selling the underlying which will incur a loss if the underlying price rises instead of falling. But if he buys a *put* instead, the upside risk will be mitigated.
- specific view: we will see later that a combination of call and put options allows to take advantage of any market view for the considered horizon.

8.1 Option Features

8.1.1 Option Style

There are two styles related to options:

European Option: an option that may be exercised only at the maturity of the option

American Option: an option that may be exercised at any time before the maturity date.

8.1.2 Mark-to-Market Price, Intrinsic Value and Time Value

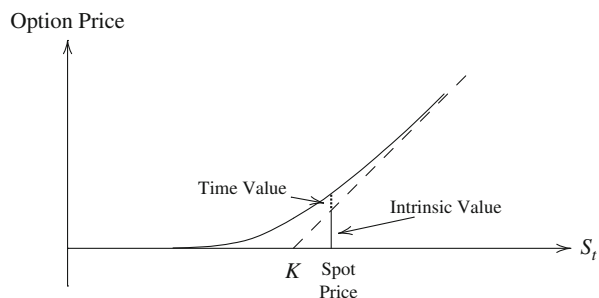
The *mark-to-market price (MtM)*, or simply option price, is the price at which the option can be bought or sold in the market before its expiry. It is made up of two components:

intrinsic value: the payoff that would be received if the underlying asset is at its current level S_t when the option expires. It is defined to be $\max\{S_t - K, 0\}$ for a call and $\max\{K - S_t, 0\}$ for a put.

time value: the difference between the MtM price and the intrinsic value. It will be zero at maturity.

See Fig. 8.1 for an illustration with call option.

Fig. 8.1 Illustration of Intrinsic value and Time value



8.1.3 Moneyness of an Option

An option at time t is said to be

in-the-money (ITM): if its intrinsic value is positive at t ; It means $K < S_t$ for a call and $K > S_t$ for a put where S_t is the underlying price.

at-the-money (ATM): if its intrinsic value is zero and that S_t is at the limit of being ITM; For a call or put option, it means $K = S_t$.

out of-the-money (OTM): if it is neither ATM nor ITM. It means $K > S_t$ for a call and $K < S_t$ for a put.

We say that the *strike* of an option is ATM (or ATMS, at-the-money spot) if the strike price is the initial price of the underlying. The strike price is called ATMF (at-the-money forward) if it is equal to the forward price of the underlying for the tenor of the option at its inception.

8.1.4 Notional Amount Definition

Each option contract has a notional amount used to calculate the payoff. There are two methods to define the option notional amount:

1. in number of units of the underlying (e.g. number of shares): For instance, the maturity payoff of an ATMS call option is

$$N \times \max\{S_T - S_0, 0\},$$

where N represents the number of shares.

2. in monetary unit (e.g. EUR amount): this method is used in structured products. For instance, the maturity payoff of an ATMS call is written as

$$N \times \max\left\{\frac{S_T}{S_0} - 1, 0\right\},$$

where N is in *EUR*.

We often omit the notional amount N when it is not important for the analysis.

8.2 Vanilla Options

Vanilla options refer to the most liquid and basic options traded in the market which are call or put options. Any option that is not a vanilla option is an exotic option. In this section, we will introduce the vanilla options for the main tradable asset classes, namely, equities, interest rates, currencies and commodities. The market

quotations for Vanilla options are based on BSM (Black–Scholes–Merton) type of models (which will be introduced in Sect. 8.2.5).

8.2.1 Equity Options

Vanilla options linked to equities are call and put options of either European or American style on a single stock or an index. For *Payoff*, there are respectively $\text{Call} = \max\{S_T - K, 0\}$ or $\text{Put} = \max\{K - S_T, 0\}$, where S_t is the time- t price of the underlying stock or equity index.

Equity index options are actively traded in the market. Many exchanges have listed equity index options which are traded with decent volume. For single stock options, even though listed at some of the exchanges (e.g. CBOE–Chicago Board Options Exchange since 1973), they are still traded mainly in OTC market with a far lower volume compared to index options. Single stock options listed at CBOE have American style and have physical settlement at expiration (securities are delivered), whereas the index options have cash settlement at expiration, and often are larger sized contracts.

Example 8.1 The *S&P500* option at CBOE is $\$100 \times \max\{\text{Final Index} - \text{Strike}, 0\}$. If the value of the index is 2100 at expiry and the strike price is 2000, the option will be exercised. The option holder will receive $\$100 \times (2100 - 2000) = \$10,000$.

8.2.2 Foreign Exchange Options

The vanilla FX options are the call/put options giving the right to buy/sell the base currency with the counter currency. Let X_t be the exchange rate. For *Payoff*, there are $\text{FX Call} = \max\{0, X_T - K\}$ and $\text{FX Put} = \max\{0, K - X_T\}$.

One particular feature of an FX option (known as *FX Option Duality*, see Sect. 18.3) is that for a pair Ccy1/Ccy2 , a call option on Ccy1 (the right to buy Ccy1 with Ccy2) is actually the put option on Ccy2 (the right to sell Ccy2 against Ccy1) with nominal amount of KX_0 . That is why practitioners use the language like: *buying USD50m, 1-month, at-the-money forward dollar Call, yen Put*.

8.2.3 Commodity Options

Depending on the commodity, the vanilla option is linked to either the spot price (e.g. Gold), or the most nearby futures contract's price (e.g. WTI) at expiration of the option. For *Payoff*, there are $\text{Call} = \max\{0, S_T - K\}$ or $\text{Put} = \max\{0, K - S_T\}$, where S_T is the underlying price at maturity. At some of the exchanges (e.g. CME, ICE), both European and American options on futures contracts are listed.

8.2.4 Interest Rate Options

For interest rate underlying, there are two types of Vanilla options: Cap/Floor and Swaption.

Cap/Floor

A European call option on a floating interest rate (e.g. 3M USD Libor) for a considered period (T_{i-1}, T_i) is called a *caplet* which takes the form

$$\text{Caplet Payoff} = N \times \delta_i \times \max\{R_i - K, 0\},$$

where N , δ_i , R_i represent respectively, the notional amount, the day count fraction and the reference interest rate covering (T_{i-1}, T_i) . The caplet buyer is entitled to receive a payment at the end of the period if the interest rate observed at the beginning of the period exceeds the agreed strike price.

An *Interest Rate Cap* is a sum of consecutive caplets in which the buyer receives payments at the end of each period where the interest rate exceeds the agreed strike price common for all caplets:

$$\text{Cap Payoff} = N \sum_{i=1}^m \delta_i \max\{R_i - K, 0\}.$$

Example 8.2 Someone obtained a USD 1m, 10 year mortgage from a bank with monthly interest payments at 1 month USD Libor +2%. Concerned by the risk of interest rate increase in the future, he decided to buy a USD 1m, 10Y Cap with the strike of 5%. Now, his effective monthly interest cost will be limited by 5% because the impact from the 1 month USD Libor above 3% will be compensated by the short party of the Cap.

Similarly, the European put option on interest rate is known as a *floorlet* when it is applied to a specific period. An *Interest Rate Floor* is a sum of floorlets in which the buyer receives payments at the end of each period where the interest rate is below the agreed strike price:

$$\text{Floor Payoff} = N \sum_{i=1}^m \delta_i \max\{K - R_i, 0\}.$$

Swaption

A *swaption* is an option giving the option holder the right but not the obligation to enter into an IRS (Interest Rate Swap) which starts at T_0 and matures at T_n , at a pre-agreed swap rate K at the option expiry date T_0 . The option's tenor is from today t_0 until the expiry date T_0 . The swaption's final payoff at T_0 is

$$\text{Payoff} = N \times A(T_0, T_n) \max\{s(T_0, T_n) - K, 0\},$$

where

$s(T_0, T_n)$ is the market swap rate at T_0 of the IRS;

$A(T_0, T_n)$ is the *annuity factor* for the IRS.

Example 8.3 A 1Y, 2.5% strike call swaption linked to 10Y USD Swap rate gives its holder the right but not the obligation for entering into the Swap with the short party at the fixed rate of 2.5% at the expiry date in one year.

8.2.5 Option Pricing, Hedging and Execution

Option Hedging/Pricing

The price of an option is the cost of replicating the payoff. In other terms, option price is its hedging cost. As seen from studies in the modelling part of the book, the replication/hedging in *complete market*¹ is based on the sensitivity of the option price (V_t) with respect to the underlying's price (S_t) change which is called *delta* ($\Delta_t = \frac{dV_t}{dS_t}$). The option hedging consists of dynamically buying or selling the underlying according to Δ_t , which is called *delta hedging*.

Black–Scholes–Merton (BSM) model² is the first and the most popular model for option hedging and pricing. Under this model, the asset's price follows the stochastic differential equation: $dS_t/S_t = \mu dt + \sigma dW_t$, where μ represents the drift, σ is called the *volatility* for the underlying asset representing the annualized standard deviation of its log return, and W_t is known as the *Brownian motion* which depicts randomness in asset prices [74].

Option Execution

The option execution is directly based on the hedging of the option. The trade order is said to be “done” (or “filled”) when the initial delta is executed by the option writer or issuer in the market. The execution of the delta means the buying or selling of the underlying according to the quantity indicated by the *delta*. The price at which the delta is executed will serve as *initial spot reference* in the option contract. For placing a limit order for a trade with more than one underlying, the limit will be set for one underlying and the initial delta for other underlyings will be executed at Market.

¹Roughly speaking, in a complete market, all contingent claims can be replicated with (dynamic) positions of underlying assets.

²In fact, BSM model has been adapted for different asset classes which carry different names: Garman–Kohlhagen model for currencies and Black model for interest rates.

Vanilla Option Pricing

For vanilla options of European style, the market participants use mainly the so-called Black&Scholes (B&S) formula³ derived in BSM model:

$$c_0 = S_0 N(d_1) - e^{-rT} K N(d_2), \quad p_0 = e^{-rT} K N(-d_2) - S_0 N(-d_1),$$

where c_0 and p_0 are the call and put option prices with maturity T and strike K , r is the continuous interest rate, S_0 is the spot price of the underlying asset with volatility σ ,

$$d_1 = \frac{\ln \frac{S_0}{K} + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \quad \text{and} \quad d_2 = d_1 - \sigma\sqrt{T}$$

$N(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{-\frac{x^2}{2}} dx$ is the standard Normal cumulative distribution function

Apart from the option features such as its tenor and strike price, other factors impacting the option price in BSM model are the volatility of the underlying and the interest rate. Volatility is the most important parameter for an option trader or risk manager as the impact from interest rate is relatively small. From Black&Scholes formula, we can see the one-to-one relationship between the option price and the volatility σ of the underlying with the other parameters fixed.

Option Quotation

Equity options are quoted by strike and by tenor in the market. The strike is expressed as a percentage of the initial spot reference which depends on the type of the order. FX options are quoted by *delta* (Δ) and tenor. In FX option market, people prefer to quote the vanilla options by its *delta* (calculated with B&S formula) instead of its *strike*.⁴ The benchmark market quotes are ATM 50% Δ , 25% Δ , 10% Δ . Bisesti et al. [9] gives a good overview of market practice for FX options. For Cap/Floor, the market quotes are available by strike and by tenor. For Swaptions, they are quoted in the market by strike, option tenor and the underlying swap tenor.

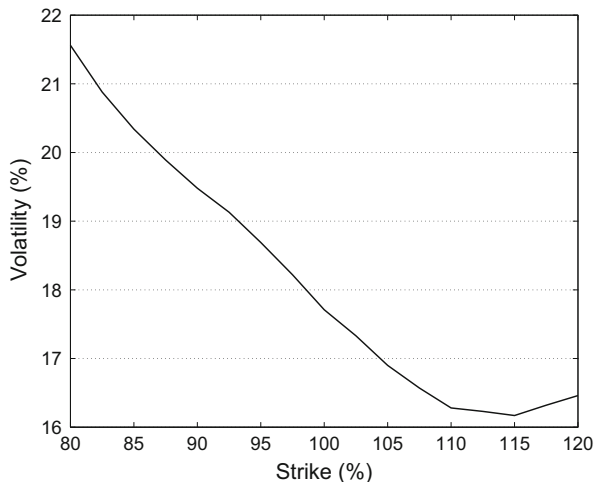
8.2.6 Implied Volatility

The *implied volatility* of an option is the volatility value derived from the option price quoted in the market. Even though the volatility has a unique value in BSM model given the underlying, the implied volatility may be different for options with

³The Black&Scholes formula presented here is for non dividend paying stocks with zero repo rate. Similar formulas can be derived for the general case as well as other asset classes.

⁴Given the price of an option, there is a one-to-one relationship between *delta* and *strike* using the B&S formula.

Fig. 8.2 Example of volatility smile for equity options



different tenors and strikes. For the same expiration date, the implied volatilities from the options of different strike prices show a pattern which is called *volatility smile* or *volatility skew*. For equity options, the smile normally takes the shape shown in Fig. 8.2.

There are various reasons for explaining the volatility smile for equity underlying, including

1. imperfection of the BSM model: BSM model implies log normal distribution for the return of the underlying. The historical data confirm that the “real” distribution shows fatter tails (kurtosis) and steeper asymmetry (skewness). Market participants use the adjusted volatility for matching the “correct” option price observed in the market.
2. demand/offer: the demand/offer for options of different strike prices influences the market prices. For instance, institutional investors such as insurance companies and fund managers tend to buy OTM equity index put options for hedging the long exposure on equities. This results in higher prices for OTM put options implying higher implied volatility compared to the implied volatility for strike prices above the current spot.

The volatility for a future period $[t_1, t_2]$, $0 = t_0 < t_1 < t_2$, can be derived with the additivity of variance: $\sigma(t_1, t_2)^2(t_2 - t_1) = \sigma_2^2 t_2 - \sigma_1^2 t_1$. It may be used for pricing forward starting options.

8.2.7 Put-Call Parity

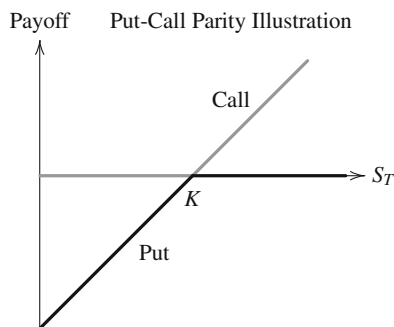
Proposition 8.1 (Put-Call Parity) *Let c_t , p_t and F_t be respectively the time- t prices for the European call, European put and forward with the same maturity T and strike K . Then, we have*

$$c_t(T, K) - p_t(T, K) = F_t(T, K), \quad \forall t \leq T.$$

Proof It is obvious that $\max\{S_T - K, 0\} - \max\{K - S_T, 0\} = S_T - K, \forall T > 0$. If *Put-Call Parity* is not verified, we then have an arbitrage opportunity by selling the over valued leg and buying the other two legs with lower price. The payoff at maturity will be 0 for the sum of all the legs. $\square$

For a non-dividend-paying stock with zero repo rate, we have $F_t = S_t - e^{-r(T-t)}K$ (See the remark in Proposition 3.1 in Chap. 3) leading to

$$c_t(T, K) - p_t(T, K) = S_t - e^{-r(T-t)}K.$$



Remark 8.1

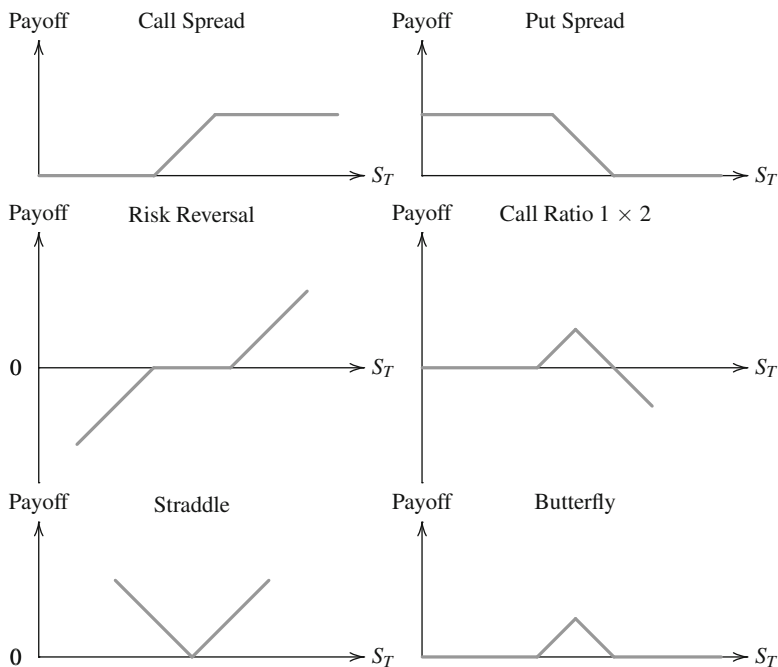
- This arbitrage forces the vanilla call and put options with the same strike level to have the same implied volatility value.
- It illustrates an interesting case that the combination of two options is no more an option.

8.2.8 Popular Strategies with European Options

European options can be used to take advantage of specific views on the future prices of the underlying asset. It can be proved that any option payoff of European style can be represented as the combination of European call and put options (cf. Sect. 19.4 “Static Hedging for European Style Options” in Chap. 19) of the same maturity.⁵

⁵The call and put payoffs are half-lines with slope ± 1 (related to 45° angle). For the graphical representation based on a fixed notional, the slope will change if we change the exposure (effective notional) to the option. It is not difficult to imagine that any payoff at a given maturity date can be presented by a combination of call and put options with different levels of exposure.

Without loss of generality, we assume that the underlying asset is a stock, but similar patterns can be obtained for any other underlying such as a foreign exchange rate, a stock index, and a futures contract. Below is a list of well-known option strategies:



Call Spread (also called Vertical Bull Spread): It is a position of long 1 call at K_1 , short 1 call at K_2 ($K_1 < K_2$). It is a cheap way to take advantage of bullish view. The price of the whole package is sensitive to the Skew instead of volatility if K_1 and K_2 are close.

Put Spread: It is a position of long 1 put at K_1 , short 1 put at K_2 ($K_1 > K_2$). It is a cheap way to take advantage of bearish view.

Risk Reversal: long 1 OTM call at K_{call} and short 1 OTM put at K_{put} ($K_{put} < K_{call}$) such that the option premium at inception is close to zero.

Butterfly: long call at strike K_1 , short 2 calls at strike K_2 and long call at strike K_3 where $K_1 < K_3$ and $K_2 = (K_1 + K_3)/2$. A butterfly spread leads to a profit if the stock price stays close to K_2 .

Collar: the position of being long share, long OTM put option and short OTM call option

Straddle: long call and long put with same strike (usually ATM) and the same maturity

Strangle: long OTM call and long OTM put with the same maturity

Call Ratio: a popular strategy used by hedge funds is one-by-two (1×2) i.e. long one call at strike K_1 and short two calls at strike K_2 where $K_1 < K_2$

Covered call (also called *Buy/Write*): a strategy where the investor has a long position of the underlying asset and a short position of an OTM call on the asset. In a covered call, the underlying asset of the investor may serve as collateral for his short call position, which mitigates the counterparty credit risk for the long party (usually an investment bank). It is a popular strategy for monetizing the stock holding when the view on the underlying is not strongly bullish for the short term.

Calendar Spread: it is formed by options with different expiration dates. Since we usually use x -axis to denote time, it is also called *horizontal spread*. It can be used for forward volatilities arbitrage.

Hedge funds often trade slightly OTM call/put options for getting the directional exposure by paying less time value. Very often they unwind the option before expiry. That is why hedge funds tend to trade more liquid options instead of customized exotic options which are more difficult to unwind in the market.

8.2.9 American Options

An *American option* is one that may be exercised at *any time* prior to expiry by its holder. If the option can be exercised only at discrete time points (e.g. every month), it is called a Bermuda option. But sometimes Bermuda options are called American options as well.

An American option holder must assess when it is the best time to exercise the option. Actually, as we will see in the modelling part, the optimal decision is when the exercise payoff is higher than the price of the remaining American option if the holder does not exercise.

Let $c_t(K, T)$ and $p_t(K, T)$ denote the time- t price of the European call option and put option with strike K and expiry T , respectively while $C_t(K, T)$ and $P_t(K, T)$ denote that of the American call option and put option, respectively. Due to its exercise right at the discretion of the option holder, an American option's price should be at least the price of its counterpart European option, i.e.,

Proposition 8.2 $C_t(K, T) \geq c_t(K, T)$ and $P_t(K, T) \geq p_t(K, T)$.

However, there is a particular case where the American call is worth the same as the European call.

Proposition 8.3 *The price of an American call is equal to the price of the corresponding European call if*

- zero dividend and zero repo rate for the underlying during the tenor of the option are assumed, and
- the interest rate r is nonnegative.

Proof We first show that

$$c_t(K, T) \geq \max\{0, S_t - e^{-r(T-t)}K\}, \quad \forall t < T.$$

Define two portfolios at date t :

Portfolio A: Long $c_t(K, T)$ + a bond with the value of $e^{-r(T-t)}K$

Portfolio B: Long one share S_t

At maturity T , we have

Portfolio A: $\max\{0, S_T - K\} + K = \max\{S_T, K\}$

Portfolio B: S_T (no dividend payment)

Hence, we should have $A \geq B$, for any time until maturity, i.e. $c_t(K, T) + e^{-r(T-t)}K \geq S_t$, or

$$c_t(K, T) \geq \max\{0, S_t - e^{-r(T-t)}K\} \quad (\text{because } c_t(K, T) \geq 0).$$

Consequently, we have the following inequalities

$$C_t(K, T) \geq c_t(K, T) \geq \max\{0, S_t - e^{-r(T-t)}K\} \geq \max\{0, S_t - K\} \quad (\text{as } r \geq 0).$$

Because the early exercise price is $\max\{0, S_t - K\}$, it is not optimal for the option holder to exercise the option before the maturity (the option holder should sell the option instead of exercising it). Therefore,

$$C_t(K, T) = c_t(K, T). \quad \square$$

Proposition 8.3 implies that one should not exercise an American call option on a non-dividend paying stock with zero repo rate before its expiry, if the interest rate is nonnegative. If the option holder thinks that the stock is currently overpriced and wants to lock in his gain, he is better off selling the option rather than exercising it.⁶

For investors, with the same option premium, it is always more advantageous to hold the American option instead of the European one because, if ever he wants to early terminate the option, he has the certainty of receiving the intrinsic value. With the European option, the option seller usually charges an unwinding fee which may lead to an unwind value lower than the intrinsic value.

8.3 Exotic Options

Any option that is not a vanilla option is an exotic option. Each exotic option offers a specific exposure to the performance of the underlying. The exotic options are

⁶Nevertheless, it may be optimal to exercise an American put option on a non-dividend-paying stock before its maturity.

often used as part of a structured product or a customized solution. If the payoff is determined by more than one price of a single underlying in the future, the option is called *path-dependent*.

Let us introduce a few notations that will be used in the following context:

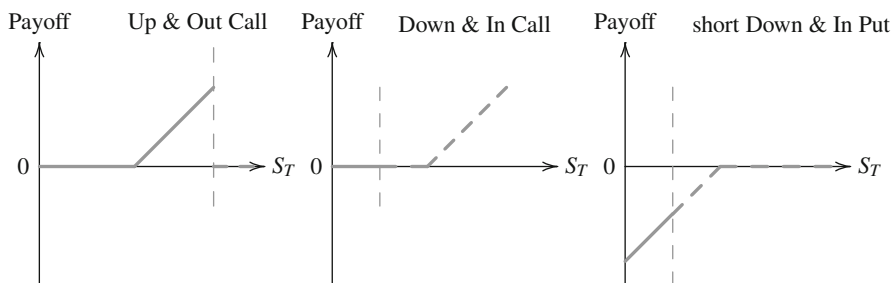
$1_{\text{condition}}$ is the *indicator function* which returns 1 if *condition* = true, else 0

$\max_{t \in [t_1, t_2]} S_t$ is the maximum value of S_t , $\forall t \geq t_1$ and $t \leq t_2$

$\min_{t \in [t_1, t_2]} S_t$ is the minimum value of S_t , $\forall t \geq t_1$ and $t \leq t_2$.

8.3.1 Barrier Option

The barrier option is a kind of call or put option which either comes into existence (“knock-in”) or becomes worthless (“knock-out”) based on a certain condition linked to the performance of the underlying. The terms such as *down-and-in*, *down-and-out*, *up-and-in*, *up-and-out* are used for indicating the condition. For instance, the payoff of *down-and-in* put is $\max\{K - S_T, 0\} 1_{\min_{t \in [0, T]} S_t \leq L}$ and the payoff of *up-and-out* call is $\max\{S_T - K, 0\} 1_{\max_{t \in [0, T]} S_t < H}$.



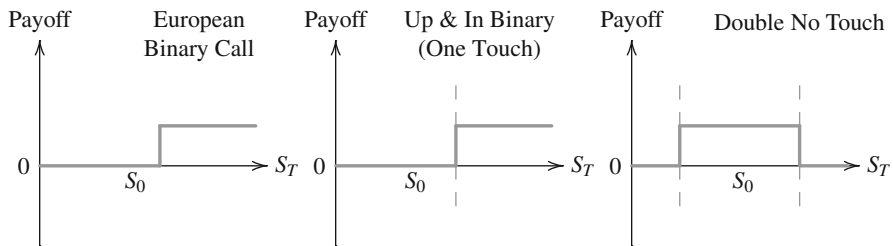
Because of the extra condition, either Knock-in or Knock-out, the barrier options are cheaper than the underlying vanilla options which explains the popularity in practice. It is easily seen from *in-out parity*; see Sect. 8.3.3.

Example 8.4 One popular derivative product listed at the HK Stock Exchange is the “callable bull/bear contracts” (CBBC). Consider a Bull CBBC with exercise price \$80, barrier (known as the *call price*) \$83 and the underlying stock price of \$100 on the issue date. If the stock price breached call price at any time before the expiration date, then the contract is *called* (i.e. terminated). The residual value of the Bull CBBC at the call is $\$83 - \$80 = \$3$. If the Bull CBBC has never been called, and if at expiry, the 5-day average closing price is \$120, then the value of the Bull CBBC at expiry is $\$120 - \$80 = \$40$. It is actually a “down-and-out” call option with ITM strike.

8.3.2 Binary (or Digital) Option

A binary option pays off a fixed amount (either cash or asset) if a certain condition is met; otherwise, 0. There are two main types of binary options:

- European binary option: the condition depends on the final price of the underlying. For instance, the payoff of a binary call option is $\mathbf{1}_{S_T \geq H}$. The payoff of a binary put is $\mathbf{1}_{S_T \leq L}$.
- American binary option: the condition depends on the maximum or minimum price of the underlying observed over a period. It is also called a *touch* option. For instance, the payoff of a *one-touch* option is $\mathbf{1}_{\max_{t \in [0, T]} S_t \geq H, S_0 < H}$. The payoff of a *double-no-touch* is $\mathbf{1}_{\min_{t \in [0, T]} S_t \geq L \cap \max_{t \in [0, T]} S_t \leq H, L < S_0 < H}$. In the FX option market, the touch options are actively traded and considered as vanilla options for the major currency pairs.



8.3.3 In-Out Parity

Similar to put-call parity for European options, there is *in-out parity* for barrier options which states that a vanilla option is equivalent to the sum of a *knock-in* option and a *knock-out* option. Actually, from $1 = \mathbf{1}_{\min_{t \in [0, T]} S_t \geq L} + \mathbf{1}_{\min_{t \in [0, T]} S_t < L}$, we obtain

$$\text{Payoff}(T) = \text{Payoff}(T) \cdot \mathbf{1}_{\min_{t \in [0, T]} S_t \geq L} + \text{Payoff}(T) \cdot \mathbf{1}_{\min_{t \in [0, T]} S_t < L}.$$

For instance, a *down-and-in* put option is equivalent to a *vanilla* put option minus a *down-and-out* put option. Following the same rationale, an *up-and-out* call option with rebate c (paying out c when the call option is knocked out) can be seen as an *up-and-out* call (without rebate) plus an *up-and-in* binary option paying c (i.e. $\text{call} \cdot \mathbf{1}_{\max_{t \in [0, T]} S_t < H} + c \cdot \mathbf{1}_{\max_{t \in [0, T]} S_t \geq H}$).

8.3.4 Asian Option or Average Option

An Asian option is like a vanilla option except that the final underlying price and/or the strike price is replaced by the averaged value of the underlying over a predefined period. For instance, the payoff of an Asian call option with averaging at the end (we say Asian Tail or Asian Out) is $\max \left\{ \frac{1}{m} \sum_{i=n-m+1}^n S_{t_i} - K, 0 \right\}$. When the averaged price of the underlying is used as strike, the option is called Asian strike option, e.g.

$$\max \left\{ S_{t_n} - \frac{1}{m} \sum_{i=0}^{m-1} S_{t_i}, 0 \right\}.$$

The Asian option offers a smoother payoff compared to a European Vanilla option which depends entirely on the single price of the underlying at maturity. Another application of Asian option is for managing the risk of “striking” and “expiry” that will be explained further in Chap. 16.

Example 8.5 Warrants in the HK Stock Exchange use the arithmetic average of the closing prices in the final 5 trading days as the final price reference when computing the payoff of the warrant.

8.3.5 Lookback Option

In this type of option, the maximum or minimum asset price sampled during a certain period is used as either the performance (*lookback spot*) or as the initial reference (*lookback strike*) in the calculation of the option payoff. For instance, a call option paying off

$$\max \left\{ \frac{S_T}{S_0^*} - 1, 0 \right\}, \quad S_0^* = \min\{S_{t_1}, S_{t_2}, \dots, S_{t_m}\}, \quad t_m \leq T$$

is called a *lookback strike* call. A call option paying off

$$\max \left\{ \frac{S_T^*}{S_0} - 1, 0 \right\}, \quad S_T^* = \max\{S_{t_{n-m+1}}, \dots, S_{t_{n-1}}, S_{t_n}\}, \quad t_n = T$$

is called a *lookback spot* call.

8.3.6 Quanto Option and Composite Option

When the payoff currency is different from the underlying’s currency, there exist two well-known options called Quanto option and Composite option.

The term *quanto* is used to indicate that the performance from the underlying is only measured in its domestic currency without taking into account any impact from the currency exchange rate. For instance, the payoff for a USD5m, USD Quanto call option linked to BNP Paribas stock quoted in EUR is simply

$$\text{USD5m} \times \max \left\{ \frac{S_T}{S_0} - K, 0 \right\}$$

paid out in USD. Quanto option is a natural product for investors. It does not require the investor to have a view on the FX rate changes.

The *Composite option* (aka Compo) is another type of option for the situation where the payoff currency is different from the underlying's currency. In this option, the performance from the underlying is derived from the underlying price converted into payoff currency with the prevailing FX rate. For instance, the payoff for a USD5m, USD Compo call option linked to BNP Paribas stock is

$$\text{USD5m} \times \max \left\{ \frac{S_T^{\$}}{S_0^{\$}} - K, 0 \right\}$$

paid out in USD where $S_t^{\$} = S_t X_t$ with S_t being the price of BNP Paribas stock in EUR and X_t being the EUR/USD rate at time t . As only $S_t X_t$ is used in the payoff, it can be considered as the price of a “composite stock” in payoff currency. Hence, the option price is not sensitive to the future change of the interest rate of the domestic currency of the stock.

8.3.7 Basket Option, Worst-of and Best-of Options

Denote S_t^i as the price of stock i at time t . The *basket call option* takes the form

$$\text{Payoff} = \max \left\{ \frac{1}{n} \sum_{i=1}^n \frac{S_T^i}{S_0^i} - K, 0 \right\}.$$

Because of the averaging, the basket option is smoother than a single underlying option in general.

A *worst-of* option takes the minimum performance of the basket of the underlying assets. The payoff of a worst-of call option is

$$\text{Payoff} = \max \left\{ \min_i \left\{ \frac{S_T^i}{S_0^i} \right\} - K, 0 \right\}.$$

Evidently, this option price tends to be low when the size of the basket increases or the basket has low correlation. Due to its low price, worst-of options are widely used in structured solutions which often have a tight constraint on cost.

The *best-of* option takes the maximum performance of the basket of the underlying assets. A best-of call option has the payoff like

$$\text{Payoff} = \max \left\{ \max_i \left\{ \frac{S_T^i}{S_0^i} \right\} - K, 0 \right\}.$$

A *rainbow basket* takes its performance as the weighted average of the individual performance of the underlyings according to their performance ranking. For instance, one can define a rainbow basket taking 70% of the worst, 30% of the second worst and 0% for the rest of the underlyings. Worst-of's and Best-of's are special cases of *rainbow* options.

8.4 Derivatives Modelling Framework

8.4.1 Purpose of Derivatives Modelling

In its normal course of business, the derivatives provider (i.e. the derivatives writer) will replicate the P/L of the derivatives based on pricing models. The replication (known as hedging) is a dynamic process over the whole life of the trade. For example, the seller of a call option linked to a stock will hold a certain quantity of stocks for producing the P/L of the option due to the stock price change. The quantity of stocks to be held at any time is determined by the pricing model. We list out the main purposes of modelling

- pricing for new transactions (primary market), especially for the OTC derivatives that do not have liquid market quotations
- pricing for size increase or unwinding of an existing trade (secondary market)
- mark-to-market (MtM) price calculation
- hedging related quantities which are, as we will see, the sensitivities (also called Greeks) of the derivative product
- risk reporting
- regulatory capital requirement for financial institutions

8.4.2 Model Input Parameters and Calibration

The models require specific parameters. The determination of the model parameters is called *calibration* or *fitting*. The calibration targets to minimize the difference between the prices calculated by the model and the market quoted prices for the

liquid tradable derivatives linked to the considered assets. So, the calibration is an optimization problem if there is no direct calculation of the model parameters from market quoted prices of the derivatives.⁷ If its calculation is fast, the calibration may be done “on-the-fly”, meaning that the model is re-calibrated just before each pricing.

Generally, the liquid market quoted derivatives are vanilla options of each asset class and other basic instruments such as Forward/Futures, Swaps, and CDS. The vanilla options are normally priced with closed-form solutions in the market. For a typical financial institution offering and hedging derivatives, the vanilla products traders are responsible for the market prices of the instruments which are used for calibrating the models. The exotic option traders, structurers, sales and risk controllers⁸ are the main users of the calibrated models for pricing and hedging.

Exercises

1. Choose a trading warrant (can be found on any issuer’s website) and verify how the gearing ratio and entitlement ratio are calculated.
2. Construct a portfolio of n T -maturity call options with strike prices $0 < K_1 < K_2 < \dots < K_n$ such that its payoff replicates a *piecewise linear* pattern with line segment slope m_i between break points K_i and K_{i+1} , for $i = 1, 2, \dots, n - 1$.
3. The implied volatilities for 30 days and 60 days (assuming same strike) are 15% and 12% respectively. Calculate the *forward volatility* between days 31 and 60.
4. An investor is long a call option on a stock which will expire in a month. The strike is \$100, the current stock price is \$110.
 - (i) what is the intrinsic value of the call?
 - (ii) what is the term given to the difference between the mark-to-market price and intrinsic value?
 - (iii) Is the option premium the expected value of the payoff?
5. An investor has bought a 3-month ATM European call option for 10,000 stocks. The underlying stock does not pay any dividend and the stock lending rate is zero. Its current stock price is \$100. The interest rate in the market is 2% p.a. (continuous rate, ACT/365). The implied BSM volatility of the stock is 25%.
 - (i) What is the strike price of the option?
 - (ii) Calculate the payout for the investor for the stock price at the expiry date at \$120 and \$70.

⁷Levenberg–Marquardt algorithm is the most popular optimization method used in financial modelling.

⁸Risk controllers are also responsible for supervising the correctness of the inputs from traders.

- (iii) Use the Black–Scholes formula for Call options to calculate the initial call option price with stock price of \$100 and \$101. Deduce the approximate value of the initial delta.
- (iv) Based on the above result, how many shares (i.e. delta position) should the option seller hold for initial delta-hedging?
- (v) If the share price moves up by 1%, what is the gain from the delta position? What is the gain if the option seller holds 10,000 stocks?
- (vi) Someone thinks that it is better for the option seller to hold 10,000 stocks instead of holding the delta amount for hedging as the gain will be higher if the stock price goes up. Do you agree with this hedging strategy? Why?
- (vii) Use the Put-Call parity to deduce the European put option price with the same strike and tenor from the above result.

Part II
Stochastic Calculus and Financial
Modelling

Chapter 9

Elements of Probability



In this chapter, we introduce the basic concepts and notations in probability that will be used or referenced to in the following chapters. We refer to [42] for further studies.

9.1 Probability Space, Measure and Properties

A *sample space* Ω is a nonempty set of all the possible outcomes from a random experiment. For example, the coin toss of n times has $\Omega = \{\omega : \omega = \omega_1\omega_2 \cdots \omega_n\}$ with 2^n elements and ω_i is H (head) or T (tail), $\forall i = 1, \dots, n$. A subset of Ω , $A \subseteq \Omega$, is called an *event*, say, first toss being a head.

A *partition* of Ω is defined as a collection of nonempty sets of Ω (known as *blocks* of the partition), $\{A_1, A_2, \dots, A_n\}$, which satisfies the following:

- (i) A_i are *disjoint*: $\forall i \neq j, A_i \cap A_j = \emptyset$
- (ii) $A_1 \cup A_2 \cup \cdots \cup A_n = \Omega$.

Definition 9.1 (σ -algebra) Let $\mathcal{F}$ be a collection of subsets of Ω and it is called a σ -algebra if it satisfies:

- (i) $\mathcal{F}$ is closed under complements: $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$
- (ii) $\mathcal{F}$ is closed under countable union: $\forall A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$
- (iii) the empty set $\emptyset$ is in $\mathcal{F}$: $\emptyset \in \mathcal{F}$ (hence $\Omega \in \mathcal{F}$ as $\Omega = \emptyset^c$).

It can be shown that a σ -algebra is closed under intersections and set differences: $\forall A, B \in \mathcal{F} \Rightarrow A \cap B \in \mathcal{F}$ and $A \setminus B \in \mathcal{F}$ ($A \setminus B$ is sometimes noted as $A - B$). For finite sample spaces, the notions of partition and σ -algebra are equivalent.

In probability theory, σ -algebra is used to represent a set of events. The smallest σ -algebra generated by intervals in $\mathbb{R}$ is called **Borel** σ -algebra.

Definition 9.2 (Probability) Let $\mathcal{F}$ be a σ -algebra on Ω . A *probability measure* $\mathbb{P}$ is a function such that:

- (i) $0 \leq \mathbb{P}(A) \leq 1, \forall A \in \mathcal{F}$
- (ii) $\mathbb{P}(\Omega) = 1$
- (iii) Countable additivity: $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$, for disjoint sets $A_i \in \mathcal{F}, i = 1, 2, \dots$

The triplet $(\Omega, \mathcal{F}, \mathbb{P})$ is then called a *probability space*. If $\mathbb{P}(A) = 1$, then, we say that the event A occurs $\mathbb{P}$ -almost surely (or simply, *almost surely, a.s.*).

Example 9.1 Let $\Omega = \{1, 2, \dots\}$ be the set of all natural numbers, $A_1 \subset \Omega$ the set of all even numbers, $A_2 \subset \Omega$ the set of all odd numbers, and $\mathcal{F} = \{\emptyset, \Omega, A_1, A_2\}$ is a σ -algebra. Defining $\mathbb{P}(A_1) = \mathbb{P}(A_2) = \frac{1}{2}$, then, $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space.

Here are some properties for probability measure:

- (i) $\mathbb{P}(\emptyset) = 0, \mathbb{P}(\Omega) = 1$
- (ii) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
- (iii) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
- (iv) if A and B are disjoint, then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$
- (v) if $A \subseteq B$, then $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$
- (vi) if $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Theorem 9.1 (Law of Total Probability) Let $\{B_1, B_2, \dots, B_n\}$ be a partition of Ω . Then, $\forall A \in \mathcal{F}$,

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \cap B_i).$$

9.2 Independence and Conditional Probability

Given $(\Omega, \mathcal{F}, \mathbb{P})$, the events $A \in \mathcal{F}$ and $B \in \mathcal{F}$ are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

A collection of events $\{A_1, \dots, A_n\}$ are independent if, for every sub-collection of at least two events $\{A_{i_1}, A_{i_2}, \dots, A_{i_k}\}, k = 2, \dots, n$,

$$\mathbb{P}(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = \mathbb{P}(A_{i_1})\mathbb{P}(A_{i_2}) \dots \mathbb{P}(A_{i_k}).$$

Example 9.2 Sophie has 3 coins in her pocket, 2 Euro coins and 1 Pound coin. She takes out randomly one coin, and then takes out another one *after* putting the first one back in.

Let A be the event that the first coin is Euro and B the event that the second coin is Euro. Then,

$$\mathbb{P}(A \cap B) = \frac{4}{9} = \frac{2}{3} \times \frac{2}{3} = \mathbb{P}(A)\mathbb{P}(B).$$

The *conditional probability* of event $A \in \mathcal{F}$ given event $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$ is defined as

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Properties of conditional probability are

- (i) $0 \leq \mathbb{P}(A|B) \leq 1$
- (ii) $\mathbb{P}(\Omega|B) = 1$
- (iii) $\mathbb{P}(A^c|B) = 1 - \mathbb{P}(A|B)$
- (iv) $\mathbb{P}(A_1 \cup A_2|B) = \mathbb{P}(A_1|B) + \mathbb{P}(A_2|B) - \mathbb{P}(A_1 \cap A_2|B)$
- (v) if A_1 and A_2 are disjoint, then $\mathbb{P}(A_1 \cup A_2|B) = \mathbb{P}(A_1|B) + \mathbb{P}(A_2|B)$.

Example 9.3 From the previous example, suppose Sophie takes out the second coin *without* putting the first one back in. Let A be the event that the first coin is Euro and B the event that the second coin is Euro. Then,

$$\mathbb{P}(A) = \mathbb{P}(B) = \frac{2}{3}$$

$$\mathbb{P}(A \cap B) = \frac{1}{3} \neq \mathbb{P}(A)\mathbb{P}(B).$$

Conditional probability for first getting Euro and then Euro again: $\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \frac{1/3}{2/3} = \frac{1}{2}$.

Let $\{B_1, B_2, \dots, B_n\}$ be a partition of Ω and $\mathbb{P}(B_i) > 0, i = 1, \dots, n$. The following is a revised version of Theorem 9.1.

Theorem 9.2 (Law of Total Probability) For $\forall A \in \mathcal{F}$,

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A|B_i)\mathbb{P}(B_i).$$

Theorem 9.3 (Bayes' Rule) For $\forall A \in \mathcal{F}$ and $\forall k = 1, \dots, n$,

$$\mathbb{P}(B_k|A) = \frac{\mathbb{P}(A|B_k)\mathbb{P}(B_k)}{\sum_{i=1}^n \mathbb{P}(A|B_i)\mathbb{P}(B_i)}.$$

9.3 Random Variable and Distribution

Definition 9.3 (Random Variable) Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and B an interval¹ in $\mathbb{R}$. A (real) *random variable* is a function $X(\omega)$ (or, with simplified notation X): $\Omega \rightarrow \mathbb{R}$ such that

$$\{X(\omega) \in B\} \in \mathcal{F},$$

where $\{X(\omega) \in B\}$ is the notation for $\{\omega \in \Omega; X \in B\}$.

It is customary to denote

- $\mathbb{P}\{X = x\}$ (i.e. $\mathbb{P}\{\omega \in \Omega : X(\omega) = x\}$) by $\mathbb{P}(X = x)$ or $\mathbb{P}[X = x]$
- $\mathbb{P}\{X \in B\}$ (i.e. $\mathbb{P}\{\omega \in \Omega : X(\omega) \in B\}$) by $\mathbb{P}(X \in B)$ or $\mathbb{P}[X \in B]$.

One particular random variable is the *indicator function* for $A \in \mathcal{F}$:

$$1_A(\omega) = \begin{cases} 1, & \text{if } \omega \in A, \\ 0, & \text{if } \omega \notin A. \end{cases}$$

9.3.1 Distribution

The function $F(x) = \mathbb{P}\{X \leq x\}$ for random variable X and $x \in \mathbb{R}$ is called the *cumulative distribution function (c.d.f.)* of X . It has the following properties:

- (i) $F(x)$ is nondecreasing of x
- (ii) $0 \leq F(x) \leq 1$, $\lim_{x \rightarrow -\infty} F(x) = 0$, $\lim_{x \rightarrow \infty} F(x) = 1$
- (iii) F is right continuous: $\lim_{y \downarrow x} F(y) = F(x)$.

For a continuous random variable, there is $\mathbb{P}\{a \leq X \leq b\} = F(b) - F(a)$, $\forall a \leq b$. For a discrete random variable, $\mathbb{P}\{X = x\} = F(x) - F(x_-)$.

The integrable function f is called *probability density function (p.d.f.)* of X if

$$F(a) = \int_{-\infty}^a f(x)dx, \quad \forall a \in \mathbb{R}.$$

We have $\mathbb{P}\{a \leq X \leq b\} = \int_a^b f(x)dx$. The derivative of *c.d.f.* gives back the *p.d.f.*, $F'(x) = f(x)$. Moreover, we have $\int_{-\infty}^{\infty} f(x)dx = \mathbb{P}\{X \in \mathbb{R}\} = 1$ which is the continuous case of probability adding up to 1.

¹More generally, a Borel set on the real line.

Let $(X_1, X_2) \in \mathbb{R}^2$ be a vector of two random variables. The *joint distribution function* of (X_1, X_2) is defined as

$$F(a, b) = \mathbb{P}\{X_1 \leq a, X_2 \leq b\}.$$

If (X_1, X_2) is a continuous random vector, then, the *joint probability density function* is the function $f(x_1, x_2)$ such that

$$F(a, b) = \int_{-\infty}^b \int_{-\infty}^a f(x_1, x_2) dx_1 dx_2.$$

For (X_1, X_2) with density $f(x_1, x_2)$, the *marginal cumulative distribution function* of X_1 is defined as

$$F_{X_1}(a) = \int_{-\infty}^a f_{X_1}(x_1) dx_1,$$

where $f_{X_1}(x_1) = \int_{-\infty}^{\infty} f(x_1, x_2) dx_2$ is called the *marginal density*.

More generally, let $(X_1, X_2, \dots, X_n)$ be a vector of random variables. The *joint distribution function* of $(X_1, X_2, \dots, X_n)$ is defined as

$$F(a_1, a_2, \dots, a_n) = \mathbb{P}\{X_1 \leq a_1, X_2 \leq a_2, \dots, X_n \leq a_n\}.$$

If $(X_1, X_2, \dots, X_n)$ is a continuous random vector, then, the *joint probability density function* is the function $f(x_1, x_2, \dots, x_n)$ such that

$$F(a_1, a_2, \dots, a_n) = \int_{-\infty}^{a_n} \int_{-\infty}^{a_{n-1}} \cdots \int_{-\infty}^{a_1} f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n.$$

9.3.2 Expectation

The *expectation* of a random variable is the probability weighted average.

For a discrete random variable $X \in \{x_1, x_2, \dots, x_n\}$ on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, it is defined as

$$\mathbb{E}[X] = \sum_{i=1}^n x_i \mathbb{P}\{X = x_i\} \text{ or } \mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\omega).$$

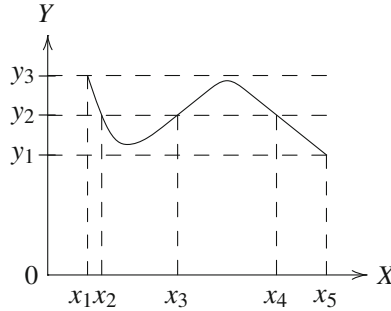
For a continuous random variable in $\mathbb{R}$ with density $f(x)$, the expectation is defined as

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} xf(x)dx.$$

Expectation of a random variable can be denoted by *Lebesgue integral*. Readers may refer to [62, 73].

For the traditional *Riemann integral* $\int_a^b f(x)dx$, x is partitioned for the interval $[a, b]$ with step size Δ_i and the integral's value is the limit $\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i)\Delta_i$. The equivalent *Lebesgue integral* consists of partitioning the value of $f(x)$ ($\dots, f_i, f_{i+1}, \dots$) in the domain $D = \{f(x), \forall x \in [a, b]\}$. The Lebesgue integral is defined as $\int_D f dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f_i \mu\{f_i\}$ where $\mu\{f_i\}$ is the *measure of f_i* which, in this case, is the sum of lengths of all the disjoint intervals $I_k^i, k = 1, 2, \dots$ such that $f_i \leq f(x) < f_{i+1}, \forall x \in I_k^i$. When both can be defined for the same problem, the two integrals will give the same value.

Example 9.4 Consider the following Lebesgue integral:



$$\int_{\mathbb{R}} y dx \approx y_1 \times [(x_3 - x_2) + (x_5 - x_4)] + y_2 \times [(x_2 - x_1) + (x_4 - x_3)].$$

Lebesgue integral is more general than the Riemann integral. When the measure is the probability measure, the Lebesgue integral gives the unified notation of expectation for a random variable which can be either discrete or continuous:

$$\mathbb{E}[X] = \int_{\Omega} X(\omega) d\mathbb{P}(\omega), \text{ or simply, } \mathbb{E}[X] = \int_{\Omega} X d\mathbb{P}.$$

In the above expression, $d\mathbb{P}(\omega)$ represents the probability measure for $X(\omega)$. For the continuous case, $d\mathbb{P}(\omega) = f(x)dx$ where $f(x)$ is the probability density function.

We have $\mathbb{E}[X\mathbf{1}_A] = \int_{\Omega} X\mathbf{1}_A d\mathbb{P} = \int_A X d\mathbb{P}$. In particular, the expectation of $\mathbf{1}_A$ can be reduced as the probability of A :

$$\mathbb{E}[\mathbf{1}_A] = \int_{\Omega} \mathbf{1}_A(\omega) d\mathbb{P}(\omega) = \int_A d\mathbb{P}(\omega) = \mathbb{P}(A).$$

9.3.3 Variance and Covariance

The *variance* of a random variable X is defined as

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2],$$

where $\sigma = \sqrt{\text{Var}(X)}$ is called the *standard deviation* of X .

Properties of variance are

- (i) $\text{Var}(c) = 0$, if c is a constant
- (ii) $\text{Var}(cX) = c^2\text{Var}(X)$, if c is a constant
- (iii) if X, Y are two *independent* random variables, then $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

The i -th moment of the random variable X is defined as $\mathbb{E}[X^i]$, $i = 1, 2, \dots$, if it exists. For instance, the second moment $\mathbb{E}[X^2]$ is involved in variance.

The *covariance* of two random variables X and Y is defined as

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

The *correlation* between X and Y is defined as

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \text{ or } \rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

and there is

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y).$$

Properties of covariance are

- (i) $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$
- (ii) $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- (iii) $\text{Cov}(aX, bY) = ab\text{Cov}(X, Y)$
- (iv) $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$

9.3.4 Independent Random Variables

The *random variables* $X_1, X_2, \dots, X_n$ are said to be independent if for Borel sets B_i ,

$$\mathbb{P}\{X_1 \in B_1, X_2 \in B_2, \dots, X_n \in B_n\} = \mathbb{P}\{X_1 \in B_1\}\mathbb{P}\{X_2 \in B_2\} \cdots \mathbb{P}\{X_n \in B_n\}.$$

For independent random variables $X_1, X_2, \dots, X_n$, the following properties hold

Joint cumulative distribution: $F_{X_1, X_2, \dots, X_n}(a_1, a_2, \dots, a_n) = F_{X_1}(a_1)F_{X_2}(a_2) \cdots F_{X_n}(a_n)$, where $F_{X_i}(a_i)$ represents the cumulative distribution function of X_i

Joint density: if $X_1, X_2, \dots, X_n$ are continuous random variables with densities $f_{X_1}, f_{X_2}, \dots, f_{X_n}$ and joint density $f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n)$, then, $f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = f_{X_1}(x_1)f_{X_2}(x_2) \cdots f_{X_n}(x_n)$

Expectation: $\mathbb{E}[X_1 X_2 \cdots X_n] = \mathbb{E}[X_1]\mathbb{E}[X_2] \cdots \mathbb{E}[X_n]$ for continuous random variables $X_1, X_2, \dots, X_n$.

9.3.5 Conditional Probability Distribution

The conditional cumulative distribution of continuous random variable X given y , $F(a|y)$ is defined by

$$F(a|y) = \mathbb{P}\{X \leq a|Y = y\} = \lim_{|Y-y| \rightarrow 0} \mathbb{P}\{X \leq a|Y \in \Delta y\}.$$

Here $Y \in \Delta y$ means the event that Y is in the small interval including y . If $f(x|y)$ is a positive function and $F(a|y) = \int_{-\infty}^a f(x|y)dx$, then, $f(x|y)$ is called the conditional probability density of X given $Y = y$. We have

$$f(x, y) = f(x|y)f(y),$$

where $f(y)$ is the marginal distribution of Y .

The conditional expectation given $Y = y$ is defined as

$$\mathbb{E}[X|Y = y] = \int_{\Omega} xf(x|y)dx.$$

For discrete random variables X and Y , the conditional expectation of X given $Y = y$ is defined as

$$\mathbb{E}[X|Y = y] = \sum_i x_i \mathbb{P}\{X = x_i|Y = y\}.$$

9.3.6 Characteristic Functions

The *characteristic function* (or *Fourier transform*) of a random variable X is defined below as, if it exists,

$$\Phi_X(t) = \mathbb{E}[e^{itX}], \quad t \in \mathbb{R}, \quad i^2 = -1.$$

For a vector of random variables $(X_1, X_2, \dots, X_n)$, the characteristic function, if it exists, is defined below for $t_k \in \mathbb{R}$, $k = 1, \dots, n$,

$$\Phi_{X_1, X_2, \dots, X_n}(t_1, t_2, \dots, t_n) = \mathbb{E}[e^{i \sum_{k=1}^n t_k X_k}].$$

The *moment-generating function* (or *Laplace transform*) of a random variable X is defined below as, if it exists,

$$\mathcal{M}_X(t) = \mathbb{E}[e^{tX}], \quad t \in \mathbb{R}.$$

For a vector of random variables $(X_1, X_2, \dots, X_n)$, the moment-generating function, if it exists, is defined below for $t_k \in \mathbb{R}$, $k = 1, \dots, n$,

$$\mathcal{M}_{X_1, X_2, \dots, X_n}(t_1, t_2, \dots, t_n) = \mathbb{E}[e^{\sum_{k=1}^n t_k X_k}].$$

As $\frac{d^n}{dt^n} \mathbb{E}[e^{tX}] \Big|_{t=0} = \mathbb{E}[X^n e^{tX}] \Big|_{t=0} = \mathbb{E}[X^n]$, the moment-generating function can be used to calculate the n -th moment.

9.3.7 Normal Distribution

A Standard Normal Distribution $X \sim \mathcal{N}(0, 1)$ satisfies

- probability density function: $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$
- cumulative distribution function: $N(a) = \int_{-\infty}^a \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$

See Fig. 9.1 for reference. We have the important property of $N(a) + N(-a) = 1$. Also $N(-\infty) = 0$, $N(0) = \frac{1}{2}$, and $N(\infty) = 1$.

The probability density function $f(x)$ is the famous “bell curve”. The mean or expected value of X is $\mathbb{E}[X] = 0$, and the variance of X is $\text{Var}(X) = 1$.

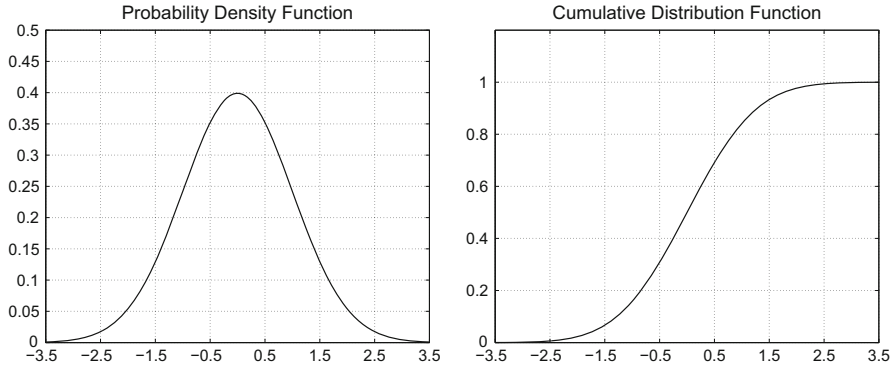


Fig. 9.1 The probability density function (left) and cumulative distribution function (right) for a standard normal random variable

The moment generating function is

$$\begin{aligned}\mathcal{M}_X(t) &= \mathbb{E}[e^{tX}] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{x^2}{2}} dx \\ &= e^{\frac{t^2}{2}} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x-t)^2} dx = e^{\frac{t^2}{2}} N(\infty) = e^{\frac{t^2}{2}}.\end{aligned}$$

It generates moments of X because its derivatives at 0 give

$$\begin{aligned}\mathcal{M}_X^{(1)} &= te^{\frac{1}{2}t^2} && \longrightarrow \mathbb{E}[X] = \mathcal{M}_X^{(1)}(0) = 0 \\ \mathcal{M}_X^{(2)} &= (1 + t^2)e^{\frac{1}{2}t^2} && \longrightarrow \mathbb{E}[X^2] = \mathcal{M}_X^{(2)}(0) = 1 \\ \mathcal{M}_X^{(3)} &= (3t + t^3)e^{\frac{1}{2}t^2} && \longrightarrow \mathbb{E}[X^3] = \mathcal{M}_X^{(3)}(0) = 0 \\ \mathcal{M}_X^{(4)} &= (3 + 6t^2 + t^4)e^{\frac{1}{2}t^2} && \longrightarrow \mathbb{E}[X^4] = \mathcal{M}_X^{(4)}(0) = 3\end{aligned}$$

and etc. In general, the n -th moment can be directly calculated [62]

$$\mathbb{E}[X^n] = \begin{cases} 0, & \text{when } n \text{ is odd,} \\ 1 \cdot 3 \cdots (n-1), & \text{when } n \text{ is even.} \end{cases} \quad (9.1)$$

A General Normal Distribution with mean μ and variance σ^2 , denoted by $Y \sim \mathcal{N}(\mu, \sigma^2)$, satisfies

- probability density function: $f(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$, $-\infty < y < \infty$
- cumulative distribution function: $F(a) = \int_{-\infty}^a \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy$.

Note that

$$X \equiv \frac{Y - \mu}{\sigma} \sim \mathcal{N}(0, 1),$$

or equivalently $Y = \mu + \sigma X$. Then, the moment-generating function can be directly obtained:

$$\mathbb{E}[e^{tY}] = \mathbb{E}[e^{t(\mu + \sigma X)}] = e^{\mu t} \mathbb{E}[e^{(t\sigma)X}] = e^{\mu t + \frac{1}{2}\sigma^2 t^2}.$$

And by (9.1),

$$\mathbb{E}[(Y - \mu)^n] = \sigma^n \mathbb{E}[X^n] = \begin{cases} 0, & \text{when } n \text{ is odd,} \\ \sigma^n 1 \cdot 3 \cdots (n-1), & \text{when } n \text{ is even.} \end{cases}$$

For $0 < \alpha < 1/2$, the 100α percentage point X_α of the standard normal distribution is defined as the number such that

$$\alpha = \mathbb{P}\{X > X_\alpha\} = \frac{1}{\sqrt{2\pi}} \int_{X_\alpha}^{\infty} e^{-\frac{x^2}{2}} dx.$$

Referring to a statistical table, we have for examples, $X_{0.050} = 1.6449$ and $X_{0.025} = 1.960$. More precisely, if $X \sim \mathcal{N}(0, 1)$, then within 95% confidence, we have $-1.96 \leq X \leq 1.96$. If $Y \sim \mathcal{N}(\mu, \sigma^2)$, we have

$$\mu - 1.96\sigma \leq Y \leq \mu + 1.96\sigma.$$

A vector-valued random variable $(X_1, X_2, \dots, X_n)$ is said to follow a *Multi-variate Normal Distribution* if every linear combination of X_i follows a Normal distribution.

For Bivariate Normal Distribution $(X_1, X_2) \sim \mathcal{N}(\mu_1, \mu_2; \sigma_1, \sigma_2; \rho)$, where ρ is the correlation,

- Probability density function $f(x_1, x_2)$ equals to

$$\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\frac{(x_1 - \mu_1)^2}{\sigma_1^2} - 2\rho \frac{(x_1 - \mu_1)(x_2 - \mu_2)}{\sigma_1\sigma_2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} \right] \right\}$$

- Cumulative distribution function:

$$F(a, b) = \int_{-\infty}^b \int_{-\infty}^a f(x_1, x_2) dx_1 dx_2.$$

Probability density function of multivariate Normal distribution is

$$f(x_1, x_2, \dots, x_n) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right),$$

where $\mathbf{x} = [x_1, x_2, \dots, x_n]^\top$ is a real n -dimensional column vector, $\boldsymbol{\mu}$ is the mean vector, and $|\Sigma|$ is the determinant of Σ , the $n \times n$ covariance matrix which is symmetric and positive definite.

9.3.8 Exponential Distribution

An exponential random variable τ ($\tau > 0$) follows the density below:

$$f(t) = \lambda e^{-\lambda t},$$

where $\lambda > 0$ and it is known as *intensity*.

Properties of exponential distribution for $\tau > 0$ and $0 < s < t$ are

Distribution: $\mathbb{P}\{\tau \leq t\} = \int_0^t f(s) ds = 1 - e^{-\lambda t}$ and $\mathbb{P}\{\tau > t\} = e^{-\lambda t}$

Expectation: $\mathbb{E}[\tau] = \int_0^\infty t f(t) dt = \frac{1}{\lambda}$

Variance: $\text{Var}(\tau) = \frac{1}{\lambda^2}$

Conditional distribution: $\mathbb{P}\{\tau > t + s | \tau > s\} = \frac{\mathbb{P}\{\tau > t+s\}}{\mathbb{P}\{\tau > s\}} = e^{-\lambda t} = \mathbb{P}\{\tau > t\}$
 (“Memorylessness property”).

9.3.9 Poisson Distribution

The Poisson random variable N is a discrete variable for counting the number of random events within a time interval. Here are some properties of Poisson distribution for the event number, an integer $k \geq 0$ and an event rate $\lambda > 0$:

Distribution: $\mathbb{P}\{N = k\} = \frac{\lambda^k}{k!} e^{-\lambda}$

Expectation: $\mathbb{E}[N] = \lambda$

Variance: $\text{Var}(N) = \lambda$.

9.4 Limit Theorems

9.4.1 Law of Large Numbers

Let $X_1, X_2, \dots$ be an infinite sequence of i.i.d. (independent and identically distributed) Lebesgue integrable random variables with expected value $\mathbb{E}[X_1] = \mathbb{E}[X_2] = \dots = \mu$. Then,

$$\frac{1}{n}(X_1 + X_2 + \dots + X_n) \xrightarrow{n \rightarrow \infty} \mu.$$

9.4.2 Central Limit Theorem

Let $X_1, X_2, \dots$ be an infinite sequence of i.i.d. random variables with finite mean μ and variance σ^2 . Denote a new sequence of random variables as

$$S_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n}\sigma}.$$

Then, S_n converges to a standard Normal random variable by

$$\mathbb{P}\{S_n \leq a\} \xrightarrow{n \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^a e^{-\frac{x^2}{2}} dx.$$

Exercises

1. Consider $X \sim \mathcal{N}(0, 2)$. Obtain the mean, variance, and probability density function of $Y = 3 + 2X$.
2. Assume $Z \sim \mathcal{N}(0, 1)$. Prove that Z and Z^2 are uncorrelated but dependent.
3. Assume $Z \sim \mathcal{N}(0, 1)$. Show that

$$\mathbb{E}[e^{-\frac{\sigma^2}{2} + \sigma Z} \mathbf{1}_{\{Z < x\}}] = N(x - \sigma),$$

where x and σ are real variables.

4. Use the RAND function to generate a number of samples in EXCEL to verify the mean of standard uniform distribution on $[0, 1]$ (equal probability for intervals of the same length) according to the law of large numbers.

Chapter 10

Stochastic Calculus Part I



10.1 Stochastic Process

The σ -algebra generated by a random variable X , denoted $\sigma(X)$, is the σ -algebra by the collection of $\{X \in B\}$ (i.e. $\{\omega : X(\omega) \in B\}$), where B is any interval in $\mathbb{R}$. Let $\mathcal{G}$ be a σ -algebra on Ω . Then X is said to be $\mathcal{G}$ -measurable if $\sigma(X) \subseteq \mathcal{G}$.

Definition 10.1 (Filtration) Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a *filtration* $\mathcal{F}_t$ is an increasing collection of σ -algebras on Ω , such that for s, t in a totally ordered set $\mathcal{T}$ (e.g. “time”),

$$\mathcal{F}_s \subseteq \mathcal{F}_t, \quad s < t.$$

Definition 10.2 (Stochastic Process) A *stochastic process* is a collection of random variables indexed by $\mathcal{T}$:

$$\{X_t : t \in \mathcal{T}\}.$$

Here X_t is short for $X_t(\omega)$. For each t , X_t is reduced as a random variable. When there is no randomness, X_t is called *deterministic*.

A stochastic process X_t on $\mathcal{T}$ is said to be *adapted to the filtration* if, for every $t \in \mathcal{T}$, X_t is $\mathcal{F}_t$ -measurable. Given a stochastic process $X = \{X_t : t \in \mathcal{T}\}$, the **natural filtration** for this process is the filtration where $\mathcal{F}_t$ is generated by all values of X_s up to $s = t$. A stochastic process is always adapted to its natural filtration.

Example 10.1 (Asset Price Example) Assume a simplistic multi-period asset price model where the price has only two possible movements for each period driven by a single economic factor with 2 status (“y” – yes or “n” – no):

- “y” drives the price up by a certain amount with probability p , or

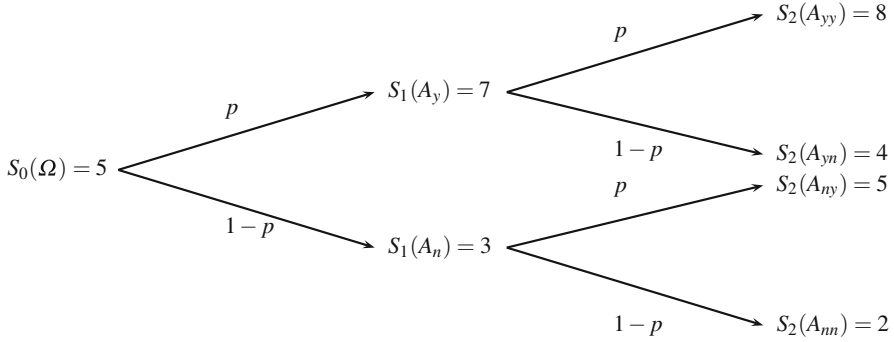


Fig. 10.1 A 2-period stock price model

- “n” drives the price down by a certain amount with probability q ($q = 1 - p$).

Consider a 2-period stock price model S_t , $t = 0, 1, 2$, represented by the tree in Fig. 10.1, where $p = 0.5$ and $q = 1 - p = 0.5$.

The sample space of S_2 is $\Omega = \{yy, yn, ny, nn\}$. The events on the tree are

$A_{yy} = \{yy\}$, $A_{yn} = \{yn\}$, $A_{ny} = \{ny\}$, $A_{nn} = \{nn\}$ (these are called “atom events”)
 $A_y = \{yy, yn\}$, $A_n = \{ny, nn\}$

We have $A_y = A_n^c$, $A_n = A_y^c$, $A_y = A_{yy} \cup A_{yn}$, $A_n = A_{ny} \cup A_{nn}$, $\Omega = A_y \cup A_n$. Probabilities for the events are

$$\mathbb{P}(A_{yy}) = 0.25, \mathbb{P}(A_{yn}) = 0.25, \mathbb{P}(A_{ny}) = 0.25, \mathbb{P}(A_{nn}) = 0.25$$

$$\mathbb{P}(A_y) = 0.5, \mathbb{P}(A_n) = 0.5$$

$$\mathbb{P}(\Omega) = 1.$$

And the Filtrations for each period are

$$\mathcal{F}_0 = \{\emptyset, \Omega\}$$

$$\mathcal{F}_1 = \{\emptyset, \Omega, A_y, A_n\}$$

$$\mathcal{F}_2 = \{\emptyset, \Omega, A_y, A_n, A_{yy}, A_{yn}, A_{ny}, A_{nn}, A_{yy}^c, A_{yn}^c, A_{ny}^c, A_{nn}^c, A_{yy} \cup A_{ny}, A_{yy} \cup A_{nn}, A_{yn} \cup A_{ny}, A_{yn} \cup A_{nn}\}.$$

10.2 Conditional Expectation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and B an event. The *conditional expectation of random variable X given event B* is defined as

$$\mathbb{E}[X|B] = \frac{1}{\mathbb{P}(B)} \int_B X(\omega) d\mathbb{P}(\omega).$$

If X is a discrete random variable, the above expression can be also written as:

$$\mathbb{E}[X|B] = \frac{1}{\mathbb{P}(B)} \sum_i x_i \mathbb{P}(\{X = x_i\} \cap B).$$

The *conditional expectation given σ -algebra $\mathcal{G}$* is the $\mathcal{G}$ -measurable random variable, denoted as $\mathbb{E}[X|\mathcal{G}]$, which satisfies

$$\int_B \mathbb{E}[X|\mathcal{G}](\omega) d\mathbb{P}(\omega) = \int_B X(\omega) d\mathbb{P}(\omega), \quad \forall B \in \mathcal{G}.$$

Note that $\mathbb{E}[X|\mathcal{G}]$ is not a single value in general. It is a variable representing the “partial averaging” for each $B \in \mathcal{G}$. It is a generalization of the single valued conditional expectation with a single event.

For illustrating the concept, we use the following less general definition of conditional expectation. Under probability space $(\Omega, \mathcal{F}, \mathbb{P})$, let $\{B_1, B_2, \dots\}$, $\forall i, B_i \in \mathcal{F}$, be a countable partition of Ω and $\mathcal{B} = \sigma(B_i, i = 1, 2, \dots)$, the σ -algebra generated by all B_i . The conditional expectation of random variable X conditional by $\mathcal{B}$ is defined as

$$\mathbb{E}[X|\mathcal{B}] = \sum_i \mathbb{E}[X|B_i] \mathbf{1}_{B_i}.$$

Example 10.2 (Conditional Expectation) In the 2-period asset price example, we have the following conditional expectation representations on the 3 σ -algebras $\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2$:

$$\mathbb{E}[S_2|\mathcal{F}_2] = 8 \cdot \mathbf{1}_{A_{yy}} + 4 \cdot \mathbf{1}_{A_{yn}} + 5 \cdot \mathbf{1}_{A_{ny}} + 2 \cdot \mathbf{1}_{A_{nn}} = S_2$$

$$\begin{aligned} \mathbb{E}[S_2|\mathcal{F}_1] &= \frac{8\mathbb{P}(A_{yy} \cap A_y) + 4\mathbb{P}(A_{yn} \cap A_y) + 5\mathbb{P}(A_{ny} \cap A_y) + 2\mathbb{P}(A_{nn} \cap A_y)}{\mathbb{P}(A_y)} \mathbf{1}_{A_y} \\ &\quad + \frac{8\mathbb{P}(A_{yy} \cap A_n) + 4\mathbb{P}(A_{yn} \cap A_n) + 5\mathbb{P}(A_{ny} \cap A_n) + 2\mathbb{P}(A_{nn} \cap A_n)}{\mathbb{P}(A_n)} \mathbf{1}_{A_n} \\ &= \frac{8\mathbb{P}(A_{yy}) + 4\mathbb{P}(A_{yn})}{\mathbb{P}(A_y)} \mathbf{1}_{A_y} + \frac{5\mathbb{P}(A_{ny}) + 2\mathbb{P}(A_{nn})}{\mathbb{P}(A_n)} \mathbf{1}_{A_n} = 6 \cdot \mathbf{1}_{A_y} + 3.5 \cdot \mathbf{1}_{A_n} \end{aligned}$$

$$\begin{aligned} \mathbb{E}[S_2|\mathcal{F}_0] &= \frac{8\mathbb{P}(A_{yy} \cap \Omega) + 4\mathbb{P}(A_{yn} \cap \Omega) + 5\mathbb{P}(A_{ny} \cap \Omega) + 2\mathbb{P}(A_{nn} \cap \Omega)}{\mathbb{P}(\Omega)} \mathbf{1}_{\Omega} \\ &= [8\mathbb{P}(A_{yy}) + 4\mathbb{P}(A_{yn}) + 5\mathbb{P}(A_{ny}) + 2\mathbb{P}(A_{nn})] \mathbf{1}_{\Omega} \\ &= \mathbb{E}[S_2] \mathbf{1}_{\Omega} = \mathbb{E}[S_2] = 4.75. \end{aligned}$$

Let $\mathcal{G}, \mathcal{H}$ be two σ -algebras and $\mathcal{G} \subset \mathcal{H}$. Then, we have the following properties:

- (1) **Independence:** If X is independent of $\mathcal{G}$, then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$
- (2) **Law of Total Expectation:** $\mathbb{E}[\mathbb{E}[X|\mathcal{G}]] = \mathbb{E}[X]$
- (3) **Taking out Known Factors:** If X is $\mathcal{G}$ -measurable, then $\mathbb{E}[XY|\mathcal{G}] = X\mathbb{E}[Y|\mathcal{G}]$
- (4) **Tower Property** (also known as **Iterated Conditioning**):
 - a. $\mathbb{E}[\mathbb{E}[X|\mathcal{H}|\mathcal{G}] = \mathbb{E}[X|\mathcal{G}]$
 - b. $\mathbb{E}[\mathbb{E}[X|\mathcal{G}|\mathcal{H}] = \mathbb{E}[X|\mathcal{G}]$

Illustration with Asset Price Example for (1) “Independence” Let $\mathcal{G}$ be a σ -algebra which is independent of S_2 . Assuming that $\{B_1, B_2\}$ is a partition of $\mathcal{G}$ (hence $\mathbf{1}_{B_1} + \mathbf{1}_{B_2} = 1$), then

$$\mathbb{E}[S_2|\mathcal{G}] = \mathbb{E}[S_2|B_1]\mathbf{1}_{B_1} + \mathbb{E}[S_2|B_2]\mathbf{1}_{B_2}.$$

The first term is

$$\begin{aligned} \mathbb{E}[S_2|B_1] &= \frac{8\mathbb{P}(A_{yy} \cap B_1) + 4\mathbb{P}(A_{yn} \cap B_1) + 5\mathbb{P}(A_{ny} \cap B_1) + 2\mathbb{P}(A_{nn} \cap B_1)}{\mathbb{P}(B_1)} \\ &= \frac{8\mathbb{P}(A_{yy})\mathbb{P}(B_1) + 4\mathbb{P}(A_{yn})\mathbb{P}(B_1) + 5\mathbb{P}(A_{ny})\mathbb{P}(B_1) + 2\mathbb{P}(A_{nn})\mathbb{P}(B_1)}{\mathbb{P}(B_1)} \\ &= 8\mathbb{P}(A_{yy}) + 4\mathbb{P}(A_{yn}) + 5\mathbb{P}(A_{ny}) + 2\mathbb{P}(A_{nn}) = \mathbb{E}[S_2]. \end{aligned}$$

We obtain the similar result $\mathbb{E}[S_2|B_2] = \mathbb{E}[S_2]$ in the second term. Hence,

$$\mathbb{E}[S_2|\mathcal{G}] = \mathbb{E}[S_2]\mathbf{1}_{B_1} + \mathbb{E}[S_2]\mathbf{1}_{B_2} = \mathbb{E}[S_2](\mathbf{1}_{B_1} + \mathbf{1}_{B_2}) = \mathbb{E}[S_2].$$

Illustration with Asset Price Example for (2) “Law of Total Expectation”

$$\begin{aligned} \mathbb{E}[S_2|\mathcal{F}_1] &= \frac{8\mathbb{P}(A_{yy}) + 4\mathbb{P}(A_{yn})}{\mathbb{P}(A_y)}\mathbf{1}_{A_y} + \frac{5\mathbb{P}(A_{ny}) + 2\mathbb{P}(A_{nn})}{\mathbb{P}(A_n)}\mathbf{1}_{A_n} \\ \mathbb{E}[\mathbb{E}[S_2|\mathcal{F}_1]] &= \frac{8\mathbb{P}(A_{yy}) + 4\mathbb{P}(A_{yn})}{\mathbb{P}(A_y)}\mathbb{E}[\mathbf{1}_{A_y}] + \frac{5\mathbb{P}(A_{ny}) + 2\mathbb{P}(A_{nn})}{\mathbb{P}(A_n)}\mathbb{E}[\mathbf{1}_{A_n}] \\ &= \frac{8\mathbb{P}(A_{yy}) + 4\mathbb{P}(A_{yn})}{\mathbb{P}(A_y)}\mathbb{P}(A_y) + \frac{5\mathbb{P}(A_{ny}) + 2\mathbb{P}(A_{nn})}{\mathbb{P}(A_n)}\mathbb{P}(A_n) \\ &= 8\mathbb{P}(A_{yy}) + 4\mathbb{P}(A_{yn}) + 5\mathbb{P}(A_{ny}) + 2\mathbb{P}(A_{nn}) = \mathbb{E}[S_2]. \end{aligned}$$

Illustration with Asset Price Example for (3) “Taking out Known Factors”

Consider $\mathbb{E}[S_1 S_2|\mathcal{F}_1]$ where $S_1 = 7 \cdot \mathbf{1}_{A_y} + 3 \cdot \mathbf{1}_{A_n}$ is $\mathcal{F}_1$ -measurable.

$$\begin{aligned}
& \mathbb{E}[S_1 S_2 | \mathcal{F}_1] \\
&= \frac{7 \cdot 8\mathbb{P}(A_{yy}) + 7 \cdot 4\mathbb{P}(A_{yn})}{\mathbb{P}(A_y)} \mathbf{1}_{A_y} + \frac{3 \cdot 5\mathbb{P}(A_{ny}) + 3 \cdot 2\mathbb{P}(A_{nn})}{\mathbb{P}(A_n)} \mathbf{1}_{A_n} \\
&= (7 \cdot \mathbf{1}_{A_y} + 3 \cdot \mathbf{1}_{A_n}) \left[\frac{8\mathbb{P}(A_{yy}) + 4\mathbb{P}(A_{yn})}{\mathbb{P}(A_y)} \mathbf{1}_{A_y} + \frac{5\mathbb{P}(A_{ny}) + 2\mathbb{P}(A_{nn})}{\mathbb{P}(A_n)} \mathbf{1}_{A_n} \right] \\
&= (7 \cdot \mathbf{1}_{A_y} + 3 \cdot \mathbf{1}_{A_n}) \mathbb{E}[S_2 | \mathcal{F}_1] = S_1 \mathbb{E}[S_2 | \mathcal{F}_1].
\end{aligned}$$

Illustration with Asset Price Example for (4.a) “Tower Property”

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[S_2 | \mathcal{F}_2] | \mathcal{F}_1] &= \mathbb{E}[8 \cdot \mathbf{1}_{A_{yy}} + 4 \cdot \mathbf{1}_{A_{yn}} + 5 \cdot \mathbf{1}_{A_{ny}} + 2 \cdot \mathbf{1}_{A_{nn}} | \mathcal{F}_1] \\
&= 8\mathbb{E}[\mathbf{1}_{A_{yy}} | \mathcal{F}_1] + 4\mathbb{E}[\mathbf{1}_{A_{yn}} | \mathcal{F}_1] + 5\mathbb{E}[\mathbf{1}_{A_{ny}} | \mathcal{F}_1] + 2\mathbb{E}[\mathbf{1}_{A_{nn}} | \mathcal{F}_1].
\end{aligned}$$

The first term is

$$8\mathbb{E}[\mathbf{1}_{A_{yy}} | \mathcal{F}_1] = 8 \frac{\mathbb{P}(A_{yy} \cap A_y)}{\mathbb{P}(A_y)} \mathbf{1}_{A_y} + 8 \frac{\mathbb{P}(A_{yy} \cap A_n)}{\mathbb{P}(A_n)} \mathbf{1}_{A_n} = 8 \frac{\mathbb{P}(A_{yy})}{\mathbb{P}(A_y)} \mathbf{1}_{A_y}.$$

We obtain similar results for the other terms. Hence,

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[S_2 | \mathcal{F}_2] | \mathcal{F}_1] &= 8 \frac{\mathbb{P}(A_{yy})}{\mathbb{P}(A_y)} \mathbf{1}_{A_y} + 4 \frac{\mathbb{P}(A_{yn})}{\mathbb{P}(A_y)} \mathbf{1}_{A_y} + 5 \frac{\mathbb{P}(A_{ny})}{\mathbb{P}(A_n)} \mathbf{1}_{A_n} + 2 \frac{\mathbb{P}(A_{nn})}{\mathbb{P}(A_n)} \mathbf{1}_{A_n} \\
&= \frac{8\mathbb{P}(A_{yy}) + 4\mathbb{P}(A_{yn})}{\mathbb{P}(A_y)} \mathbf{1}_{A_y} + \frac{5\mathbb{P}(A_{ny}) + 2\mathbb{P}(A_{nn})}{\mathbb{P}(A_n)} \mathbf{1}_{A_n} \\
&= 6 \cdot \mathbf{1}_{A_y} + 3.5 \cdot \mathbf{1}_{A_n} = \mathbb{E}[S_2 | \mathcal{F}_1].
\end{aligned}$$

Illustration with Asset Price Example for (4.b) “Tower Property”

$$\mathbb{E}[\mathbb{E}[S_2 | \mathcal{F}_1] | \mathcal{F}_2] = \mathbb{E}[6 \cdot \mathbf{1}_{A_y} + 3.5 \cdot \mathbf{1}_{A_n} | \mathcal{F}_2] = 6\mathbb{E}[\mathbf{1}_{A_y} | \mathcal{F}_2] + 3.5\mathbb{E}[\mathbf{1}_{A_n} | \mathcal{F}_2].$$

The first term is

$$\begin{aligned}
6\mathbb{E}[\mathbf{1}_{A_y} | \mathcal{F}_2] &= 6 \left[\frac{\mathbb{P}(A_y \cap A_{yy})}{\mathbb{P}(A_{yy})} \mathbf{1}_{A_{yy}} + \frac{\mathbb{P}(A_y \cap A_{yn})}{\mathbb{P}(A_{yn})} \mathbf{1}_{A_{yn}} + \frac{\mathbb{P}(A_y \cap A_{ny})}{\mathbb{P}(A_{ny})} \mathbf{1}_{A_{ny}} \right. \\
&\quad \left. + \frac{\mathbb{P}(A_y \cap A_{nn})}{\mathbb{P}(A_{nn})} \mathbf{1}_{A_{nn}} \right] \\
&= 6 \left[\frac{\mathbb{P}(A_{yy})}{\mathbb{P}(A_{yy})} \mathbf{1}_{A_{yy}} + \frac{\mathbb{P}(A_{yn})}{\mathbb{P}(A_{yn})} \mathbf{1}_{A_{yn}} \right] = 6 \cdot \mathbf{1}_{A_y}.
\end{aligned}$$

Similarly, we obtain $3.5\mathbb{E}[\mathbf{1}_{A_n} | \mathcal{F}_2] = 3.5 \cdot \mathbf{1}_{A_n}$. Hence,

$$\mathbb{E}[\mathbb{E}[S_2 | \mathcal{F}_1] | \mathcal{F}_2] = 6 \cdot \mathbf{1}_{A_y} + 3.5 \cdot \mathbf{1}_{A_n} = \mathbb{E}[S_2 | \mathcal{F}_1].$$

10.3 Martingale

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with the filtration $\mathcal{F}_t$. Let $0 \leq s \leq t$. An $\mathcal{F}_t$ -adapted stochastic process M_t is said to be

- a *martingale*, if $\mathbb{E}[M_t | \mathcal{F}_s] = M_s$
- a *supermartingale*, if $\mathbb{E}[M_t | \mathcal{F}_s] \leq M_s$
- a *submartingale*, if $\mathbb{E}[M_t | \mathcal{F}_s] \geq M_s$.

Supermartingales tend to fall and submartingales tend to rise. Processes that are both a supermartingale and a submartingale are martingales, thus they have no tendency to rise or fall. In particular, the expected value of a martingale M_t is its initial value M_0 , i.e., $\mathbb{E}[M_t] = M_0$.

Theorem 10.1 (Doob–Meyer Decomposition) *A supermartingale (submartingale) X_t admits a unique decomposition: $X_t = Y_t - A_t$ ($= Y_t + A_t$) where Y_t is a martingale and A_t is an increasing predictable¹ process.*

Proof can be found in [66].

10.4 Markov Property

The Markov property ensures the memorylessness of a process, i.e., the past does not matter when predicting the future, only the present matters. It is defined as follows.

An $\mathcal{F}_t$ -measurable stochastic process X_t is said to be a *Markov process* if, for any function² f ,

$$\mathbb{E}[f(X_t) | \mathcal{F}_s] = \mathbb{E}[f(X_t) | X_s] = \varphi(s, X_s), \quad \forall t > s.$$

A continuous Markov process is called a *diffusion*.

Proposition 10.1 *The stochastic process $\varphi(s, X_s)$ is a martingale.*

Proof Let $0 \leq s \leq t \leq T$, then

$$\mathbb{E}[\varphi(t, X_t) | \mathcal{F}_s] = \mathbb{E}[\mathbb{E}[f(X_T) | \mathcal{F}_t] | \mathcal{F}_s] = \mathbb{E}[f(X_T) | \mathcal{F}_s] = \varphi(s, X_s). \quad \square$$

10.5 Quadratic Variation

The *quadratic variation* of X_t is defined as, if it exists,

$$\langle X, X \rangle_t = \lim_{\|P\| \rightarrow 0} \sum_i (X_{s_{i+1}} - X_{s_i})^2,$$

¹Left continuous and $\mathcal{F}_t$ -adapted.

²To be specific, any Borel measurable function.

where $P = \{s_0, s_1, \dots, s_n\}$ represents a partition of the interval $[0, t]$ and $\|P\|$ is the maximum step size. The quadratic variation is also denoted as $\langle X \rangle_t$.

Remark 10.1 If X_t has a continuous derivative X'_t and $\int_0^t |X'_s|^2 ds$ is finite, then $\langle X, X \rangle_t = 0$ which is illustrated in the below:

$$\begin{aligned} \langle X, X \rangle_t &= \lim_{\|P\| \rightarrow 0} \sum_i (X_{s_{i+1}} - X_{s_i})^2 \\ &= \lim_{\|P\| \rightarrow 0} \sum_i [X'_{s_i^*} (s_{i+1} - s_i)]^2, s_i^* \in [s_i, s_{i+1}] \quad (\text{by Mean Value Theorem}) \\ &\leq \lim_{\|P\| \rightarrow 0} \|P\| \sum_i |X'_{s_i^*}|^2 (s_{i+1} - s_i) = \lim_{\|P\| \rightarrow 0} \|P\| \cdot \int_0^t |X'_s|^2 ds = 0. \end{aligned}$$

The last equality holds as $\int_0^t |X'_s|^2 ds$ is finite.

The *cross variation* (or *quadratic covariation*) of two processes, X_t and Y_t , is defined as, if it exists,

$$\langle X, Y \rangle_t = \lim_{\|P\| \rightarrow 0} \sum_i (X_{s_{i+1}} - X_{s_i})(Y_{s_{i+1}} - Y_{s_i}).$$

The differential of $\langle X, Y \rangle_t$ is denoted by $d\langle X, Y \rangle_t$ or $dX_t dY_t$ in this book.

10.6 Brownian Motion

Definition 10.3 (Brownian Motion) A continuous process $W_t(\omega)$ is said to be a *standard Brownian Motion* if it satisfies

- (1) $W_0(\omega) = 0$ for all ω
- (2) for all ω , the map $t \mapsto W_t(\omega)$ is a continuous function for $t \geq 0$
- (3) W_t has independent increments, i.e. $W_u(\omega) - W_v(\omega)$ and $W_t(\omega) - W_s(\omega)$ are independent for all $0 \leq v \leq u \leq s \leq t$
- (4) W_t has stationary increments and every increment $W_t(\omega) - W_s(\omega)$, $0 \leq s < t$, follows the Normal distribution $\mathcal{N}(0, t - s)$.

For simplicity, we will write $W_t(\omega)$ as W_t . We have

$$\mathbb{E}[W_t - W_s] = 0$$

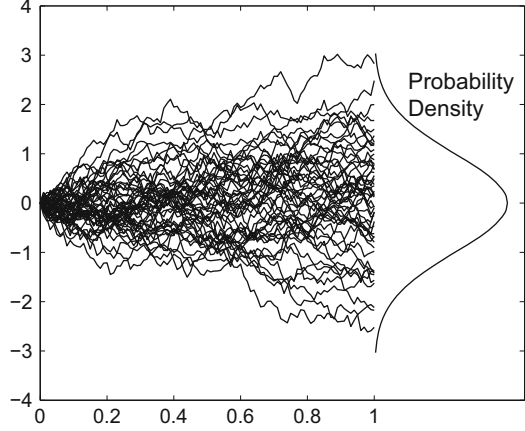
$$\text{Var}(W_t - W_s) = t - s$$

$$W_t = \epsilon \sqrt{t} \text{ where } \epsilon \sim \mathcal{N}(0, 1) \text{ (because } W_t = W_t - W_0).$$

We can see that W_t is not differentiable because

$$\lim_{\delta t \rightarrow 0} \frac{W_{t+\delta t} - W_t}{\delta t} = \lim_{\delta t \rightarrow 0} \frac{\epsilon \sqrt{\delta t}}{\delta t} = \frac{\epsilon}{\lim_{\delta t \rightarrow 0} \sqrt{\delta t}} \rightarrow \infty.$$

Fig. 10.2 Fifty Brownian paths with probability density at time 1



See Fig. 10.2 for 50 sample paths of Brownian motion.

A d -dimensional Brownian motion is defined as

$$\mathbf{W}_t = (W_t^1, W_t^2, \dots, W_t^d),$$

where $W_t^i, i = 1, \dots, d$, is a standard Brownian motion and W_t^i and W_t^j ($\forall i \neq j$) are independent.

Properties of a Brownian Motion

Independence of Brownian increments implies that $W_t - W_s$ ($s < t$) is independent of $\mathcal{F}_s = \sigma(W_u, u \leq s)$. Hence, we have

$$\mathbb{E}[f(W_t - W_s) | \mathcal{F}_s] = \mathbb{E}[f(W_t - W_s)].$$

An $\mathcal{F}_t$ -adapted Brownian Motion is called $\mathcal{F}_t$ -Brownian Motion.

Theorem 10.2 (Properties of $\mathcal{F}_t$ -Brownian Motion)

- (1) W_t is an $\mathcal{F}_t$ -martingale
- (2) $W_t^2 - t$ is an $\mathcal{F}_t$ -martingale
- (3) $e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}$ is an $\mathcal{F}_t$ -martingale
- (4) time-change or scaling property: $cW_t \stackrel{(d)}{\Longleftrightarrow} W_{c^2 t}$ where the arrow with (d) means equivalence in terms of distribution.

Proof In the following, we assume that $u \geq t$.

- (1) $\mathbb{E}[W_u - W_t | \mathcal{F}_t] = \mathbb{E}[W_u - W_t] = 0$. Hence $\mathbb{E}[W_u | \mathcal{F}_t] = \mathbb{E}[W_t | \mathcal{F}_t] = W_t$ (as W_t is $\mathcal{F}_t$ -measurable).

- (2) $\mathbb{E}[W_u^2 - W_t^2 | \mathcal{F}_t] = \mathbb{E}[(W_u - W_t)^2 + 2W_t(W_u - W_t) | \mathcal{F}_t] = \mathbb{E}[(W_u - W_t)^2 | \mathcal{F}_t] = u - t$. Hence $\mathbb{E}[W_u^2 - u | \mathcal{F}_t] = \mathbb{E}[W_t^2 | \mathcal{F}_t] - t = W_t^2 - t$ (because W_t^2 is $\mathcal{F}_t$ -measurable).
- (3) $\mathbb{E}[e^{-\frac{1}{2}\sigma^2 u + \sigma W_u} | \mathcal{F}_t] = e^{-\frac{1}{2}\sigma^2 u + \sigma W_t} \mathbb{E}[e^{\sigma(W_u - W_t)} | \mathcal{F}_t] = e^{-\frac{1}{2}\sigma^2 u + \sigma W_t} \mathbb{E}[e^{\sigma(W_u - W_t)}]$. The expectation is given by the moment generating function of a normal random variable: $\mathbb{E}[e^{\sigma(W_u - W_t)}] = e^{\frac{1}{2}\sigma^2(u-t)}$. Hence, we obtain $\mathbb{E}[e^{-\frac{1}{2}\sigma^2 u + \sigma W_u} | \mathcal{F}_t] = e^{-\frac{1}{2}\sigma^2 u + \sigma W_t} e^{\frac{1}{2}\sigma^2(u-t)} = e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}$.
- (4) can be checked by calculating the first and second order moments. $\square$

Example 10.3 Let $X_t = W_t^2$. The process X_t is a submartingale from the Jensen's inequality [79]:

$$\mathbb{E}[X_t | \mathcal{F}_s] = \mathbb{E}[W_t^2 | \mathcal{F}_s] \geq (\mathbb{E}[W_t | \mathcal{F}_s])^2 = W_s^2 = X_s.$$

The Doob–Meyer decomposition for X_t is

$$X_t = Y_t + A_t = (W_t^2 - t) + t,$$

where $Y_t = W_t^2 - t$ is a martingale and $A_t = t$ is an increasing predictable process.

Theorem 10.3 (Quadratic Variation of Brownian Motion) *Let W_t be a Brownian motion, then $\langle W, W \rangle_t = t$, almost surely. In differential form, we have the following calculation rules*

$$\begin{cases} dW_t dW_t = dt \\ dW_t dt = 0 \quad (\text{because } \mathbb{E}[\Delta W_t \Delta t] = \Delta t \mathbb{E}[\Delta W_t] = 0) \\ dt dt = 0. \end{cases}$$

Proof Let $P = \{s_0, s_1, \dots, s_n\}$ be a partition of $[0, t]$ with $\|P\| \rightarrow 0$. We want to prove that $\mathbb{E}[\langle W, W \rangle_t] = t$ and $\text{Var}(\langle W, W \rangle_t) = 0$. Firstly,

$$\begin{aligned} \mathbb{E}[\langle W, W \rangle_t] &= \mathbb{E}\left[\lim_{\|P\| \rightarrow 0} \sum_i (W_{s_{i+1}} - W_{s_i})^2\right] \\ &= \lim_{\|P\| \rightarrow 0} \sum_i \text{Var}(\Delta W_{s_i}) = \lim_{\|P\| \rightarrow 0} \sum_i \Delta s_i = t, \end{aligned}$$

where $\Delta W_{s_i} = W_{s_{i+1}} - W_{s_i}$ and $\Delta s_i = s_{i+1} - s_i$. And,

$$\begin{aligned} \text{Var}\left(\sum_i (W_{s_{i+1}} - W_{s_i})^2\right) &= \text{Var}\left(\sum_i \Delta W_{s_i}^2\right) = \sum_i \text{Var}(\Delta W_{s_i}^2) \\ &= \sum_i \mathbb{E}[(\Delta W_{s_i}^2 - \mathbb{E}[\Delta W_{s_i}^2])^2] = \sum_i \mathbb{E}[(\Delta W_{s_i}^2 - \Delta s_i)^2] \end{aligned}$$

$$\begin{aligned}
&= \sum_i \mathbb{E}[\Delta W_{s_i}^4] - 2\mathbb{E}[\Delta W_{s_i}^2]\Delta s_i + \Delta s_i^2 \\
&= \sum_i 3\Delta s_i^2 - 2\Delta s_i^2 + \Delta s_i^2 = \sum_i 2\Delta s_i^2.
\end{aligned}$$

Hence,

$$\begin{aligned}
\text{Var}(\langle W, W \rangle_t) &= \text{Var}\left(\lim_{\|P\| \rightarrow 0} \sum_i (W_{s_{i+1}} - W_{s_i})^2\right) = \lim_{\|P\| \rightarrow 0} \text{Var}\left(\sum_i (W_{s_{i+1}} - W_{s_i})^2\right) \\
&= \lim_{\|P\| \rightarrow 0} \sum_i 2\Delta s_i^2 \leq \lim_{\|P\| \rightarrow 0} \|P\| \sum_i 2\Delta s_i = \lim_{\|P\| \rightarrow 0} 2\|P\|t = 0.
\end{aligned}$$

In the above, $\mathbb{E}[\Delta W_{s_i}^4] = 3\Delta s_i^2$ is from the property of normal distribution which is proved in Sect. 9.3.7. The other two properties can be proved by showing that $\sum_i (W_{s_{i+1}} - W_{s_i})(s_{i+1} - s_i)$ and $\sum_i (s_{i+1} - s_i)^2$ directly converge to 0. $\square$

Remark 10.2 It is interesting to note that the Brownian motion is continuous everywhere but *not* of finite variation on any interval, i.e., $\sum_i |W_{s_{i+1}} - W_{s_i}| \rightarrow \infty$, which is different from continuous differentiable functions [66].

The following theorem is useful for identifying Brownian motions.

Theorem 10.4 (Lévy's Theorem) *Let M_t be a continuous martingale to its natural filtration $\mathcal{F}_t$, $M_0 = 0$, and $\langle M, M \rangle_t = t$. Then M_t is a Brownian motion.*

We refer to [44] for the proof.

Correlated Brownian Motions

Two Brownian motions W_t and $\widehat{W}_t$ are correlated by ρ if $\langle W, \widehat{W} \rangle_t = \rho t$. The parameters $\rho = 0$ and $\rho = 1$ give independency and perfect correlation respectively.

We show how to construct correlated Brownians. Consider

$$B_t = \rho W_t + \sqrt{1 - \rho^2} \widehat{W}_t, \quad -1 \leq \rho \leq 1,$$

where W_t and $\widehat{W}_t$ are *independent* standard Brownian motions (or equivalently $[W_t, \widehat{W}_t]$ a standard two dimensional Brownian motion). Lévy's Theorem shows that B_t is a Brownian motion: firstly B_t is a continuous martingale with $B_0 = 0$ and

$$d\langle B, B \rangle_t = dB_t dB_t = \rho^2 dW_t dW_t + (1 - \rho^2) d\widehat{W}_t d\widehat{W}_t = \rho^2 dt + (1 - \rho^2) dt = dt,$$

since W_t and $\widehat{W}_t$ are independent. Then $d\langle B, W \rangle_t = dB_t dW_t = \rho dt$ and the two Brownian motions B_t and W_t are correlated with ρ . Note that B_t and $\widehat{W}_t$ are also correlated with $\sqrt{1 - \rho^2}$.

10.7 Itô Integral

Definition 10.4 (Itô Integral) Let $P = \{t_0, t_1, \dots, t_n\}$ be a partition of the interval $[0, T]$, f_t be an $\mathcal{F}_t$ -adapted process and $\mathbb{E} \left[\int_0^T f_t^2 dt \right] < +\infty$, the *Itô integral* is defined as

$$\int_0^T f_t dW_t = \lim_{\|P\| \rightarrow 0} \sum_{i=0}^{n-1} f_{t_i} (W_{t_{i+1}} - W_{t_i}).$$

Note that *the value of f_t is known at the beginning of the sub interval*. This is a crucial property for the application of Itô integral in finance. For instance, the integral can represent the cumulative trading P/L where f_t acts as a betting size and is determined by the trading strategy before each time period. As only Itô integrals are used in this book, we often use the simple term stochastic integral instead.

Properties of stochastic integral $I_t = \int_0^t f_s dW_s$ include

- (1) *Martingale*: I_t is an $\mathcal{F}_t$ -martingale
- (2) *Itô-Isometry*: $\mathbb{E}[I_t^2] = \mathbb{E} \left[\int_0^t f_s^2 ds \right]$
- (3) *Quadratic Variation*: $\langle I, I \rangle_t = \int_0^t f_s^2 ds$.

Remark 10.3 (1) may be understood from the fact that, for $0 < t \leq T$,

$$\mathbb{E} \left[\int_0^T f_s dW_s \middle| \mathcal{F}_t \right] = \mathbb{E} \left[\int_0^t f_s dW_s \middle| \mathcal{F}_t \right] + \mathbb{E} \left[\int_t^T f_s dW_s \middle| \mathcal{F}_t \right].$$

The second term is actually zero because the expected value of any future Brownian increment is zero.

For (2), we notice that

$$\mathbb{E} \left[\left(\sum_{i=0}^{n-1} f_{t_i} (W_{t_{i+1}} - W_{t_i}) \right)^2 \right] = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \mathbb{E} [f_{t_i} f_{t_j} (W_{t_{i+1}} - W_{t_i})(W_{t_{j+1}} - W_{t_j})].$$

For each term of the form $i < j$:

$$\begin{aligned} \mathbb{E} [f_{t_i} f_{t_j} \Delta W_{t_i} \Delta W_{t_j}] &= \mathbb{E} [\mathbb{E} [f_{t_i} f_{t_j} \Delta W_{t_i} \Delta W_{t_j} | \mathcal{F}_{t_j}]] \\ &= \mathbb{E} [f_{t_i} f_{t_j} \Delta W_{t_i} \mathbb{E} [\Delta W_{t_j} | \mathcal{F}_{t_j}]] = 0 \text{ (as } \mathbb{E} [\Delta W_{t_j} | \mathcal{F}_{t_j}] = 0). \end{aligned}$$

For terms with $i = j$:

$$\mathbb{E}[f_{t_i}^2 \Delta W_{t_i}^2] = \mathbb{E}[\mathbb{E}[f_{t_i}^2 \Delta W_{t_i}^2 | \mathcal{F}_{t_i}]] = \mathbb{E}[f_{t_i}^2 \mathbb{E}[\Delta W_{t_i}^2 | \mathcal{F}_{t_i}]] = \mathbb{E}[f_{t_i}^2 \Delta t_i].$$

$$\text{Hence, } \mathbb{E}[(\int_0^T f_t dW_t)^2] = \lim_{\|P\| \rightarrow 0} \sum_{i=0}^{n-1} \mathbb{E}[f_{t_i}^2 \Delta W_{t_i}^2] = \lim_{\|P\| \rightarrow 0} \sum_{i=0}^{n-1} \mathbb{E}[f_{t_i}^2 \Delta t_i] = \mathbb{E}[\int_0^T f_t^2 dt].$$

$$\text{Lastly, we have } \langle I, I \rangle_t = \lim_{\|P\| \rightarrow 0} \sum_{i=0}^{n-1} (I_{t_{i+1}} - I_{t_i})^2 = \lim_{\|P\| \rightarrow 0} \sum_{i=0}^{n-1} (f_{t_i} (W_{t_{i+1}} - W_{t_i}))^2.$$

Then, following the proof for Theorem 10.3 for quadratic variation of Brownian motion, we obtain the result in property (3).

10.8 Itô Calculus

Definition 10.5 (One-dimensional Itô Process) The stochastic process X_t is called an Itô process if it follows:

$$X_t = X_0 + \int_0^t \theta_s ds + \int_0^t \phi_s dW_s,$$

where W_s is an $\mathcal{F}_t$ -Brownian motion and θ_s and ϕ_s are $\mathcal{F}_t$ -adapted processes.

The differential form of Itô process is

$$dX_t = \theta_t dt + \phi_t dW_t.$$

θ_t is known as the *drift* of X_t . It is easily seen that X_t is a martingale if its drift is zero. The below theorem for functions of an Itô process is the key component of stochastic calculus.

Theorem 10.5 (Itô–Doebelin Formula) Let $f(t, x)$ be a function for which $\frac{\partial f}{\partial t}(t, x)$, $\frac{\partial f}{\partial x}(t, x)$ and $\frac{\partial^2 f}{\partial x^2}(t, x)$ are defined and continuous. Then, for $T > 0$,

$$\left\{ \begin{array}{l} f(T, X_T) = f(0, X_0) + \int_0^T \frac{\partial f}{\partial t} dt + \int_0^T \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \int_0^T \frac{\partial^2 f}{\partial x^2} d\langle X, X \rangle_t \\ f(T, X_T) = f(0, X_0) + \int_0^T \frac{\partial f}{\partial t} dt + \int_0^T \frac{\partial f}{\partial x} \theta_t dt + \int_0^T \frac{\partial f}{\partial x} \phi_t dW_t \\ \quad + \frac{1}{2} \int_0^T \frac{\partial^2 f}{\partial x^2} \phi_t^2 dt. \end{array} \right.$$

The differential form of Itô–Doebelin Formula is written as either of the following equations:

$$\begin{cases} df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} d\langle X, X \rangle_t \\ df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} \theta_t dt + \frac{\partial f}{\partial x} \phi_t dW_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} \phi_t^2 dt. \end{cases}$$

Proof Itô–Doebelin Formula can be understood from the Taylor series of $f(t, X_t)$:

$$df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} dX_t dX_t + [other terms].$$

Applying the calculation rules of Brownian Motion:

$$dW_t dW_t = dt, \quad dW_t dt = 0, \quad dt dt = 0,$$

the value of $[other terms]$ is zero.

From $dX_t = \theta_t dt + \phi_t dW_t$, we obtain

$$dX_t dX_t = \phi_t^2 dW_t dW_t + \phi_t \theta_t dW_t dt + \theta_t \phi_t dt dW_t + \theta_t^2 dt dt = \phi_t^2 dt.$$

Hence, replacing dX_t and $dX_t dX_t$ in the Taylor series, we obtain

$$df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} \theta_t dt + \frac{\partial f}{\partial x} \phi_t dW_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} \phi_t^2 dt. \quad \square$$

Readers may refer to [60] for the rigorous proof. Note that sometimes we also write $\frac{\partial f}{\partial x}$ as $\frac{\partial f}{\partial X_t}$ and $\frac{\partial^2 f}{\partial x^2}$ as $\frac{\partial^2 f}{\partial X_t^2}$ respectively. Similarly to the one-dimensional Itô–Doebelin Formula, we can obtain the below two-dimensional Itô–Doebelin formula.

Theorem 10.6 (Two-Dimensional Itô–Doebelin Formula) *Let $f(t, x, y)$ be a function for which the required derivatives are defined and continuous, X_t and Y_t be Itô processes. Then,*

$$\begin{aligned} df(t, X_t, Y_t) &= \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{\partial f}{\partial y} dY_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} d\langle X, X \rangle_t \\ &\quad + \frac{\partial^2 f}{\partial x \partial y} d\langle X, Y \rangle_t + \frac{1}{2} \frac{\partial^2 f}{\partial y^2} d\langle Y, Y \rangle_t. \end{aligned}$$

Applying the Two-Dimensional Itô–Doebelin Formula to $f(t, X_t, Y_t) = X_t Y_t$ we obtain the very useful result below.

Corollary 10.1 (Itô’s Product Rule) *Let X_t and Y_t be Itô processes, then*

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + d\langle X, Y \rangle_t.$$

Notice that $\langle X, Y \rangle_t$ is an additional term compared to Riemann calculus. The integral form of the Itô’s product rule can also be regarded as the integration by parts formula [60].

Example 10.4 In this example we calculate $d(e^{-\int_0^t r_s ds} X_t)$ using the Itô's product rule, where r_t is deterministic and X_t is an adapted process. As a result, the covariation term in Itô's product rule is zero and we have

$$\begin{aligned} d(e^{-\int_0^t r_s ds} X_t) &= X_t d(e^{-\int_0^t r_s ds}) + e^{-\int_0^t r_s ds} dX_t + d\langle e^{-\int_0^t r_s ds}, X_t \rangle \\ &= -r_t e^{-\int_0^t r_s ds} X_t dt + e^{-\int_0^t r_s ds} dX_t \\ &= e^{-\int_0^t r_s ds} (-r_t X_t dt + dX_t). \end{aligned}$$

The above equation will frequently appear in martingale option pricing framework later. For instance, it is used later for solving the so-called Ornstein-Uhlenbeck process. In particular, when r_t is a degenerate process $r_t = r$, we have

$$d(e^{-rt} X_t) = e^{-rt} (dX_t - r X_t dt).$$

If X_t represents an asset value, the right hand side of the equation is the discounted value for the increment of self-financing investment.

10.9 Poisson Process

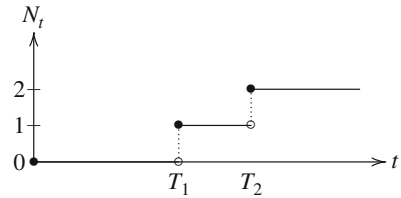
Let $\tau_k, k = 1, 2, \dots$ be *i.i.d.* random variables following the exponential distribution. Then, τ_k can be used to define an increasing sequence of random times by

$T_n = \sum_{k=1}^n \tau_k$, which is known as the arrival time of the n -th event. The arrival times are known to follow the gamma density, see [79].

The *Poisson process* N_t is a right continuous process which counts the number of events up to and including t . See Fig. 10.3. It is defined as

$$N_t = \sum_{n \geq 1} \mathbf{1}_{\{t \geq T_n\}}.$$

Fig. 10.3 A sample path for Poisson process N_t



For a right continuous process, there exists a left continuous version and the value prior to the jump at time t is denoted by “ $t-$ ”. For instance, the first jump of Poisson process happens at $t = T_1$, and hence $N_{t-} = 0$, $N_t = 1$ with $\delta N_t \equiv N_t - N_{t-} = 1$.

Here are some properties of Poisson processes for integer $k \geq 0$ and $0 \leq s < t$ that can be proved with straightforward calculation where $e^x = \sum_{j=0}^{\infty} \frac{x^j}{j!}$ is sometimes used (cf. [79]):

- (1) Distribution: $\mathbb{P}\{N_t = k\} = \frac{(\lambda t)^k}{k!} e^{-\lambda t}$
- (2) Mean: $\mathbb{E}[N_t - N_s] = \lambda(t - s)$
- (3) Variance: $\text{Var}(N_t - N_s) = \lambda(t - s)$
- (4) Martingale Property: $N_t - \lambda t = N_t - \mathbb{E}[N_t]$ is a martingale
- (5) Independent and stationary increments:

$$\mathbb{P}\{N_t - N_s = k\} = \frac{\lambda^k (t-s)^k}{k!} e^{-\lambda(t-s)} = \mathbb{P}\{N_{t-s} = k\}$$

- (6) Moment generating function: $\mathbb{E}[e^{uN_t}] = e^{\lambda t(e^u - 1)}$.

The parameter λ is the *intensity* in the exponential distribution for the arrival time. The independence feature for the increments of Poisson processes can be understood from the memorylessness of the exponential distribution. The other properties can be obtained by direct calculation.

Note that

$$\mathbb{P}(\tau \leq t + \Delta t | \tau > t) = \frac{\mathbb{P}(t < \tau \leq t + \Delta t)}{\mathbb{P}(\tau > t)} = \frac{e^{-\lambda t} - e^{-\lambda(t+\Delta t)}}{e^{-\lambda t}} = 1 - e^{-\lambda \Delta t}.$$

Hence, $\lim_{\Delta t \rightarrow 0} \mathbb{P}(\tau \leq t + \Delta t | \tau > t) = \lambda dt$ using the Taylor expansion of $e^{-\lambda \Delta t}$. In financial modelling, λdt is often used as the probability of an event in $(t, t + dt)$, conditioned by no such event happened before t . It can be proved that there can be at most one event arrival within an infinitesimally small interval $(t, t + dt)$ [66]. So, in differential notation, the Poisson process is often written as

$$dN_t = \begin{cases} 0, & \text{with probability } 1 - \lambda dt, \\ 1, & \text{with probability } \lambda dt. \end{cases}$$

Exercises

1. Let Ω be a infinite coin toss space

$$\Omega = \{\omega = \omega_1 \omega_2 \cdots ; \omega_i = y \text{ or } n, i = 1, 2, \dots\}.$$

where y stands for a head and n stands for a tail with equal probability. Define the random variables

$$Y_i(\omega) = \begin{cases} 1, & \text{if } \omega_i = y, \\ -1, & \text{if } \omega_i = n. \end{cases}$$

Then, the random walk is defined as a discrete stochastic process $X_k = \sum_{i=1}^k Y_i$ with initial value $X_0 = 0$. Find $\sigma(X_2)$, the σ -algebra generated by the random walk at time 2.

2. Consider an Itô process $dX_t = W_t^2 dt + W_t^2 dW_t$, or

$$X_t = X_0 + \int_0^t W_s^2 ds + \int_0^t W_s^2 dW_s.$$

Derive $\mathbb{E}[X_t]$.

3. Let σ_t^1 and σ_t^2 be non-zero adapted processes and we have two correlated Brownian motions given by $\langle W, \widehat{W} \rangle_t = \rho t$. Show that

$$B_t = \int_0^t \frac{\sigma_s^1 dW_s + \sigma_s^2 d\widehat{W}_s}{\sqrt{(\sigma_s^1)^2 + (\sigma_s^2)^2 + 2\rho\sigma_s^1\sigma_s^2}}$$

is a Brownian motion.

Chapter 11

Black–Scholes–Merton Model for Option Pricing



In this chapter, we start off the discussion of option pricing with the pioneering work by Black, Scholes and Merton who proposed the first hedging (replication) framework in 1973. Their work laid the foundation for the rapid growth of derivative products. In recognition of their contributions, Scholes and Merton received the 1997 Nobel Prize in Economics.¹

The option pricing and hedging framework of *Black–Scholes–Merton Model* was originally derived with PDE (Partial Differential Equation) approach. In later years, researchers have generalized the model and developed a rigorous and practical modelling framework with probabilistic approach that will be presented in the next two chapters.

11.1 The Black–Scholes–Merton Model

The option price is sensitive to the change of the underlying's price. Take the example of a call option on a certain stock as underlying asset. If the stock price goes up, the price of the call option will increase, which implies a gain to the option holder and a loss to the option writer. But if the writer of the call has held a certain quantity of the stock, the increase of the stock price will result in a gain which may mitigate or offset perfectly the loss on the option position.

This intuitive analysis gives rise to the question of whether this strategy can perfectly hedge the option and what quantity of the stock the option writer needs to hold at a given point in time. The answer was given by Black, Scholes and Merton in the seminal work on option pricing published in 1973 [10, 55]. The authors demonstrated that investors can create a *riskless* hedging portfolio where all market

¹Unfortunately, Black passed away in 1995.

risks are eliminated under the model now known as the *BSM model* (*Black–Scholes–Merton Model*).

Under the Black–Scholes–Merton model, the price of a non-dividend paying asset follows the so called *geometric Brownian motion*:

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad (11.1)$$

where μ and $\sigma > 0$ are constants, representing the asset's drift and volatility respectively.

Proposition 11.1 *Let S_t be the process defined in (11.1). Then,*

$$S_t = S_0 e^{(\mu - \frac{\sigma^2}{2})t + \sigma W_t}. \quad (11.2)$$

Proof Define $f(t, x) = \ln x$. Then

$$\frac{\partial f}{\partial x} = \frac{1}{x}, \quad \frac{\partial^2 f}{\partial x^2} = -\frac{1}{x^2}, \quad \frac{\partial f}{\partial t} = 0.$$

The Itô's lemma gives

$$\begin{aligned} d \ln(S_t) &= df(t, S_t) = 0 \cdot dt + \frac{\partial f}{\partial x}(t, S_t) dS_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, S_t) (dS_t)^2 \\ &= \frac{1}{S_t} dS_t + \frac{1}{2} \left(-\frac{1}{S_t^2} \right) (dS_t)^2 = (\mu - \frac{\sigma^2}{2})dt + \sigma dW_t, \end{aligned}$$

where (11.1) and $(dS_t)^2 = \sigma^2 S_t^2 dt$ are used.

Integrating from time 0 to t leads to the solution. $\square$

In (11.2), if we neglect the Brownian motion W_t , then S_t changes geometrically with a common factor $e^{(\mu - \frac{\sigma^2}{2})}$. This is why S_t is called a *geometric Brownian motion*.

Corollary 11.1 *The logarithm of S_t is an arithmetic Brownian motion*

$$\ln S_t = \ln S_0 + (\mu - \frac{\sigma^2}{2})t + \sigma W_t \sim \mathcal{N} \left(\ln S_0 + (\mu - \frac{\sigma^2}{2})t, \sigma^2 t \right).$$

Thus, the distribution of S_t is said to be *log-normal*. Furthermore,

$$\mathbb{E}[S_t] = S_0 e^{\mu t} \quad \text{and} \quad \text{Var}(S_t) = S_0^2 e^{2\mu t} (e^{\sigma^2 t} - 1).$$

Proof The normal distribution of $\ln S_t$ follows from taking the logarithm on both sides of (11.2). Note that $W_t \sim \mathcal{N}(0, t)$, then $W_t = \sqrt{t}\epsilon$ with $\epsilon \sim \mathcal{N}(0, 1)$. By the moment generating function, we have

$$\mathbb{E}[S_t] = \mathbb{E} \left[S_0 e^{(\mu - \frac{\sigma^2}{2})t + \sigma W_t} \right] = S_0 e^{(\mu - \frac{\sigma^2}{2})t} \mathbb{E}[e^{\sigma \sqrt{t} \epsilon}] = S_0 e^{(\mu - \frac{\sigma^2}{2})t} e^{\frac{\sigma^2}{2}t} = S_0 e^{\mu t}.$$

Similarly,

$$\mathbb{E}[S_t^2] = \mathbb{E} \left[S_0^2 e^{(2\mu - \sigma^2)t + 2\sigma W_t} \right] = S_0^2 e^{(2\mu - \sigma^2)t} \mathbb{E}[e^{2\sigma \sqrt{t} \epsilon}] = S_0^2 e^{(2\mu + \sigma^2)t}.$$

$$\text{Hence } \text{Var}(S_t) = \mathbb{E}[S_t^2] - \mathbb{E}[S_t]^2 = S_0^2 e^{2\mu t} (e^{\sigma^2 t} - 1).$$

□

Equivalently, the lognormal distribution can be written as

$$\ln \left(\frac{S_t}{S_0} \right) \sim \mathcal{N} \left(\left(\mu - \frac{\sigma^2}{2} \right) t, \sigma^2 t \right).$$

11.2 Derivation of the Black–Scholes Equation

The basic version of the model assumes a “perfect market” in which

- there is no arbitrage possibility
- the asset can be bought or sold at any time for any quantity (including a fraction of the asset)
- there is no short selling restriction
- there is no transaction cost
- there exists a constant risk-free interest rate r for borrow and lending

Suppose that we have an option V of European style linked to asset S . Actually, V can be the value of a whole portfolio of a certain asset and European options linked to that asset. But for simplicity, and without loss of generality, we can think of a single option linked to a non-dividend paying stock S in the following.

Let $V_t = V(t, S_t)$ denote the option value process. Using Itô’s lemma, we obtain V_t as an Itô process

$$dV_t = \left(\mu S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + \frac{\partial V}{\partial t} \right) dt + \sigma S_t \frac{\partial V}{\partial S} dW_t. \quad (11.3)$$

Note that $\frac{\partial V}{\partial t}$, $\frac{\partial V}{\partial S}$ and $\frac{\partial^2 V}{\partial S^2}$ are abbreviated for convenience. They represent the processes $\frac{\partial V}{\partial t}(t, S_t)$, $\frac{\partial V}{\partial S}(t, S_t)$ and $\frac{\partial^2 V}{\partial S^2}(t, S_t)$ respectively.

Next, we construct a portfolio Π_t consisting of a long position of one option and a short position of Δ_t units² of the underlying asset at time t .

The value of this portfolio is

$$\Pi_t = V_t - \Delta_t S_t. \quad (11.4)$$

The key idea is to find the value of Δ_t so that the portfolio becomes instantaneously deterministic, i.e. without the term dW_t representing the stochastic movement of the stock price. The change in the value of this portfolio in one time-step dt is

$$d\Pi_t = dV_t - \Delta_t dS_t. \quad (11.5)$$

Substituting (11.1) and (11.3) into (11.5), we find

$$d\Pi_t = \left(\mu S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + \frac{\partial V}{\partial t} - \mu \Delta_t S_t \right) dt + \sigma S_t \left(\frac{\partial V}{\partial S} - \Delta_t \right) dW_t. \quad (11.6)$$

The first term on the right hand side is “deterministic” while the second term is stochastic as it involves W_t . By setting $\Delta_t = \frac{\partial V}{\partial S}(t, S_t)$, (11.6) becomes

$$d\Pi_t = \left(\frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + \frac{\partial V}{\partial t} \right) dt. \quad (11.7)$$

Under the assumptions stated earlier, we should have $d\Pi_t = r\Pi_t dt$, or

$$dV_t = \Delta_t dS_t + r(V_t - \Delta_t S_t) dt.$$

In fact, if $d\Pi_t$ were greater than $r\Pi_t dt$, an arbitrager could make a riskless profit by borrowing an amount Π_t at rate r to invest in the portfolio. Conversely, $d\Pi_t$ were less than $r\Pi_t dt$, then the arbitrager would short the portfolio and invest Π_t at the risk free rate r . Combining with (11.7), we obtain

$$\left(\frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + \frac{\partial V}{\partial t} \right) dt = r\Pi_t dt.$$

Now replace Π_t by $V_t - \Delta_t S_t$ as given in (11.4), and replace Δ_t by $\partial V / \partial S$. We arrive at

²Here $\Delta_t < 0$ together with the minus sign in front implies a long position in underlying asset. For instance, if the option is a put, we may want to buy certain amount of stock to offset the risk. In that case, $\Delta_t < 0$.

$$\frac{\partial V_t}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V_t}{\partial S_t^2} + r S_t \frac{\partial V_t}{\partial S_t} - r V_t = 0. \quad (11.8)$$

This is the famous *Black–Scholes partial differential equation*. Any derivative security whose price depends only on the current value of S_t and on t must satisfy the Black–Scholes equation.

Note that the Black–Scholes equation (11.8) does not contain the drift parameter μ of the underlying asset. The price will depend on the risk-free interest rate r and the volatility σ however. A consequence of this is that two people may have quite different views on μ , yet still agree on the value of an option.

For the Black–Scholes equation in (11.8), which is a backward parabolic equation, we must specify final and boundary conditions, for having a unique solution. For the moment we restrict our attention to a *vanilla European call* $c(t, S)$, with exercise price K and expiry date T .

The final condition of a call is just its payoff at T :

$$c(T, S_T) = \max\{S_T - K, 0\}. \quad (11.9)$$

Our ‘spatial’ or asset-price boundary conditions are applied at zero asset price, $S_t = 0$, and at $S_t \rightarrow \infty$. We can see from (11.1) that if S_t is ever zero, then dS_t is also zero, and therefore S_t can never change. Since if $S_t = 0$ at expiry, the payoff will be zero. Thus on $S_t = 0$ we have

$$c(t, 0) = 0 \quad \text{for all } t \geq 0. \quad (11.10)$$

Finally as $S_t \rightarrow \infty$, it becomes ever more likely that the option will be exercised. Thus, as $S_t \rightarrow \infty$, the value of the option becomes that of the asset minus the exercise price. Hence we have for all $t > 0$,

$$c(t, S_t) \sim S_t - K e^{-r(T-t)}, \quad \text{as } S_t \rightarrow \infty. \quad (11.11)$$

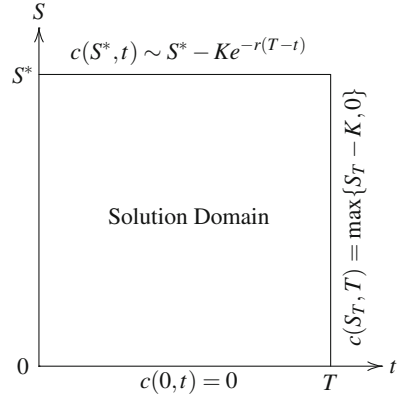
For a European call option, Black–Scholes equation (11.8) together with the boundary conditions (11.9)–(11.11) can be solved exactly to give its value. In Fig. 11.1, we give the solution domain (the domain where we want to solve the call option value) and the boundary conditions.

For a *vanilla European put* option $p(t, S_t)$, the final condition is the payoff

$$p(T, S_T) = \max\{K - S_T, 0\}.$$

For the boundary conditions, we have already mentioned that if S_t is ever zero, then it must remain zero. In this case the final payoff for a put is known with certainty to be K . To determine $p(t, 0)$, we simply have to calculate the present value of an amount K received at time T . Assuming that interest rates are constant we find the boundary condition at $S_t = 0$ to be

Fig. 11.1 Solution domain of a European call option and the boundary conditions. Here S^* is a sufficiently large value



$$p(t, 0) = Ke^{-rt}, \quad \text{for all } t \geq 0.$$

As $S_t \rightarrow \infty$, the option is unlikely to be exercised and so for $t > 0$, we have

$$p(t, S_t) \rightarrow 0, \quad \text{as } S_t \rightarrow \infty.$$

11.3 Black–Scholes Formulas for Vanilla Options

By solving the PDE, closed-form formulas are obtained for European call and put options [10]. In this section, we only verify that the formulas are solutions of the Black–Scholes equation.

Theorem 11.1 (Black–Scholes Formula) *The time- t value of the vanilla European call with strike price K and maturity T is given by*

$$c(t, S_t) = S_t N(d_1) - Ke^{-r(T-t)} N(d_2), \quad (11.12)$$

where $N(x)$ is the standard normal cumulative distribution function,

$$d_1 = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} \quad \text{and} \quad d_2 = d_1 - \sigma\sqrt{T-t}. \quad (11.13)$$

Proof Firstly, $c(t, S_t)$ in (11.12) satisfies the Black–Scholes equation (11.8). This can be checked by direct calculation of partial derivative terms $\frac{\partial c}{\partial t}$, $\frac{\partial c}{\partial S}$, $\frac{\partial^2 c}{\partial S^2}$ in (11.8).

We note the below for calculating the partial derivatives

$$\frac{\partial N(d_i)}{\partial x} = \frac{\partial N(d_i)}{\partial d_i} \frac{\partial d_i}{\partial x} = \frac{e^{-\frac{d_i^2}{2}}}{\sqrt{2\pi}} \frac{\partial d_i}{\partial x}, \quad i = 1, 2.$$

For the boundary condition (11.10), we first note that by (11.13), $d_1, d_2 \rightarrow -\infty$ as $S_t \rightarrow 0$. Obviously $N(-\infty) = 0$. Hence

$$c(t, 0) = 0 \times N(-\infty) - K e^{-r(T-t)} N(-\infty) = 0.$$

For the boundary condition (11.11), we note again that $d_1, d_2 \rightarrow \infty$ as $S_t \rightarrow \infty$ where $N(\infty) = 1$. Hence

$$c(t, S_t) \rightarrow S_t N(\infty) - K e^{-r(T-t)} N(\infty) \sim S_t - K e^{-r(T-t)}, \quad \text{as } S_t \rightarrow \infty.$$

Finally, we consider the final condition (11.9). At $t = T$, if $S_T > K$, then $d_1, d_2 \rightarrow \infty$. Hence $c(T, S_T) = S_T - K$. If $S_T < K$, then $d_1, d_2 \rightarrow -\infty$. Hence $c(T, S_T) = 0$. If $S_T = K$, by continuity, $c(T, S_T) = 0$. $\square$

Next we give the formula for European put options.

Theorem 11.2 *The value of the vanilla European put is given by*

$$p(t, S_t) = K e^{-r(T-t)} N(-d_2) - S_t N(-d_1), \quad (11.14)$$

where d_1 and d_2 are given in (11.13).

Proof Instead of checking the PDE and boundary, final conditions as we did in the proof of Theorem 11.1, we can derive (11.14) immediately by using the put-call parity formula (see Proposition 8.1)

$$c(t, S_t) - p(t, S_t) = S_t - K e^{-r(T-t)},$$

Theorem 11.1, and the equality $N(d) + N(-d) \equiv 1$ for any d . $\square$

We remark that although (11.12) and (11.14) seem to be closed-form solutions for the vanilla options, one still has to compute the integral $N(d_i)$ numerically by quadrature rules such as Simpson's rule or Gaussian rule.

By straightforward derivation, we can obtain the deltas of vanilla calls and puts:

$$\Delta_c(t, S_t) = N(d_1) \quad \text{and} \quad \Delta_p(t, S_t) = N(d_1) - 1.$$

A Useful Approximation Formula

For at-the-money forward (ATMF) call/put options, we have the below approximation formula

$$c(0, S_0) = p(0, S_0) \approx 0.4S_0\sigma\sqrt{T}.$$

It is useful to practitioners for a quick estimation of the option price without using a pricer.

Proof The strike of an ATMF option is $K = e^{rT}S_0$, where S_0 is the spot level at time $t = 0$. In this case, the European call and put options have the same price according to put-call parity.

Let us check the approximation formula for the ATMF call option. Firstly,

$$S_0 = e^{-rT}K, \quad d_1 = \frac{\sigma\sqrt{T}}{2} \quad \text{and} \quad d_2 = -\frac{\sigma\sqrt{T}}{2}.$$

From Taylor expansion, we have

$$N(x) = N(0) + N'(0)x + \frac{1}{2}N''(0)x^2 + O(x^3).$$

Using $N'(0) = \frac{1}{\sqrt{2\pi}} \approx 0.4$, we obtain

$$\begin{aligned} c(0, S_0) &= S_0N(d_1) - e^{-rT}KN(d_2) = S_0 \left[N\left(\frac{\sigma\sqrt{T}}{2}\right) - N\left(-\frac{\sigma\sqrt{T}}{2}\right) \right] \\ &\approx S_0N'(0)\sigma\sqrt{T} \approx 0.4S_0\sigma\sqrt{T}. \end{aligned} \quad \square$$

11.4 Discussion on Hedging Issues

11.4.1 Sensitivities (Greeks)

Consider a portfolio Π which is composed of options linked to a single asset S and a quantity of the asset itself. In Black–Scholes–Merton model, the price of the portfolio Π_t depends on the stock price, time to maturity, volatility and risk-free interest rate: $\Pi_t = \Pi(T - t, S_t, \sigma, r)$. We denote these sensitivities with respect to these parameters by Greek letters³:

- Delta : $\Delta_t = \frac{\partial \Pi_t}{\partial S_t}$
- Theta : $\Theta_t = \frac{\partial \Pi_t}{\partial t}$
- Rho : $\rho_t = \frac{\partial \Pi_t}{\partial r}$
- Gamma : $\Gamma_t = \frac{\partial^2 \Pi_t}{\partial S_t^2}$
- Vega : $\mathcal{V}_t = \frac{\partial \Pi_t}{\partial \sigma}$

³ Actually, vega $\mathcal{V}_t$ is not a Greek letter.

We can write the Black–Scholes PDE $\frac{\partial \Pi_t}{\partial t} + r S_t \frac{\partial \Pi_t}{\partial S_t} + \frac{1}{2} \frac{\partial^2 \Pi_t}{\partial S_t^2} \sigma^2 S_t^2 = r \Pi_t$ as

$$\Theta_t + r S_t \Delta_t + \frac{1}{2} \Gamma_t \sigma^2 S_t^2 = r \Pi_t.$$

If Π_t is delta-neutral ($\Delta_t = 0$), then

$$\Theta_t + \frac{1}{2} \Gamma_t \sigma^2 S_t^2 = r \Pi_t.$$

It shows that Gamma and Theta should have opposite signs if their values are large and $r \Pi_t$ is small:

$$\Theta_t \approx -\frac{1}{2} \Gamma_t \sigma^2 S_t^2.$$

This is known as **Gamma-Theta relationship**. The decision on the amount of Theta and Gamma for the portfolio is one of the major concerns for option traders. From Itô–Doebelin formula, we observe that the price change of an option can be expressed by the Greeks of the option:

$$d\Pi_t = \Theta_t dt + \Delta_t dS_t + \frac{1}{2} S_t^2 \sigma^2 \Gamma_t dt.$$

As each Greek represents a certain risk exposure for the portfolio, the derivative product providers (i.e. hedging party) always apply limits in dollar amount on the major sensitivity indicators such as Vega, Gamma and (residual) Delta.

See Fig. 11.2 and Fig. 11.3 for illustrations of call option price, Delta, Gamma, and Vega.

Delta – Δ_t

The delta of a portfolio or an option measures the value change w.r.t. the price change of the underlying. The delta-hedging consists of holding (long or short depending on the sign of delta) Δ_t units of the underlying asset (called delta amount or, simply, delta) at any time t . If Δ_t is positive, a loan of $\Delta_t S_t$ is necessary for financing the delta position. If Δ_t is negative, the underlying assets are borrowed and sold to the market. As the asset price changes, a P/L will be produced by the delta. According to BSM model, the cumulative P/L from the delta-hedging for a single option combined with the option premium will be exactly the option price for any time t . In particular, when t is the expiry time, the option price is the payoff amount.

Let us take the example of the hedging of a call sold by an option writer who borrowed $\Delta_0 S_0$ for the delta at inception. The initial portfolio is composed of the

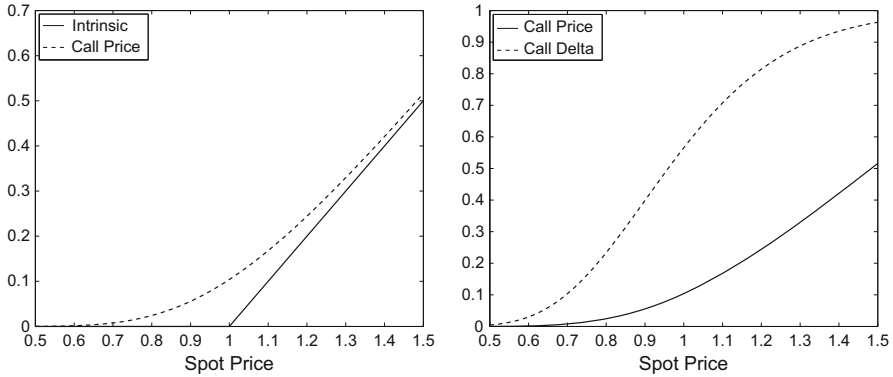


Fig. 11.2 Price and Delta of a call option with $K = 1$, $r = 0.01$, $\sigma = 0.25$ and $T = 1$

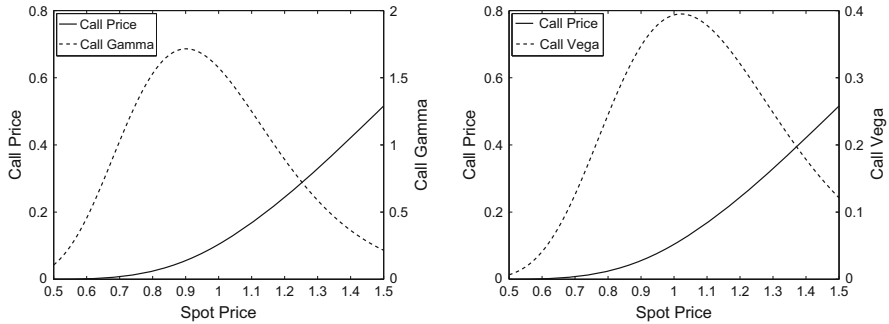


Fig. 11.3 Gamma and Vega of a call option with $K = 1$, $r = 0.01$, $\sigma = 0.25$ and $T = 1$

call option (short), option premium (long), Δ_0 units of the asset and a loan of $\Delta_0 S_0$. As the option premium offsets the value of the short option, we have $\Pi_0 = 0$ and it is self-financed. During the life of the trade, Δ_t units of the underlying asset should be maintained at any time t . The cash in the portfolio is invested in money market.

We will revisit delta-hedging analysis in the chapter of risk-neutral pricing.

Gamma – Γ_t

Gamma measures the change of delta w.r.t. the price change of underlying. If Gamma is zero, the delta amount is constant throughout the entire considered tenor, i.e. static hedging. In particular, the Gamma of a portfolio of underlying assets is zero. A non-zero Gamma portfolio implies dynamic hedging instead of static hedging. Gamma is an important factor for risk management which will be explained in the discrete hedging section. The Gamma of a portfolio can be reduced or eliminated by adding other options having the opposite Gamma amount. Obviously, the resulting additional delta can be hedged by the assets.

Theta – Θ_t

Theta measures the value change of portfolio or option with time. It is also called *time decay*. For a delta-neutral portfolio, Theta is compensated by the Gamma term in Black–Scholes PDE.

Vega – $\mathcal{V}_t$

In BSM model, the volatility is a constant which is a simplistic assumption. In practice, it is a parameter that is difficult to estimate, especially when applied to the future period until the option's expiry.

The Vega measures the impact of volatility. As for the gamma position for a portfolio, the vega position can be hedged with other options having the opposite amount of Vega.

Rho – ρ_t

In BSM model, the risk-free interest rate is assumed to be constant. The parameter Rho measures the sensitivity of the portfolio's value w.r.t. the change of interest rate level. In practice, the risk-free rate is replaced by the borrow/lending rate available for hedging transactions. This rate needs to have little or zero credit risk such as the internal funding rate from the same institution.

The assumptions made previously in Black–Scholes–Merton analysis are over simplistic. We highlight a few issues related to hedging in practice.

11.4.2 Practical Issues in Hedging**Discrete Hedging**

Delta hedging can only be done at discrete times (one or a few times a day at most). If the hedging is not adjusted continuously (or very frequently), a negative Gamma position will make the portfolio under-hedged while a positive Gamma position will make the portfolio over-hedged when the underlying price makes large moves.

Recall that for a delta hedged portfolio, Gamma and Theta have opposite signs when their values are significant which roughly offset each other.

- For $\Gamma > 0$, earn from large market moves but lose from $\Theta < 0$
- For $\Gamma < 0$, lose from large market moves but earn from $\Theta > 0$.

The Gamma-Theta ratio is an important factor for managing a trading book.

Example 11.1 Consider a long call position with ATMF strike. Thus, we have $\Gamma > 0$. At inception, the hedging is done by shorting Δ_0 shares at spot price S_0 , where $\Delta_0 \approx 0.5$. Then, the next day,

Scenario	Hedging	Result
S_t drops and $\Delta_t \rightarrow 0.45$	Need to short less, buy back at cheaper price	Buy low
S_t rises and $\Delta_t \rightarrow 0.55$	Need to short more, sell at higher price	Sell high

Therefore, the $\Gamma > 0$ position is beneficial to call option holder when market moves. However, this is counterbalanced by $\Theta < 0$ as the call is worth less.

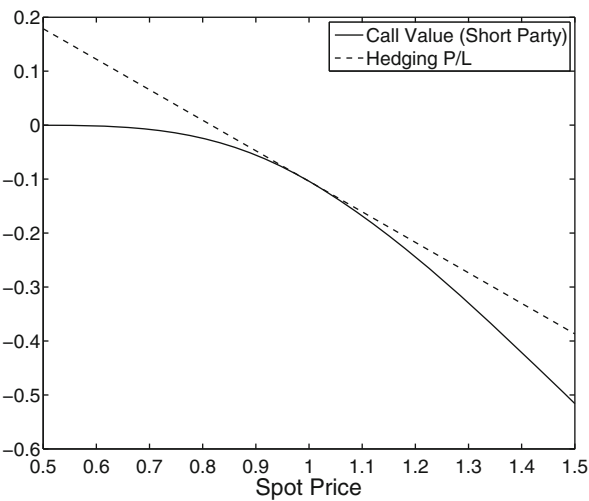
The positive Gamma of long call can be explained in Fig. 11.4, because it stands for the difference between the curve c_t (option value to the short call party) and the delta-hedging line at one point in time with slope Δ_t .

Hedging P/L

Volatility is the most important parameter for the valuation of an option. From the price of a market traded Vanilla option (i.e. call or put), we can reverse calculate the so-called *implied volatility* from Black–Scholes formula.

The realized volatility during the life of an option trade may differ from the implied volatility used for pricing and hedging. Let us analyze the P/L impact of the realized volatility assuming a constant implied volatility for pricing and hedging. For in-depth analysis of hedging P/L under various models, we refer to [8].

Fig. 11.4 Delta-hedging



Denote

- σ_H , the constant “hedging volatility” which is used for calculating the Greeks and the option mark-to-market price;
- σ , as realized volatility during the life of the trade.

Then we have the following important result on option hedging.

Proposition 11.2 *The P/L impact at maturity of the inaccurate volatility used for hedging is*

$$\frac{1}{2} \int_0^T e^{r(T-t)} \Gamma_t S_t^2 (\sigma^2 - \sigma_H^2) dt.$$

Proof Applying Itô–Doebelin formula to V and assuming that the underlying S_t follows the geometric Brownian motion with realized volatility σ ,

$$dV_t = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \sigma^2 S_t^2 dt.$$

From the PDE of BSM model, which is related to the hedging volatility σ_H ,

$$\frac{\partial V}{\partial t} = -(r S_t \frac{\partial V}{\partial S} + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \sigma_H^2 S_t^2 - r V_t).$$

Substituting $\frac{\partial V}{\partial t}$ in the formula for dV_t , we get

$$\begin{aligned} dV_t &= r V_t dt - r S_t \frac{\partial V}{\partial S} dt + \frac{\partial V}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} S_t^2 (\sigma^2 - \sigma_H^2) dt \\ dV_t - r V_t dt &= \Delta_t (-r S_t dt + dS_t) + \frac{1}{2} \Gamma_t S_t^2 (\sigma^2 - \sigma_H^2) dt \\ d(e^{-rt} V_t) &= \Delta_t d(e^{-rt} S_t) + \frac{1}{2} e^{-rt} \Gamma_t S_t^2 (\sigma^2 - \sigma_H^2) dt \\ V_T &= \left[e^{rT} V_0 + \int_0^T e^{r(T-t)} \Delta_t (dS_t - r S_t dt) \right] \\ &\quad + \frac{1}{2} \int_0^T e^{r(T-t)} \Gamma_t S_t^2 (\sigma^2 - \sigma_H^2) dt. \end{aligned}$$

The first bracketed term on RHS is the hedging P/L with σ_H as the volatility for hedging and initial pricing (cf. Chap. 13, Sect. 6). The second term is the missing P/L which is the result to be proved. $\square$

Transaction Cost

The framework of option pricing with transaction costs presented here was proposed by Leland [48]. Let k be the round trip transaction cost per dollar value of the stock. The self-financing hedging portfolio with no transaction cost satisfies

$$\delta V \approx \Delta \delta S + r(V - \Delta S)\delta t,$$

where δt is a small discrete change in time. The cost for adjusting the Δ position during the time interval δt is

$$\delta K = \frac{k}{2} S |\delta \Delta| \approx \frac{k}{2} S |\Gamma \sigma S \delta W|. \quad \left(\Gamma = \frac{\delta \Delta}{\delta S}, \delta S \approx \sigma S \delta W \right)$$

In this analysis, $\mathbb{E}[|\delta W|]$ is applied instead of $|\delta W|$. By direct integration and the fact that $\delta W \sim \mathcal{N}(0, \delta t)$, we can obtain

$$\begin{aligned} \mathbb{E}[|\delta W|] &= \int_{-\infty}^{\infty} \frac{|x|}{\sqrt{2\pi\delta t}} e^{-\frac{x^2}{2\delta t}} dx = 2 \int_0^{\infty} \frac{x}{\sqrt{2\pi\delta t}} e^{-\frac{x^2}{2\delta t}} dx \\ &= \sqrt{\frac{2\delta t}{\pi}} \int_0^{\infty} e^{-\frac{x^2}{2\delta t}} d\left(\frac{x^2}{2\delta t}\right) = \sqrt{\frac{2\delta t}{\pi}} \int_0^{\infty} e^{-y} dy = \sqrt{\frac{2\delta t}{\pi}}. \end{aligned}$$

Defining Leland number, $\text{Le} = \sqrt{\frac{2}{\pi}} \left(\frac{k}{\sigma \sqrt{\delta t}} \right)$, we get $\delta K = \frac{1}{2} \sigma^2 S^2 |\Gamma| \text{Le} \delta t$. Adding in the cost for rebalancing the delta δK , we obtain

$$\delta V \approx \Delta \delta S + r(V - \Delta S)\delta t - \frac{1}{2} \sigma^2 S^2 |\Gamma| \text{Le} \delta t.$$

From the Itô–Doebelin formula, we have $dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} dt$. Therefore,

$$\frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} dt = \Delta dS + r(V - \Delta S)dt - \frac{1}{2} \sigma^2 S^2 |\Gamma| \text{Le} dt.$$

Arranging the terms, we obtain the modified BSM model with transaction cost:

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + \frac{1}{2} \sigma^2 S^2 \left| \frac{\partial^2 V}{\partial S^2} \right| \text{Le} + rS \frac{\partial V}{\partial S} - rV = 0.$$

If $1 + \text{sgn}(\Gamma) \text{Le} \geq 0$, we have

$$\frac{\partial V}{\partial t} + \frac{1}{2} \tilde{\sigma}^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

with adjusted volatility

$$\tilde{\sigma}^2 = \sigma^2(1 + \text{sgn}(\Gamma)\text{Le}).$$

The impact on option price for small $(\tilde{\sigma} - \sigma)$ is roughly $\text{Vega} \times (\tilde{\sigma} - \sigma)$ with $\tilde{\sigma} - \sigma \approx \frac{k}{\sqrt{2\pi}\delta t} \text{sgn}(\Gamma)$ ($\tilde{\sigma}^2 - \sigma^2 = \sigma^2 \text{sgn}(\Gamma)\text{Le}$, $\tilde{\sigma}^2 - \sigma^2 = (\tilde{\sigma} + \sigma)(\tilde{\sigma} - \sigma) \approx 2\sigma(\tilde{\sigma} - \sigma)$). We can see that the more frequent the rebalancing (smaller δt), the higher will be the transaction cost.

Exercises

1. Without using the put-call parity, verify that the Black–Scholes put option formula (11.14) satisfies the Black–Scholes equation (11.8) and the corresponding boundary conditions and terminal condition.
2. Let $N'(d)$ denote the derivative of the standard normal c.d.f.. Show that

$$S_t N'(d_1) = K e^{-r(T-t)} N'(d_2)$$

and use this equality to derive the Delta of a vanilla call.

3. Transform the Black–Scholes equation with log price by making the variable change $x = \ln S$, and obtain the corresponding boundary conditions and terminal condition.
4. Find the value of Black–Scholes call price when volatility approaches ∞ . What does the corresponding Delta imply?

Chapter 12

Stochastic Calculus Part II



12.1 Change of Probability

Changing probability measure plays an important role in risk-neutral option pricing. We first review the following important theorem.

Theorem 12.1 (Radon–Nikodym) *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. There exists a unique $Z > 0$, $\mathbb{E}[Z] = 1$, $\mathbb{P}$ -a.s., such that a new probability measure $\tilde{\mathbb{P}}$ can be defined*

$$\tilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega), \quad \forall A \in \mathcal{F}.$$

Furthermore,

$$\tilde{\mathbb{E}}[X] = \mathbb{E}[ZX] \quad \text{and} \quad \tilde{\mathbb{E}}\left[\frac{X}{Z}\right] = \mathbb{E}[X],$$

Z is called the Radon–Nikodym derivative of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$. We often note it as $Z = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}$. Also, $\tilde{\mathbb{P}}$ is equivalent to $\mathbb{P}$ (noted as $\tilde{\mathbb{P}} \sim \mathbb{P}$), i.e., they agree on the probability zero sets.

The proof can be found in real analysis or probability textbooks such as [73].

We next define the Radon–Nikodym derivative process from the Radon–Nikodym derivative.

Lemma 12.1 *Let Z be the Radon–Nikodym derivative of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$. If we define $Z_t = \mathbb{E}[Z | \mathcal{F}_t]$, then Z_t is a $\mathbb{P}$ -martingale and*

$$\tilde{\mathbb{E}}[Y] = \mathbb{E}[YZ_t],$$

where Y is an $\mathcal{F}_t$ -measurable random variable for any t .

Proof For any $s < t$, the following arguments show that Z_t is a martingale:

$$\mathbb{E}[Z_t | \mathcal{F}_s] = \mathbb{E}[\mathbb{E}[Z | \mathcal{F}_t] | \mathcal{F}_s] = \mathbb{E}[Z | \mathcal{F}_s] = Z_s.$$

For the expectation under $\tilde{\mathbb{P}}$, the Radon–Nikodym theorem leads to

$$\tilde{\mathbb{E}}[Y] = \mathbb{E}[YZ] = \mathbb{E}[\mathbb{E}[YZ | \mathcal{F}_t]] = \mathbb{E}[Y\mathbb{E}[Z | \mathcal{F}_t]] = \mathbb{E}[YZ_t]. \quad \square$$

Z_t is called a Radon–Nikodym derivative process and is noted as $\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} \Big|_{\mathcal{F}_t}$. As $\mathbb{E}[Z_t | \mathcal{F}_s] = Z_s$ implies $\mathbb{E}[\frac{Z_t}{Z_s} | \mathcal{F}_s] = 1$, it is natural to question the equivalence between $\tilde{\mathbb{E}}[Y | \mathcal{F}_s]$ and $\mathbb{E}[Y \frac{Z_t}{Z_s} | \mathcal{F}_s] (= \frac{1}{Z_s} \mathbb{E}[YZ_t | \mathcal{F}_s])$ which seems to be the “conditioned version” of Radon–Nikodym theorem. This is confirmed by the following which is a very useful formula when we calculate the conditional expectation under the new measure $\tilde{\mathbb{P}}$.

Proposition 12.1 (Bayes’ Formula) *Let $0 \leq s \leq t$, Y be an $\mathcal{F}_t$ -measurable random variable, $Z_t = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} \Big|_{\mathcal{F}_t}$ be a Radon–Nikodym derivative process, then*

$$\tilde{\mathbb{E}}[Y | \mathcal{F}_s] = \frac{1}{Z_s} \mathbb{E}[YZ_t | \mathcal{F}_s].$$

Proof We only need to check the partial averaging property of conditional expectation. For any $A \in \mathcal{F}_s$,

$$\begin{aligned} \int_A \frac{1}{Z_s} \mathbb{E}[YZ_t | \mathcal{F}_s] d\tilde{\mathbb{P}} &= \tilde{\mathbb{E}} \left[\mathbf{1}_A \frac{1}{Z_s} \mathbb{E}[YZ_t | \mathcal{F}_s] \right] = \mathbb{E}[\mathbf{1}_A \mathbb{E}[YZ_t | \mathcal{F}_s]] \\ &= \mathbb{E}[\mathbb{E}[\mathbf{1}_A Y Z_t | \mathcal{F}_s]] = \mathbb{E}[\mathbf{1}_A Y Z_t] = \tilde{\mathbb{E}}[\mathbf{1}_A Y] = \int_A Y d\tilde{\mathbb{P}}. \quad \square \end{aligned}$$

We introduce here a special type of martingale known as the exponential martingale. It is an important tool for applying the upcoming Girsanov Theorem.

Proposition 12.2 (Exponential Martingale) *Let θ_t be an $\mathcal{F}_t$ -adapted process.¹ The stochastic process*

$$Z_t = \exp \left\{ \int_0^t \theta_s dW_s - \frac{1}{2} \int_0^t \theta_s^2 ds \right\}$$

is an $\mathcal{F}_t$ -martingale and $\mathbb{E}[Z_t] = Z_0 = 1$.

¹Which satisfies the so-called Novikov’s condition.

Proof Let $f(x) = e^x$ and $X_t = \int_0^t \theta_s dW_s - \frac{1}{2} \int_0^t \theta_s^2 ds$, which implies

$$dX_t = \theta_t dW_t - \frac{1}{2} \theta_t^2 dt.$$

By Itô–Doebelin formula, we have

$$\begin{aligned} dZ_t &= df(X_t) = f'(X_t) dX_t + \frac{1}{2} f''(X_t) dX_t dX_t \\ &= e^{X_t} \left(\theta_t dW_t - \frac{1}{2} \theta_t^2 dt \right) + \frac{1}{2} e^{X_t} \theta_t^2 dt = \theta_t Z_t dW_t. \end{aligned}$$

Hence Z_t is an $\mathcal{F}_t$ -martingale and $\mathbb{E}[Z_t] = Z_0 = 1$. $\square$

Theorem 12.2 (Girsanov Theorem) *Let W_t be a Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$ and θ_t be an adapted process. Then*

$$Z_t = \exp \left\{ - \int_0^t \theta_s dW_s - \frac{1}{2} \int_0^t \theta_s^2 ds \right\} \quad (12.1)$$

is an exponential martingale such that $dZ_t = -\theta_t Z_t dW_t$. Taking Z_t as a Radon–Nikodym derivative process $\left. \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} \right|_{\mathcal{F}_t}$, and define

$$\tilde{W}_t = W_t + \int_0^t \theta_s ds,$$

then $\tilde{W}_t$ is a Brownian motion under the probability measure $\tilde{\mathbb{P}}$, assuming that $\mathbb{E}[\int_0^T \theta_s^2 ds] < \infty$.

Note that we write indifferently $dZ_t = -\theta_t Z_t dW_t$ with $Z_0 = 1$ or (12.1) in the following context.

Proof It can be proved by the direct application of the upcoming general form of Girsanov Theorem, i.e., Theorem 12.3 with $dZ_t = -\theta_t Z_t dW_t$. $\square$

Let us make an intuitive analysis with constant $\theta_t (= \theta)$, $W_t = \sqrt{t}\epsilon$ ($\epsilon \sim \mathcal{N}(0, 1)$) and $\tilde{W}_t = W_t + \theta t$. Let $\tilde{\epsilon}$ be a random variable such that $\tilde{W}_t = \sqrt{t}\tilde{\epsilon}$. Then, we can check that

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{\epsilon^2}{2}} = \frac{1}{\sqrt{2\pi}} e^{-\frac{\tilde{\epsilon}^2}{2}} \left(e^{-\frac{1}{2}\theta^2 t - \theta\sqrt{t}\epsilon} \right) = \frac{1}{\sqrt{2\pi}} e^{-\frac{\tilde{\epsilon}^2}{2}} Z_t.$$

Due to the appearance of a drift term $\int_0^t \theta_s ds$, $\tilde{W}_t$ is not a martingale under the original probability measure $\mathbb{P}$ in general. However, the Girsanov Theorem states that $\tilde{W}_t$ becomes a martingale under $\tilde{\mathbb{P}}$. As a result, $\tilde{W}_t$ enjoys all the Brownian motion properties under $\tilde{\mathbb{P}}$, for instance, $\mathbb{E}[\tilde{W}_t] = 0$ (note that $\mathbb{E}[W_t] \neq 0$).

Next we introduce a more general form of Girsanov theorem, which does not rely on the explicit form of the Radon–Nikodym derivative.

Lemma 12.2 *Suppose $\tilde{\mathbb{P}} \sim \mathbb{P}$ with Radon–Nikodym derivative process $Z_t = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} \Big|_{\mathcal{F}_t}$. Then, $\tilde{X}_t$ is a $\tilde{\mathbb{P}}$ -martingale if and only if $\tilde{X}_t Z_t$ is a $\mathbb{P}$ -martingale.*

Proof Let $\tilde{X}_t$ be a $\tilde{\mathbb{P}}$ -martingale. From Bayes' formula, we have

$$\tilde{X}_s = \tilde{\mathbb{E}}[\tilde{X}_t | \mathcal{F}_s] = \frac{\mathbb{E}[Z_t \tilde{X}_t | \mathcal{F}_s]}{Z_s}, \quad \forall 0 < s < t,$$

i.e. $Z_s \tilde{X}_s = \mathbb{E}[Z_t \tilde{X}_t | \mathcal{F}_s]$. Conversely, let $\tilde{X}_t Z_t$ be a $\mathbb{P}$ -martingale, then we have $Z_s \tilde{X}_s = \mathbb{E}[Z_t \tilde{X}_t | \mathcal{F}_s]$, i.e.

$$\tilde{X}_s = \frac{\mathbb{E}[Z_t \tilde{X}_t | \mathcal{F}_s]}{Z_s} = \tilde{\mathbb{E}}[\tilde{X}_t | \mathcal{F}_s],$$

by Bayes' formula again. □

Theorem 12.3 (Girsanov–Meyer Theorem) *Suppose $\tilde{\mathbb{P}} \sim \mathbb{P}$ with Radon–Nikodym derivative process Z_t , which is continuous ($\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} \Big|_{\mathcal{F}_t} = Z_t$). Moreover, X_t is a continuous $\mathbb{P}$ -martingale. Define $\tilde{X}_t$ by*

$$d\tilde{X}_t = dX_t - \frac{1}{Z_t} d\langle X, Z \rangle_t.$$

Then $\tilde{X}_t$ is a continuous $\tilde{\mathbb{P}}$ -martingale.

Proof First, both X_t and Z_t are continuous $\mathbb{P}$ -martingales. The quadratic covariation $\langle X, Z \rangle_t$ is a process without the Itô integral term (actually a finite variation process) [43] and hence $d\langle X, Z \rangle_t = 0$. Then,

$$d\langle \tilde{X}, Z \rangle_t = \left(dX_t - \frac{1}{Z_t} d\langle X, Z \rangle_t \right) dZ_t = d\langle X, Z \rangle_t - \frac{1}{Z_t} d\langle \langle X, Z \rangle, Z \rangle_t = d\langle X, Z \rangle_t.$$

Now, according to the Lemma 12.2, it is enough to check that $\tilde{X}_t Z_t$ is $\mathbb{P}$ -martingale. Note that

$$\begin{aligned} d(\tilde{X}_t Z_t) &= Z_t d\tilde{X}_t + \tilde{X}_t dZ_t + d\langle \tilde{X}, Z \rangle_t = Z_t d\tilde{X}_t + \tilde{X}_t dZ_t + d\langle X, Z \rangle_t \\ &= Z_t dX_t - d\langle X, Z \rangle_t + \tilde{X}_t dZ_t + d\langle X, Z \rangle_t = Z_t dX_t + \tilde{X}_t dZ_t. \end{aligned}$$

Hence $\tilde{X}_t Z_t$ is a $\mathbb{P}$ -martingale. □

This above theorem is useful when Z_t is correlated to the $\mathbb{P}$ -martingale X_t , e.g., Z_t and X_t are driven by different but correlated Brownian motions.

12.2 Predictable Martingale Representation

An $\mathcal{F}_t$ -adapted process is said to be *predictable* if it is left continuous.

Theorem 12.4 *Let W_t be a Brownian with $\mathcal{F}_t$ as its natural filtration and X_t be an $\mathcal{F}_t$ -measurable continuous martingale, then, there is a unique predictable process ζ_s , such that*

$$X_t = X_0 + \int_0^t \zeta_s dW_s.$$

The proof can be found in [44] and [66].

The following illustrations may help to understand the theorem:

- if $\langle X, X \rangle_t = t$ for all t , then, $X_t = X_0 + \int_0^t dW_s$
- if $\langle X, X \rangle_t = ct$ for all t , then, $X_t = X_0 + \int_0^t \sqrt{c} dW_s$
- if $\frac{d\langle X, X \rangle_t}{dt} = \lambda_t$ exists for all t , then, $X_t = X_0 + \int_0^t \sqrt{\lambda_s} dW_s$.

More generally, let $\mathcal{F}_t^Y$ be the natural filtration of a martingale set Y . The necessary and sufficient condition for X_t to admit the predictable representation with the integral of Y_t ($X_t = X_0 + \int_0^t \zeta_s \cdot dY_s$) is that there is a unique probability measure $\mathbb{P}$ under which X_t is an $\mathcal{F}_t^Y$ -martingale [66]. It can be shown that $\mathbb{P}$ is unique if Y is a set of Brownians.

12.3 Stochastic Differential Equations

Consider Itô processes of the form:

$$dX_t = b(t, X_t)dt + \phi(t, X_t)dW_t,$$

where $b(t, x)$ is considered as the drift coefficient and $\phi(t, x)$ the diffusion coefficient. The above equation is a kind of stochastic differential equation (SDE).

Suppose $X_t = x$ is the initial condition at time $t \geq 0$, $x \in \mathbb{R}$. Then, the solution of the SDE is a stochastic process X_T , $T > t$, such that

$$\begin{cases} X_t = x, \\ X_T = X_t + \int_t^T b(s, X_s)ds + \int_t^T \phi(s, X_s)dW_s. \end{cases} \quad (12.2)$$

Theorem 12.5 (Existence and uniqueness theorem for one dimensional SDE)

Let b and ϕ be continuous functions such that, there exists a $K < +\infty$, and

1. $|b(t, x) - b(t, y)| + |\phi(t, x) - \phi(t, y)| \leq K|x - y|$
2. $|b(t, x)| + |\phi(t, x)| \leq K|1 + x|$

for all $t < T$. Then, (12.2) admits a unique solution.

Readers are referred to [60].

The following processes are frequently used in financial modelling.

Arithmetic Brownian Motion

The following SDE is used in Bachelier stock price model [5]

$$dS_t = \mu dt + \sigma dW_t,$$

which gives normally distributed $S_T = x + \mu(T - t) + \sigma(W_T - W_t)$ with $S_t = x$.

Geometric Brownian Motion

Consider the following SDE used in Black–Scholes–Merton model,

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where μ and $\sigma > 0$ are constants. Here $b(t, x) = \mu x$ and $\phi(x, t) = \sigma x$. With $S_t = x$, the unique solution is the lognormally distributed geometric Brownian motion

$$S_T = x e^{(\mu - \frac{1}{2}\sigma^2)(T-t) + \sigma(W_T - W_t)}.$$

Ornstein–Uhlenbeck Process

The OU (Ornstein–Uhlenbeck) process is the unique solution of

$$dX_t = -cX_t dt + \sigma dW_t, \quad X_0 = x.$$

Let $Y_t = X_t e^{ct}$. Applying Itô's product rule, we get $dY_t = \sigma e^{ct} dW_t$. Hence

$$X_t = x e^{-ct} + \sigma e^{-ct} \int_0^t e^{cs} dW_s,$$

which is a Gaussian process² with mean $\mathbb{E}[X_t] = x e^{-ct}$. Using Itô isometry,

²For any $t_i, i = 1, \dots, n$, the random vector $(X_{t_1}, X_{t_2}, \dots, X_{t_n})$ follows the multivariate normal distribution. For instance, the Brownian motion is a Gaussian process.

$$\begin{aligned}\text{Var}(X_t) &= \mathbb{E}[(X_t - \mathbb{E}[X_t])^2] = \sigma^2 e^{-2ct} \mathbb{E} \left[\left(\int_0^t e^{cs} dW_s \right)^2 \right] \\ &= \sigma^2 e^{-2ct} \mathbb{E} \left[\int_0^t e^{2cs} ds \right] = \sigma^2 \frac{1 - e^{-2ct}}{2c}.\end{aligned}$$

Cox–Ingersoll–Ross Square Root Process

CIR (Cox–Ingersoll–Ross) process is the unique solution of

$$dX_t = (a - bX_t)dt + \sigma\sqrt{X_t}dW_t, \quad X_t \in [0, \infty), \quad X_0 = x > 0, a > 0, b > 0.$$

It is a positive process with mean reversion behavior which means that X_t tends to rise when $X_t < \frac{a}{b}$ and tends to fall when $X_t > \frac{a}{b}$. The parameter b characterizes the speed of the mean reversion.

Let $\tau_0 = \inf\{t \geq 0 | X_t = 0\}$ be the first time of hitting 0 (more generally, a *stopping time*; see Chap. 15). The properties of CIR process are given below

- if $a \geq \frac{\sigma^2}{2}$, $\mathbb{P}\{\tau_0 = \infty\} = 1$
- if $0 \leq a < \frac{\sigma^2}{2}$, $\mathbb{P}\{\tau_0 < \infty\} = 1$.

It is also known that given the current value, the future values of a CIR process follow a non-central Chi-squared distribution extended to non-integer degrees of freedom, which can be calculated in closed-form.

The references for the proof are given in Appendix B where we show that the CIR process can be represented with a Squared Bessel process which is well studied for its properties. The relevant properties of Squared Bessel are also introduced in Appendix B.

Proposition 12.3 (Markov Property of SDE Solutions) *Let X_u , $u \geq 0$ be a solution to the SDE with initial condition given at time 0. Then for any function h ,*

$$\mathbb{E}[h(X_T) | \mathcal{F}_t] = \mathbb{E}_t^{X_t}[h(X_T)],$$

where $\mathbb{E}_t^{X_t}[\cdot]$ denotes an expectation evaluated with initial value X_t at time t .

Proof can be found at [60]. We see that the value of X_T should depend on two things, the initial value X_t , which is $\mathcal{F}_t$ -measurable, and the increments of X_u , $t \leq u \leq T$, which, in turn, are determined by increments of W_u . As increments of W_u are independent of $\mathcal{F}_t$, it seems natural to drop the conditioning of $\mathcal{F}_t$ in the expectation.

The below theorem links the expectation computation from an SDE related problem with a PDE problem.

Theorem 12.6 (Feynman–Kac Theorem) *Let X_t follow the following SDE*

$$dX_t = b(t, X_t)dt + \phi(t, X_t)dW_t.$$

Assume that h satisfies $\mathbb{E}_t^x[|h(X_T)|] < \infty$. For any $x \in \mathbb{R}$ and $t < T$, let

$$g(t, x) = \mathbb{E}_t^x[h(X_T)].$$

Then, $g(t, x)$ satisfies the below PDE

$$\frac{\partial g}{\partial t}(t, x) + b(t, x) \frac{\partial g}{\partial x}(t, x) + \frac{1}{2} \phi^2(t, x) \frac{\partial^2 g}{\partial x^2}(t, x) = 0,$$

and the terminal condition $g(T, x) = h(x)$.

Proof We apply the Itô's formula and obtain

$$\begin{aligned} dg(t, X_t) &= \frac{\partial g}{\partial t} dt + b \frac{\partial g}{\partial x} dt + \phi \frac{\partial g}{\partial x} dW_t + \frac{1}{2} \phi^2 \frac{\partial^2 g}{\partial x^2} dt \\ &= \left(\frac{\partial g}{\partial t} + b \frac{\partial g}{\partial x} + \frac{1}{2} \phi^2 \frac{\partial^2 g}{\partial x^2} \right) dt + \phi \frac{\partial g}{\partial x} dW_t. \end{aligned}$$

As $g(t, X_t)$ is a martingale (Proposition 10.1), the dt term should be zero. We obtain

$$\frac{\partial g}{\partial t}(t, X_t) + b(t, X_t) \frac{\partial g}{\partial x}(t, X_t) + \frac{1}{2} \phi^2(t, X_t) \frac{\partial^2 g}{\partial x^2}(t, X_t) = 0$$

along every path of X . Therefore,

$$\frac{\partial g}{\partial t}(t, x) + b(t, x) \frac{\partial g}{\partial x}(t, x) + \frac{1}{2} \phi^2(t, x) \frac{\partial^2 g}{\partial x^2}(t, x) = 0$$

at every point (t, x) that can be reached by (t, X_t) . □

Exercises

1. Consider a stochastic integral $I_t = \int_0^t f_s dW_s$ where f_t is *deterministic*, i.e., a non-random function of t .

(i) Show that $\mathbb{E}[I_t] = 0$ and $\text{Var}(I_t) = \int_0^t f_s^2 ds$.

(ii) Show that the moment generating function of I_t for any $a \in \mathbb{R}$ is

$$\mathbb{E}[e^{aI_t}] = \exp \left\{ \frac{1}{2} a^2 \int_0^t f_s^2 ds \right\} = \exp \left\{ a \mathbb{E}[I_t] + \frac{1}{2} a^2 \text{Var}(I_t) \right\},$$

i.e., it uniquely determines that I_t is normally distributed with mean 0 and variance $\int_0^t f_s^2 ds$.

2. Let $X_t = W_t^2 - t$ and recall that it is an $\mathcal{F}_t$ -measurable martingale. Find the unique predictable process ζ_s as in the predictable martingale representation such that

$$X_t = X_0 + \int_0^t \zeta_s dW_s.$$

3. Find the value of $\mathbb{E}_t^{W_t}[W_T^2]$.
 4. The following assumes the same notations as in the Feynman–Kac theorem. Let r be a constant. Define f as a function of t and x with

$$f(t, x) = e^{-r(T-t)} \mathbb{E}_t^x[h(X_T)],$$

assuming $\mathbb{E}_t^x[|h(X_T)|] < \infty$, $\forall x$ and $t < T$. Show that

- (i) $e^{-rt} f(t, X_t)$ is a martingale
 (ii) $f(t, x)$ satisfies the PDE

$$\frac{\partial f}{\partial t}(t, x) + \theta(t, x) \frac{\partial f}{\partial x}(t, x) + \frac{1}{2} \phi^2(t, x) \frac{\partial^2 f}{\partial x^2}(t, x) = rf(t, x)$$

and the terminal condition $f(T, x) = h(x)$.

Chapter 13

Risk-Neutral Pricing Framework



The risk-neutral pricing framework is about the analysis and techniques for derivatives hedging and pricing. The pioneer work of Black, Scholes and Merton marked the beginning of the development of both theory and practice. The no-arbitrage hedging techniques of cash-and-carry strategy and delta-hedging strategy of BSM model are examples of risk-neutral hedging.

13.1 Money Market Account

Definition 13.1 (Money Market Account Process) Let r_t be the interest rate which may be deterministic or stochastic. The money market account process is defined as $M_t = e^{\int_0^t r_s ds}$, with the properties

$$dM_t = r_t M_t dt \quad \text{and} \quad d\frac{1}{M_t} = -\frac{r_t}{M_t} dt.$$

The money market process M_t can be considered as the value of a default-free cash deposit account and $\frac{1}{M_t} = e^{-\int_0^t r_s ds}$ is the discount factor for cashflows at time t . The discounting of an asset price S_t can be written as $\frac{S_t}{M_t}$ which demonstrates the price of S_t in the number of units of the money market account M_t . The role of the money market account is the so called **numéraire** which will be revisited in later chapters as it facilitates the financial modelling for derivatives hedging/pricing.

r_t represents the instantaneous interest rate for borrow/lending which may be stochastic. In this book, if there is no specific statement, we generally assume that

the correlation between r_t and the considered asset is zero. In particular, when r_t is a constant r , we get the familiar notations:

$$M_t = e^{rt} \quad \text{and} \quad \frac{1}{M_t} = e^{-rt}.$$

13.2 Risk-Neutral Probability Measure

Definition 13.2 (Risk-Neutral Probability) A risk-neutral probability measure $\tilde{\mathbb{P}}$, is defined as the one under which the discounted price of any tradable non-dividend paying asset S_t is a martingale:

$$\tilde{\mathbb{E}} \left[\frac{S_t}{M_t} \middle| \mathcal{F}_0 \right] = \frac{S_0}{M_0}.$$

For a portfolio of tradable assets, as its price Π_t is a linear combination of the asset prices, $\frac{\Pi_t}{M_t}$ must be a martingale under the risk-neutral probability $\tilde{\mathbb{P}}$.

In this chapter, we make continuous-time modelling analysis on non-dividend paying assets following the stochastic differential equations as the below one:

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t, \quad \text{under } \mathbb{P},$$

where μ_t and σ_t are $\mathcal{F}_t$ -adapted processes. We have assumed the one-factor model (i.e. with one stochastic variable) for simplifying the presentation. The results can be generalized to the multi-dimensional case.

Proposition 13.1 *There exists a risk-neutral probability measure $\tilde{\mathbb{P}}$ under which $\frac{S_t}{M_t}$ is a martingale and*

$$d \frac{S_t}{M_t} = \sigma_t \frac{S_t}{M_t} d\tilde{W}_t,$$

or equivalently in the more popular notation,

$$dS_t = r_t S_t dt + \sigma_t S_t d\tilde{W}_t.$$

Proof By Itô's product rule,

$$\begin{aligned} d \frac{S_t}{M_t} &= \frac{1}{M_t} dS_t + S_t d \frac{1}{M_t} = \frac{1}{M_t} (\mu_t S_t dt + \sigma_t S_t dW_t) - r_t \frac{S_t}{M_t} dt \\ &= (\mu_t - r_t) \frac{S_t}{M_t} dt + \sigma_t \frac{S_t}{M_t} dW_t = \sigma_t \frac{S_t}{M_t} \left(dW_t + \frac{\mu_t - r_t}{\sigma_t} dt \right). \end{aligned}$$

Applying Girsanov theorem with $Z_t = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} \Big|_{\mathcal{F}_t} = \exp \left\{ -\int_0^t \theta_s dW_s - \frac{1}{2} \int_0^t \theta_s^2 ds \right\}$, where $\theta_t = (\mu_t - r_t)/\sigma_t$, we obtain $d\tilde{W}_t = dW_t + \theta_t dt$ such that

$$d \frac{S_t}{M_t} = \sigma_t \frac{S_t}{M_t} d\tilde{W}_t,$$

which is an exponential martingale under $\tilde{\mathbb{P}}$.

From $d \frac{S_t}{M_t} = \frac{1}{M_t} (dS_t - r_t S_t dt)$, we obtain

$$dS_t = r_t S_t dt + \sigma_t S_t d\tilde{W}_t. \quad \square$$

13.3 Self-Financing Portfolio

The discounted asset price is a risk-neutral martingale by Proposition 13.1. Next we study this property for a portfolio of assets and money market account, i.e., $\Pi_t = \Delta_t S_t + \zeta_t M_t$ where Δ_t, ζ_t (may be negative) represent respectively the number of assets and the units in money market account.

Definition 13.3 (Self-financing Portfolio) A *self-financing portfolio* is defined as the one satisfying

$$d\Pi_t = \Delta_t dS_t + \zeta_t dM_t.$$

In the discrete form, it is equivalent to

$$S_{t+\delta t} \Delta_{t+\delta t} + M_{t+\delta t} \zeta_{t+\delta t} = S_{t+\delta t} \Delta_t + M_{t+\delta t} \zeta_t,$$

which implies that when one readjusts the portions Δ_t and ζ_t in the asset and money markets respectively, the costs should cancel each other such that no money is brought in or taken out of the portfolio (see Chapter 4 Exercise 10 in [79]).

We then have the following important theorem.

Theorem 13.1 A portfolio $\Pi_t = \Delta_t S_t + \zeta_t M_t$ is self-financing if and only if

$$\frac{\Pi_t}{M_t} = \frac{\Pi_0}{M_0} + \int_0^t \Delta_u d \frac{S_u}{M_u}.$$

Hence, if $\frac{S_t}{M_t}$ is a $\tilde{\mathbb{P}}$ -martingale, $\frac{\Pi_t}{M_t}$ is also a $\tilde{\mathbb{P}}$ -martingale with

$$\tilde{\mathbb{E}} \left[\frac{\Pi_t}{M_t} \Big| \mathcal{F}_s \right] = \frac{\Pi_s}{M_s}, \quad s \leq t.$$

Proof If Π_t is self-financed, then

$$\begin{aligned} d\frac{\Pi_t}{M_t} &= -r_t \frac{\Pi_t}{M_t} dt + \frac{1}{M_t} d\Pi_t = -r_t \frac{\Pi_t}{M_t} dt + \frac{1}{M_t} (\Delta_t dS_t + \zeta_t dM_t) \\ &= -r_t \frac{\Pi_t}{M_t} dt + \frac{1}{M_t} [\Delta_t dS_t + r_t (\Pi_t - \Delta_t S_t) dt] \\ &= \Delta_t \frac{1}{M_t} (dS_t - r_t S_t dt) = \Delta_t d\frac{S_t}{M_t}. \end{aligned}$$

Hence, we proved the necessary condition.

Now, assuming $\Pi_t = \Delta_t S_t + \zeta_t M_t$ and $\frac{\Pi_t}{M_t} = \frac{\Pi_0}{M_0} + \int_0^t \Delta_u d\frac{S_u}{M_u}$, we first have

$$d\frac{\Pi_t}{M_t} = \Delta_t d\frac{S_t}{M_t} = \Delta_t \left(S_t d\frac{1}{M_t} + \frac{1}{M_t} dS_t \right).$$

On the other hand, Itô's product rule implies

$$d\frac{\Pi_t}{M_t} = \Pi_t d\frac{1}{M_t} + \frac{1}{M_t} d\Pi_t = (\Delta_t S_t + \zeta_t M_t) d\frac{1}{M_t} + \frac{1}{M_t} d\Pi_t.$$

Equating the two and then rearranging, we obtain

$$d\Pi_t = \Delta_t dS_t + \zeta_t dM_t. \quad \square$$

13.3.1 Discounting, Excess Return and Self-Financing

In finance, the discounting, excess return and self-financing are intimately related concepts. The interpretation of the term $d\frac{S_t}{M_t} = d(e^{-\int_0^t r_u du} S_t) = e^{-\int_0^t r_u du} (dS_t - r_t S_t dt)$ can be equivalently expressed as

- increment of discounted asset price for the time interval dt
- present value of the excess return from the asset for the time interval dt
- present value of the return from the self-financing investment in the asset over time dt .

We highlight that such return corresponds to practical implementation in the financial market.

13.4 Fundamental Theorems of Asset Pricing

Definition 13.4 An arbitrage is a self-financing portfolio Π_t such that, with $\Pi_0 = 0$, it satisfies for some $T > 0$:

$$\mathbb{P}\{\Pi_T \geq 0\} = 1 \quad \text{and} \quad \mathbb{P}\{\Pi_T > 0\} > 0.$$

Theorem 13.2 (First Fundamental Theorem of Asset Pricing) *If a market model has a risk-neutral probability measure, then, it does not admit arbitrage.*

Proof Let $\tilde{\mathbb{P}}$ be the risk-neutral probability measure that is equivalent to $\mathbb{P}$. Let $\Pi_0 = 0$ and $\mathbb{P}\{\Pi_T \geq 0\} = 1$. As $\mathbb{P}\{\Pi_T \geq 0\} = 1$, we have necessarily $\mathbb{P}\{\Pi_T < 0\} = 0$ which implies $\tilde{\mathbb{P}}\{\Pi_T < 0\} = 0$ because $\tilde{\mathbb{P}} \sim \mathbb{P}$.

By Proposition 13.1, we have

$$\tilde{\mathbb{E}}\left[\frac{\Pi_t}{M_t}\right] = \frac{\Pi_0}{M_0} = 0.$$

Hence, we have $\tilde{\mathbb{P}}\{\Pi_T > 0\} = 0$ (otherwise, we would get a contradiction $\tilde{\mathbb{E}}[\frac{\Pi_t}{M_t}] > 0$). As $\tilde{\mathbb{P}} \sim \mathbb{P}$, we get $\mathbb{P}\{\Pi_T > 0\} = 0$, i.e., no arbitrage exists. $\square$

Definition 13.5 A market model is called *complete* if every derivative security can be hedged.

Theorem 13.3 (Second Fundamental Theorem of Asset Pricing) *The model is complete if and only if the risk-neutral probability measure is unique.*

We refer to [79] for the proof.

For hedging a contingent claim on tradable assets, we wish to replicate the contingent claim by a self-financing portfolio of the assets. Under the risk-neutral probability $\tilde{\mathbb{P}}$, the discounted value of the portfolio is a martingale. The uniqueness of $\tilde{\mathbb{P}}$ is the necessary and sufficient condition for the predictable martingale representation property which implies the existence of the self-financing portfolio process for replicating the contingent claim. Actually, the integral part in the martingale representation constitutes the dynamic hedging process.

13.5 Hedging and Pricing in Complete Market

Proposition 13.2 *In a complete market, the self-financing hedging process for a short position of a contingent claim paying out V_T at T is*

$$\frac{V_t}{M_t} = \frac{V_0}{M_0} + \int_0^t \Delta_u d\frac{S_u}{M_u}, \quad 0 \leq t \leq T \quad (M_0 = 1),$$

where Δ_u is a unique predictable process and $\frac{S_u}{M_u}$ is a $\tilde{\mathbb{P}}$ -martingale.

Proof We replicate (hedge) the contingent claim by a self-financing portfolio Π_t with terminal value $\Pi_T = V_T$. In a complete market, $\tilde{\mathbb{P}}$ is unique and $\frac{\Pi_T}{M_T}$ is a $\tilde{\mathbb{P}}$ -martingale. Hence, by the predictable martingale representation property, there exists a unique predictable process ϕ_u , such that

$$\frac{\Pi_t}{M_t} = \frac{\Pi_0}{M_0} + \int_0^t \phi_u d\tilde{W}_u, \quad 0 \leq t \leq T.$$

Defining the unique predictable process as $\Delta_u = \phi_u / (\sigma_u \frac{S_u}{M_u})$, we obtain

$$\frac{\Pi_t}{M_t} = \frac{\Pi_0}{M_0} + \int_0^t \Delta_u \sigma_u \frac{S_u}{M_u} d\tilde{W}_u = \frac{\Pi_0}{M_0} + \int_0^t \Delta_u d\frac{S_u}{M_u} \quad (\text{Proposition 13.1}).$$

By Theorem 13.1, Π_t is a self-financing portfolio. Setting $V_t = \Pi_t$, $\forall 0 \leq t \leq T$ finishes the proof. $\square$

13.5.1 General Pricing Formula

The general pricing formula for a European option paying out V_T at maturity can be obtained by taking conditional expectation for time-0 (hence $M_0 = 1$):

$$V_0 = \tilde{\mathbb{E}} \left[\frac{V_T}{M_T} \middle| \mathcal{F}_0 \right].$$

Now, we take conditional expectation for time- t :

$$\begin{aligned} \tilde{\mathbb{E}} \left[\frac{V_T}{M_T} \middle| \mathcal{F}_t \right] &= \frac{V_0}{M_0} + \tilde{\mathbb{E}} \left[\int_0^T \Delta_u d\frac{S_u}{M_u} \middle| \mathcal{F}_t \right] \\ &= \frac{V_0}{M_0} + \tilde{\mathbb{E}} \left[\int_0^t \Delta_u d\frac{S_u}{M_u} \middle| \mathcal{F}_t \right] + \tilde{\mathbb{E}} \left[\int_t^T \Delta_u d\frac{S_u}{M_u} \middle| \mathcal{F}_t \right]. \end{aligned}$$

The last term is $\tilde{\mathbb{E}} \left[\int_t^T \Delta_u d\frac{S_u}{M_u} \middle| \mathcal{F}_t \right] = \tilde{\mathbb{E}} \left[\int_t^T \Delta_u \sigma_u \frac{S_u}{M_u} d\tilde{W}_u \middle| \mathcal{F}_t \right] = 0$. On the other hand, Proposition 13.2 implies that $\frac{V_0}{M_0} + \tilde{\mathbb{E}} \left[\int_0^t \Delta_u d\frac{S_u}{M_u} \middle| \mathcal{F}_t \right] = \frac{V_0}{M_0} + \int_0^t \Delta_u d\frac{S_u}{M_u} = \frac{V_t}{M_t}$. Hence, V_T/M_T is a martingale under $\tilde{\mathbb{P}}$. We obtain the general formula for the mark-to-market price (MtM) of the derivative contract at any time t before its expiry:

$$V_t = M_t \widetilde{\mathbb{E}} \left[\frac{V_T}{M_T} \middle| \mathcal{F}_t \right]. \quad (13.1)$$

13.5.2 Determination of Hedging Parameter Δ_t

Itô–Doebelin formula and Itô’s product rule give the differential of $\frac{V_t}{M_t}$ as

$$\begin{aligned} d \frac{V_t}{M_t} &= \frac{1}{M_t} \left(-r_t V_t dt + \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} d\langle S, S \rangle_t \right) \\ &= \frac{1}{M_t} \left(-r_t V_t + \frac{\partial V}{\partial t} + r_t S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \frac{\partial V}{\partial S} \sigma_t \frac{S_t}{M_t} d\widetilde{W}_t. \end{aligned} \quad (13.2)$$

The risk-neutral SDE $dS_t = r_t S_t dt + \sigma_t S_t d\widetilde{W}_t$ in Proposition 13.1 is used. We also know that $\frac{V_t}{M_t}$ is a $\widetilde{\mathbb{P}}$ -martingale from self-financing replication. The drift term in (13.2) equals to 0 and the remaining term is then

$$d \frac{V_t}{M_t} = \frac{\partial V}{\partial S} \sigma_t \frac{S_t}{M_t} d\widetilde{W}_t \quad \text{or} \quad \frac{V_T}{M_T} = \frac{V_0}{M_0} + \int_0^T \frac{\partial V}{\partial S} d \frac{S_t}{M_t}.$$

On the other hand, from Proposition 13.2, we have

$$\frac{V_T}{M_T} = \frac{V_0}{M_0} + \int_0^T \Delta_t d \frac{S_t}{M_t}.$$

Comparing the above equations, we obtain the delta amount

$$\Delta_t = \frac{\partial V}{\partial S}(t, S_t).$$

13.6 Discussion on Hedging, Pricing and Risk-Neutral Framework

The key objective of modelling for derivatives is to determine the self-financing hedging process and the pricing methodology. The hedging of a European style option (not limited to vanilla options) with expiry T consists of setting up a self-financing portfolio for the option with initial value representing the option premium at inception. By Proposition 13.2, the portfolio will have its final value equal to the option payoff at expiry. The hedging process consists of holding a quantity of Δ_t of underlying assets through self-financing: if $\Delta_t \geq 0$, the cash amount of $\Delta_t S_t$

is borrowed for financing the purchase of Δ_t units of underlying asset; otherwise, $|\Delta_t|$ units of the asset are borrowed and sold at the prevailing asset price S_t with the selling proceeds $|\Delta_t|S_t$ lent to money market. This describes the dynamic hedging for any time $t < T$. In particular, expanding $d\frac{S_t}{M_t}$ with $M_t = e^{rt}$ in Proposition 13.2 leads to

$$V_T = e^{rT} V_0 + \int_0^T e^{r(T-t)} \Delta_t (dS_t - rS_t dt),$$

where the first term is the accrued option premium and the second term is the accrued hedging profit and loss (P/L). We recall that the term $-rS_t dt$ can be regarded as the financing cost per asset unit and $(dS_t - rS_t dt)$ is the return from the self-financing investment into one asset. Hence, $\Delta_t (dS_t - rS_t dt)$ is the P/L of holding Δ_t units of underlying assets with self-financing.

The option price at any time $t < T$ is simply defined as the portfolio's value V_t , knowing that at expiry time T , the portfolio's value V_T is the option's payoff. The discounted price of the portfolio is a martingale because the discounted prices of its components (i.e. underlying assets and money market account position) are martingales. This leads to the general pricing formula (13.1):

$$V_t = M_t \tilde{\mathbb{E}} \left[\frac{V_T}{M_T} \middle| \mathcal{F}_t \right].$$

By Theorem 13.2, the existence of risk-neutral probability measure ensures no arbitrage. Risk-neutral framework is characterized by the drift r_t for the process of any tradable non-dividend paying asset when money market account is the numéraire and is used for financing. The intuitive illustration of the rationale, apart from arbitrage issue, is that with any other choice for the drift, say $\mu \neq r$, one would expect a “hedging process” like $V_T = e^{\mu T} V_0 + \int_0^T e^{\mu(T-t)} \Delta_t (dS_t - \mu S_t dt)$, which does not correspond to the money market account accrual and self-financing implementation under current financial/economic system. On the other hand, the risk-neutral framework provides the naturally implementable self-financing hedging process as previously explained.

Example 13.1 (Discrete delta-hedging) Consider the discretized self-financing portfolio and see how the portfolio approximates the option value. Expanding $d\frac{\Pi_t}{M_t} = \Delta_t d\frac{S_t}{M_t}$ with $M_t = e^{rt}$, we have

$$d\Pi_t = \Delta_t dS_t + r(\Pi_t - \Delta_t S_t) dt.$$

Then, the discrete case corresponds to

$$\Pi_{t+\delta t} \approx \Pi_t + \Delta_t (S_{t+\delta t} - S_t) + r(\Pi_t - \Delta_t S_t) \delta t = (1 + r\delta t)(\Pi_t - \Delta_t S_t) + \Delta_t S_{t+\delta t}.$$

This can be understood by owning Δ_t units of asset and $(\Pi_t - \Delta_t S_t)$ in the money market at time t . At time $t + \delta t$, the asset value becomes $S_{t+\delta t}$ and is worth $\Delta_t S_{t+\delta t}$, while the cash position grows by the factor $(1 + r\delta t)$ in the discrete sense.

13.7 Black–Scholes–Merton Model Revisited

Recall that the geometric Brownian motion models the asset price and the corresponding SDE is

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where σ is the constant volatility and μ is the drift of the asset in its real probability $\mathbb{P}$. Let r be the constant money market rate. By Proposition 13.1, we have $d\frac{S_t}{M_t} = \sigma \frac{S_t}{M_t} d\tilde{W}_t$, where $Z_t = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}\Big|_{\mathcal{F}_t} = \exp\left\{-\theta W_t - \frac{1}{2}\theta^2 t\right\}$, and $\theta = (\mu - r)/\sigma$. Under $\tilde{\mathbb{P}}$, $\frac{S_t}{M_t}$ is a martingale. Furthermore, the SDE for the asset price process is

$$dS_t = r S_t dt + \sigma S_t d\tilde{W}_t.$$

Let $V(t, S)$ be the derivative paying out $h(S_T)$ at maturity T and $V_t = V(t, S_t)$ be its mark-to-market value. Since $\frac{V_t}{M_t}$ is a $\tilde{\mathbb{P}}$ -martingale by self-financing replication, the dt term (drift) in (13.2) should be zero which leads to the so-called Black–Scholes PDE:

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0. \quad (13.3)$$

13.7.1 Closed-Form Solution for Call Option

To obtain the price, one can solve the governing PDE (13.3) or resort to the fact that the discounted option value is a risk-neutral martingale, thus,

$$V(t, S_t) = \tilde{\mathbb{E}}[e^{-r(T-t)} h(S_T) | \mathcal{F}_t].$$

For simplicity and without loss of generality, we first consider pricing a European call option with $t = 0$ under the Black–Scholes model.

Since the asset price follows the log-normal distribution, the solution for the asset price is the geometric Brownian motion

$$S_T = S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma \tilde{W}_T}.$$

The pricing of a European call option paying $c_T = (S_T - K)^+$ at maturity T is

$$\begin{aligned}
 c_0 &= \tilde{\mathbb{E}} \left[\frac{c_T}{M_T} \middle| \mathcal{F}_0 \right] = \tilde{\mathbb{E}} \left[\frac{(S_T - K)^+}{e^{rT}} \middle| \mathcal{F}_0 \right] \\
 &= \tilde{\mathbb{E}}[e^{-rT} (S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma \tilde{W}_T} - K)^+ | \mathcal{F}_0] \\
 &= \tilde{\mathbb{E}}[e^{-rT} (S_T - K)^+] \quad (\tilde{W}_T - \tilde{W}_0 \text{ is independent of } \mathcal{F}_0) \\
 &= \tilde{\mathbb{E}}[e^{-rT} S_T \mathbf{1}_{\{S_T > K\}}] - \tilde{\mathbb{E}}[e^{-rT} K \mathbf{1}_{\{S_T > K\}}] \equiv A - B.
 \end{aligned}$$

The condition $S_T > K$ is equivalent to $\tilde{W}_T > -d_2\sqrt{T}$, where we define

$$d_2 = \left[\ln \frac{S_0}{K} + (r - \frac{1}{2}\sigma^2)T \right] / (\sigma\sqrt{T})$$

and then

$$A = S_0 \tilde{\mathbb{E}} \left[e^{-\frac{1}{2}\sigma^2 T + \sigma \tilde{W}_T} \mathbf{1}_{\{\tilde{W}_T > -d_2\sqrt{T}\}} \right] = S_0 \tilde{\mathbb{E}} \left[Z_T \mathbf{1}_{\{\tilde{W}_T > -d_2\sqrt{T}\}} \right].$$

The term $Z_T = e^{-\frac{1}{2}\sigma^2 T + \sigma \tilde{W}_T}$ is an exponential martingale. It can be eliminated by applying Girsanov Theorem with $Z_t = \frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_t}$ as Radon–Nikodym derivative process for defining a new probability $\mathbb{Q}$. Then

$$\begin{aligned}
 A &= S_0 \tilde{\mathbb{E}}[Z_T \mathbf{1}_{\{\tilde{W}_T > -d_2\sqrt{T}\}}] = S_0 \tilde{\mathbb{E}}^{\mathbb{Q}}[\mathbf{1}_{\{\tilde{W}_T^{\mathbb{Q}} + \sigma T > -d_2\sqrt{T}\}}] \\
 &= S_0 \mathbb{Q}\{\tilde{W}_T^{\mathbb{Q}} > -d_2\sqrt{T} - \sigma T\} = S_0 \mathbb{Q}\{\epsilon > -d_2 - \sigma\sqrt{T}\} \\
 &\quad (\epsilon \text{ is standard normal variable}) \\
 &= S_0 \mathbb{Q}\{\epsilon < d_2 + \sigma\sqrt{T}\} \quad (\text{symmetry of normal distribution}) \\
 &= S_0 N(d_2 + \sigma\sqrt{T}) = S_0 N \left(\frac{\ln \frac{S_0}{K} + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \right) \equiv S_0 N(d_1).
 \end{aligned}$$

For the second term B , the expectation turns out to be a probability

$$\begin{aligned}
 B &= \tilde{\mathbb{E}}[e^{-rT} K \mathbf{1}_{\{S_T > K\}}] = e^{-rT} K \tilde{\mathbb{E}}[\mathbf{1}_{\{\tilde{W}_T > -d_2\sqrt{T}\}}] = e^{-rT} K \tilde{\mathbb{P}}\{\tilde{W}_T > -d_2\sqrt{T}\} \\
 &= e^{-rT} K \tilde{\mathbb{P}}\{\epsilon > -d_2\} = e^{-rT} K \tilde{\mathbb{P}}\{\epsilon < d_2\} = e^{-rT} K N(d_2).
 \end{aligned}$$

Hence $c_0 = A - B = S_0 N(d_1) - e^{-rT} K N(d_2)$, where

$$d_1 = \frac{\ln \frac{S_0}{K} + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \quad \text{and} \quad d_2 = d_1 - \sigma\sqrt{T}.$$

In the same way, we can obtain $c_t = S_t N(d_1) - e^{-r(T-t)} K N(d_2)$, where

$$d_1 = \frac{\ln \frac{S_t}{K} + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} \quad \text{and} \quad d_2 = d_1 - \sigma\sqrt{T-t}.$$

From the derivation, we see that the term $N(d_2)$ is the risk-neutral probability of a call option ending up in-the-money:

$$N(d_2) = \tilde{\mathbb{E}}[\mathbf{1}_{\{S_T > K\}}] = \tilde{\mathbb{P}}\{S_T > K\}.$$

On the other hand, the delta $N(d_1)$ (cf. the exercise on delta calculation in Chap. 11) is the amount of assets needed for hedging. From the relationship $d_2 = d_1 - \sigma\sqrt{T-t}$, there are two scenarios when $N(d_1)$ and $N(d_2)$ are close to each other, i.e., either $\sigma \rightarrow 0$ or $t \rightarrow T$.

13.8 Dividend Modelling

Many assets, such as equities, pay out dividends. The price of an option on an asset that pays dividends is affected by the payments. When we model dividend payments, we need to consider two issues: (1) When, and how often, are dividend payments made? (2) How large are the payments?

There are mainly three types of dividend modelling:

Cash dividend dividends are paid as fixed cash amount at discrete times. This is the most common dividend form.

Proportional dividend dividend amount of $q_t S_t$ are paid at discrete time $t_1, t_2, t_3, \dots$

Continuous dividend

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t - q_t S_t dt,$$

where q_t is the dividend yield and the term $q_t S_t dt$ can be understood as the value reduction in asset price after dividend. The continuous dividend is the easiest to model. So, it is applied for options where this approximation does not have significant impact.

Furthermore we define *price return* of an asset as the return without adding back the already paid dividends, and *total return* reflects the return by reinvesting all the paid dividends in the same asset.

Fixed cash dividend for long term is not a viable dividend modelling (a stock of \$100 paying \$3 dividend will not still pay \$3 even when its price drops to \$10). In practice, people often use:

- cash dividends for the short term (e.g. ≤ 2 years)

- proportional dividends for long term (e.g. ≥ 5 years)
- mixture of cash dividends and proportional dividends in between.

13.8.1 Risk-Neutral SDE with Continuous Dividend

The asset price model is

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t - q_t S_t dt. \quad (13.4)$$

Let $S_t^* = S_t e^{\int_0^t q_s ds}$ denote the total price of the asset with dividends reinvested, its discounted value S_t^*/M_t should be a martingale under the risk-neutral probability $\tilde{\mathbb{P}}$:

$$\begin{aligned} d \frac{S_t^*}{M_t} &= d \left(e^{-\int_0^t r_s ds} e^{\int_0^t q_s ds} S_t \right) = d \left(e^{\int_0^t (q_s - r_s) ds} S_t \right) \\ &= e^{\int_0^t (q_s - r_s) ds} [(q_t - r_t) S_t dt + dS_t] \\ &= e^{\int_0^t (q_s - r_s) ds} S_t [(q_t - r_t) dt + (\mu_t - q_t) dt + \sigma_t dW_t] = \sigma_t \frac{S_t^*}{M_t} d\tilde{W}_t, \end{aligned}$$

where $d\tilde{W}_t = \frac{\mu_t - r_t}{\sigma_t} dt + dW_t$. Equating $e^{\int_0^t (q_s - r_s) ds} [(q_t - r_t) S_t dt + dS_t] = \sigma_t \frac{S_t^*}{M_t} d\tilde{W}_t$, we obtain the risk-neutral SDE,

$$dS_t = (r_t - q_t) S_t dt + \sigma_t S_t d\tilde{W}_t. \quad (13.5)$$

Next, consider $r_t = r$, $q_t = q$, and $\sigma_t = \sigma$. For a contingent claim V replicated by a self-financing portfolio,

$$\begin{aligned} d \frac{V_t}{M_t} &= e^{-rt} \left(-r V_t dt + \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} d\langle S, S \rangle_t \right) \\ &= e^{-rt} \left[-r V_t + \frac{\partial V}{\partial t} + (r - q) S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right] dt + e^{-rt} \frac{\partial V}{\partial S} \sigma S_t d\tilde{W}_t. \end{aligned}$$

Since $\frac{V_t}{M_t}$ is a $\tilde{\mathbb{P}}$ -martingale, the drift term equals to zero and gives the governing equation

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r - q) S \frac{\partial V}{\partial S} - r V = 0.$$

13.8.2 Vanilla Option Pricing Formula

Let us consider pricing European call and put options $c(t, S_t)$ and $p(t, S_t)$ with continuous dividend q . The first way is to use the risk-neutral pricing formula with

$$S_T = S_t e^{(r-q-\frac{1}{2}\sigma^2)(T-t)+\sigma(W_T-W_t)}.$$

Theorem 13.4 *Under continuous dividend yield q ,*

$$c(t, S_t) = e^{-q(T-t)} S_t N(d_1) - K e^{-r(T-t)} N(d_2),$$

$$p(t, S_t) = K e^{-r(T-t)} N(-d_2) - e^{-q(T-t)} S_t N(-d_1),$$

where

$$d_1 = \frac{\ln(S_t/K) + (r - q + \frac{1}{2}\sigma^2)(T - t)}{\sigma\sqrt{T - t}}, \quad d_2 = d_1 - \sigma\sqrt{T - t}.$$

In PDE approach, the boundary conditions are modified. For the European call option, the final condition and the boundary condition at $S_t = 0$ are the same as the non-dividend paying case. The only change is the boundary at $S_t \rightarrow \infty$ given by $c(t, S_t) \sim S_t e^{-q(T-t)} - K e^{-r(T-t)}$, which can be verified by no arbitrage argument.

The call and put option prices give the modified version of put-call parity:

$$S_t e^{-q(T-t)} + p(t, S_t) - c(t, S_t) = K e^{-r(T-t)},$$

which can also be verified by a no-arbitrage argument.

13.8.3 Option Adjustment for Exceptional Dividend and Right Issuance

In option pricing, assumptions are made for the known and regular future dividends. The parties of the option trade take dividend risk that the effective paid dividends may differ from the assumed levels.

However, if there will be an exceptional dividend or right issuance after the inception of the option, the common market practice consists of adjusting the size of the option as well as the initial spot related parameters such as strike and barrier of the option so that the option price remains unchanged. This type of adjustment is covered in the termsheet and agreed by both parties of the option transaction.

The adjustment factor α is illustrated by the case for a Call option:

$$\begin{aligned}
N\tilde{\mathbb{E}}[(S_T - K)^+ | \mathcal{F}_t] &= N\tilde{\mathbb{E}}[(S_t^{cumdiv} e^{(r - \frac{1}{2}\sigma^2)(T-t) + \sigma\tilde{W}_{T-t}} - K)^+ | \mathcal{F}_t] \\
&= \frac{N}{\alpha} \tilde{\mathbb{E}}[((1 - div)S_t^{cumdiv} e^{(r - \frac{1}{2}\sigma^2)(T-t) + \sigma\tilde{W}_{T-t}} - \hat{K})^+ | \mathcal{F}_t],
\end{aligned}$$

where $\alpha = 1 - div$ and $\hat{K} = K\alpha$.

Hence, the adjustment consists of changing the notional to N/α and Strike to $K\alpha$.

Similar methods can be deduced for the adjustment factors for other type of options, such as Asian option which has already started with past known fixings.

Let us take the example of a \$2m, 1Y, 110% strike call option linked to a single stock. The initial spot is \$80 and the strike price is \$88. A few days after the option has started, the price of stock is \$90 and an exceptional dividend of \$4.5 (5%) is paid on the stock. The adjustment factor will be $\alpha = (100\% - 5\%)$. The strike price of the option contract will be adjusted to $88 \times (100\% - 5\%) = \83.6 and the notional amount of the option will be adjusted to $\$2\text{ m} / (100\% - 5\%) = \2.105 m .

13.9 Collateralized Derivative Pricing and FVA

As introduced in Chap. 1, an OTC derivative may be traded under a master agreement such as ISDA/CSA where the CSA (Credit Support Annex) agreement sets out the collateral posting terms and conditions. The derivative pricing framework presented so far has not considered the impact from collateral. In the following, we will adjust the standard derivative pricing framework with collateral posting.

We consider the situation for an OTC derivative where the CSA agreement between the two counterparties has the following terms:

- zero threshold for posting collateral
- zero MTA (Minimum Transfer Amount)
- collateral is cash in the same currency as the denomination for the derivative transaction
- the interest rate applied to the collateral is r^c (e.g. OIS based rate, cf. Chap. 2)

Under such condition, the party having the positive mark-to-market V_t (say, party A) will receive the cash collateral from the other party (say, party B) and will pay the interest for the cash collateral at the rate of r^c . If the normal financing rate for party A is r (which could be internal financing rate or unsecured funding rate), then the cash collateral with interest rate r^c implies a PnL of $(r_t - r_t^c) \times V_t \times \delta_t$ for the small period δ_t to party A. Recall that V_t represents the value of the self-financing portfolio replicating the payoff of the derivative. For taking into account this PnL impact from r^c , we set $V_t^* = V_t e^{\int_0^t (r_u - r_u^c) du}$ which is the “total price” of the derivative. It implies that any interest received/paid will be invested in/divested from V_t . Under $\tilde{\mathbb{P}}$, its discounted value V_t^*/M_t should be a martingale. Hence, by martingale representation property, we have

$$\frac{V_T^*}{M_T} = \frac{V_0^*}{M_0} + \int_0^T \phi_u d\tilde{W}_u,$$

where ϕ_u is a predictable process, $\frac{V_T^*}{M_T} = \frac{V_T e^{\int_0^T (r_u - r_u^c) du}}{M_T} = V_T e^{-\int_0^T r_u^c du}$ and $\frac{V_0^*}{M_0} = V_0$.

Taking conditional expectation, we obtain

$$V_0 = \tilde{\mathbb{E}} \left[e^{-\int_0^T r_u^c du} V_T \mid \mathcal{F}_0 \right].$$

We highlight that r^c appears only in the discounting term. The interest rate in the risk-neutral SDE for the underlying is still the financing rate r , i.e. for BSM model, $dS_t = rS_t dt + \sigma S_t d\tilde{W}_t$.

This pricing framework is known as the method of *CSA discounting* or *OIS discounting*. Its application needs to take into account the CSA terms and conditions on the collateral type, threshold/MTA and netting, etc. There is an abundance of literature on this topic after the financial crisis of 2008, e.g. [65].

For counterparties with CSA, the **FVA (Funding Value Adjustment)** is caused by the difference between the interest rate for hedging and the interest rate applied on the collateral for the trade's MtM exposure. The value of FVA is actually the expected funding P/L over the life of the trade. It can be calculated as the difference between the pricing with CSA discounting and the pricing without CSA discounting on the future cashflows of the transaction.

Exercises

1. Consider the Black–Scholes–Merton model for a non-dividend-paying stock. Show that the price of a forward start ATM call option (future stock price as strike) equals to the value of a European ATM call option with the same life.
2. Given the asset process

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t,$$

where μ_t and σ_t are non-random functions of t . The risk-free interest rate r_t is also a deterministic function of t .

Derive the SDE and PDE for European options and obtain the closed-form solution for the time- t price of a vanilla put option paying out $p_T = (K - S_T)^+$ at maturity T .

3. The futures price with maturity T at time t is defined as $F_t(T) = \tilde{\mathbb{E}}[S_T | \mathcal{F}_t]$, despite the modelling of S_t .

- (i) Show that the futures price $F_t(T)$ is a martingale under the risk-neutral probability measure.
- (ii) Assume by the martingale representation theorem that $dF_t(T) = \sigma_t F_t(T) d\tilde{W}_t$, where σ_t is deterministic. For $0 \leq s \leq t$, show the *Black's formula*:

$$\tilde{\mathbb{E}}[(F_t(T) - K)^+ | \mathcal{F}_s] = F_s(T)N(d_1) - KN(d_2)$$

where

$$d_1 = \frac{\ln(\frac{F_s(T)}{K}) + \frac{1}{2} \int_s^t \sigma_u^2 du}{\sqrt{\int_s^t \sigma_u^2 du}} \quad \text{and} \quad d_2 = d_1 - \sqrt{\int_s^t \sigma_u^2 du}.$$

Chapter 14

Numerical Methods for Option Pricing



Closed-form solutions (e.g. Black–Scholes formula) are accurate and fast to calculate. However, only very few options have had the closed-form solutions found under simple models such as the Black–Scholes–Merton. Therefore, numerical methods in the following are necessary for option pricing.

- *Numerical integration*, which is an efficient method. However, it is only applicable when the asset's probability of transition is known.
- *Tree methods*, which assume two or more possible moves for asset price for each small time step. A tree can be constructed for all possible asset prices until maturity date of the considered option. The option price can be calculated from the tree. Tree methods are easy to implement but are limited to options involving less than two random variables in general.
- *PDE methods*, which make use of the link between SDE and PDE. Numerical methods for PDEs are well studied in mathematics and physics. PDE methods are efficient for options with no more than two underlying assets.
- *Monte Carlo method*, which is the simplest but the most universal method for pricing all types of options (excluding American options which require special treatment). It consists of simulating the random paths and then calculating the averaged discounted payoff value as the option price.

In this chapter, we will focus on the methods of binomial tree, Monte Carlo, and finite difference for PDE.

14.1 The Binomial Tree

Cox, Ross and Rubinstein pioneered the binomial tree method in 1979 [25]. In this section, we introduce the binomial tree method for the Black–Scholes–Merton model under risk-neutral probability $\mathbb{P}$:

$$dS_t = rS_t dt + \sigma S_t d\tilde{W}_t.$$

In a binomial tree, we assume that, over a small time step δt , the asset price S_t has only two possible moves: “up” by a proportion of u with risk-neutral probability p , or “down” by a proportion of d with risk-neutral probability $1 - p$. Intuitively, u and d should be related to σ as we will verify later.

14.1.1 Tree Construction

We first construct an m -step binomial tree of possible asset prices by dividing $[0, T]$ into $0 = t_0 < t_1 < \dots < t_m = T$, $t_j = j\delta t$, $j = 0, 1, \dots, m$ and $\delta t = T/m$. Under Black–Scholes–Merton model, the binomial tree must satisfy the martingale property (i.e. $S_{t_j} = \mathbb{E}[e^{-r\delta t} S_{t_{j+1}} | \mathcal{F}_{t_j}]$) and the variance of $S_{t_{j+1}} - S_{t_j}$.

Let S_j^i denote the i -th asset price at time t_j . At time $t = 0$ we know the asset price S_0 . Then at the next time-step δt there are two possible asset prices, $S_1^1 = dS_0$ and $S_1^2 = uS_0$. At final time $m\delta t$, there will be $m + 1$ possible asset prices S_m^i , $i = 1, 2, \dots, m + 1$. The condition $u = 1/d$ is often applied to maintain the symmetric structure of the tree. See Fig. 14.1.

Next we apply the payoff function for the option on S_m^i , the last layer of the tree. For instance, in case of a call or put option, we have,

$$V_m^i = \max\{S_m^i - K, 0\} \quad \text{or} \quad V_m^i = \max\{K - S_m^i, 0\}, \quad i = 1, \dots, m + 1,$$

where K is the exercise price and V_m^i denotes the i -th possible value of the call/put at the last time-step m . We then evaluate the option prices in between, V_j^i , $j =$

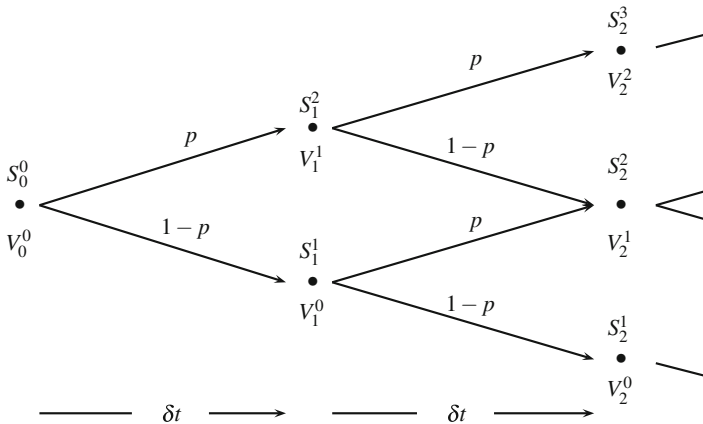


Fig. 14.1 m -step binomial method

$0, \dots, m - 1$, backward one step at a time using the transition probabilities p and $1 - p$. In particular, the time-0 price V_0 is the option premium.

Under risk-neutral probability, discounted asset price is a martingale. As we know the probability for the price move between two successive time steps (p for moving up and $1 - p$ for moving down), we have, for a European option,

$$V_j^i = e^{-r\delta t} [pV_{j+1}^{i+1} + (1 - p)V_{j+1}^i], \quad i = 1, \dots, j + 1.$$

Example 14.1 Let $S_0 = 20$, $u = 1.1$, $d = 0.9$, $r = 2\%$ and $p = 0.525$. Consider a European call option with $K = 21$ and $T = 0.5$ (6 months). We break $[0, T]$ into two time-steps with $\delta t = 0.25$ and construct a two-step binomial tree. See Fig. 14.2.

Since we have the stock prices for the nodes D, E and F at the expiry date, we can obtain the payoffs of the option at these nodes. Then, we compute the value of the option at node B:

$$e^{-0.02 \times 0.25} (0.525 \times 3.2 + 0.475 \times 0) = 1.6716.$$

Similarly, option values at nodes E and F give the option value at node C.

Now we repeat the computation once again, using the value of the options at nodes B and C, we get the option price at node A:

$$e^{-0.02 \times 0.25} (0.525 \times 1.6716 + 0.475 \times 0) = 0.8732.$$

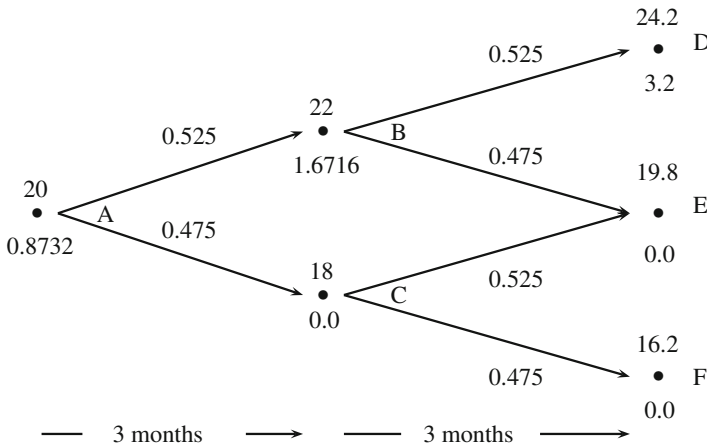


Fig. 14.2 Two-step binomial method

14.1.2 Determination of the Parameters

Since the moves u and d are independent at each level, the martingale property $S_{t_j} = \tilde{\mathbb{E}}[e^{-r\delta t} S_{t_{j+1}} | \mathcal{F}_{t_j}]$ can be written as $S = \tilde{\mathbb{E}}[e^{-r\delta t} S_{\delta t}]$ where S is any tree node and $S_{\delta t}$ is its value after δt . Under the risk-neutral assumption, $Se^{r\delta t} = \tilde{\mathbb{E}}[S_{\delta t}] = pSu + (1-p)Sd$, or

$$e^{r\delta t} = pu + (1-p)d. \quad (14.1)$$

From Corollary 11.1, we have $\tilde{\text{Var}}(S_{\delta t}) = S^2 e^{2r\delta t} (e^{\sigma^2 \delta t} - 1)$. Since $\tilde{\text{Var}}(S_{\delta t}) = \tilde{\mathbb{E}}[S_{\delta t}^2] - (\tilde{\mathbb{E}}[S_{\delta t}])^2$, and $\tilde{\mathbb{E}}[S_{\delta t}^2] = p(uS)^2 + (1-p)(dS)^2$, it follows that

$$S^2 e^{2r\delta t} (e^{\sigma^2 \delta t} - 1) = pu^2 S^2 + (1-p)d^2 S^2 - S^2 (pu + (1-p)d)^2,$$

or

$$e^{2r\delta t + \sigma^2 \delta t} = pu^2 + (1-p)d^2. \quad (14.2)$$

Equations (14.1) and (14.2) impose two conditions on p , u and d . As discussed previously, the third condition could be [25]

$$u = \frac{1}{d}. \quad (14.3)$$

Solving (14.1), (14.2) and (14.3), we obtain

$$p = \frac{e^{r\delta t} - d}{u - d}, \quad u = A + \sqrt{A^2 - 1}, \quad d = A - \sqrt{A^2 - 1},$$

where

$$A = \frac{1}{2} \left(e^{-r\delta t} + e^{(r+\sigma^2)\delta t} \right).$$

In [25], the choice of the binomial tree of Cox, Ross and Rubinstein is

$$p = \frac{e^{r\delta t} - d}{u - d}, \quad u = e^{\sigma\sqrt{\delta t}}, \quad d = \frac{1}{u}.$$

Note that the first three terms of the Taylor expansion of $u = A + \sqrt{A^2 - 1}$ coincides with those of $u = e^{\sigma\sqrt{\delta t}}$, i.e., $u = 1 + \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t + O((\sqrt{\delta t})^3)$. We remark that the choice is not unique. For instance, instead of applying $u = \frac{1}{d}$, we may choose $p = \frac{1}{2}$, which leads to $u = e^{r\delta t} (1 + \sqrt{e^{\sigma^2 \delta t} - 1})$ and $d = e^{r\delta t} (1 - \sqrt{e^{\sigma^2 \delta t} - 1})$.

It can be shown that with $\delta t \rightarrow 0$, the result obtained from binomial tree converges to that of Black–Scholes equation. Interested readers may refer to [45] for more details of tree methods for option pricing.

The binomial method is memory efficient with only one underlying. The largest memory requirement is at the last time-step m , where $2(m + 1)$ memory locations are required for the stock and option prices. Memory in the $(j + 1)$ -th time-step can be reused for the j -th time-step. Each node requires $O(1)$ calculations and there are $O(m^2)$ nodes, therefore the execution time grows quadratically as $O(m^2)$.

For d underlyings, the binomial method will require $O(m^d)$ memory locations and $O(m^{d+1})$ calculations. Thus, it is computationally heavy when d is large.

14.2 Monte Carlo Method

14.2.1 Monte Carlo Simulation for Option Pricing

Risk-neutral pricing suggests that the European option price can be calculated by discounting the expected payoff,

$$V_0 = \tilde{\mathbb{E}}[e^{-rT} V_T].$$

The Monte Carlo method relies on the **Strong Law of Large Numbers**, i.e., assuming i.i.d. (independent and identically distributed) random variables $X_1, X_2, \dots$, and $\mathbb{E}[|h(X)|] < +\infty$, then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n h(X_i) = \mathbb{E}[h(X)] \quad \text{a.s.}$$

Hence, the expected value of the payoff of a European option can be estimated as the arithmetic average of payoff values $V_T^i, i = 1, 2, \dots, n$, i.e.

$$V_0 = \tilde{\mathbb{E}}[e^{-rT} V_T] \approx e^{-rT} \frac{1}{n} \sum_{i=1}^n V_T^i.$$

The implementation of Monte Carlo method consists of simulating the underlying price paths under risk-neutral measure in order to get the option payoff values $V_T^i, i = 1, 2, \dots, n$. In the following sections, we will present a method for generating sample paths for one-dimensional case followed by the simulation for multi-dimensional problems. We refer to [37] for a comprehensive presentation of Monte Carlo implementation for financial engineering.

14.2.2 Generating One-Dimensional Random Variates

We first introduce how to generate a random variate with cumulative distribution function F .

Denote $\mathcal{U}[0, 1]$ for the uniform distribution over $[0, 1]$. By definition,

$$\mathbb{P}\{u < z\} = \begin{cases} 0, & z < 0, \\ z, & 0 \leq z \leq 1, \\ 1, & z > 1. \end{cases}$$

Consider a random variable X generated as follows:

1. generate u from $\mathcal{U}[0, 1]$
2. calculate $X = F^{-1}(u)$.

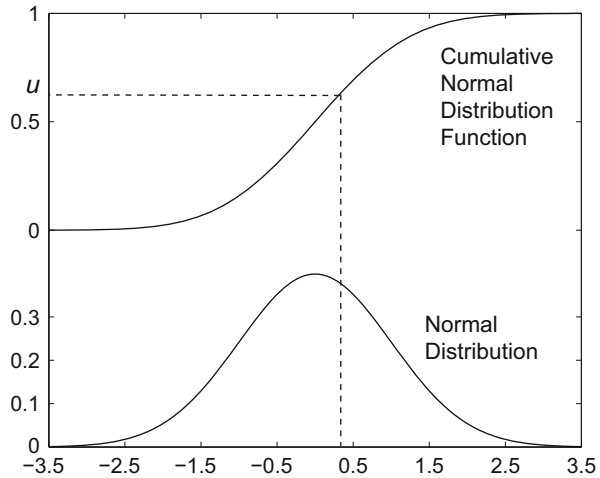
Then X follows the distribution of F :

$$\mathbb{P}\{X < x\} = \mathbb{P}\{F^{-1}(u) < x\} = \mathbb{P}\{u < F(x)\} = F(x) \text{ (} u \text{ follows } \mathcal{U}[0, 1]\text{)}.$$

Note that $F^{-1}(u) < x$ if and only if $u < F(x)$ because $F(x)$ is a strictly increasing function of x . Consequently, the second equality in the above holds true because the events for $\{F^{-1}(u) < x\}$ and $\{u < F(x)\}$ coincide for all u and x . In the following, we mainly use $F = N$ for generating normally distributed $\epsilon = N^{-1}(u)$. See Fig. 14.3 for reference. The Monte Carlo simulation for Poisson process based jump models will be presented in Chap. 20.

Now, we consider the general Itô process below

Fig. 14.3 Graphical illustration of normal inverse transform with $u \in \mathcal{U}[0, 1]$



$$dS_t = \theta(t, S_t)dt + \phi(t, S_t)dW_t,$$

where the solution of S_T , $t < T$ is unknown for a direct simulation from S_t . This is generally the case when the pricing model has non constant parameters. A sample path with intermediary time points is then necessary and can be generated as described below:

1. Discretize the time period $[t, T]$ into m time intervals: $t = t_0 < t_1 < t_2 < \dots < t_m = T$. The intervals depend on the option payoff and the model parameters. It is not necessary to have equally spaced intervals.
2. Generate u_j from $\mathcal{U}[0, 1]$, $j = 1, 2, \dots, m$
3. Calculate the normal variate $\epsilon_j = N^{-1}(u_j)$
4. Generate a Brownian increment $\delta W_j = \epsilon_j \sqrt{\delta t_j}$ where $\delta t_j = t_j - t_{j-1}$.
5. Calculate the sample path with *Euler–Maruyama scheme*, i.e.,

$$S_{t_j} = S_{t_{j-1}} + \theta(t_{j-1}, S_{t_{j-1}})\delta t_j + \phi(t_{j-1}, S_{t_{j-1}})\delta W_j.$$

The Euler–Maruyama scheme¹ is applicable to all different models.

For Black–Scholes–Merton model, there is no need to simulate with small time steps with an Euler–Maruyama type of scheme. This is because the model parameters are constant (i.e. volatility and interest rate) and that the future price S_T has a solution under risk-neutral measure

$$S_T = S_t e^{(r - \sigma^2/2)(T-t) + \sigma \epsilon \sqrt{T-t}},$$

where S_t is the current asset price and $\epsilon \sim \mathcal{N}(0, 1)$. So, the simulation can be done with one time step from t to T .

If we need to simulate some asset prices between t and T (e.g. for pricing path dependant options), the above equation can be used between two consecutive required dates. The i -th path from t to T is generated by:

$$S_{t_j}^i = S_{t_{j-1}}^i e^{(r - \sigma^2/2)\delta t_j + \sigma \epsilon_j^i \sqrt{\delta t_j}}, \quad j = 1, \dots, m, \quad i = 1, \dots, n, \quad (14.4)$$

where ϵ_j^i are independent random numbers distributed as $\mathcal{N}(0, 1)$. The number of paths n usually goes from 10,000 to 1,000,000, depending on the accuracy we want. In Fig. 14.4, we show $n = 10$ simulated stock price paths using (14.4), with $S_t = 100$, $r = 0.01$, $\sigma = 0.25$, $t = 0$, $T = 1$, and equal time-step $\delta t_j = 0.01$.

If we are only interested in evaluating European options, there is no need to know what the stock prices are in between t and T . We only need to know the stock price at T . We can modify (14.4) to get it directly:

¹There exist other schemes, e.g. Milstein scheme which contains a second order term for increasing the accuracy.

Fig. 14.4 Ten simulated paths of stock prices under Black–Scholes model

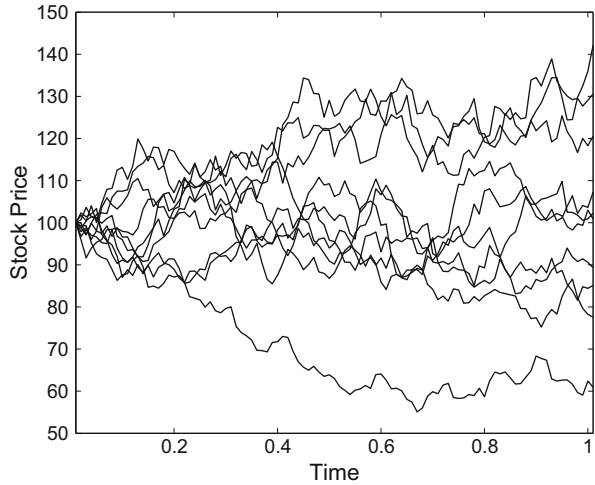


Table 14.1 Monte Carlo method for European put with $K = 10$, $S_0 = 5$, $r = 0.01$ and $\sigma = 0.25$

$T - t$	True	$n = 10$	$n = 10^2$	$n = 10^3$	$n = 10^4$	$n = 10^5$
0.25	4.9750	4.4913	4.9211	5.0018	4.9727	4.9751
0.50	4.9501	4.8818	4.9796	4.9510	4.9597	4.9495
0.75	4.9256	5.1352	4.7105	4.9073	4.9094	4.9284
1.00	4.9022	4.4807	4.8380	4.9072	4.8957	4.9015

$$S_T^i = S_t^i e^{(r-\sigma^2/2)(T-t) + \sigma \epsilon^i \sqrt{T-t}}, \quad i = 1, \dots, n,$$

i.e. for each ϵ^i sampled, we can generate one sample of S_T^i . From each of them, a payoff for the option can be calculated at expiry. For example, if it is a European call option, we can compute $V_T^i = \max\{S_T^i - K, 0\}$, $i = 1, \dots, n$. Then, the option price is approximated by $e^{-r(T-t)} \frac{1}{n} \sum_{i=1}^n V_T^i$.

Example 14.2 Consider a European put with $K = 10$, $S_0 = 5$, $r = 0.01$ and $\sigma = 0.25$. Table 14.1 shows the comparison of the Monte Carlo method and the true solution of the Black–Scholes equation for different expiry time and numbers of paths. We see that the results are more accurate when we increase the number of paths n .

Consider the n simulations V_T^i , $i = 1, \dots, n$ and denote their sample mean by $\bar{\mu}$ and standard deviation by $\bar{\sigma}$. The Central Limit Theorem states that the option price V should be distributed as $\mathcal{N}(\bar{\mu}, \frac{\bar{\sigma}^2}{n})$. Hence a 95% confidence interval for V is given by

$$\bar{\mu} - \frac{1.96\bar{\sigma}}{\sqrt{n}} < V < \bar{\mu} + \frac{1.96\bar{\sigma}}{\sqrt{n}}.$$

See Sect. 9.3.7. This shows that the uncertainty about the option value is inversely proportional to $\sqrt{n}$. To halve the confidence interval, we must quadruple the number of simulations; and to increase the accuracy by a factor of 10, the number of simulations must be increased by roughly a factor of 100.

To enhance Monte Carlo method's convergence rate of $O(\frac{1}{\sqrt{n}})$, we can turn to useful variance reduction techniques like doubling the simulation trials with the negation of normal variates [37]. We also highlight that the quasi-Monte Carlo with Sobol sequences² is widely used in practice as a general method.

Monte Carlo simulation tends to be numerically more efficient than other procedures when there are three or more underlying assets. This is because the time taken to carry out a Monte Carlo simulation increases approximately linearly with the number of unknown variables. In contrast, the time taken for most other procedures increases exponentially with the number of unknown variables.

Monte Carlo simulation has the following advantages:

- numerically efficient for high-dimensional case because the complexity increases linearly with the number of underlyings
- provides standard error for the estimates
- can accommodate complex payoffs and complex stochastic processes.

For pricing American options, special techniques need to be employed because all the intermediate stock prices in all the paths are involved for estimating the *continuation price* and deciding to hold or not. We will return to American options in the next chapter.

14.2.3 Generating Multivariate Randoms

For multi-asset products or multi-factor models which require simulation of multiple assets, we also need to take the correlation among assets into consideration. For example, it may involve generating correlated Brownian motions, or essentially a normally distributed vector $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ which is symmetric and positive semidefinite:

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_{nn} \end{bmatrix}$$

²Quasi-Monte Carlo method uses deterministic but more evenly dispersed sequence of numbers instead of randomly generated numbers. Illustratively, a controlled sequence like $\{1, 2, 3, 4, 5, 6, 1, 2, 3, 4, 5, 6, \dots\}$ may converge faster than $\{1, 6, 2, 4, 3, 6, 3, 5, 1, \dots\}$ for estimating the expectation of casting a die, which is 3.5.

where $\sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$ for $i \neq j$ and $\sigma_{ii} = \sigma_i^2$. The correlation matrix for $\mathbf{X}$ is

$$\mathbf{C} \equiv \begin{bmatrix} 1 & \rho_{12} & \cdots & \rho_{1n} \\ \rho_{21} & 1 & \cdots & \rho_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{n1} & \rho_{n2} & \cdots & 1 \end{bmatrix}$$

and $\mathbf{\Sigma} = \mathbf{C}\mathbf{D}\mathbf{C}^\top$ where $\mathbf{D} = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_n)$ is an $n \times n$ diagonal matrix.

We have the following result for generating a normal vector from a standard normal vector.

Proposition 14.1 *The normally distributed $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{\Sigma})$ can be generated through either one of the below methods*

- (i) $\mathbf{X} = \boldsymbol{\mu} + \mathbf{A}\boldsymbol{\epsilon}$
- (ii) $\mathbf{X} = \boldsymbol{\mu} + \mathbf{D}\mathbf{L}\boldsymbol{\epsilon}$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is an $n \times 1$ vector, $\mathbf{0}$ is an $n \times 1$ vector of all zeros, and $\mathbf{I}$ is an $n \times n$ identity matrix, i.e., entries of $\boldsymbol{\epsilon}$ are independent standard normals. The matrices $\mathbf{A}$ and $\mathbf{L}$ are $n \times n$ matrices that decompose, respectively, the covariance matrix $\mathbf{\Sigma} = \mathbf{A}\mathbf{A}^\top$ and correlation matrix $\mathbf{C} = \mathbf{L}\mathbf{L}^\top$.

Proof By the Linear Transformation Property of Multivariate Normal Vector (see Proposition A.3 in Appendix),

$$\mathbf{A}\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{A}\mathbf{0}, \mathbf{A}\mathbf{I}\mathbf{A}^\top) \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}).$$

Therefore $\mathbf{X} = \boldsymbol{\mu} + \mathbf{A}\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{\Sigma})$.

Similarly, we have

$$\mathbf{D}\mathbf{L}\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{D}\mathbf{L}\mathbf{0}, \mathbf{D}\mathbf{L}\mathbf{L}^\top\mathbf{D}^\top) \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}).$$

and $\mathbf{X} = \boldsymbol{\mu} + \mathbf{D}\mathbf{L}\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{\Sigma})$. □

Suppose there exists an estimate for the covariance matrix $\mathbf{\Sigma}$ or the correlation matrix $\mathbf{C}$. The following two methods find the decomposition.

Cholesky Factorization

The Cholesky factorization decomposes $\mathbf{\Sigma}$ or $\mathbf{C}$ into the product of a lower triangular matrix and its transpose. Consider the positive definite covariance matrix $\mathbf{\Sigma}$,

$$\Sigma = \mathbf{A}\mathbf{A}^\top = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ 0 & a_{22} & \cdots & a_{n2} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn} \end{bmatrix}.$$

Given the entries σ_{ij} of Σ , we can solve $\mathbf{A}$ recursively through the following system of equations:

$$\begin{cases} a_{11}^2 = 1, \\ a_{11}a_{21} = \sigma_{12}, \\ \vdots \\ a_{n1}a_{11} = \sigma_{1n}, \\ a_{21}^2 + a_{22}^2 = 1, \\ \vdots \\ a_{n1}^2 + a_{n2}^2 + \cdots + a_{nn}^2 = 1. \end{cases}$$

Example 14.3 The Cholesky decomposition of a two-dimensional correlation matrix $\mathbf{C}$ is

$$\mathbf{C} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{bmatrix} \begin{bmatrix} 1 & \rho \\ 0 & \sqrt{1-\rho^2} \end{bmatrix} = \mathbf{L}\mathbf{L}^\top.$$

To generate two dependent normal variates such that

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}\right),$$

we can carry out³

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{bmatrix} \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \end{bmatrix}$$

where ϵ_1 and ϵ_2 are independent standard normal variates. Readers can compare this example to the correlated Brownian motions in Chap. 10 by setting $\mu_1 = \mu_2 = 0$ and $\sigma_1 = \sigma_2 = 1$.

³Note that it is not necessary to implement the calculations in matrix form.

Eigenvector Factorization

The Cholesky decomposition is not the only way to have $\Sigma = \mathbf{A}\mathbf{A}^\top$. According to Spectral Theorem in linear algebra, a symmetric and real valued matrix such as Σ can be diagonalized as

$$\Sigma = \mathbf{U}\mathbf{A}\mathbf{U}^\top,$$

where $\mathbf{A}$ is a diagonal matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, and $\mathbf{U}$ is an orthogonal matrix ($\mathbf{U}\mathbf{U}^\top = \mathbf{I}$). Setting

$$\mathbf{A} = \mathbf{U}\mathbf{A}^{1/2} = \mathbf{U} \begin{bmatrix} \sqrt{\lambda_1} & & & \\ & \sqrt{\lambda_2} & & \\ & & \ddots & \\ & & & \sqrt{\lambda_n} \end{bmatrix},$$

we have $\mathbf{A}\mathbf{A}^\top = \mathbf{U}\mathbf{A}^{1/2}\mathbf{A}^{1/2}\mathbf{U}^\top = \mathbf{U}\mathbf{A}\mathbf{U}^\top = \Sigma$.

One of the variance reduction techniques (principal components method) is based on the eigenvector factorization.

14.3 PDE Method with Finite Difference Approximation

Besides the Black–Scholes equation, in practice, there are other PDEs derived from other models that govern option prices. However, not many of them have closed-form solutions like Black–Scholes formula for call/put options. In this section, we discuss how to solve PDEs by finite difference approach generally. Finite difference approaches consist of approximating the differential operators in the differential equation by difference operators.

Given a function $u(x)$, using Taylor's expansion, we have

$$\begin{aligned} u(x + \delta x) &= u(x) + \delta x \cdot u'(x) + \frac{\delta x^2}{2} u''(x) + \frac{\delta x^3}{6} u'''(x) + O(\delta x^4) \\ u(x - \delta x) &= u(x) - \delta x \cdot u'(x) + \frac{\delta x^2}{2} u''(x) - \frac{\delta x^3}{6} u'''(x) + O(\delta x^4). \end{aligned} \quad (14.5)$$

If δx is small, we can approximate $u'(x)$ at x by the *first order central difference*

$$u'(x) = \frac{u(x + \delta x) - u(x - \delta x)}{2\delta x} + O(\delta x^2), \quad (14.6)$$

which is of second order accuracy. From (14.5), we can also approximate the second order derivative $u''(x)$ at x by the *second order central difference*:

$$u''(x) = \frac{u(x + \delta x) - 2u(x) + u(x - \delta x)}{(\delta x)^2} + O(\delta x^2). \quad (14.7)$$

The *forward difference* refers to

$$u'(x) = \frac{u(x + \delta x) - u(x)}{\delta x} + O(\delta x). \quad (14.8)$$

It is usually used for the differential w.r.t. time in problems of physics.

Suppose we are to solve the Black–Scholes equation. In this case, as it is a log-normal model, it is more efficient to work with the case after the variable change $x = \ln S$:

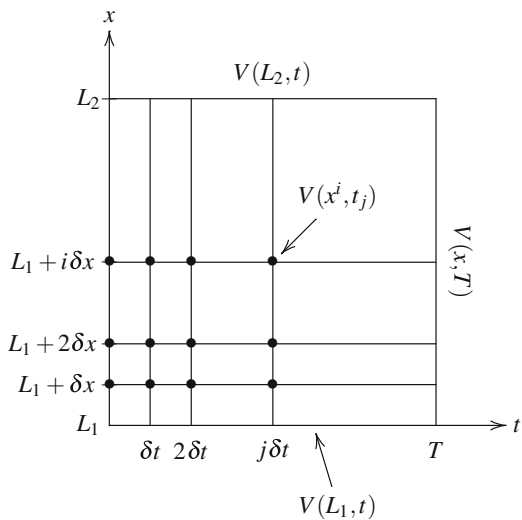
$$\frac{\partial V(x, t)}{\partial t} + \frac{1}{2}\sigma^2 \frac{\partial^2 V(x, t)}{\partial x^2} + \left(r - \frac{\sigma^2}{2}\right) \frac{\partial V(x, t)}{\partial x} - rV(x, t) = 0, \quad (14.9)$$

with European-type payoff $V(x, T)$ for $-\infty < x < \infty$. Since it is impossible to work with infinity numerically, we first truncate the spatial domain as $x \in [L_1, L_2]$. Then we replace the boundary conditions $V(-\infty, t)$ by $V(L_1, t)$ and $V(\infty, t)$ by $V(L_2, t)$.

Next we partition the solution domain $[L_1, L_2] \times [0, T]$ by grid lines: $[L_1, L_2]$ into n equal sub-intervals, each of length δx and $[0, T]$ into m equal sub-intervals, each of length δt , see Fig. 14.5. Denote $x^i = L_1 + i\delta x$ and $t_j = j\delta t$ for $0 \leq i \leq n$ and $0 \leq j \leq m$.

Let $V^i(t)$ denote the *approximate* values to $V(x_i, t)$ for $i = 1, 2, \dots, n-1$. The boundary conditions can usually be approximated, e.g. for a European call option $V(x^0, t) = V(L_1, t) \approx 0$ and $V(x^n, t) = V(L_2, t) \approx e^{L_2} - Ke^{-r(T-t)}$.

Fig. 14.5 Solution domain of European options and the grid



Let $\mathbf{V}(t) = [V^1(t), \dots, V^{n-1}(t)]^\top$. Applying the second-order accurate finite difference schemes (14.6) and (14.7) for $\frac{\partial V(x, t)}{\partial x}$ and $\frac{\partial^2 V(x, t)}{\partial x^2}$ in (14.9) gives

$$\begin{aligned} \frac{\partial V^i(t)}{\partial t} + \frac{1}{2}\sigma^2 \frac{V^{i+1}(t) - 2V^i(t) + V^{i-1}(t)}{\delta x^2} \\ + \left(r - \frac{\sigma^2}{2}\right) \frac{V^{i+1}(t) - V^{i-1}(t)}{2\delta x} - rV^i(t) = 0 \end{aligned}$$

which can be written as a system of equations

$$\begin{bmatrix} \partial_t V^1(t) \\ \partial_t V^2(t) \\ \vdots \\ \partial_t V^{n-2}(t) \\ \partial_t V^{n-1}(t) \end{bmatrix} = \begin{bmatrix} \mathbf{b} & \mathbf{c} & & & \\ \mathbf{a} & \mathbf{b} & \mathbf{c} & & \\ & \ddots & \ddots & \ddots & \\ & & \mathbf{a} & \mathbf{b} & \mathbf{c} \\ & & & \mathbf{a} & \mathbf{b} \end{bmatrix} \begin{bmatrix} V^1(t) \\ V^2(t) \\ \vdots \\ V^{n-2}(t) \\ V^{n-1}(t) \end{bmatrix} + \begin{bmatrix} \mathbf{a}V(x^0, t) \\ 0 \\ \vdots \\ 0 \\ \mathbf{c}V(x^n, t) \end{bmatrix},$$

or

$$\frac{\partial \mathbf{V}(t)}{\partial t} = \mathbf{D}\mathbf{V}(t) + f(t), \quad (14.10)$$

where $\mathbf{a} = \frac{r - \sigma^2/2}{2\delta x} - \frac{\sigma^2}{2(\delta x)^2}$, $\mathbf{b} = r + \frac{\sigma^2}{(\delta x)^2}$, $\mathbf{c} = -\frac{r - \sigma^2/2}{2\delta x} - \frac{\sigma^2}{2(\delta x)^2}$, and $f(t)$ is a known vector at time t owing to boundary conditions. The matrix $\mathbf{D}$ has constant diagonals and it is known as a *Toeplitz matrix*. Its special structure enjoys a number of fast numerical algorithms [17, 18].

The following illustration will assume $f(t) = 0$ and the non-zero case is straightforward for temporal discretization. Actually, we have $f(t) = 0$ if we apply the zero gamma boundary condition, which is a general boundary condition for all options with natural smoothness ($\partial_{xx}V = 0$). It applies to both upper and lower bounds, i.e., $V(x^0, t) = 2V^1(t) - V^2(t)$ and $V(x^n, t) = 2V^{n-1}(t) - V^{n-2}(t)$.

Let $\mathbf{V}_j$ be the *approximate* value for the option value $\mathbf{V}(t_j)$. Applying forward difference (14.8) for $\frac{\partial \mathbf{V}(t)}{\partial t}$ in (14.10) with different treatment on the RHS results in

Explicit scheme:

$$\frac{\mathbf{V}_j - \mathbf{V}_{j-1}}{\delta t} = \mathbf{D}\mathbf{V}_j$$

Implicit scheme:

$$\frac{\mathbf{V}_j - \mathbf{V}_{j-1}}{\delta t} = \mathbf{D}\mathbf{V}_{j-1}$$

Crank–Nicolson scheme:

$$\frac{\mathbf{V}_j - \mathbf{V}_{j-1}}{\delta t} = \frac{1}{2}(\mathbf{D}\mathbf{V}_j + \mathbf{D}\mathbf{V}_{j-1}).$$

14.3.1 Explicit Scheme

The explicit scheme is equivalent to, for $j = 0, 1, \dots, m$,

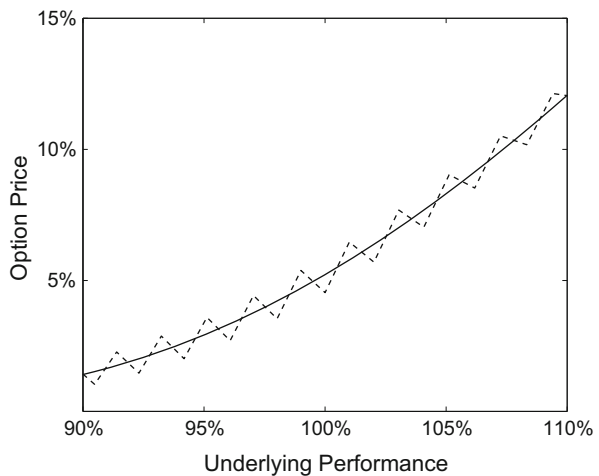
$$\mathbf{V}_{j-1} = (\mathbf{I} - \delta t \mathbf{D})\mathbf{V}_j,$$

where $\mathbf{I}$ is an identity matrix of size $n-1$. Since $\mathbf{V}_m = [V(x^1, t_m), \dots, V(x^{n-1}, t_m)]$ are known values at maturity, a time-marching operation is performed to obtain $\mathbf{V}_{m-1}$, $\mathbf{V}_{m-2}$, and after m times, $\mathbf{V}_0$ at time 0.

The accuracy of explicit scheme is $O(\delta t)$ and $O(\delta x^2)$. One can show that $\alpha \equiv \delta t / (\delta x)^2$ has to be less than a certain constant α_0 for convergence. For some problems, the condition $\alpha < \alpha_0$ will lead to a small δt (hence a large m) and the method may be too slow to find the solution.

Example 14.4 Consider pricing a European call option by explicit scheme with $K = 1$, $T = 0.25$, $r = 0.02$, $\sigma = 0.25$, $L_1 = -0.4$, $L_2 = 0.3$. We fix $n = 70$ and test $m = 154$ or $m = 200$, corresponding to $\alpha = 16.23$ or $\alpha = 12.5$ respectively. The finite difference solutions with the two settings are plotted against underlying performance $S = e^x$ in Fig. 14.6. We see that the method diverges with $\alpha = 16.23$ and the solution becomes oscillatory.

Fig. 14.6 Option price (close-up around strike) obtained from explicit scheme with $\alpha = 12.5$ (solid line); oscillatory when $\alpha = 16.23$ (dashed line)



14.3.2 Implicit Scheme

The implicit scheme is equivalent to, for $j = 0, 1, \dots, m$,

$$(\mathbf{I} + \delta t \mathbf{D}) \mathbf{V}_{j-1} = \mathbf{V}_j.$$

The time-marching process starts from the known $\mathbf{V}_m$ and a tridiagonal linear system needs to be solved at each time step, which requires computational cost of $O(n)$ by Thomas algorithm in numerical linear algebra.

The accuracy of implicit scheme is $O(\delta t)$ and $O(\delta x^2)$. Compared to the explicit method before, the implicit method is more robust. It does not suffer from the choice of δt and δx , which is called unconditional stability in numerical PDE approaches.

14.3.3 Crank-Nicolson Scheme

The Crank–Nicolson scheme [28] is the balance between the explicit and implicit schemes. It is equivalent to

$$\left(\mathbf{I} + \frac{\delta t}{2} \mathbf{D} \right) \mathbf{V}_{j-1} = \left(\mathbf{I} - \frac{\delta t}{2} \mathbf{D} \right) \mathbf{V}_j,$$

and a tridiagonal system is solved at each time-step during time-marching. The scheme is second order accurate in both space and time and also unconditionally stable.

However, the Crank–Nicolson scheme is known to suffer from the non-smoothness of payoff function, and a technique called Rannacher time stepping [67] is usually used for the initial time-steps.

14.3.4 Alternating Direction Implicit (ADI) Method

For options with two underlyings, the PDE is in 2-D and the ADI method is commonly used. Here we illustrate the situation without cross derivative term (correlations between underlying assets). Readers are referred to [27] for the general case. A half-timestep is incorporated as

$$\begin{cases} \frac{\mathbf{V}_{j+\frac{1}{2}} - \mathbf{V}_j}{\delta t/2} = (\mathbf{D}_1 \mathbf{V}_{j+\frac{1}{2}} + \mathbf{D}_2 \mathbf{V}_j) \\ \frac{\mathbf{V}_{j+1} - \mathbf{V}_{j+\frac{1}{2}}}{\delta t/2} = (\mathbf{D}_1 \mathbf{V}_{j+\frac{1}{2}} + \mathbf{D}_2 \mathbf{V}_{j+1}). \end{cases}$$

Here $\mathbf{D}_1$ and $\mathbf{D}_2$ are the matrices resulting from finite difference schemes for Asset 1 and Asset 2 respectively. Spatial directions are alternately handled at each half-timestep.

The finite difference method is extremely efficient for options with only one underlying asset: its computational cost is of $O(nm)$. However, it also has the *curse of dimensionality* in that the cost grows like $O(mn^d)$ for an option whose price depends on d assets.

Exercises

1. Suppose the assumption $p = 0.5$ is used in the binomial model.

(i) Prove that in a risk-neutral world,

$$u = e^{r\delta t} \left(1 + \sqrt{e^{\sigma^2 \delta t} - 1} \right) \quad \text{and} \quad d = e^{r\delta t} \left(1 - \sqrt{e^{\sigma^2 \delta t} - 1} \right).$$

(ii) Find the upper bound for δt such that the binomial model with $p = 0.5$ is feasible.

(iii) Show that $ud \rightarrow 1$ as $\delta t \rightarrow 0$.

2. Under BSM model for a non-dividend-paying stock, let $S_0 = 1$, $r = 0.01$, $\sigma = 0.2$. Calculate the price of a 1-year ATM call option with the below methods and compare the results:

(i) Black–Scholes formula

(ii) Monte Carlo based on the direct integration:

$$S_T = S_0 e^{(r - \frac{\sigma^2}{2})T + \sigma \tilde{W}_T}$$

(iii) Euler–Maruyama scheme (step size $\delta t = 0.005$):

$$S_{t_i} = S_{t_{i-1}} + r S_{t_{i-1}} \delta t + \sigma S_{t_{i-1}} \tilde{W}_{\delta t}.$$

3. Consider pricing a European call option using finite difference method with explicit scheme. The parameters are $K = 1$, $T = 0.25$, $r = 0.02$, $\sigma = 0.25$, $L_1 = -0.5$, $L_2 = 0.5$, and number of time-steps $m = 5000$. The zero gamma boundary conditions are applied. Let c_1 and c_2 denote the finite difference solutions at $x = 0$ ($S = 1$) using $n = 50$ and $n = 100$ respectively. Compare the error of c_1 and c_2 respectively against the Black–Scholes formula at $x = 0$. How fast is the error reducing? Explain this phenomenon.

Chapter 15

American Options



In this chapter, the American option modelling is introduced in a simple way through Bermudan option which has discrete exercise times. Then, the major applicable numerical methods are presented for Black–Scholes–Merton model. The general Partial Differential Inequality is derived and the boundary conditions for American Put option are analyzed as an illustration example for the methodology.

15.1 Stopping Time

A stopping time τ is a *random variable* taking values in $[0, \infty]$ and satisfying $\{\tau \leq t\} \equiv \{\omega \in \Omega; \tau(\omega) \leq t\} \in \mathcal{F}_t$ for all $t \geq 0$.

Theorem 15.1 (Optional Sampling) *A martingale (supermartingale, submartingale) X_t stopped at a stopping time, i.e., $X_{\min\{t, \tau\}}$ is a martingale (supermartingale, submartingale).*

15.2 Backward Pricing

An *American option* gives the option holder the right to early terminate the option before its maturity. Upon the option exercise, the option holder will be paid the intrinsic value or other predefined payout. The early termination time is a stopping time.

An American option with discrete exercise dates is called *Bermudan option*. For illustrating the general results on American options, we consider only Bermudan options under a *complete market*. Let $\{t_1, t_2, \dots, t_n\}$ be the exercise dates with t_n as the expiry of the option. Denote

H_{t_i} : for the exercise payoff at t_i (H_{t_n} for the final payoff if no early exercise happened, for example: $H_{t_n} = (K - S_{t_n})^+$ for put)

V_{t_i} : for the option value at time t_i , $0 \leq t_i \leq t_n$, if the option is not exercised before t_i

M_{t_i} : for the money market numéraire from 0 to t_i .

At maturity t_n , we have $V_{t_n} = H_{t_n}$. At the preceding exercise date t_{n-1} , there can be two cases:

- (i) if the option is exercised at t_{n-1} , the option value is $H_{t_{n-1}}$ (the payout);
- (ii) if the option is not exercised at t_{n-1} , the option value $V_{t_{n-1}}$ will become V_{t_n} at time t_n . Because there is no exercise date between t_{n-1} and t_n , $\frac{V_t}{M_t}$ ($t_{n-1} \leq t \leq t_n$) should be a martingale under complete market with the risk-neutral probability $\tilde{\mathbb{P}}$ and there is a self-financing strategy, following which, the amount of $M_{t_{n-1}} \tilde{\mathbb{E}}[\frac{V_{t_n}}{M_{t_n}} | \mathcal{F}_{t_{n-1}}]$ at time t_{n-1} , will become V_{t_n} at time t_n . A rational option holder will make his exercise decision for getting the maximum value:

$$V_{t_{n-1}} = \max \left\{ H_{t_{n-1}}, M_{t_{n-1}} \tilde{\mathbb{E}} \left[\frac{V_{t_n}}{M_{t_n}} | \mathcal{F}_{t_{n-1}} \right] \right\}.$$

Denoting the discount processes $\hat{V}_t = \frac{V_t}{M_t}$, $\hat{H}_t = \frac{H_t}{M_t}$, we get

$$\hat{V}_{t_{n-1}} = \max \{ \hat{H}_{t_{n-1}}, \tilde{\mathbb{E}}[\hat{V}_{t_n} | \mathcal{F}_{t_{n-1}}] \}.$$

Doing similar analysis recursively backward, we obtain the general result for all $0 \leq i < n$:

$$\hat{V}_{t_i} = \max \{ \hat{H}_{t_i}, \tilde{\mathbb{E}}[\hat{V}_{t_{i+1}} | \mathcal{F}_{t_i}] \},$$

and V_{t_i} is then the option price at time t_i . For a given exercise date t_i , we call

- $M_{t_i} \tilde{\mathbb{E}} \left[\frac{V_{t_{i+1}}}{M_{t_{i+1}}} | \mathcal{F}_{t_i} \right]$ the *continuation value*, and
- H_{t_i} the *exercise value*.

The problem cannot be solved explicitly as there exists no closed-form solution. One has to use numerical schemes, such as numerical PDE methods or the binomial tree method in Chap. 14, to find the solution.

15.3 Methods for Pricing American Options

So far, no closed-form solution is known for American options under BSM model. In general, backward pricing methods like tree or PDE approach are used for American options involving one or two dimensions. The standard Monte Carlo method is

a forward pricing method starting from time-0. It is not applicable for American option pricing because of the early termination decision which depends on the continuation value that is not available. However, special Monte Carlo methods have been proposed and used for American options which will be presented in the last part of this chapter.

In the following sections, we assume BSM model for presenting the methodologies that can be extended to other models.

15.3.1 Binomial Tree

The tree is constructed in the same way as the European option but the option calculation is different. With the backward induction, we calculate, at each node:

- the *continuation value*: the option value
- the *exercise value*: the payout of the option with early exercise
- the (conditional) *option price* at the node will be

$$\max\{\text{Exercise Value}, \text{Continuation Value}\}.$$

Consider the situation at time-step j and at asset price S_{j+1}^i . The option can be exercised prior to expiry to yield a profit determined by the payoff function, e.g., call and put respectively

$$H_j^i = \max\{S_j^i - K, 0\} \quad \text{or} \quad H_j^i = \max\{K - S_j^i, 0\}, \quad i = 1, \dots, j+1.$$

If the option is retained, its value is, as in the European case,

$$e^{-r\delta t} \left(pV_{j+1}^{i+1} + (1-p)V_{j+1}^i \right), \quad i = 1, \dots, j+1.$$

The value of the American option is the maximum of two choices: the choice of exercising the option and the choice of keeping the option, i.e.

$$V_j^i = \max \left\{ H_j^i, e^{-r\delta t} (pV_{j+1}^{i+1} + (1-p)V_{j+1}^i) \right\}, \quad i = 1, \dots, j+1.$$

Example 15.1 Let us compute an American put by a 2-step binomial method. We will use the same data as given in Chap. 14, i.e. $S_0 = 20$, $K = 21$, $T = 0.5$, $r = 0.02$, $\delta t = 0.25$, and $u = 1.1$ and $d = 0.9$ which gives $p = 0.525$.

The binomial tree is given in Fig. 15.1. The option prices at the expiry nodes D, E, and F are clear. Next consider node B. If we exercise, we get 0. If we do not exercise, then the option price is the same as the European one:

$$e^{-0.02 \times 0.25} (0.525 \times 0 + 0.475 \times 1.2) = 0.5672.$$

Clearly, in this case, we do not exercise, and the option price is 0.5672. We illustrate this in Fig. 15.1 by underlining the maximum of the two values: 0 and 0.5672.

It is similar for node C. If we exercise, we get 3. If we do not exercise, then the option price is the same as the European one:

$$e^{-0.02 \times 0.25} (0.525 \times 1.2 + 0.475 \times 4.8) = 2.8955.$$

Clearly, in this case, we exercise, and the option price is 3. We illustrate this in Fig. 15.1 by underlining the maximum of the two values: 3 and 2.8955. Finally at $t = 0$, the option price is given by

$$e^{-0.02 \times 0.25} (0.525 \times 0.5672 + 0.475 \times 3) = 1.7142.$$

15.3.2 Partial Differential Inequality

The pricing of American options can be transformed into a partial differential inequality problem for which numerical methods are available. We first study the properties of American option price process and then derive the partial differential inequality.

Properties of American Option Price Process

Consider a *finite time horizon* $[t_0, t_n]$ and *discrete stopping times* taking values in $\{t_0, t_1, \dots, t_n\}$. Let h_t be an $\mathcal{F}_t$ -adapted process. Define a process v_t , starting from v_{t_0} , as follows:

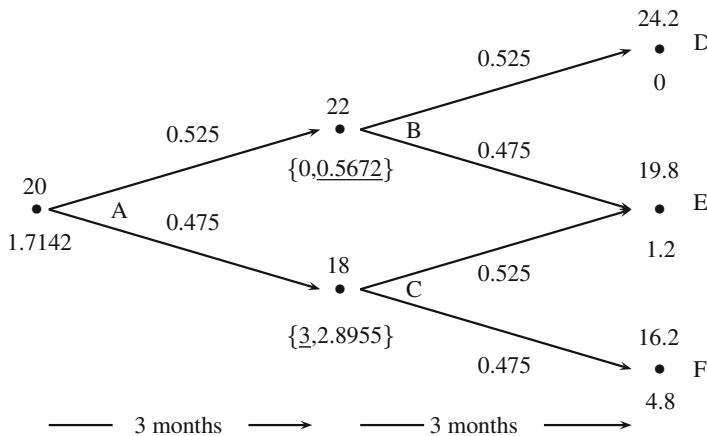


Fig. 15.1 Two-step binomial method for an American option

$$\begin{cases} v_{t_n} = h_{t_n}, \\ v_{t_i} = \max\{h_{t_i}, \mathbb{E}[v_{t_{i+1}} | \mathcal{F}_{t_i}]\}, & \forall 0 \leq i < n, \\ v_t = \mathbb{E}[v_{t_i} | \mathcal{F}_t], & \forall t_{i-1} \leq t < t_i, \quad 1 \leq i < n. \end{cases}$$

Proposition 15.1 *The process v_t is a supermartingale, i.e., $\mathbb{E}[v_t | \mathcal{F}_s] \leq v_s$, $\forall t_0 \leq s < t \leq t_n$.*

Proof First, we show that $\mathbb{E}[v_{t_i} | \mathcal{F}_t] \geq \mathbb{E}[v_{t_m} | \mathcal{F}_t]$, $\forall t \leq t_i \leq t_m$. From definition, we get

$$v_{t_i} = \max\{h_{t_i}, \mathbb{E}[v_{t_{i+1}} | \mathcal{F}_{t_i}]\} \geq \mathbb{E}[v_{t_{i+1}} | \mathcal{F}_{t_i}].$$

Taking conditional expectation on both sides with $\mathcal{F}_t$, we get

$$\mathbb{E}[v_{t_i} | \mathcal{F}_t] \geq \mathbb{E}[\mathbb{E}[v_{t_{i+1}} | \mathcal{F}_{t_i}] | \mathcal{F}_t] = \mathbb{E}[v_{t_{i+1}} | \mathcal{F}_t].$$

By iteration, we prove that $\mathbb{E}[v_{t_i} | \mathcal{F}_t] \geq \mathbb{E}[v_{t_m} | \mathcal{F}_t]$.

Now, $\forall t_0 \leq s < t \leq t_n$, we find j, k such that $t_{j-1} \leq s \leq t_j$ and $t_{k-1} \leq t \leq t_k$. We have

$$\mathbb{E}[v_t | \mathcal{F}_s] = \mathbb{E}[\mathbb{E}[v_{t_k} | \mathcal{F}_t] | \mathcal{F}_s] = \mathbb{E}[v_{t_k} | \mathcal{F}_s] \leq \mathbb{E}[v_{t_j} | \mathcal{F}_s] = v_s. \quad \square$$

Proposition 15.2 *Let $0 \leq t \leq t_n$ and τ be a stopping time defined as*

$$\tau = \min\{t_i \in \{t_1, t_2, \dots, t_n\} | v_{t_i} = h_{t_i}\}.$$

The stopped process $v_{t \wedge \tau}$ is a martingale.

Proof We observe that for $\forall 0 \leq i < j$ such that $t_j \leq \tau$,

$$v_{t_i} = \max\{h_{t_i}, \mathbb{E}[v_{t_{i+1}} | \mathcal{F}_{t_i}]\} \text{ becomes } v_{t_i} = \mathbb{E}[v_{t_{i+1}} | \mathcal{F}_{t_i}],$$

or otherwise if $v_{t_i} = h_{t_i}$ then $\tau = t_i$ is a contradiction. Hence,

$$v_{t_i} = \mathbb{E}[v_{t_{i+1}} | \mathcal{F}_{t_i}] = \mathbb{E}[\mathbb{E}[v_{t_{i+2}} | \mathcal{F}_{t_{i+1}}] | \mathcal{F}_{t_i}] = \mathbb{E}[v_{t_{i+2}} | \mathcal{F}_{t_i}], \quad \text{if } i+2 \leq j.$$

Continuing the above analysis until t_j , we obtain $v_{t_i} = \mathbb{E}[v_{t_j} | \mathcal{F}_{t_i}]$.

Now, $\forall t_0 \leq s < t \leq \tau$, we find j, k such that $t_{j-1} \leq s \leq t_j$ and $t_{k-1} \leq t \leq t_k \leq \tau$,

$$\mathbb{E}[v_t | \mathcal{F}_s] = \mathbb{E}[\mathbb{E}[v_{t_k} | \mathcal{F}_t] | \mathcal{F}_s] = \mathbb{E}[v_{t_k} | \mathcal{F}_s] = \mathbb{E}[\mathbb{E}[v_{t_k} | \mathcal{F}_{t_j}] | \mathcal{F}_s] = \mathbb{E}[v_{t_j} | \mathcal{F}_s] = v_s. \quad \square$$

Proposition 15.3 *Let $\mathcal{T}_{t_1, t_n} = \{t_1, t_2, \dots, t_n\}$. Then,*

$$v_t = \max_{\tau \in \mathcal{T}_{t_1, t_n}} \mathbb{E}[h_\tau | \mathcal{F}_t], \quad t \leq t_1.$$

Proof Let τ be the first time that $v_\tau = h_\tau$ as defined in Proposition 15.2. According to Proposition 15.1, v_t is a supermartingale. The stopped process $v_{t \wedge \tau}$ is also a supermartingale by *Optional Sampling Theorem*. And

$$v_t \geq \mathbb{E}[v_{t_n \wedge \tau} | \mathcal{F}_t] = \mathbb{E}[v_\tau | \mathcal{F}_t] = \mathbb{E}[h_\tau | \mathcal{F}_t], \quad \forall \tau \in \{t_1, t_2, \dots, t_n\}.$$

But from Proposition 15.2, the stopped process $v_{t \wedge \tau}$ is martingale,

$$v_t = \mathbb{E}[v_{t_n \wedge \tau} | \mathcal{F}_t] = \mathbb{E}[v_\tau | \mathcal{F}_t] = \mathbb{E}[h_\tau | \mathcal{F}_t].$$

Hence, necessarily,

$$v_t = \max_{\tau \in \mathcal{T}_{t_1, t_n}} \mathbb{E}[h_\tau | \mathcal{F}_t]. \quad \square$$

Comparing the definitions of v_t and $\widehat{V}_t$ in Sect. 15.2, we can see that v_t is applicable for pricing Bermudan options by setting $v_t = \widehat{V}_t$, $h_t = \widehat{H}_t$ and $\mathbb{E}$ as $\widetilde{\mathbb{E}}$.

It can be shown that, for an American option with continuous time exercise, the discounted price of the option is

$$\widehat{V}_t = \max_{\tau \in \mathcal{T}_{t, T}} \mathbb{E}[\widehat{H}_\tau | \mathcal{F}_t],$$

where $\mathcal{T}_{t, T}$ is the set of stopping times taking values in $[t, T]$. The process $\widehat{V}_t$ is a supermartingale and the stopped process $\widehat{V}_{t \wedge \tau}$ is a martingale. For a deep analysis on American options, we refer to [47].

Partial Differential Inequality

Because of the early exercise feature, the Black–Scholes PDE does not apply in general for American options under BSM model. However, the price of an American option satisfies a system of inequality as described in the below with a heuristic explanation.

From Itô–Doebelin formula, the price of the option with exercise payoff of $H_t = H(t, S_t)$ ¹ gives $\widehat{V}_t = \widehat{V}(t, S_t) = \max_{\tau \in \mathcal{T}_{t, T}} \mathbb{E}[\widehat{H}_\tau | \mathcal{F}_t]$ which satisfies

¹For American call or put options, $H_t = H(t, S_t)$; for path dependent options, $H_t = H(t, S_t, Y_t)$ where Y_t is an additional state variable.

$$\begin{aligned} \frac{V_T}{M_T} &= \frac{V_0}{M_0} + \int_0^T \frac{1}{M_t} \left[\frac{\partial V}{\partial t} + r S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} - r V \right] dt \\ &\quad + \int_0^T \frac{1}{M_t} \frac{\partial V}{\partial S} \sigma S_t d\tilde{W}_t. \end{aligned}$$

As $\frac{V_t}{M_t}$ is a supermartingale, the term before dt should be ≤ 0 . We obtain the partial differential system for American options:

$$\begin{cases} \frac{\partial V}{\partial t} + r S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} - r V \leq 0, \\ V \geq H, \\ \left(\frac{\partial V}{\partial t} + r S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} - r V \right) (V - H) = 0, \\ V_T = H_T. \end{cases}$$

The third equation, called linear complementarity problem, implies that

$$\frac{\partial V}{\partial t} + r S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} - r V = 0$$

in the continuation region $V_t > H_t$ with strict inequality.

Free Boundary Condition for American Put Option

The boundary condition for an American option needs to be analyzed according to the type of option. In this part, we take the vanilla American put option $P_t(S_t)$ with continuous exercise as an example to illustrate the boundary condition analysis. We assume that the interest rate $r > 0$. See Fig. 15.2 for the comparison between European and American puts.

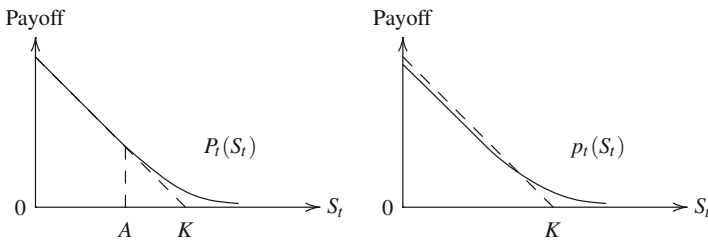


Fig. 15.2 Payoff functions of American put (left) and European put (right) at time $t < T$. Note that $P_t(S_t) \geq \max\{K - S_t, 0\}$ and it is not optimal to exercise for $S_t > A$

With European options, we know the boundary conditions on the rectangular domain $\{(S, t) : 0 \leq S < \infty, 0 \leq t \leq T\}$, see Fig. 15.3 (left).

With American options, we do not know the a priori *exercise boundary* S_t^f which is in fact part of the solution that we need to solve together with $P_t(S_t)$, see Fig. 15.3 (right). For every given t , any put option must be exercised if $S_t = 0$. This is because the immediate exercise will allow the option holder to earn the accrued $e^{r(T-t)}(K - 0)$ at time T instead of $(K - 0)$ which is the payoff at maturity. Hence, we can assume that there exists $0 \leq S_t^f < +\infty$ such that S_t^f is the optimal exercise point for the American put option at time t , i.e., from the previous analysis, that first time τ such that $V_\tau = H_\tau$ is the optimal exercise point. So, if the American put option has time value at inception, S_t^f should be the highest among all S_t that the option should be exercised and

$$P_t(S_t^f) = K - S_t^f > 0, \text{ for all } 0 \leq t \leq T. \quad (15.1)$$

All the $S_t^f, 0 \leq t \leq T$ constitute a boundary condition. As S_t^f is not known a priori and it is part of the whole problem to solve, we call S_t^f a *free boundary* and the whole problem a *free boundary problem*.

Let us now consider the properties of $P_t(S_t)$ in the two separate regions: exercise region ($S_t \leq S_t^f$) and continuation region ($S_t > S_t^f$).

For the continuation region ($S_t > S_t^f$), as there is no exercising, it is equivalent to the European option problem. Hence, the Black–Scholes equation holds:

$$P > K - S_t, \quad \frac{\partial P}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 P}{\partial S_t^2} + r S_t \frac{\partial P}{\partial S_t} - r P = 0.$$

The boundary conditions are the same as for the European option:

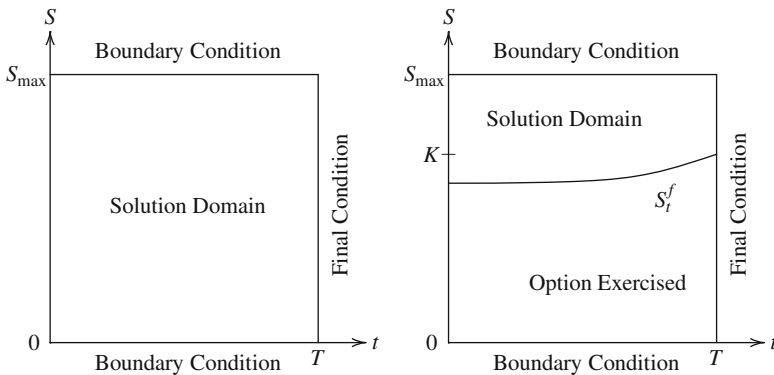


Fig. 15.3 Solution domains of European options (left) and American put options (right). Here $S_{\max}$ is a truncation point approximating ∞ in spatial direction for computational feasibility

$$P_T(S_T) = \max\{K - S_T, 0\}, \text{ for all } 0 \leq S_T < \infty, \quad \lim_{S_t \rightarrow \infty} P_t(S_t) = 0, \text{ for all } 0 \leq t \leq T.$$

For the exercise region ($S_t \leq S_t^f$),

$$P = K - S_t, \quad \frac{\partial P}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 P}{\partial S_t^2} + r S_t \frac{\partial P}{\partial S_t} - r P < 0.$$

We need an extra condition on the unknown boundary S_t^f . It is given by the following proposition and is called the *high contact* or *tangential* condition.

Proposition 15.4 *On the boundary S_t^f of the exercise region, we have*

$$\left. \frac{\partial P_t(S_t)}{\partial S_t} \right|_{S_t=S_t^f} = -1, \quad \text{for all } 0 \leq t < T. \quad (15.2)$$

Proof If we denote the solution of the Black–Scholes PDE with a *given* boundary b_t by $P_t^{BS}(S_t; b_t)$, then we should have

$$P_t(S_t) = \max_{b_t} P_t^{BS}(S_t; b_t).$$

The optimal point is $b_t = S_t^f$,

$$P_t(S_t) = P_t^{BS}(S_t; S_t^f) \quad \text{for all } 0 \leq t \leq T, S_t^f \leq S_t < \infty. \quad (15.3)$$

The first order condition implies

$$\left. \frac{\partial P_t^{BS}(S_t; b_t)}{\partial b_t} \right|_{b_t=S_t^f} = 0, \quad \text{for all } 0 \leq t \leq T, S_t^f \leq S_t < \infty. \quad (15.4)$$

Since b_t is the exercise boundary for the option $P_t^{BS}(S_t; b_t)$, by definition we have

$$P_t^{BS}(b_t; b_t) = K - b_t, \quad \text{for all } b_t.$$

Hence

$$\frac{\partial P_t^{BS}(b_t; b_t)}{\partial b_t} = -1, \quad \text{for all } b_t.$$

However, by the chain rule for multivariate functions, we also have for all b_t ,

$$-1 = \frac{\partial P_t^{BS}(b_t; b_t)}{\partial b_t} = \left\{ \frac{\partial P_t^{BS}(S_t; b_t)}{\partial S_t} \frac{\partial S_t}{\partial b_t} + \frac{\partial P_t^{BS}(S_t; b_t)}{\partial b_t} \right\} \Big|_{S_t=b_t}$$

$$= \frac{\partial P_t^{BS}(S_t; b_t)}{\partial S_t} \Big|_{S_t=b_t} + \frac{\partial P_t^{BS}(S_t; b_t)}{\partial b_t} \Big|_{S_t=b_t}. \quad (15.5)$$

Note that (15.5) is true for all b_t , in particular $b_t = S_t^f$. Then using (15.4), we obtain

$$\frac{\partial P_t^{BS}(S_t; b_t)}{\partial b_t} \Big|_{S_t=b_t, b_t=S_t^f} = 0.$$

Using (15.3),

$$\frac{\partial P_t^{BS}(S_t; b_t)}{\partial S_t} \Big|_{S_t=b_t, b_t=S_t^f} = \frac{\partial P_t(S_t)}{\partial S_t} \Big|_{S_t=S_t^f},$$

and hence we have (15.2). $\square$

Note that the condition (15.2) is not obtained directly by differentiating $P_t(S_t^f) = K - S_t^f > 0$ w.r.t S_t^f . Since S_t^f is itself an unknown variable, if we directly differentiate (15.1), the term $\partial S_t / \partial S_t^f$ will be unaccounted for. By Proposition 15.4, we also have

Corollary 15.1 *Across the exercise boundary S_t^f ,*

$$P_t(S_t) \quad \text{and} \quad \frac{\partial P_t(S_t)}{\partial S_t} \quad \text{are continuous.}$$

Proof In the exercise region, $P_t(S_t) \equiv K - S_t$. Therefore

$$\lim_{S_t \rightarrow (S_t^f)^-} P_t(S_t) = K - S_t^f \quad \text{and} \quad \lim_{S_t \rightarrow (S_t^f)^-} \frac{\partial P_t(S_t)}{\partial S_t} = -1,$$

see Fig. 15.2 (left). But comparing with $P_t(S_t^f) = K - S_t^f > 0$ and (15.2), these values are exactly the same as the corresponding values in continuation region. $\square$

In summary, the American put problem is written as a free boundary problem.

Theorem 15.2 *For each time t , the stock price axis is divided into two distinct regions. The first, $0 \leq S_t \leq S_t^f$, is where early exercise is optimal and*

$$P = K - S_t, \quad \frac{\partial P}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 P}{\partial S_t^2} + r S_t \frac{\partial P}{\partial S_t} - r P < 0.$$

In the other region, $S_t^f < S_t < \infty$, early exercise is not optimal and

$$P > K - S_t, \quad \frac{\partial P}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 P}{\partial S_t^2} + r S_t \frac{\partial P}{\partial S_t} - r P = 0.$$

The boundary conditions are

$$P_t(S_t^f) = \max\{K - S_t^f, 0\}, \quad \left. \frac{\partial P_t(S_t)}{\partial S_t} \right|_{S_t=S_t^f} = -1, \quad \text{and} \quad \lim_{S_t \rightarrow \infty} P_t(S_t) = 0,$$

for all $0 \leq t \leq T$, whereas the final condition at expiry is given by

$$P_T(S_T) = \max\{K - S_T, 0\}, \quad \text{for all } 0 \leq S_T < \infty.$$

Actually, we can combine the equations and the inequalities in the two regions $0 \leq S_t \leq S_t^f$ and $S_t^f < S_t < \infty$ into one equality, where the free boundary S_t^f is eliminated. For all $0 \leq S_t < \infty$ and $0 \leq t \leq T$, we have

$$(K - S_t - P) \left(\frac{\partial P}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 P}{\partial S_t^2} + r S_t \frac{\partial P}{\partial S_t} - r P \right) = 0,$$

with

$$K - S_t - P \leq 0 \quad \text{and} \quad \left(\frac{\partial P}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 P}{\partial S_t^2} + r S_t \frac{\partial P}{\partial S_t} - r P \right) \leq 0.$$

The above formulation is said to be in the *linear complementarity form*. Since the free boundary S_t^f does not appear explicitly, it will not interfere with the solution process, and it can be recovered from the solution after it has been found. When solving the linear complementarity form, we are solving the problem in the whole domain $0 \leq S_t < \infty$, $0 \leq t \leq T$ which include the free boundary S_t^f .

Similarly, we can derive the free boundary problem for American call option with the high contact condition

$$\left. \frac{\partial C_t(S_t)}{\partial S_t} \right|_{S_t=S_t^f} = 1.$$

It is almost impossible to find a useful explicit solution to any given free boundary problem, and so, efficient and robust numerical methods are needed for the computation.

15.3.3 Least-Square American Monte Carlo Method

For multi-dimensional problems, it is necessary to price with Monte Carlo. The usual Monte Carlo method for European options is no more applicable because,

Table 15.1 Simulated stock price paths

Path	$t = 0$	$t = 1$	$t = 2$	$t = 3$
1	1	1.09	1.08	1.34
2	1	1.16	1.26	1.54
3	1	1.22	1.07	1.03
4	1	0.93	0.97	0.92
5	1	1.11	1.56	1.52
6	1	0.76	0.77	0.90
7	1	0.92	0.84	1.01
8	1	0.88	1.22	1.34

Table 15.2 Cash-flow Matrix at Year 3 (Conditional on No Exercise before Year 3)

Path	$t = 1$	$t = 2$	$t = 3$
1			
2			
3			0.07
4			0.18
5			
6			0.20
7			0.09
8			

for a single simulated path of the asset price, it is impossible to make the exercise decision at an exercise date without knowing its continuation value.

Special Monte Carlo algorithms have been proposed for pricing American options, among which, the *Least-Square Method* (LSM), proposed by Longstaff and Schwartz [50], is the most popular and widely used one. The option pricing with LSM is realized through the following steps:

1. generate the paths of the underlying assets
2. determine the optimal exercise decision on each generated path
 - for the exercise decision, we need to compare exercise value and continuation value
 - in LSM, the continuation value is estimated with the least-square method
3. based on the exercise decisions for each generated path fixed in the precedent step, the option price can be simulated with the normal Monte Carlo.

The method is explained through the following example from the original paper [50].

Example 15.2 Consider a 3-year put option with 2 exercise dates (year 1 and year 2) with *initial spot* = 1, *strike* = 1.1 and *one year discount factor* = 0.94176.

Table 15.1 shows 8 simulated paths at inception and the end of each year. We can see the option payoff at year 3 conditional on no early exercise in Table 15.2.

Let X denote the stock price and Y be the discounted cash flow received at the considered year, conditional on no exercise before that year. In this example, it is

Table 15.3 Regression at year 2

Path	Y	X
1	0×0.94176	1.08
2		
3	0.07×0.94176	1.07
4	0.18×0.94176	0.97
5		
6	0.20×0.94176	0.77
7	0.09×0.94176	0.84
8		

Table 15.4 Exercise decision at year 2

Path	Exercise	Continuation
1	0.02	0.0369
2		
3	0.03	0.0461
4	0.13	0.1176
5		
6	0.33	0.1520
7	0.26	0.1565
8		

Table 15.5 Cash-flow matrix at year 2 (Conditional on no early exercise)

Path	$t = 1$	$t = 2$	$t = 3$
1			
2			
3			0.07
4		0.13	
5			
6		0.33	
7		0.26	
8			

assumed that $\widetilde{\mathbb{E}}[Y|X] = a + bX + cX^2$, where X is used as a state variable for the regression. It is up to the practitioner to choose the functional and state variable(s) according to the model/option, for instance, also using the volatility as another state variable.

The regression variables at year 2 are listed in Table 15.3. The dependent variable Y contains the discounted values from Table 15.2. Note that for X , only the in-the-money paths where $X < 1.1$ are used for the regression. It allows to improve the efficiency of the algorithm.

From Table 15.3, the regression result is $\widetilde{\mathbb{E}}[Y|X] = -1.070 + 2.983X - 1.813X^2$. It will be used for calculating the continuation values in Table 15.4 for determining the exercise decisions after comparing with the exercise values. For instance, Path 7 implies exercising as $0.26 > 0.1565$ in Table 15.4; see all the exercise decisions at year 2 in Table 15.5.

Table 15.6 Regression at year 1

Path	Y	X
1	0×0.94176	1.09
2		
3		
4	0.13×0.94176	0.93
5		
6	0.33×0.94176	0.76
7	0.26×0.94176	0.92
8	0×0.94176	0.88

Table 15.7 Exercise decision at year 1

Path	Exercise	Continuation
1	0.01	0.0139
2		
3		
4	0.17	0.1092
5		
6	0.34	0.2866
7	0.18	0.1175
8	0.22	0.1533

Table 15.8 Final stopping rule

Path	$t = 1$	$t = 2$	$t = 3$
1			
2			
3			Y
4	Y		
5			
6	Y		
7	Y		
8	Y		

We can repeat the process at year 1 similarly to year 2. Table 15.6 shows the independent variables X which are in-the-money and also the dependent variables Y . The corresponding regression result is $\widetilde{\mathbb{E}}[Y|X] = 2.038 - 3.335X + 1.356X^2$. It is used for determining exercise decisions at year 1 in Table 15.7.

With the stopping rule matrix in Table 15.8 and the final option cash flow matrix in Table 15.9, we can apply the normal Monte Carlo method for pricing the option using the same simulated paths.

We remark that LSM gives a lower bound of the real option price because the exercise decision might not be 100% optimal.

- If the option hedger has the exercise right (callable structure), he/she tends to under-price the option value (conservative for the option hedger).

Table 15.9 Final option cash-flow matrix

Path	$t = 1$	$t = 2$	$t = 3$
1			
2			
3			0.07
4	0.17		
5			
6	0.34		
7	0.18		
8	0.22		

- If the option hedger is short of the exercise right (puttable structure), he (she) has the risk of selling the exercise right at a too low price.

Methods have been proposed for giving the upper bound of American options, e.g., [69]. But they are often not easy to implement in a general way.

Exercises

1. Let Ω be the infinite coin toss space and $\mathcal{F}_k$ be the natural filtration generated by the random walk; see Exercise 1 in Chap. 10.

(i) Define τ as the random variable

$$\tau(\omega) = \begin{cases} 1, & \text{if } \omega = y \cdots, \\ 2, & \text{if } \omega = n \cdots. \end{cases}$$

Show that τ is a stopping time.

(ii) Modify τ to be

$$\tau(\omega) = \begin{cases} 1, & \text{if } \omega = y \cdots \text{ or } \omega = ny \cdots, \\ 2, & \text{if } \omega = nn \cdots. \end{cases}$$

Show that τ is no longer a stopping time.

2. Compute an American call by a 2-step binomial method with $S_0 = 20$, $K = 21$, $T = 0.5$, $r = 0.02$, $\delta t = 0.25$, $u = 1.1$ and $d = \frac{1}{u}$. Show that at each node of the tree it is also not optimal to early exercise the American call since the stock does not give dividends.

3. Derive the free boundary problem for American call option with the high contact condition

$$\left. \frac{\partial C_t(S_t)}{\partial S_t} \right|_{S_t=S_t^f} = 1,$$

where the notation follows the ones used in this chapter.

Chapter 16

Exotic Options Pricing and Hedging



In this chapter, we will first introduce some Brownian motion related distributions that are useful for pricing exotic options. Then, we will discuss the pricing and hedging of exotic options presented in Part I, Chap. 8 including binary option, barrier option, and Asian option. Some explicit pricing formulas will be derived under BSM model. The last section is dedicated to the hedging practice for the typical risks in exotic options and structured products.

16.1 Further Properties of Brownian Motion

For exotic options, their payoff structures are often path-dependent, and therefore we need more properties of Brownian motion.

16.1.1 Covariance of Brownian Motions

Proposition 16.1 For any $s, t \geq 0$, $\text{Cov}(W_s, W_t) = \mathbb{E}[W_s W_t] = \min\{s, t\}$.

Proof We only prove the case $\mathbb{E}[W_s W_t] = s$ when $s < t$ because the other case is symmetric. With the independence between $W_t - W_s$ and W_s ,

$$\begin{aligned}\mathbb{E}[W_s W_t] &= \mathbb{E}[W_s(W_t - W_s + W_s)] = \mathbb{E}[W_s(W_t - W_s)] + \mathbb{E}[W_s^2] \\ &= \mathbb{E}[W_s]\mathbb{E}[W_t - W_s] + s = s.\end{aligned}\quad \square$$

The result can be used in a property of Asian options.

16.1.2 First Passage Time

Let x be any real number. The first passage time to level x , also called the first hitting time, describes the time when the Brownian motion hits the level x for the first time. It is defined as the following *random variable*:

$$\tau_x = \min\{t \geq 0; W_t = x\},$$

which is a stopping time. If the Brownian motion never reaches the level x , we set $\tau_x = \infty$. Different sample paths of W_t firstly hit x at different moments, making it a random variable $\tau_x(\omega)$ for each ω (sample path). See two realizations of τ_x shown in Fig. 16.1 (left).

16.1.3 Extremum to Date

The *maximum to date* for Brownian motion starting at 0 is defined as the following stochastic process:

$$M_t = \max_{0 \leq s \leq t} W_s.$$

Note that M_t is a non-decreasing process, and visually consists of a considerable amount of flat lines. See Fig. 16.1 (right) for reference. Also $M_t \geq W_0 = 0$.

The *minimum to date* for Brownian motion starting from 0 is defined as the following stochastic process:

$$m_t = \min_{0 \leq s \leq t} W_s.$$

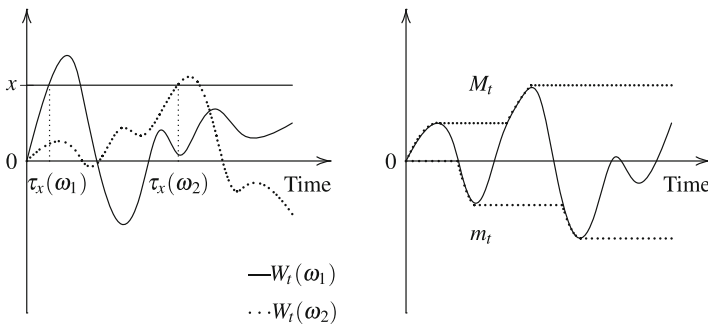


Fig. 16.1 First passage time and maximum/minimum to date

It is also easy to see that m_t is a non-increasing process with a lot of flat lines. See Fig. 16.1 (right) for reference. Moreover, $m_t \leq W_0 = 0$. Together, the processes M_t and m_t form some sort of envelope around W_t .

16.1.4 Reflection Principle

The Brownian motion $W_t \sim \mathcal{N}(0, t)$ is symmetric, i.e., it has an equal chance of reaching a positive number or its negation at a future time (starting from time 0). This property can further be elevated and reveals how the Brownian motion has a mirror image *after* it hits a certain value; see Fig. 16.2.

This can be mathematically described as follows. Suppose $x > 0$ and time is t . The set $\{\tau_x \leq t\}$ implies that the Brownian motion has already touched x before. Then, the probability of W_t having retraced to a lower level $y \leq x$, should be equal to the probability of W_t rising above y 's mirror image about x , i.e., $2x - y$. Thus, we first have the heuristic observation below without proof.

Theorem 16.1 (Reflection Principle)

$$\mathbb{P}\{\tau_x \leq t, W_t \leq y\} = \mathbb{P}\{W_t \geq 2x - y\}, \quad y \leq x, \quad x > 0, \quad W_0 = 0.$$

The special case when $y = x$ is also useful:

$$\mathbb{P}\{\tau_x \leq t, W_t \leq x\} = \mathbb{P}\{W_t \geq x\}, \quad x > 0, \quad W_0 = 0.$$

Corollary 16.1 We have $\mathbb{P}\{\tau_x \leq t\} = 2\mathbb{P}\{W_t \geq x\}$.

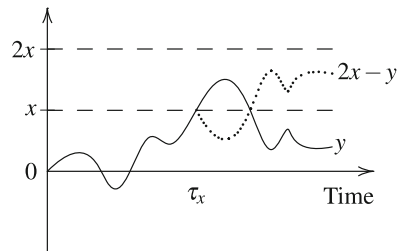
Proof By the special case of reflection principal, we have

$$\mathbb{P}\{\tau_x \leq t, W_t \leq x\} = \mathbb{P}\{W_t \geq x\}.$$

Note that if $W_t \geq x$, then $\tau_x \leq t$ so that $\mathbb{P}\{\tau_x \leq t, W_t \geq x\} = \mathbb{P}\{W_t \geq x\}$. Therefore

$$\mathbb{P}\{\tau_x \leq t\} = \mathbb{P}\{\tau_x \leq t, W_t \leq x\} + \mathbb{P}\{\tau_x \leq t, W_t \geq x\} = 2\mathbb{P}\{W_t \geq x\}. \quad \square$$

Fig. 16.2 The reflection principle: Brownian motion (solid line) and its reflection with respect to x (dotted line)



There is also another version of reflection principle that involves the maximum to date M_t . The important thing is to realize that the set $\{\tau_x \leq t\}$ is equivalent to $\{M_t \geq x\}$. Then,

$$\mathbb{P}\{M_t \geq x, W_t \leq y\} = \mathbb{P}\{\tau_x \leq t, W_t \leq y\} = \mathbb{P}\{W_t \geq 2x - y\}, \quad y \leq x, x > 0,$$

where the second quality is from the reflection principle. From the corollary, we also have the following useful result:

$$\mathbb{P}\{M_t \geq x\} = \mathbb{P}\{\tau_x \leq t\} = 2\mathbb{P}\{W_t \geq x\}.$$

16.1.5 Distribution of First Passage Time

The first passage time w.r.t. x is a random variable when we consider a fixed x . It is interesting to study the distribution of this random variable.

Proposition 16.2 (First Passage Time Distribution) *Suppose $t > 0$ and $\forall x \neq 0$. The random variable τ_x has the following cumulative distribution function and probability density function*

$$\mathbb{P}\{\tau_x \leq t\} = \frac{2}{\sqrt{2\pi}} \int_{\frac{|x|}{\sqrt{t}}}^{\infty} e^{-\frac{z^2}{2}} dz \quad \text{and} \quad f_{\tau_x}(t) = \frac{|x|}{t\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}.$$

Proof We first consider the case $x > 0$. By the Reflection Principle's corollary, and the normal density of Brownian motion, we have

$$\mathbb{P}\{\tau_x \leq t\} = 2\mathbb{P}\{W_t \geq x\} = \frac{2}{\sqrt{2\pi t}} \int_x^{\infty} e^{-\frac{u^2}{2t}} du.$$

Defining $z = \frac{u}{\sqrt{t}}$, we get $\mathbb{P}\{\tau_x \leq t\} = \frac{2}{\sqrt{2\pi}} \int_{\frac{|x|}{\sqrt{t}}}^{\infty} e^{-\frac{z^2}{2}} dz$. For $x < 0$, we observe that τ_x and $\tau_{|x|}$ have the same distribution, and we have the same result. For obtaining the probability density function, we recall that $f_{\tau_x}(t) = \frac{d}{dt} \mathbb{P}\{\tau_x \leq t\}$ and direct differentiation will give the result. $\square$

16.1.6 Joint Distribution with Extremum of Brownian Motion

The reflection principle also allows us to derive the joint density of the Brownian motion and its maximum or minimum.

Proposition 16.3 *Let $t > 0$. The joint density of (M_t, W_t) is*

$$f_{M_t, W_t}(x, y) = \frac{2(2x - y)}{t\sqrt{2\pi t}} e^{-\frac{(2x-y)^2}{2t}} \mathbf{1}_{x \geq \max\{y, 0\}}.$$

The joint density of (m_t, W_t) is

$$f_{m_t, W_t}(\bar{x}, \bar{y}) = \frac{2(\bar{y} - 2\bar{x})}{t\sqrt{2\pi t}} e^{-\frac{(\bar{y}-2\bar{x})^2}{2t}} \mathbf{1}_{\bar{x} \leq \min\{\bar{y}, 0\}}.$$

Proof Note that by definition of distribution function

$$\mathbb{P}\{M_t \geq x, W_t \leq y\} = \int_x^\infty \int_{-\infty}^y f_{M_t, W_t}(u, v) dv du$$

and the LHS is equal to $\mathbb{P}\{W_t \geq 2x - y\}$ by the reflection principle with M_t . Moreover, by the normal density of W_t we also have

$$\mathbb{P}\{W_t \geq 2x - y\} = \frac{1}{\sqrt{2\pi t}} \int_{2x-y}^\infty e^{-\frac{z^2}{2t}} dz.$$

We get the proof by differentiating the right-hand side of both equations w.r.t. x and y .

The joint density of (m_t, W_t) can be deduced from (M_t, W_t) . Defining

$$\bar{W}_t = -W_t, \quad \bar{x} = -x \quad \text{and} \quad \bar{y} = -y,$$

we have

$$\bar{m}_t = \min_{0 \leq s \leq t} \bar{W}_s = -\max_{0 \leq s \leq t} (-\bar{W}_s) = -\max_{0 \leq s \leq t} W_s = -M_t.$$

The following equivalence

$$\mathbb{P}\{\bar{m}_t \leq \bar{x}, \bar{W}_t \geq \bar{y}\} = \mathbb{P}\{M_t \geq x, W_t \leq y\}$$

would lead to $f_{m_t, W_t}(\bar{x}, \bar{y}) = f_{\bar{m}_t, \bar{W}_t}(\bar{x}, \bar{y}) = f_{M_t, W_t}(x, y)$ which is the result to prove after inserting $x = -\bar{x}$ and $y = -\bar{y}$ in $f_{M_t, W_t}(x, y)$. $\square$

Note that $M_t \geq 0$ and $m_t \leq 0$ and therefore their state variables inherit the same ranges. For the state variable of W_t , it can not exceed that of M_t or stay under that of m_t . Therefore the domain of the joint density is not the whole 2D plane.

Let the Brownian motion with drift be defined as $\widehat{W}_t = W_t + \mu t$, $0 \leq t \leq T$. Evidently, $\widehat{W}_t$ follows the normal distribution with mean μt and variance t . Its density is

$$f_{\widehat{W}_t}(y) = f_{W_t}(y - \mu t) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(y-\mu t)^2}{2t}} = \frac{1}{\sqrt{2\pi t}} e^{\mu y - \frac{1}{2}\mu^2 t} e^{-\frac{y^2}{2t}}.$$

The maximum of Brownian motion with drift is defined as $\widehat{M}_T = \max_{0 \leq t \leq T} \widehat{W}_t$. The minimum of Brownian motion with drift is defined as $\widehat{m}_T = \min_{0 \leq t \leq T} \widehat{W}_t$.

Proposition 16.4 *The joint density of $(\widehat{M}_T, \widehat{W}_T)$ is*

$$f_{\widehat{M}_T, \widehat{W}_T}(x, y) = \frac{2(2x - y)}{T\sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{(2x-y)^2}{2T}} \mathbf{1}_{x \geq \max\{y, 0\}}.$$

The joint density of $(\widehat{m}_T, \widehat{W}_T)$ is

$$f_{\widehat{m}_T, \widehat{W}_T}(\bar{x}, \bar{y}) = \frac{2(\bar{y} - 2\bar{x})}{T\sqrt{2\pi T}} e^{\mu \bar{y} - \frac{1}{2}\mu^2 T - \frac{(\bar{y}-2\bar{x})^2}{2T}} \mathbf{1}_{\bar{x} \leq \min\{\bar{y}, 0\}}.$$

The distribution functions for First Passage Time and Maximum of Brownian motion allow the derivation of closed-form solutions for some of the exotic options.

Proof We apply Girsanov Theorem for removing the drift. Defining $\widehat{Z}_t = e^{-\mu \widehat{W}_t + \frac{1}{2}\mu^2 t}$ and $\widehat{\mathbb{P}} = \int_A \widehat{Z}_T d\widehat{\mathbb{P}}$, then, $\widehat{W}_t$ is a Brownian motion with zero drift under $\widehat{\mathbb{P}}$. Hence, for $x \geq \max\{y, 0\}$,

$$\begin{aligned} \widehat{\mathbb{P}}\{\widehat{M}_T \leq x, \widehat{W}_T \leq y\} &= \widehat{\mathbb{E}}\left[\mathbf{1}_{\{\widehat{M}_T \leq x, \widehat{W}_T \leq y\}}\right] = \widehat{\mathbb{E}}\left[\frac{1}{\widehat{Z}_T} \mathbf{1}_{\{\widehat{M}_T \leq x, \widehat{W}_T \leq y\}}\right] \\ &= \widehat{\mathbb{E}}\left[e^{\mu \widehat{W}_T - \frac{1}{2}\mu^2 T} \mathbf{1}_{\{\widehat{M}_T \leq x, \widehat{W}_T \leq y\}}\right] \\ &= \int_{-\infty}^y \int_{-\infty}^x e^{\mu v - \frac{1}{2}\mu^2 T} \left[\frac{2(2u - v)}{T\sqrt{2\pi T}} e^{-\frac{(2u-v)^2}{2T}} \right] dudv, \end{aligned}$$

where the last equality comes from Proposition 16.3. Differentiating w.r.t. y and x , we get

$$f_{\widehat{M}_T, \widehat{W}_T}(x, y) = \frac{2(2x - y)}{T\sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{(2x-y)^2}{2T}} \mathbf{1}_{x \geq \max\{y, 0\}}.$$

Similarly, we can prove the joint density of $(\widehat{m}_T, \widehat{W}_T)$. □

Proposition 16.5 (Conditional Distribution of $\widehat{M}_t$ and $\widehat{m}_t$) *The conditional distribution of $(\widehat{M}_t, \widehat{W}_t)$ is*

$$\begin{aligned} f_{\widehat{M}_t | \widehat{W}_t}(x|y) &= \frac{2(2x - y)}{t} e^{-\frac{2x(x-y)}{t}} \mathbf{1}_{x \geq \max\{y, 0\}}, \\ \mathbb{P}\{\widehat{M}_t \leq x | \widehat{W}_t = y\} &= (1 - e^{-\frac{2x(x-y)}{t}}) \mathbf{1}_{x \geq \max\{y, 0\}}. \end{aligned}$$

The conditional distribution of $(\widehat{m}_t, \widehat{W}_t)$ is

$$f_{\hat{m}_t|\hat{W}_t}(\bar{x}|\bar{y}) = \frac{2(\bar{y} - 2\bar{x})}{t} e^{-\frac{2\bar{x}(\bar{x}-\bar{y})}{t}} \mathbf{1}_{\bar{x} \leq \min\{\bar{y}, 0\}},$$

$$\mathbb{P}\{\hat{m}_t \leq \bar{x} | \hat{W}_t = \bar{y}\} = e^{-\frac{2\bar{x}(\bar{x}-\bar{y})}{t}} \mathbf{1}_{\bar{x} \leq \min\{\bar{y}, 0\}}.$$

Proof The conditional density $f_{\hat{M}_t|\hat{W}_t}(x|y)$ can be obtained with $f_{\hat{M}_t|\hat{W}_t}(x|y) = \frac{f_{\hat{M}_t, \hat{W}_t}(x, y)}{f_{\hat{W}_t}(y)}$. Then,

$$\mathbb{P}\{\hat{M}_t \leq x | \hat{W}_t = y\} = \int_0^x f_{\hat{M}_t|\hat{W}_t}(u, y) du = (1 - e^{-\frac{2x(x-y)}{t}}) \mathbf{1}_{x \geq \max\{y, 0\}}.$$

Similarly, we can obtain the conditional distribution for $(\hat{m}_t, \hat{W}_t)$. $\square$

This proposition is useful for Monte Carlo simulations where both the terminal value and the extremum are needed for the considered period. The simulation of the pair can be done with two steps: (1) generating the terminal value, and (2) generating the extremum using $\mathbb{P}^{-1}$ (e.g. $\hat{m}_t = \frac{\bar{y}}{2} - \sqrt{\frac{\bar{y}^2}{4}} - \frac{t \ln u}{2}$, $u \in \mathcal{U}[0, 1]$ is a uniform random variate).

We refer to [7] for generating the triplet $(\hat{m}_t, \hat{M}_t, \hat{W}_t)$ with Monte Carlo.

16.2 Pricing of Exotic Options Under BSM Model

In this section we focus on binary, barrier and Asian options. We refer to [45, 79] for the closed-form solutions under BSM for some other path dependent options.

16.2.1 European Binary Options

The binary option is the simplest option with a discontinuous payoff. The contract rewards the buyer assets or cash if the underlying ends up in-the-money, or nothing if out-of-the money. Here we consider the normalized cash-or-nothing call option, and its payoff at maturity T is given by

$$\mathbf{1}_{\{S_T > K\}}.$$

There is a closed-form formula for theoretically pricing the cash-or-nothing call option under the BSM model. In reality, one can quickly realize that the desired price appears as the second term in the derivation of the Black–Scholes formula,

$$\tilde{\mathbb{E}} \left[\frac{1}{M_T} \mathbf{1}_{\{S_T > K\}} \right] = e^{-rT} \tilde{\mathbb{P}}\{S_T > K\} = e^{-rT} N(d_2),$$

where we recall that $d_1 = \frac{\ln \frac{S_0}{K} + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$ and $d_2 = d_1 - \sigma\sqrt{T}$. As a side note, the price of an asset-or-nothing call option with payoff $S_T \cdot \mathbf{1}_{\{S_T > K\}}$ is given by

$$\tilde{\mathbb{E}} \left[\frac{1}{M_T} S_T \cdot \mathbf{1}_{\{S_T > K\}} \right] = S_0 N(d_1).$$

Together they form a typical European call option. This can also be verified by no-arbitrage argument.

Another interesting property regarding binary options is the relation between the cash-or-nothing ATMS call option with payoff $\mathbf{1}_{\{S_T > S_0\}}$ and the cash-or-nothing ATMS put option with payoff $\mathbf{1}_{\{S_T \leq S_0\}}$. We know that their prices are

$$e^{-rT} \tilde{\mathbb{P}}\{S_T > S_0\} \text{ and } e^{-rT} \tilde{\mathbb{P}}\{S_T \leq S_0\}$$

respectively. Assuming interest rate is zero, the forward price of a non-dividend paying underlying will be equal to its initial price. One may expect the aforementioned probabilities will be equal. Actually, since S_T is log-normally distributed in the BSM model, its probability density function is skewed to the left. So, $\tilde{\mathbb{P}}\{S_T \leq S_0\}$ is higher than $\tilde{\mathbb{P}}\{S_T > S_0\}$, and hence the binary put is worth more than the binary call.

Numerical Approaches

Both PDE and Tree methods are directly applicable to binary options. Note that the discontinuity around K at maturity is known to cause trouble for various finite difference schemes. For instance, the Crank–Nicolson method will suffer from oscillatory results and high order differencing techniques in spatial direction will fail to achieve the supposed accuracy.

Monte Carlo applies directly in the same way as for the case of European Vanilla options.

The discontinuity creates issues for hedging in practice. We will discuss about it in the last section of this chapter.

16.2.2 American Binary and Barrier Options

Barrier options are widely used path-dependent options with payoff related to the asset price history during the option life. There are two main categories of barrier options. A knock-in (also known as kick-in) option becomes existent when the underlying price crosses a certain barrier level, while a knock-out option becomes void when the underlying asset price crosses a predetermined level before maturity. The “in” options and “out” options respect the *in-out parity* such that one can induce the other (cf. Sect. 8.3.3). Therefore, we only take the up-and-out call option to

illustrate the pricing and hedging of barrier options. The payoff at maturity T can be written as

$$(S_T - K)^+ \mathbf{1}_{\{\max_{0 \leq t \leq T} S_t < H\}},$$

where $H (> K)$ is the fixed barrier level. Notice that

$$(S_T - K)^+ \mathbf{1}_{\{\max_{0 \leq t \leq T} S_t < H\}} \leq (S_T - K)^+.$$

Therefore, the up-and-out call option is cheaper than the vanilla European call option, which is attractive to investors who believe in limited upside price moves.

The risk-neutral pricing formula for an up-and-out call is

$$V_0 = \mathbb{E}[e^{-rT} \max\{S_T - K, 0\} \mathbf{1}_{\max_{t \in [0, T]} S_t \leq H}].$$

Let $\widehat{W}_T = \widetilde{W}_T + \mu T$ where $\mu = \frac{1}{\sigma}(r - \frac{1}{2}\sigma^2)$. Then, $S_t = S_0 e^{\sigma \widehat{W}_t}$. The joint distribution of S_T and $\max_{t \in [0, T]} S_t$ can be represented by the joint distribution of $\widehat{W}_T, \widehat{M}_T$.

The region for the integration implied by the expectation is $S_T \leq \max_{t \in [0, T]} S_t \leq H$, $K \leq S_T \leq H$ which is equivalent to

$$\max\{\widehat{W}_T, 0\} \leq \widehat{M}_T \leq h \quad \text{and} \quad l \leq \widehat{W}_T \leq h$$

where $l = \ln \frac{K}{S_0}/\sigma$ and $h = \ln \frac{H}{S_0}/\sigma$. Hence,

$$\begin{aligned} V_0 &= \mathbb{E}[e^{-rT} \max\{S_T - K, 0\} \mathbf{1}_{\max_{t \in [0, T]} S_t \leq H}] \\ &= \int_l^h \int_{\max\{y, 0\}}^h e^{-rT} (S_0 e^{\sigma y} - K) \frac{2(2x - y)}{T \sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{(2x - y)^2}{2T}} dx dy \\ &= \int_l^h e^{-rT} (S_0 e^{\sigma y} - K) \frac{1}{\sqrt{2\pi T}} [-e^{\mu y - \frac{1}{2}\mu^2 T - \frac{(2x - y)^2}{2T}}]_{\max\{y, 0\}}^h dy \\ &= A + B + C + D, \end{aligned}$$

where

$$\begin{aligned} A &= \int_l^h e^{-rT} S_0 e^{\sigma y} \frac{1}{\sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{y^2}{2T}} dy \\ B &= - \int_l^h e^{-rT} K \frac{1}{\sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{y^2}{2T}} dy \end{aligned}$$

$$C = - \int_l^h e^{-rT} S_0 e^{\sigma y} \frac{1}{\sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{(2h-y)^2}{2T}} dy$$

$$D = \int_l^h e^{-rT} K \frac{1}{\sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{(2h-y)^2}{2T}} dy.$$

The first term is

$$A = S_0 \int_l^h e^{-rT + \sigma y} \frac{1}{\sqrt{2\pi T}} e^{\mu y - \frac{1}{2}\mu^2 T - \frac{y^2}{2T}} dy = S_0 \int_l^h \frac{1}{\sqrt{2\pi T}} e^{-\frac{[y - (\sigma + \mu)T]^2}{2T}} dy$$

$$= S_0 \int_{\frac{l - (\sigma + \mu)T}{\sqrt{T}}}^{\frac{h - (\sigma + \mu)T}{\sqrt{T}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = S_0 [N(a_1) - N(a_2)],$$

where $z = \frac{y - (\sigma + \mu)T}{\sqrt{T}}$, $a_1 = \frac{\ln \frac{S_0}{K} + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$, $a_2 = \frac{\ln \frac{S_0}{H} + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$ by some algebraic operations and the property $N(-x) = 1 - N(x)$.

Similarly, we can obtain B, C, D which leads to the formula for an up-and-out call option:

$$V_0 = S_0 [N(a_1) - N(a_2)] - K e^{-rT} [N(a_3) - N(a_4)]$$

$$- H \left(\frac{S_0}{H} \right)^{-\frac{2r}{\sigma^2}} [N(a_5) - N(a_6)] + K e^{-rT} \left(\frac{S_0}{H} \right)^{-\frac{2r}{\sigma^2} + 1} [N(a_7) - N(a_8)],$$

where $a_3 = a_1 - \sigma\sqrt{T}$, $a_4 = a_2 - \sigma\sqrt{T}$, $a_5 = \frac{\ln \frac{H^2}{KS_0} + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$, $a_6 = \frac{\ln \frac{H}{S_0} + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$, $a_7 = a_5 - \sigma\sqrt{T}$ and $a_8 = a_6 - \sigma\sqrt{T}$.

Numerical Approaches

Both PDE and tree methods are applicable by condition related to the knock-out barrier. Consider the up-and-out call with PDE approach. The spatial direction of the solution domain is now reduced to $[0, H]$ and the spatial truncation naturally follows. The RHS boundary condition is also updated as $c = 0$ at H when there is no rebate upon hitting the barrier. Together with $c = 0$ at 0, we have the so-called homogeneous boundary conditions (values are all zeros).

From “in-out” parity, we only need to solve the Black–Scholes equation for an in-option or an out-option only. Usually we solve for the out-options with zero boundary conditions in the spatial direction, when the option has no rebate.

For Monte Carlo simulation, there are two cases:

- if the barrier observation is discrete (daily, monthly, quarterly, or etc.), we need to set the time steps accordingly when generating asset price paths. Such treatment is for getting the maximum level achieved on these discrete dates
- if the barrier event observation is continuous, Proposition 16.5 can be used for simulating the extremum between two time points.

The hedging in practice is tricky for Barrier options due to the discontinuity. We will discuss about it in the last section as well.

16.2.3 Asian Options

There are two kinds of averaging, namely the arithmetic average and geometric average. Their mathematical representations take the following forms for $0 < t_1 < t_2 < \dots < t_m = T$:

- Arithmetic average: $Y_T = \frac{1}{m} \sum_{j=1}^m S_{t_j}$
- Geometric average: $Y_T = \sqrt[m]{S_{t_1} S_{t_2} \cdots S_{t_m}}$.

The continuous-time approximation will be

- Arithmetic average: $Y_T = \frac{1}{T} \int_0^T S_t dt$
- Geometric average: $Y_T = \exp(\frac{1}{T} \int_0^T \ln S_t dt)$.

In Part I, we have discussed the purpose of Asian options in practice. One thing worth noticing is that the averaging mechanism helps to control the delta at the sampling dates because of less volatility. However, the averaging does not help much to control the gamma.

The closed-form formulas of Asian options are not yet known for most cases except when the payoff uses geometric average and the underlying follows the log-normal distribution. The pricing utilizes the fact that the product of two log-normal variates is still log-normal. On the other hand, the arithmetic average makes use of addition and the sum of log-normal variates does not keep the log-normal distribution. Therefore, numerical approaches are used for these cases.

Numerical Approaches

Monte Carlo approach is simple and flexible for pricing Asian options. However, the time steps for generating the paths need to cover all the sampling points specified by the option.

Things are complicated for PDE and binomial tree methods. Due to the path-dependency feature of the payoff, the option price at time t depends not only on (t, S_t) but also on $\{S_u, u < t\}$, leading to a non-Markovian problem. We need to add a new spatial dimension with the variable $Y_t = \int_0^t S_u du$ ($\frac{1}{t} \int_0^t S_u du$ is known as

running average) so that the option price will be dependent only on (t, S_t, Y_t) and the problem becomes Markovian. Then, 2-D PDE or binomial tree can be applied.

The hedging in practice for Asian options is similar to vanilla options.

Practitioners often make a quick estimation of Asian option price with the pricer of Vanilla options by applying the volatility of $\frac{\sigma}{\sqrt{3}}$. This is based on the below property of Brownian motion.

Proposition 16.6 Consider a process X_t satisfying $dX_u = \sigma dW_u$. Then,

$$\text{Var} \left(\frac{1}{t-s} \int_s^t (X_u - X_s) du \right) = \frac{\sigma^2(t-s)}{3}.$$

Proof It is equivalent to prove

$$\text{Var} \left(\int_s^t (W_u - W_s) du \right) = \frac{(t-s)^3}{3}.$$

Covariance of Brownian motion and direct calculations give

$$\begin{aligned} \text{Var} \left(\int_s^t (W_u - W_s) du \right) &= \mathbb{E} \left[\int_s^t \int_s^t (W_u - W_s)(W_v - W_s) dudv \right] \\ &= \int_s^t \int_s^t \mathbb{E}[(W_u - W_s)(W_v - W_s)] dudv = \int_s^t \int_s^t (\min\{u, v\} - s) dudv \\ &= \int_s^t \int_s^t (u - s) \mathbf{1}_{\{u \leq v\}} dudv + \int_s^t \int_s^t (v - s) \mathbf{1}_{\{u > v\}} dudv \\ &= 2 \int_s^t \int_s^y (x - s) dx dy = 2 \int_s^t \int_0^{y-s} z dz dy = \int_s^t (y - s)^2 dy = \frac{(t-s)^3}{3}. \quad \square \end{aligned}$$

Let Y_t denote arithmetic average of X_t with continuous sampling, i.e., $Y_t = \frac{1}{t} \int_0^t X_u du$ with $Y_0 = 0$. Since $\text{Var}(X_t) = \sigma^2 t$, we know from above that variance of Y_t becomes one third of the original, i.e.,

$$\text{Var}(Y_t) = \text{Var} \left(\frac{1}{t} \int_0^t X_u du \right) = \frac{\sigma^2 t}{3},$$

by taking $s = 0$. Equivalently, the volatility of Y_t for the time period is $\frac{\sigma}{\sqrt{3}}$.

16.2.4 Basket Options, Worst-of, Best-of Options

Numerical Approaches

For options linked to multiple assets, it is difficult to obtain closed-form formulas. For problems with a dimension of more than two, the PDE or tree methods start to face issues as well. So, Monte Carlo method is usually applied for such options.

In the market, implied volatilities that are quoted can be hedged for a large range of assets. However, it is not the case for the correlation between the asset prices. Practitioners have to take risk of correlation which will be further analyzed in the next section.

16.3 Hedging Issues and Practical Solutions

16.3.1 Gap Risk

Some exotic options and structured products offer different payoff levels contingent on the price level of the underlying assets. This often creates discontinuity in the price of the product. The delta at the discontinuity point may be very large as it tends to infinity theoretically. The execution of a delta amount of more than 20% may start to impact the market price of the underlying. This problem is known as the gap risk.

The below techniques are usually applied in practice for mitigating the gap risk:

- call/put spread smoothing: when the barrier is observed at fixed time point (European Barrier); See Fig. 16.3
- barrier shift: when the barrier is observed continuously (American Barrier).

For a European barrier, the Dirac¹ behaviour of delta can be resolved by applying a call/put spread. A call spread smoothing means that instead of using

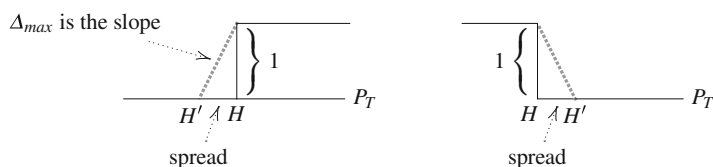


Fig. 16.3 Smoothing gap risk by call or put spread

¹A mathematical concept in which infinity is achieved at one point but zero otherwise.

$N \times \mathbf{1}_{\{P_T > H\}}$, $P_T = \frac{S_T}{S_0}$ is the performance of the underlying

as payout for hedging, the trader uses

$$N \times \min\{1, \max\{0, \Delta_{max} \times (P_T - H')\}\},$$

where

$N \times \Delta_{max}$ is the maximum delta amount the hedger can accept which is normally less than 20% of the average daily trading volume (ADTV) of the underlying spread = $1/\Delta_{max}$ and $H' = H - \text{spread}$.

Let H' be set at a level either **higher** or **lower** than H so that the pricing is *always* on the conservative side for the product hedging party (i.e. more expensive to the investor). This is called *over-hedging*. We note that the product terms for the investor are not changed by the over-hedging. If the final performance ends up at a level between H and H' , the “over-hedged payout” will belong to the hedging party, but not to the investor.

The Put Spread smoothing case can be deduced following the same rationale.

For Worst-of or Best-of observations, the call/put spread can be applied to the worst/best performance.

Example 16.1 Figure 16.4 shows the over-hedging with call spread on a European binary call.

For large size trades with basket underlyings of low liquidity, the following trick may be considered for obtaining more aggressive pricing without losing the control on Δ_{max} . The idea is to transform the payoff conditioned by one event (on worst or best performance) into a payoff linked to the sum payoffs— performance of all underlyings.

Consider a contract payoff $\mathbf{1}_{\{\min_i \{P_T^i\} > H\}}$. We apply the transformation

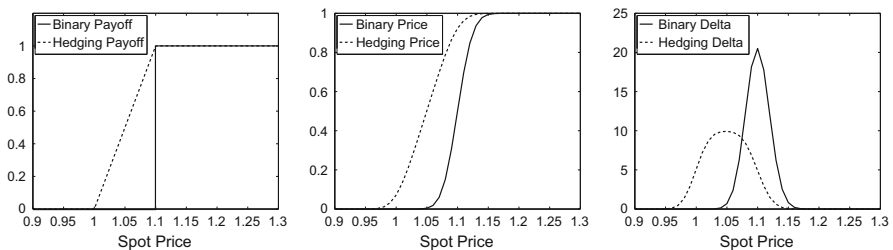


Fig. 16.4 Delta-hedging a binary call and a call spread

$$1_{\{\min_i \{P_T^i\} > H\}} = 1 - 1_{\{\min_i \{P_T^i\} < H\}} = 1 - \min \left\{ 1, \sum_{i=1}^n 1_{\{P_T^i < H\}} \right\}.$$

For each $1_{\{P_T^i < H\}}$, we apply put spread for smoothing:

$$\text{Smoothed Payoff} = 1 - \min \left\{ 1, \sum_{i=1}^n \text{Put Spread}(i) \right\},$$

where

$$\text{Put Spread}(i) = \max \{ 0, \min \{ 1, \Delta_{max}^i \times (H - P_T^i) \} \}.$$

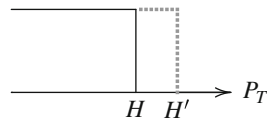
Let us assume that the worst performance has very poor liquidity, but the second worst underlying has very high liquidity. The poor liquidity of the worst results in a large spread which makes the price quite expensive. However, the second worst has a very tight spread thanks to its liquidity. In the normal smoothing method, only the worst performance will be used for the pricing. But with our method, the second worst performance may play a role in pricing which cheapens the price of the product.

For smoothing the Gamma/Delta near an American barrier, the technique of barrier shift may be applied which consists of hedging/pricing the product with a barrier that is not at the contract barrier described in the product terms. The barrier shift should put the barrier further from the spot price so that the option will be knocked out by the contract barrier away from the shifted barrier level used in hedging. See Fig. 16.5 for reference.

For example, the hedging party selling an Up-and-Out Binary option with contract barrier at 135% of initial price may shift the barrier to 140% for hedging/pricing. The pricing will be on the conservative side to the hedging party.

However, if the hedging party is selling the Up-and-In Binary option, the hedging barrier can not be simply shifted closer to the current price because this will not smooth out Gamma/Delta at the barrier. In this case, the hedging party may apply call/put spread on the continuous extremum (i.e. P_T^i replaced by the continuous extremum for the period, either minimum or maximum according to the product). Another almost equivalent method consists of breaking the trade into n smaller size trades over a series of shifted barrier levels between the current price and the contract barrier. See Fig. 16.6 for reference.

Fig. 16.5 Barrier shift for smoothing gamma or delta



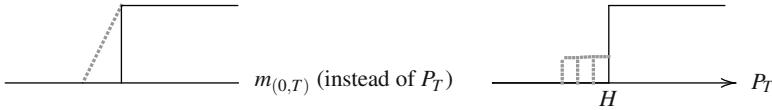


Fig. 16.6 Hedging up-and-in binary option ($m_{(0,T)}$ is the minimum performance in the period $[0, T]$)

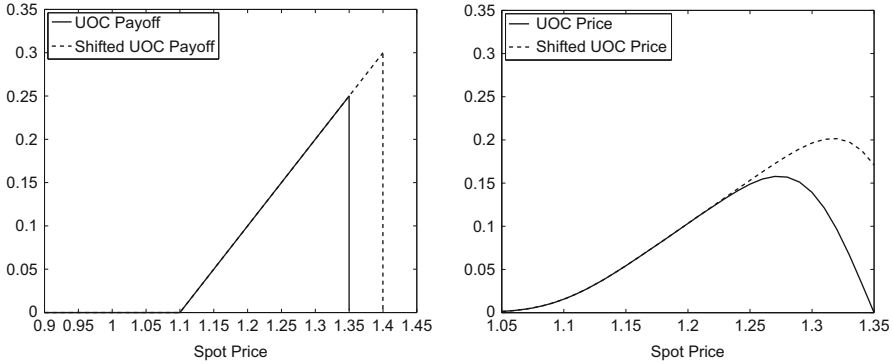


Fig. 16.7 Delta-hedging up-and-out calls with two different barriers

Example 16.2 Figure 16.7 illustrates “barrier shift” applied to an up-and-out call with barrier H on the contract. The option writer used H' ($H' > H$) for pricing and hedging.

Notice that over-hedging with a shifted barrier does not bound the delta as the case in European binary option. The delta and gamma may be still large near H' , but it does not matter any more because the contract is already void after breaking H and that the hedging stops.

16.3.2 Large Delta Issue

For large Delta amount (compared to the liquidity of the underlying) at payoff expiration or at initial hedging time, the execution of the Delta amount in the market is exposed to slippage risk. Averaging is often used for mitigating such risk. It consists of taking the averaged price over a few days (e.g. 3 days) as reference (either final reference price or initial reference price) to determine the payoff amount. We note that averaging reduces the Delta at expiration but it does not help much for reducing the Gamma before expiration unless the averaging is applied for a significantly long period.

16.3.3 Correlation Risk

In general, the correlation risk is difficult to hedge in the market. Sometimes, correlation exposure may be obtained through outperformance options, correlation Swaps, etc. on liquid names in the broker market. But the correlation risks from structured business are not transferred to the market. For the majority of the structured products, the trading books are selling correlation, i.e. the basket of underlyings with higher correlations will make the product more expensive.

The trading books take reserves on correlation by conservatively shifting the correlation levels in pricing. If the shifted correlations are also used for producing the Greeks for hedging, this reserve will be gradually released through the time decay.

The correlation parameter is estimated from historical correlation over one or more time windows, e.g. 3 months or 6 months window rolling over a period of 3 years. Weekly returns are often used for assets from different time zones. For shifting the correlation parameters, the simple method used by practitioners consists of interpolating between the estimated historical correlation matrix $\Sigma_{\text{Historical}}$ and the perfect correlation matrix $\mathbf{I}$, i.e. the matrix with value 1 for every element:

$$\text{Shifted Correlation} = \alpha \Sigma_{\text{Historical}} + (1 - \alpha) \mathbf{I}, \quad \alpha \in [0, 1].$$

This method keeps naturally the positive semi-definite property for the shifted correlation matrix.

Exercises

1. Assume the underlying follows the BSM with no dividend. Let c denote the time- t Black–Scholes European call price with strike price K and maturity T . Show that

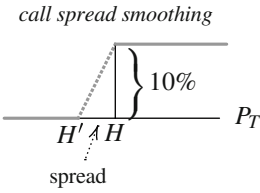
$$-\frac{\partial c}{\partial K} = e^{-r(T-t)} N(d_2).$$

Explain how a bullish call spread and a binary cash-or-nothing call option are related through the above equation.

2. Assume the underlying follows the BSM with no dividend. Derive the time- t delta for a binary cash-or-nothing call option with payoff $\mathbf{1}_{\{S_T > K\}}$ where S_T is the asset price at maturity T and K is the strike. Show that the delta approaches $+\infty$ when $t \rightarrow T$.
3. Consider a stock with ADTV (Average Daily Trading Volume) being USD80m. A client of an investment bank wants to buy a 3 month, 105% Strike Binary Call

option (European style) which pays a coupon of 10% of the notional amount of USD10m.

The trader needs to control the maximum delta to be at or below 20% of ADTV by applying a call-spread type of smoothing as illustrated below:



Determine the width of *spread* in percentage (%) of the initial stock price.

Chapter 17

Numéraires and the Pricing of Vanilla Interest Rate Options



17.1 Introduction of Numéraire

In this chapter, we will introduce the concept of *numéraire* and its application in the modelling and pricing of derivatives. In financial mathematics, a numéraire is the unit of account for denominating the assets. Any positive and non-dividend paying process can be used as a numéraire. The risk-neutral measure for a given numéraire is the probability measure under which any non-dividend paying tradable asset denominated in this numéraire is a martingale. The techniques related to numéraire lead to simplification and clarity in financial derivatives modelling.

Actually, we have already encountered a numéraire in previous content, which is the money market account:

$$M_t = e^{\int_0^t r_s ds}.$$

It is a positive process which has been used as numéraire. Its value at t is the return of investing \$1 with continuous interest rate r_s . Furthermore, under the “money market” risk-neutral probability $\tilde{\mathbb{P}}$, all non-dividend paying asset price denominated in units of M_t , that is, $S_t/M_t = e^{-\int_0^t r_s ds} S_t$ is a martingale under $\tilde{\mathbb{P}}$.

17.2 Change of Numéraire

Now, we consider how the change of probability is affected when the original numéraire is changed to another one. Let N be a numéraire and $\tilde{\mathbb{P}}^N$ be the probability measure under which all non-dividend paying assets denominated in the units of N (i.e. S_t/N_t) are $\tilde{\mathbb{P}}^N$ -martingales.

Let U be a strictly positive non-dividend paying asset, which can be potentially another numéraire. Denoting

$$Z_t = \frac{U_t/U_0}{N_t/N_0}, \quad (17.1)$$

we have the following from the martingale property:

$$\tilde{\mathbb{E}}^N[Z_t] = \frac{N_0}{U_0} \tilde{\mathbb{E}}^N \left[\frac{U_t}{N_t} \right] = 1.$$

Then, we define $\tilde{\mathbb{P}}^U$ with $\left. \frac{d\tilde{\mathbb{P}}^U}{d\tilde{\mathbb{P}}^N} \right|_{\mathcal{F}_t} = Z_t$ as Radon–Nikodym derivative process.

We call the below formula set as **the toolkit for numéraire change**:

$$\begin{cases} Z_t = \frac{U_t/U_0}{N_t/N_0} \\ d\tilde{W}_t^U = d\tilde{W}_t^N - \frac{d\langle Z, \tilde{W}^N \rangle_t}{Z_t} & \text{(Girsanov Theorem)} \\ \tilde{\mathbb{E}}^U[Y|\mathcal{F}_s] = \frac{1}{Z_s} \tilde{\mathbb{E}}^N[Y Z_t | \mathcal{F}_s] & \text{(Bayes' Formula)} \end{cases}$$

for all $0 \leq s \leq t$ and any $\mathcal{F}_t$ -measurable variable Y .

Suppose a pricing problem requires evaluation under the measure $\tilde{\mathbb{P}}^N$, as in the right hand side in Bayes' formula. There will be one less stochastic term Z_t inside the expectation $\tilde{\mathbb{E}}^U$, after changing the measure from $\tilde{\mathbb{P}}^N$ to $\tilde{\mathbb{P}}^U$. This is helpful for the resolution and may imply a simpler Monte Carlo simulation or an easier numerical integration.

Example 17.1 (Stock Numéraire) The stock numéraire (non-dividend-paying) has been implicitly used when deriving the Black–Scholes formulas. Specifically, it is used for calculating the first term in Black–Scholes formula for call option, i.e., $\tilde{\mathbb{E}}[S_T \mathbf{1}_{\{S_T > K\}} | \mathcal{F}_t]$ (or equivalently pricing an asset-or-nothing call option).

Our aim is to take S_T out of the risk-neutral expectation with a measure change, and thus simplifying the pricing formula. Let Z_t be the Radon–Nikodym derivative for the numéraire change

$$Z_t = \frac{S_t/S_0}{M_t/M_0} (= e^{-\frac{1}{2}\sigma^2 t + \sigma \tilde{W}_t}).$$

Then, under the new measure $\tilde{\mathbb{P}}^S$ (or $\mathbb{Q}$ in Chap. 13),

$$\begin{aligned} V_t &= M_t \tilde{\mathbb{E}} \left[\frac{S_T \mathbf{1}_{\{S_T > K\}}}{M_T} \middle| \mathcal{F}_t \right] = S_t \frac{M_t}{S_t} \tilde{\mathbb{E}} \left[\frac{S_T \mathbf{1}_{\{S_T > K\}}}{M_T} \middle| \mathcal{F}_t \right] \\ &= S_t \frac{1}{Z_t} \tilde{\mathbb{E}}[Z_T \mathbf{1}_{\{S_T > K\}} | \mathcal{F}_t] = S_t \tilde{\mathbb{E}}^S[\mathbf{1}_{\{S_T > K\}} | \mathcal{F}_t]. \end{aligned}$$

17.3 Self-Financing Condition

Previously we have seen that the changes in the discounted self-financing portfolio Π_t/M_t are related to the changes in the discounted stock price S_t/M_t .

In the following, we prove the general case for a self-financing portfolio denominated in any given numéraire N .

Theorem 17.1 (Self-financing condition under a given numéraire) *Let N be a numéraire. A portfolio $\Pi_t = \Delta_t S_t + \zeta_t N_t$ is self-financing, i.e.,*

$$d\Pi_t = \Delta_t dS_t + \zeta_t dN_t,$$

if and only if

$$d\frac{\Pi_t}{N_t} = \Delta_t d\frac{S_t}{N_t}.$$

Hence, if $\frac{S_t}{N_t}$ is a $\widetilde{\mathbb{P}}^N$ -martingale, $\frac{\Pi_t}{N_t}$ is also a $\widetilde{\mathbb{P}}^N$ -martingale such that

$$\widetilde{\mathbb{E}}^N \left[\frac{\Pi_t}{N_t} \middle| \mathcal{F}_u \right] = \frac{\Pi_u}{N_u}.$$

Proof Suppose we have a self-financing portfolio Π_t . Then

$$\begin{aligned} d\frac{\Pi_t}{N_t} &= \Pi_t d\frac{1}{N_t} + \frac{1}{N_t} d\Pi_t + d\langle \Pi, \frac{1}{N} \rangle_t \\ &= (\Delta_t S_t + \zeta_t N_t) d\frac{1}{N_t} + \frac{1}{N_t} (\Delta_t dS_t + \zeta_t dN_t) + \Delta_t d\langle S, \frac{1}{N} \rangle_t + \zeta_t d\langle N, \frac{1}{N} \rangle_t \\ &= \zeta_t \left(N_t d\frac{1}{N_t} + \frac{1}{N_t} dN_t + d\langle N, \frac{1}{N} \rangle_t \right) + \Delta_t \left(S_t d\frac{1}{N_t} + \frac{1}{N_t} dS_t + d\langle S, \frac{1}{N} \rangle_t \right) \\ &= \zeta_t d(N_t \frac{1}{N_t}) + \Delta_t d(S_t \frac{1}{N_t}) = \Delta_t d\frac{S_t}{N_t}. \end{aligned}$$

Here we have made use of the fact that

$$d(\frac{1}{N_t} N_t) = d(1) = 0.$$

Conversely, suppose we have $d\frac{\Pi_t}{N_t} = \Delta_t d\frac{S_t}{N_t}$ where $\Pi_t = \Delta_t S_t + \zeta_t N_t$, then it leads to

$$d\langle \frac{\Pi}{N}, N \rangle_t = \Delta_t d\langle \frac{S}{N}, N \rangle_t.$$

Applying Itô's Product Rule to $(\frac{S_t}{N_t} N_t) (= S_t)$, we have

$$d\left(\frac{S_t}{N_t} N_t\right) = \frac{S_t}{N_t} dN_t + N_t d\frac{S_t}{N_t} + d\left\langle \frac{S}{N}, N \right\rangle_t = \frac{S_t}{N_t} dN_t + N_t d\frac{S_t}{N_t} + \frac{1}{\Delta_t} d\left\langle \frac{\Pi}{N}, N \right\rangle_t.$$

Then,

$$d\frac{S_t}{N_t} = \frac{1}{N_t} \left(dS_t - \frac{S_t}{N_t} dN_t - \frac{1}{\Delta_t} d\left\langle \frac{\Pi}{N}, N \right\rangle_t \right).$$

Now, applying Itô's product rule to $(\frac{\Pi_t}{N_t} N_t) = \Pi_t$, we get

$$d\left(\frac{\Pi_t}{N_t} N_t\right) = \frac{\Pi_t}{N_t} dN_t + N_t d\frac{\Pi_t}{N_t} + d\left\langle \frac{\Pi}{N}, N \right\rangle_t.$$

Using $\Pi_t = \Delta_t S_t + \zeta_t N_t$, $d\frac{\Pi_t}{N_t} = \Delta_t d\frac{S_t}{N_t}$ and the result above, we get

$$\begin{aligned} d\Pi_t &= \frac{\Delta_t S_t + \zeta_t N_t}{N_t} dN_t + \Delta_t \left(dS_t - \frac{S_t}{N_t} dN_t - \frac{1}{\Delta_t} d\left\langle \frac{\Pi}{N}, N \right\rangle_t \right) + d\left\langle \frac{\Pi}{N}, N \right\rangle_t \\ &= \Delta_t dS_t + \zeta_t dN_t, \end{aligned}$$

i.e., Π_t is a self-financing portfolio. □

Remark 17.1 The theorem can be extended to a portfolio with n assets ($n \geq 2$), i.e., $\Pi_t = \sum_{i=1}^n \Delta_t^i S_t^i + \zeta_t N_t$. The proof is similar by replacing the S_t -related parts with a linear combination.

17.4 Usual Numéraires and the Pricing of Vanilla Interest Rate Options

In the following, we introduce some widely used numéraires and the corresponding measures. The application of numéraires helps to derive explicit formulas for vanilla interest rate options under Black models (lognormal model).

17.4.1 Money Market Account

The numéraire is

$$M_t = e^{\int_0^t r_s ds},$$

where r_s is the instantaneous interest rate which may be deterministic or stochastic. The corresponding measure is the risk-neutral measure used so far in this book. The pricing of a European derivative paying V_T at T is

$$V_t = M_t \tilde{\mathbb{E}} \left[\frac{V_T}{M_T} \middle| \mathcal{F}_t \right].$$

17.4.2 Zero Coupon Bond

A zero coupon bond paying 1 at maturity T is noted as $P_t(T)$ for its value at time t . The probability measure with zero coupon bond as numéraire is called *T-forward measure*. The pricing framework under *T-forward measure* admits a simple form which does not contain a discounting term inside the expectation operator.

Proposition 17.1 *The time- t price of a European-style derivative with maturity T is*

$$V_t = P_t(T) \tilde{\mathbb{E}}^T [V_T | \mathcal{F}_t],$$

where $\tilde{\mathbb{E}}^T$ is the expectation under the *T-forward measure*.

Proof We first note that $P_T(T) = 1$. Then,

$$\begin{aligned} V_t &= M_t \tilde{\mathbb{E}} \left[\frac{V_T}{M_T} \middle| \mathcal{F}_t \right] = M_t \tilde{\mathbb{E}} \left[\frac{P_T(T) V_T}{M_T} \middle| \mathcal{F}_t \right] \\ &= P_t(T) \frac{M_t}{P_t(T)} \tilde{\mathbb{E}} \left[\frac{P_T(T) V_T}{M_T} \middle| \mathcal{F}_t \right] \\ &= P_t(T) \frac{1}{Z_t} \tilde{\mathbb{E}} [Z_T V_T | \mathcal{F}_t] = P_t(T) \tilde{\mathbb{E}}^T [V_T | \mathcal{F}_t], \end{aligned}$$

where Z_t is the Radon–Nikodym derivative for the numéraire change

$$Z_t = \frac{P_t(T)/P_0(T)}{M_t/M_0}. \quad \square$$

The *T-forward measure* is also closely related to derivatives written on Libor rates. Suppose we have the tenor structure

$$0 \leq T_0 < T_1 < T_2 < \cdots < T_{n-1} < T_n.$$

Denote $L_t(T_i)$ for the Libor rate seen at t for the period $\Delta_i = T_{i+1} - T_i$. From the arbitrage relationship seen in FRA rate calculation, we have

$$L_t(T_i) = \frac{P_t(T_i) - P_t(T_{i+1})}{\Delta_i P_t(T_{i+1})}.$$

Proposition 17.2 *The Libor rate seen at time t , $L_t(T_i)$ is a martingale under $\tilde{\mathbb{P}}^{T_{i+1}}$.*

Proof Let $0 < s < t$. Note that $\frac{P_s(T_i) - P_s(T_{i+1})}{\Delta_i}$ is a portfolio of non-dividend paying assets. We then have

$$\tilde{\mathbb{E}}^{T_{i+1}}[L_t(T_i) | \mathcal{F}_s] = \tilde{\mathbb{E}}^{T_{i+1}} \left[\frac{P_t(T_i) - P_t(T_{i+1})}{\Delta_i P_t(T_{i+1})} \middle| \mathcal{F}_s \right] = \frac{P_s(T_i) - P_s(T_{i+1})}{\Delta_i P_s(T_{i+1})} = L_s(T_i).$$

□

Pricing of Cap or Floor

An interest rate cap contract is a strip of European call options (each option is called a caplet) on Libor rate paying $\Delta_i (L_{T_i}(T_i) - K)^+$ at time T_{i+1} . Normally the payoff is further multiplied by a notional but it is omitted here.

If $L_t(T_i)$ is always positive, then, under $\tilde{\mathbb{P}}^{T_{i+1}}$, $L_t(T_i)$ is a positive martingale and we have

$$dL_t(T_i) = L_t(T_i) \sigma_i(t) d\tilde{W}_t^{T_{i+1}},$$

in which the volatility $\sigma_i(t)$ is assumed to be deterministic.

From Proposition 17.1, the pricing of a caplet can be performed under the T_{i+1} -forward measure, which can be solved by Black's formula (similar to applying the Black–Scholes formula when the drift of riskless rate is absent):

$$V_i(t) = P_t(T_{i+1}) \tilde{\mathbb{E}}^{T_{i+1}}[\Delta_i (L_{T_i}(T_i) - K)^+ | \mathcal{F}_t] = \Delta_i P_t(T_{i+1}) (L_t(T_i) N(d_1) - K N(d_2)),$$

with

$$d_1 = \frac{\ln \frac{L_t(T_i)}{K} + \frac{1}{2} \int_t^{T_i} \sigma_i^2(u) du}{\sqrt{\int_t^{T_i} \sigma_i^2(u) du}} \quad \text{and} \quad d_2 = \frac{\ln \frac{L_t(T_i)}{K} - \frac{1}{2} \int_t^{T_i} \sigma_i^2(u) du}{\sqrt{\int_t^{T_i} \sigma_i^2(u) du}}.$$

Similar results can be obtained for the interest rate floor contract which is a strip of European put options on Libor rate.

17.4.3 Annuity Factor

We recall from Chap. 2 Sect. 2.8 that Annuity factor is defined as

$$A_t = \sum_{i=0}^{n-1} \Delta_i P_t(T_{i+1}),$$

where T_i 's represent the Libor fixing/payment dates. It is a positive linear combination of zero coupon bonds and hence can be used as numéraire.

For a standard swap contract, the swap rate s_t is

$$s_t = \frac{P_t(T_0) - P_t(T_n)}{A_t}.$$

Proposition 17.3 *The swap rate s_t is a martingale under $\tilde{\mathbb{P}}^A$.*

Proof Note that $P_u(T_0) - P_u(T_n)$ is a portfolio of non-dividend paying assets. Then,

$$\tilde{\mathbb{E}}^A[s_t | \mathcal{F}_u] = \tilde{\mathbb{E}}^A \left[\frac{P_t(T_0) - P_t(T_n)}{A_t} \middle| \mathcal{F}_u \right] = \frac{P_u(T_0) - P_u(T_n)}{A_u} = s_u. \quad \square$$

Pricing of Swaption

Here we consider pricing a basic swaption, which is essentially a European call option on a swap. Assume T_0 is the time when the swaption expires and the underlying swap starts. Its payoff function is

$$A_{T_0}(s_{T_0} - K)^+.$$

We observe that the swap rate is denominated by the annuity factor A_t , which can be used as a numéraire. The pricing of a swaption could be simplified when the risk-neutral measure is changed to a new measure with A_t as numéraire. Let T_0 be the start date of the underlying swap. Then, for $t < T_0$,

$$V_t = M_t \tilde{\mathbb{E}} \left[\frac{A_{T_0}(s_{T_0} - K)^+}{M_{T_0}} \middle| \mathcal{F}_t \right] = A_t \tilde{\mathbb{E}}^A[(s_{T_0} - K)^+ | \mathcal{F}_t]$$

where the numéraire change is done with $Z_t = \frac{A_t/A_0}{M_t/M_0}$.

Now it comes down to the distribution of s_{T_0} under $\tilde{\mathbb{P}}^A$. The swap rate is a martingale under $\tilde{\mathbb{P}}^A$ and now we assume it is positive. By martingale representation theorem, we have

$$ds_t = s_t \sigma_t d\tilde{W}_t^A.$$

Here once again σ_t is modelled as deterministic. And then, s_t is lognormal.

Thus, the key term $\tilde{\mathbb{E}}^A[(s_{T_0} - K)^+ | \mathcal{F}_t]$ in the pricing formula can be obtained again by Black's formula:

$$\tilde{\mathbb{E}}^A[(s_{T_0} - K)^+ | \mathcal{F}_t] = s_t N(d_1) - K N(d_2)$$

with

$$d_1 = \frac{\ln \frac{s_t}{K} + \frac{1}{2} \int_t^{T_0} \sigma_u^2 du}{\sqrt{\int_t^{T_0} \sigma_u^2 du}} \quad \text{and} \quad d_2 = \frac{\ln \frac{s_t}{K} - \frac{1}{2} \int_t^{T_0} \sigma_u^2 du}{\sqrt{\int_t^{T_0} \sigma_u^2 du}}.$$

Therefore the swaption price at time t is

$$V_t = A_t(s_t N(d_1) - K N(d_2)).$$

Remark 17.2 As negative interest rates for certain currencies are present around 2010s, the modelling requirement for positive interest rate is no more mandatory. For vanilla interest rate options (caps, floors, swaptions), some practitioners now use Bachelier model ($dL_t = \mu dt + \sigma dW_t$) which assumes normal distribution.

17.4.4 SABR Model for Vanilla Interest Rate Options

Similar to the equity options, the market implied volatilities (using Black model or Bachelier model) for swaptions and caps/floors vary with strike levels. Among the models allowing to take into account the volatility skew, we mention the SABR (Stochastic Alpha Beta Rho) model which is a *CEV (Constant Elasticity of Variance)* type model with stochastic volatility:

$$\begin{cases} dL_t = \sigma_t L_t^\beta d\tilde{W}_t, \\ d\sigma_t = \alpha \sigma_t d\tilde{W}_t^\sigma, \end{cases}$$

where $d\tilde{W}_t d\tilde{W}_t^\sigma = \rho dt$, $\alpha \geq 0$, and $0 < \beta \leq 1$.

The volatility skew can be produced by both β and ρ .

Exercises

1. Assume we have the following two non-dividend paying asset processes:

$$dX_t = rX_t dt + \sigma_X X_t d\tilde{W}_t \quad \text{and} \quad dY_t = rY_t dt + \sigma_Y Y_t d\tilde{W}_t.$$

Let X_t be a numéraire and $\tilde{W}_t^X$ its corresponding Brownian motion.

- (i) Use the Girsanov Theorem to derive the relation between $\tilde{W}_t^X$ and $\tilde{W}_t$.
- (ii) Show that

$$d\left(\frac{Y_t}{X_t}\right) = (\sigma_Y - \sigma_X) \frac{Y_t}{X_t} d\tilde{W}_t^X.$$

- 2. Let S_t be a non-dividend paying asset and $P_t(T)$ be a zero coupon bond with maturity T . Define $\text{For}_t(T) = S_t/P_t(T)$ as the forward price.

- (i) Show that $\text{For}_t(T) = \tilde{\mathbb{E}}^T[S_T|\mathcal{F}_t]$.
- (ii) Recall that the futures price is defined as $\tilde{\mathbb{E}}[S_T|\mathcal{F}_t]$, a similar structure but under the basic risk-neutral measure. When will the forward and futures prices coincide?

- 3. Let $L_t(T_i)$ be modelled as a positive martingale under $\tilde{\mathbb{P}}^{T_{i+1}}$,

$$dL_t(T_i) = L_t(T_i)\sigma_i(t)d\tilde{W}_t^{T_{i+1}},$$

where $\sigma_i(t)$ is deterministic. Derive the time- t value of a floorlet paying $\Delta_i(K - L_{T_i}(T_i))^+$ at time T_{i+1} .

Chapter 18

Foreign Exchange Modelling



In this chapter, we study the modelling of foreign exchange rate for FX options as well as other derivatives with FX risk exposure. Quanto and composite options are analyzed in detail. Lastly, we discuss the hedging of FX risk for derivatives products.

18.1 Stochastic Model for Foreign Exchange Rate

Assume constant foreign and domestic interest rates r^f and r^d for simplifying the notation in the analysis. From the domestic investor's point of view, foreign currency rate X_t can be considered as an asset paying continuous dividend and $e^{r^f t} X_t$ is similar to a total asset return process. Hence, under the **domestic risk-neutral measure** $\tilde{\mathbb{P}}^d$ (denoted as $\tilde{\mathbb{P}}$ in previous context), the process

$$\frac{e^{r^f t} X_t}{e^{r^d t}} = e^{-(r^d - r^f)t} X_t$$

is a martingale.

Assuming that X_t follows a geometric Brownian motion, its risk-neutral SDE will be

$$dX_t = (r^d - r^f)X_t dt + \sigma_X X_t d\tilde{W}_t^{d,X}, \quad (18.1)$$

which is obtained from (13.5).

This model is known as the *Garman–Kohlhagen model*. The notation $\tilde{W}_t^{d,X}$ is nothing but a Brownian motion under $\tilde{\mathbb{P}}^d$. The superscript X is used to specify that the Brownian motion is about X_t as there will be other $\tilde{\mathbb{P}}^d$ -Brownians in this chapter.

As “domestic” and “foreign” are only relative concepts, the same analysis as the above leads to

$$d\frac{1}{X_t} = (r^f - r^d)\frac{1}{X_t}dt - \sigma_X \frac{1}{X_t}d\tilde{W}_t^{f, \frac{1}{X}},$$

where $\tilde{\mathbb{P}}^f$ is the **foreign risk-neutral measure**.

It can be checked by applying Itô–Doebelin formula to $\frac{1}{X_t}$ in $\tilde{\mathbb{P}}^d$ and the Radon–Nikodym derivative process for the numéraire change from $\tilde{\mathbb{P}}^d$ to $\tilde{\mathbb{P}}^f$ is

$$Z_t = \frac{e^{r^f t} X_t / X_0}{e^{r^d t}}. \quad (18.2)$$

Cross Rate Volatility

Suppose two foreign exchange rates are given as X_t^1 and X_t^2 , say USD/JPY and USD/CHF, under Garman–Kohlhagen model. The cross rate CHF/JPY, which is equal to $X_t = X_t^1 / X_t^2$, satisfies

$$\ln X_t = \ln X_t^1 - \ln X_t^2.$$

From the normality of $\ln X_t^1$ and $\ln X_t^2$ it is easy to find out that the volatility σ of the cross rate is

$$\sigma^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2,$$

where σ_1 and σ_2 are the volatilities for X_t^1 and X_t^2 and ρ is their correlation.

18.2 Pricing Formulas for Vanilla Options and FX Option Duality

Consider simple call or put options on the FX rate X_t under Garman–Kohlhagen model. Their payoffs at maturity T are

$$N^f(X_T - K)^+ \quad \text{or} \quad N^f(K - X_T)^+$$

respectively. Here N^f is the notional amount in *foreign currency* written on the contract, and the payoff is settled in domestic currency. We will drop N^f in the following because it is just a constant. The closed-form solutions can be derived in the same way as the BSM formula for stocks paying continuous dividends. They are also known as the Garman–Kohlhagen formulas:

$$\begin{aligned}
c_{\text{FX}} &= e^{-r^d T} \tilde{\mathbb{E}}^d[(X_T - K)^+ | \mathcal{F}_0] = e^{-r^f T} X_0 N(d_1) - e^{-r^d T} K N(d_2), \\
p_{\text{FX}} &= e^{-r^d T} \tilde{\mathbb{E}}^d[(K - X_T)^+ | \mathcal{F}_0] = e^{-r^d T} K N(-d_2) - e^{-r^f T} X_0 N(-d_1),
\end{aligned}$$

where

$$d_1 = \frac{\ln \frac{X_0}{K} + \left(r^d - r^f + \frac{1}{2}\sigma_X^2\right)T}{\sigma_X \sqrt{T}} \quad \text{and} \quad d_2 = d_1 - \sigma_X \sqrt{T}.$$

FX Option Duality

The call option on Ccy1/Ccy2 is actually the put option on Ccy2/Ccy1. This phenomenon is known as FX Option Duality, which can be verified with the numéraire change (assuming $K \neq 0$):

$$\begin{aligned}
&\tilde{\mathbb{E}}^d[e^{-r^d T} (X_T - K)^+] = \tilde{\mathbb{E}}^d\left[e^{-r^d T} X_T \left(1 - \frac{K}{X_T}\right)^+\right] \\
&= \tilde{\mathbb{E}}^d\left[X_0 e^{-r^f T} \frac{e^{r^f T} X_T / X_0}{e^{r^d T}} \left(1 - \frac{K}{X_T}\right)^+\right] \\
&= K X_0 e^{-r^f T} \tilde{\mathbb{E}}^d\left[Z_T \left(\frac{1}{K} - \frac{1}{X_T}\right)^+\right] \\
&= K X_0 e^{-r^f T} \tilde{\mathbb{E}}^f\left[\left(\frac{1}{K} - \frac{1}{X_T}\right)^+\right]. \tag{18.3}
\end{aligned}$$

18.3 SDE for Foreign Asset Under Domestic Measure

For pricing certain options linked to foreign assets, it is often necessary to use the SDE of the foreign assets under the domestic risk-neutral probability measure $\tilde{\mathbb{P}}^d$. For instance, for options linked to a basket of assets containing foreign assets, the option pricing should be using a *common* risk-neutral measure which is often chosen as the domestic one. In this section, we will make the analysis and provide the SDE followed by a foreign asset under $\tilde{\mathbb{P}}^d$.

Let S_t be a foreign asset process following the below SDE under the foreign risk-neutral probability measure $\tilde{\mathbb{P}}^f$:

$$dS_t = r^f S_t dt + \sigma_S S_t d\tilde{W}_t^{f,S}, \tag{18.4}$$

where $\tilde{W}_t^{f,S}$ is a $\tilde{\mathbb{P}}^f$ -Brownian motion. From Girsanov Theorem, there exists θ such that

$$\tilde{W}_t^{d,S} = \tilde{W}_t^{f,S} + \theta t \quad (18.5)$$

is a $\tilde{\mathbb{P}}^d$ -Brownian. So, it suffices to determine θ for obtaining the SDE of S_t under $\tilde{\mathbb{P}}^d$.

For the foreign exchange process X_t , we assume (18.1) and $d\tilde{W}_t^{d,S}d\tilde{W}_t^{d,X} = \rho dt$.

Proposition 18.1 *Under $\tilde{\mathbb{P}}^d$, the process $Y_t = X_t S_t$, which can be regarded as a domestic asset, is given by*

$$dY_t = Y_t(r^d dt + \sigma_Y d\tilde{W}_t^{d,Y}),$$

where $\sigma_Y = \sqrt{\sigma_X^2 + \sigma_S^2 + 2\rho\sigma_X\sigma_S}$ and $\tilde{W}_t^{d,Y}$ is a $\tilde{\mathbb{P}}^d$ -Brownian.

Proof By Itô's product rule,

$$\begin{aligned} dY_t &= S_t dX_t + X_t dS_t + d\langle X, S \rangle_t \\ &= S_t X_t [(r^d - r^f)dt + \sigma_X d\tilde{W}_t^{d,X} + r^f dt + \sigma_S d\tilde{W}_t^{f,S} + \rho\sigma_S\sigma_X dt] \\ &= Y_t [r^d dt + \sigma_X d\tilde{W}_t^{d,X} + \sigma_S (d\tilde{W}_t^{d,S} - \theta dt) + \rho\sigma_S\sigma_X dt] \\ &= Y_t [(r^d + \rho\sigma_S\sigma_X - \theta\sigma_S)dt + \sigma_X d\tilde{W}_t^{d,X} + \sigma_S d\tilde{W}_t^{d,S}]. \end{aligned}$$

Under $\tilde{\mathbb{P}}^d$, Y_t should earn the domestic interest rate r^d . From the drift term, we obtain

$$r^d + \rho\sigma_S\sigma_X - \theta\sigma_S = r^d. \quad (18.6)$$

Furthermore,

$$\sigma_X \tilde{W}_t^{d,X} + \sigma_S \tilde{W}_t^{d,S} = \sqrt{\sigma_X^2 + \sigma_S^2 + 2\rho\sigma_X\sigma_S} \tilde{W}_t^{d,Y} = \sigma_Y \tilde{W}_t^{d,Y}.$$

Then, we obtain the result. □

Corollary 18.1 *The drift θ is determined by*

$$\theta = \rho\sigma_X. \quad (18.7)$$

This is obtained directly from (18.6).

Now we are ready to give the SDE of foreign asset under $\tilde{\mathbb{P}}^d$.

Proposition 18.2 *Under the domestic risk-neutral measure $\tilde{\mathbb{P}}^d$, the SDE of the foreign asset S_t is*

$$dS_t = (r^f - \rho\sigma_S\sigma_X)S_t dt + \sigma_S S_t d\tilde{W}_t^{d,S}. \quad (18.8)$$

Proof From (18.4), (18.5) and (18.7), we have

$$\begin{aligned} dS_t &= r^f S_t dt + \sigma_S S_t d\tilde{W}_t^{f,S} = r^f S_t dt + \sigma_S S_t d(\tilde{W}_t^{d,S} - \rho\sigma_X dt) \\ &= (r^f - \rho\sigma_S\sigma_X)S_t dt + \sigma_S S_t d\tilde{W}_t^{d,S}. \end{aligned} \quad \square$$

Example 18.1 For options involving X_T and S_T , Monte Carlo method may be considered. The simulation of these two processes under $\tilde{\mathbb{P}}^d$ will be based on Proposition 18.2 and the correlation term embedded. To be specific,

$$\begin{bmatrix} \ln S_t \\ \ln X_t \end{bmatrix} = \begin{bmatrix} \ln S_0 + (r^f - \rho\sigma_S\sigma_X - \frac{1}{2}\sigma_S^2)t \\ \ln X_0 + (r^d - r^f - \frac{1}{2}\sigma_X^2)t \end{bmatrix} + \begin{bmatrix} \sigma_S\sqrt{t} & 0 \\ 0 & \sigma_X\sqrt{t} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{bmatrix} \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \end{bmatrix}$$

where ϵ_1 and ϵ_2 are independently drawn from a standard normal distribution.

18.4 Composite Option

In a composite option, $X_T S_T$ is in the payoff formula. For instance, the composite call option has the payoff $(X_T S_T - K)^+$ at maturity T . Here the strike price K is in domestic currency.

Obviously, the SDE of $Y_t = S_t X_t$ as described in Proposition 18.1 can be used for option pricing. For example,

$$c_{\text{compo}} = e^{-r^d T} \tilde{\mathbb{E}}^d[(X_T S_T - K)^+ | \mathcal{F}_0] = X_0 S_0 N(d_1) - e^{-r^d T} K N(d_2),$$

where

$$d_1 = \frac{\ln \frac{X_0 S_0}{K} + (r^d + \frac{1}{2}\sigma_Y^2)T}{\sigma_Y \sqrt{T}} \quad \text{and} \quad d_2 = d_1 - \sigma_Y \sqrt{T}.$$

One can notice from the composite call option formula that there is no r^f term. Hence there is no foreign interest rate risk in composite options. This is an important feature for making options on the foreign underlying when there is no liquid financial instruments in the market for hedging the foreign interest rate risk.

18.5 Quanto Option

In a Quanto option, the payoff is linked to the relative move of the underlying but paid in domestic currency.

For instance, the quanto call option has the payoff $N^d \times \left(\frac{S_T}{S_0} - K \right)^+$ at maturity. For the pricing of the call option, we make use of the stock price process under $\tilde{\mathbb{P}}^d$ as described in Proposition 18.2. The following assumes $N^d = S_0 = 1$.

$$\begin{aligned}
 c_{\text{quanto}} &= e^{-r^d T} \tilde{\mathbb{E}}^d[(S_T - K)^+ | \mathcal{F}_0] = e^{-r^d T} \tilde{\mathbb{E}}^d[(S_T - K)^+] \\
 &= e^{-r^d T} \tilde{\mathbb{E}}^d[(S_0 e^{(r^f - \rho \sigma_X \sigma_S)T - \frac{1}{2} \sigma_S^2 T + \sigma_S \tilde{W}_T^{d,S}} - K)^+] \\
 &= e^{(-r^d + r^f - \rho \sigma_X \sigma_S)T} \tilde{\mathbb{E}}^d[e^{-(r^f - \rho \sigma_X \sigma_S)T} (S_0 e^{(r^f - \rho \sigma_X \sigma_S)T - \frac{1}{2} \sigma_S^2 T + \sigma_S \tilde{W}_T^{d,S}} - K)^+] \\
 &= e^{(-r^d + r^f - \rho \sigma_X \sigma_S)T} [S_0 N(d_1) - e^{-(r^f - \rho \sigma_X \sigma_S)T} K N(d_2)] \\
 &= e^{-(r^d - r^f)T} (e^{-\rho \sigma_X \sigma_S T} S_0 N(d_1) - e^{-r^f T} K N(d_2)),
 \end{aligned}$$

where

$$d_1 = \frac{\ln \frac{S_0}{K} + (r^f - \rho \sigma_X \sigma_S + \frac{1}{2} \sigma_S^2)T}{\sigma_S \sqrt{T}} \quad \text{and} \quad d_2 = d_1 - \sigma_S \sqrt{T}.$$

18.6 Discussions on Hedging

For understanding the hedging of an option involving FX risk, we may consider that the option is linked to a basket of assets and FX rate(s) is part of the underlying basket. So, the general analysis on basket option should be applicable. Nevertheless, we provide the below specific analysis due to the importance of such issues.

In the presence of FX risk, the hedging of a short position of a derivative $V(t, S_t, X_t)$ is realized through the following self-financing portfolio $\Pi(t, S_t, X_t)$:

- holding Δ_t^S units of the asset (stock) S_t
- holding Δ_t^X units of the foreign currency
- financing the position with domestic interest rate r^d
- investing the portfolio value $\Pi(t, S_t, X_t)$ in money market with rate r^d .

In the following, we simplify the notations by dropping the subscript t . Firstly, the differential of the self-financing portfolio is

$$d\Pi = r^d \Pi dt + \Delta^S d(XS) + \Delta^X dX + r^f \Delta^X X dt - r^d (\Delta^S XS + \Delta^X X) dt,$$

where the fourth term is the carry return from the holding of foreign currency and the last term represents the financing cost in domestic currency. Then

$$\begin{aligned} d\Pi &= r^d \Pi dt + \Delta^S [d(XS) - r^d XS dt] + \Delta^X [dX - (r^d - r^f)X dt] \\ &= r^d \Pi dt + e^{r^d t} \Delta^S d(e^{-r^d t} XS) + e^{(r^d - r^f)t} \Delta^X d(e^{-(r^d - r^f)t} X) \end{aligned}$$

by reverse-engineering the Itô's product rule. Hence, denoting $\widehat{\Pi} = e^{-r^d t} \Pi$ and once again using Itô's product rule, we obtain

$$d\widehat{\Pi} = \Delta^S d(e^{-r^d t} XS) + e^{-r^f t} \Delta^X d(e^{-(r^d - r^f)t} X). \quad (18.9)$$

We observe that the discounted portfolio $\widehat{\Pi}$ is a martingale because the terms on the right hand side are martingales under $\widetilde{\mathbb{P}}^d$. From the risk-neutral pricing analysis, the value of the derivative paying out $V(T, S_T, X_T)$ can be delta-hedged by setting $V(t, S_t, X_t) = \Pi(t, S_t, X_t)$.

To obtain Δ^S and Δ^X , we apply Itô–Doebelin formula to $\widehat{V} = e^{-r^d t} V$ and get

$$d\widehat{V} = \frac{\partial \widehat{V}}{\partial S} dS + \frac{\partial \widehat{V}}{\partial X} dX + [\dots]dt = e^{-r^d t} \left(\frac{\partial V}{\partial S} dS + \frac{\partial V}{\partial X} dX \right) + [\dots]dt, \quad (18.10)$$

where the drift term is not our concern and omitted. From (18.9), we have

$$\begin{aligned} d\widehat{\Pi} &= e^{-r^d t} \Delta^S [d(XS) - r^d XS dt] + e^{-r^d t} \Delta^X [dX - (r^d - r^f)X dt] \\ &= e^{-r^d t} \Delta^S (XdS + SdX) + e^{-r^d t} \Delta^X dX + [\dots]dt, \end{aligned}$$

or,

$$d\widehat{\Pi} = e^{-r^d t} [\Delta^S X dS + (\Delta^X + \Delta^S S) dX] + [\dots]dt. \quad (18.11)$$

From the martingale properties of $\widehat{\Pi}$ and $\widehat{V}$, the dt terms should be zero. Equating the dS and dX terms in (18.10) and (18.11), we get:

$$\Delta_t^S = \frac{1}{X_t} \frac{\partial V}{\partial S_t} \quad \text{and} \quad \Delta_t^X = \frac{\partial V}{\partial X_t} - \frac{S_t}{X_t} \frac{\partial V}{\partial S_t}.$$

Exercises

1. Assume that the FX rate X_t follows the Garman–Kohlhagen model (18.1) and the foreign asset process S_t follows (18.8). Show that for $t < T$,

$$\widetilde{\mathbb{E}}^d[X_T S_T | \mathcal{F}_t] = \widetilde{\mathbb{E}}^d[X_T | \mathcal{F}_t] \widetilde{\mathbb{E}}^d[S_T | \mathcal{F}_t] e^{\rho \sigma_X \sigma_S (T-t)}.$$

- 2. Consider a domestic call option with payoff $X_T(S_T - K)^+$ at maturity T (here strike price K in foreign currency). Show that its time-0 price equals to the time-0 price of the foreign stock call option multiplied by spot rate X_0 .
- 3. Consider a EUR-denominated stock S^1 , and a USD-denominated stock S^2 . EUR/USD represents the exchange rate (units of USD per EUR).

Market data			
Name	S^1	S^2	EUR/USD
Volatility	30%	25%	10%
EUR rate (continuous)		0.2%	
USD rate (continuous)		0.4%	
Correlation $S^1 \sim S^2$		60%	
Correlation $S^1 \sim \text{EUR/USD}$		-20%	

S^1 and S^2 follow Black–Scholes–Merton model. Assume the below independent standard normal variates in the following tables.

	Z^1	Z^2
Trial 1	1.02	0.12
Trial 2	-0.50	-0.10
Trial 3	0.6	0.70

Construct 3 paths in Monte Carlo simulation to price the following 1Y tenor, USD10m notional amount (N) options:

- (i) Call Option: $N \times \max \left\{ \frac{S_T^2}{S_0^2} - 1, 0 \right\}$
- (ii) Quanto Call Option: $N \times \max \left\{ \frac{S_T^1}{S_0^1} - 1, 0 \right\}$
- (iii) Quanto Worst-of Call Option: $N \times \max \left\{ \min \left\{ \frac{S_T^1}{S_0^1}, \frac{S_T^2}{S_0^2} \right\} - 1, 0 \right\}$.

Part III

Extensions to Financial Modelling

Chapter 19

Local, Stochastic Volatility Models, Static Hedging and Variance Swap



19.1 Introduction

In this chapter, we first introduce the Kolmogorov equations that describe the transition probabilities, i.e., how the probability density changes from one time to another. Next, we talk about asset price models that allow more flexibility in the volatility term. At last, we discuss how to replicate a European-style option with call and put options of different strikes. From there, a special contract called the variance swap will be covered, together with its valuation and hedging. We refer to [8] for the in-depth analysis on hedging for local volatility model and stochastic volatility models.

Kolmogorov Equations

Consider the general diffusion process

$$dX_s = \theta(s, X_s)ds + \phi(s, X_s)dW_s.$$

Let $p(t, T; x, y)$ ($0 \leq t \leq T$) be the transition probability density of X_s . The forward case assumes t and x “fixed” with $X_t = x$ and $p(t, T; x, y)$ is the probability density function with state variable y and a future time T . Conversely, the backward case assumes T and y “fixed” with $X_T = y$ and shows the behaviour of $p(t, T; x, y)$ backwards in time t with state variable x .

The *Kolmogorov forward equation*, also known as the *Fokker–Planck equation*, governs the forward transition probability with t and x fixed.

Theorem 19.1 (Kolmogorov Forward Equation)

$$\frac{\partial}{\partial T} p(t, T; x, y) + \frac{\partial}{\partial y} (\theta(T, y) p(t, T; x, y)) - \frac{1}{2} \frac{\partial^2}{\partial y^2} (\phi^2(T, y) p(t, T; x, y)) = 0.$$

Proof Let $h(y)$ be any C^2 function with compact support (such a function vanishes outside a finite interval).¹ From Itô–Doeblin formula,

$$\begin{aligned} h(X_T) &= h(X_t) + \int_t^T \theta(s, X_s) h'(X_s) ds + \int_t^T \phi(s, X_s) h'(X_s) dW_s \\ &\quad + \frac{1}{2} \int_t^T \phi^2(s, X_s) h''(X_s) ds. \end{aligned}$$

Taking expectation with initial condition $X_t = x$, we have

$$\mathbb{E}_t^x[h(X_T)] = h(x) + \int_t^T \mathbb{E}_t^x \left[\theta(s, X_s) h'(X_s) + \frac{1}{2} \phi^2(s, X_s) h''(X_s) \right] ds.$$

On the other hand, the transition probability is the density that gives

$$\mathbb{E}_t^x[h(X_T)] = \int_{-\infty}^{\infty} h(y) p(t, T; x, y) dy.$$

Equating the above leads to

$$\begin{aligned} &\int_{-\infty}^{\infty} h(y) p(t, T; x, y) dy \\ &= h(x) + \int_t^T \left[\int_{-\infty}^{\infty} \left(\theta(s, y) h'(y) + \frac{1}{2} \phi^2(s, y) h''(y) \right) p(t, s; x, y) dy \right] ds. \end{aligned}$$

From integration by parts and the fact that h and h' vanish at ∞ :

$$\begin{aligned} \int_{-\infty}^{\infty} h(y) p(t, T; x, y) dy &= h(x) - \int_t^T \left[\int_{-\infty}^{\infty} h(y) \frac{\partial}{\partial y} (\theta(s, y) p(t, s; x, y)) dy \right] ds \\ &\quad + \frac{1}{2} \int_t^T \left[\int_{-\infty}^{\infty} h(y) \frac{\partial^2}{\partial y^2} (\phi^2(s, y) p(t, s; x, y)) dy \right] ds. \end{aligned}$$

¹For example, the following function $h(y)$ vanishes for $y \notin (0, b)$: $h(y) = \begin{cases} -\frac{1}{1 - \left(\frac{2y-b}{b}\right)^2}, & 0 < y < b, \\ 0, & \text{otherwise.} \end{cases}$

Differentiating w.r.t. T , we obtain that

$$\int_{-\infty}^{\infty} h(y) \left[\frac{\partial}{\partial T} p(t, T; x, y) + \frac{\partial}{\partial y} (\theta(T, y) p(t, T; x, y)) - \frac{1}{2} \frac{\partial^2}{\partial y^2} (\phi^2(T, y) p(t, T; x, y)) \right] dy = 0.$$

From the arbitrariness of $h(y)$, we get

$$\frac{\partial}{\partial T} p(t, T; x, y) + \frac{\partial}{\partial y} (\theta(T, y) p(t, T; x, y)) - \frac{1}{2} \frac{\partial^2}{\partial y^2} (\phi^2(T, y) p(t, T; x, y)) = 0.$$

□

The following is the backward case with future time T and state y fixed.

Theorem 19.2 (Kolmogorov Backward Equation)

$$\frac{\partial}{\partial t} p(t, T; x, y) + \theta(t, x) \frac{\partial}{\partial x} p(t, T; x, y) + \frac{1}{2} \phi^2(t, x) \frac{\partial^2}{\partial x^2} p(t, T; x, y) = 0.$$

Proof Let $h(y)$ be any C^2 function with compact support. From Feynman–Kac formula, the transition function $g(t, x) = \mathbb{E}_t^x[h(X_T)] = \int_{-\infty}^{\infty} h(y) p(t, T; x, y) dy$ satisfies

$$\frac{\partial g}{\partial t}(t, x) + \theta(t, x) \frac{\partial g}{\partial x}(t, x) + \frac{1}{2} \phi^2(t, x) \frac{\partial^2 g}{\partial x^2}(t, x) = 0,$$

and hence

$$\int_{-\infty}^{\infty} h(y) \left[\frac{\partial p}{\partial t}(t, T; x, y) + \theta(t, x) \frac{\partial p}{\partial x}(t, T; x, y) + \frac{1}{2} \phi^2(t, x) \frac{\partial^2 p}{\partial x^2}(t, T; x, y) \right] dy = 0.$$

From the arbitrariness of $h(y)$, we obtain

$$\frac{\partial p}{\partial t}(t, T; x, y) + \theta(t, x) \frac{\partial p}{\partial x}(t, T; x, y) + \frac{1}{2} \phi^2(t, x) \frac{\partial^2 p}{\partial x^2}(t, T; x, y) = 0. \quad \square$$

Theorem 19.3 (Breedén–Litzenberger formula) *Let $c(T, K)$ be the European call price with strike K and maturity T . Then*

$$\frac{\partial^2 c}{\partial K^2}(T, K) = e^{-rT} f(T, K),$$

where $f(T, K)$ is the forward transition probability density, i.e., $f(t, x) = p(0, t; S_0, x)$.

Proof From the risk-neutral representation of the call option price,

$$c(T, K) = e^{-rT} \int_0^\infty (x - K)^+ f(T, x) dx$$

and hence

$$\begin{aligned} c(T, K) &= e^{-rT} \int_0^\infty (x - K)^+ f(T, x) dx = e^{-rT} \int_K^\infty (x - K) f(T, x) dx \\ &= e^{-rT} \int_K^\infty f(T, x) \int_K^x dy dx = e^{-rT} \int_K^\infty \int_y^\infty f(T, x) dx dy. \end{aligned}$$

Differentiating w.r.t. K gives

$$\frac{\partial c}{\partial K}(T, K) = -e^{-rT} \int_K^\infty f(T, x) dx,$$

and differentiating again leads to

$$\frac{\partial^2 c}{\partial K^2}(T, K) = e^{-rT} f(T, K). \quad \square$$

19.2 Local Volatility Model

Compared to BSM model, the *local volatility model* takes into account the skew and kurtosis of the distribution of the underlying asset by applying $\sigma(t, S_t)$ instead of a constant σ . Local volatility is widely used for pricing equity derivatives. The SDE for local volatility model is

$$dS_t = (r - q)S_t dt + \sigma(t, S_t)S_t d\tilde{W}_t,$$

under risk-neutral probability measure $\tilde{\mathbb{P}}$, and q is the sum of continuous dividend yield and repo rate.

Theorem 19.4 (Dupire's formula) *For the local volatility model,*

$$\sigma^2(T, K) = 2 \frac{\partial_T c(T, K) + qc(T, K) + (r - q)K \partial_K c(T, K)}{K^2 \partial_{KK}^2 c(T, K)}.$$

Proof For notational convenience, we change the notations T and K to t and x respectively, and hence from Breeden–Litzenberger formula

$$f(t, x) = e^{rt} \frac{\partial^2 c}{\partial x^2}(t, x).$$

Direct differentiation with respect to t gives

$$\frac{\partial f}{\partial t} = r e^{rt} \frac{\partial^2 c}{\partial x^2} + e^{rt} \frac{\partial}{\partial t} \left(\frac{\partial^2 c}{\partial x^2} \right).$$

To further ease the notation, let σ^2 denote $\sigma^2(t, x)$ for the time being. The transition density f satisfies the Kolmogorov forward equation:

$$\frac{\partial f}{\partial t} + \frac{\partial}{\partial x}((r - q)xf(t, x)) - \frac{1}{2} \frac{\partial^2}{\partial x^2}(x^2 \sigma^2 f(t, x)) = 0$$

or

$$\frac{\partial f}{\partial t} = \frac{1}{2} \frac{\partial^2}{\partial x^2}(x^2 \sigma^2 f) - (r - q) \left(x \frac{\partial f}{\partial x} + f \right).$$

Replacing f , we get

$$r e^{rt} \frac{\partial^2 c}{\partial x^2} + e^{rt} \frac{\partial}{\partial t} \left(\frac{\partial^2 c}{\partial x^2} \right) = e^{rt} \frac{1}{2} \frac{\partial^2}{\partial x^2} \left(x^2 \sigma^2 \frac{\partial^2 c}{\partial x^2} \right) - (r - q) \left(x e^{rt} \frac{\partial^3 c}{\partial x^3} + e^{rt} \frac{\partial^2 c}{\partial x^2} \right).$$

Simplifying,

$$r \frac{\partial^2 c}{\partial x^2} + \frac{\partial^2}{\partial x^2} \left(\frac{\partial c}{\partial t} \right) = \frac{1}{2} \frac{\partial^2}{\partial x^2} \left(x^2 \sigma^2 \frac{\partial^2 c}{\partial x^2} \right) - (r - q) \frac{\partial}{\partial x} \left(x \frac{\partial c}{\partial x} - c \right).$$

Integrating twice w.r.t x ,

$$rc + \frac{\partial c}{\partial t} = \frac{1}{2} x^2 \sigma^2(t, x) \frac{\partial^2 c}{\partial x^2} - (r - q) \left(x \frac{\partial c}{\partial x} - c \right) + \alpha_t x + \beta_t.$$

Assuming that all the terms except $\alpha_t x + \beta_t$ in the above go to zero as x goes to infinity (as the strike price goes to infinity), we get $\lim_{x \rightarrow \infty} (\alpha_t x + \beta_t) = 0, \forall t$. Hence $\alpha_t = \beta_t = 0$. We obtain

$$\frac{1}{2}x^2\sigma^2(t, x)\frac{\partial^2 c}{\partial x^2}(t, x) = \frac{\partial c}{\partial t}(t, x) + qc + (r - q)x\frac{\partial c}{\partial x}(t, x),$$

which is essentially Dupire's formula. $\square$

Local Volatility in Terms of Implied Volatility and its Derivatives

The Dupire's formula requires the numerical calculation of the derivatives of call prices. Actually, Dupire's formula can be transformed into a function involving the calculation with implied volatilities instead of the calculation with call option prices [36, 82, 83]. The calculation with implied volatilities has the advantage of being more stable.

For calibrating the local volatilities surface with the implied volatilities, it is convenient to work on an alternative new process defined as $\widehat{S}_t = \frac{S_t}{F_t}$ where $F_t = S_0 e^{(r-q)t}$. It is easy to see that the SDE for $\widehat{S}_t$ is

$$d\widehat{S}_t = \hat{\sigma}(t, \widehat{S}_t)\widehat{S}_t d\widetilde{W}_t,$$

where $\hat{\sigma}(t, \widehat{S}_t) = \sigma(t, F_t \widehat{S}_t)$ and σ is the local volatility for the process S_t . Let c denote the call option with payoff $\max\{0, S_T - K\}$ at maturity T , and $\hat{c}$ the call option with payoff $\frac{1}{F_T} \max\{0, S_T - K\} \equiv \max\{0, \widehat{S}_T - \widehat{K}\}$ where $\widehat{K} = \frac{K}{F_T}$ is the "moneyness" of the option. The time-0 prices of c and $\hat{c}$ are then related by $c(T, K) = F_T e^{-rT} \hat{c}(T, \widehat{K})$. The Dupire's formula with $\hat{c}$ is

$$\hat{\sigma}^2(T, \widehat{K}) = \frac{2\partial_T \hat{c}(T, \widehat{K})}{\widehat{K}^2 \partial_{\widehat{K}}^2 \hat{c}(T, \widehat{K})}.$$

In the following, we use t and x to represent T and $\widehat{K}$ respectively for brevity. There exists an implied volatility function $\Sigma(t, x)$ in terms of maturity and moneyness such that the call price $\hat{c}(t, x)$ satisfies the Black–Scholes formula,

$$\hat{c}(t, x) = N(d_1(t, x)) - xN(d_2(t, x)),$$

where $d_1(t, x) = \frac{-\ln x + \frac{1}{2}\Sigma(t, x)^2 t}{\Sigma(t, x)\sqrt{t}}$ and $d_2(t, x) = d_1(t, x) - \Sigma(t, x)\sqrt{t}$. Note that $\Sigma(t, x)$ equals to the implied volatility parameterized by strike K .

The call price $\hat{c}$ can be regarded as a function of t , x , and $\Sigma(t, x)$. Then, the Dupire's formula can be revised as $\hat{\sigma}^2(t, x) = 2 \frac{d\hat{c}}{dt} \bigg/ x^2 \frac{d^2 \hat{c}}{dx^2}$, where the operator d stands for total differentiation. The total derivative w.r.t. t is given by

$$\frac{d\hat{c}}{dt} = \frac{\partial \hat{c}}{\partial t} + \frac{\partial \hat{c}}{\partial \Sigma} \frac{\partial \Sigma}{\partial t}$$

and the second-order total derivative w.r.t. x is

$$\begin{aligned} \frac{d^2 \hat{c}}{dx^2} &= \frac{d}{dx} \left(\frac{\partial \hat{c}}{\partial x} + \frac{\partial \hat{c}}{\partial \Sigma} \frac{\partial \Sigma}{\partial x} \right) = \frac{d}{dx} \left(\frac{\partial \hat{c}}{\partial x} \right) + \frac{d}{dx} \left(\frac{\partial \hat{c}}{\partial \Sigma} \right) \frac{\partial \Sigma}{\partial x} + \frac{\partial \hat{c}}{\partial \Sigma} \frac{\partial^2 \Sigma}{\partial x^2} \\ &= \frac{\partial^2 \hat{c}}{\partial x^2} + \frac{\partial^2 \hat{c}}{\partial x \partial \Sigma} \frac{\partial \Sigma}{\partial x} + \frac{\partial^2 \hat{c}}{\partial \Sigma \partial x} \frac{\partial \Sigma}{\partial x} + \frac{\partial^2 \hat{c}}{\partial \Sigma^2} \left(\frac{\partial \Sigma}{\partial x} \right)^2 + \frac{\partial \hat{c}}{\partial \Sigma} \frac{\partial^2 \Sigma}{\partial x^2} \\ &= \frac{\partial^2 \hat{c}}{\partial x^2} + 2 \frac{\partial^2 \hat{c}}{\partial x \partial \Sigma} \frac{\partial \Sigma}{\partial x} + \frac{\partial^2 \hat{c}}{\partial \Sigma^2} \left(\frac{\partial \Sigma}{\partial x} \right)^2 + \frac{\partial \hat{c}}{\partial \Sigma} \frac{\partial^2 \Sigma}{\partial x^2}. \end{aligned}$$

The partial derivatives of $\hat{c}$ can be obtained from the option formula $\hat{c}(t, x) = N(d_1) - xN(d_2)$, e.g.,

$$\frac{\partial \hat{c}}{\partial t} = N'(d_1) \left(\frac{\Sigma}{2\sqrt{t}} + \frac{\partial d_2}{\partial t} \right) - xN'(d_2) \frac{\partial d_2}{\partial t} = N'(d_1) \frac{\Sigma}{2\sqrt{t}},$$

where $N'(d_1) = xN'(d_2)$ (see Exercise 2 in Chap. 11) is used. Similarly

$$\begin{aligned} \frac{\partial \hat{c}}{\partial \Sigma} &= N'(d_1) \left(\sqrt{t} + \frac{\partial d_2}{\partial \Sigma} \right) - xN'(d_2) \frac{\partial d_2}{\partial \Sigma} = N'(d_1) \sqrt{t}, \\ \frac{\partial \hat{c}}{\partial x} &= N'(d_1) \frac{\partial d_1}{\partial x} - N(d_2) - xN'(d_2) \frac{\partial d_2}{\partial x} = -N(d_2). \end{aligned}$$

And the second-order derivatives are

$$\begin{aligned} \frac{\partial^2 \hat{c}}{\partial \Sigma^2} &= N''(d_1) \frac{\partial d_1}{\partial \Sigma} \sqrt{t} = -d_1 N'(d_1) \frac{\partial d_1}{\partial \Sigma} \sqrt{t} = \frac{d_1 d_2 N'(d_1) \sqrt{t}}{\Sigma}, \\ \frac{\partial^2 \hat{c}}{\partial x \partial \Sigma} &= -N'(d_2) \frac{\partial d_2}{\partial \Sigma} = -\frac{N'(d_1)}{x} \left(-\frac{d_1}{\Sigma} \right) = \frac{d_1 N'(d_1)}{x \Sigma}, \\ \frac{\partial^2 \hat{c}}{\partial x^2} &= -N'(d_2) \frac{\partial d_2}{\partial x} = -\frac{N'(d_1)}{x} \left(-\frac{1}{x \Sigma \sqrt{t}} \right) = \frac{N'(d_1)}{x^2 \Sigma \sqrt{t}}. \end{aligned}$$

Replacing the partial derivatives of $\hat{c}$ in $\frac{d\hat{c}}{dt}$ and $\frac{d^2 \hat{c}}{dx^2}$, we have

$$\begin{aligned} \frac{d\hat{c}}{dt} &= N'(d_1) \frac{\Sigma}{2\sqrt{t}} + N'(d_1) \sqrt{t} \frac{\partial \Sigma}{\partial t} = \frac{N'(d_1)}{2\Sigma\sqrt{t}} \left(\Sigma^2 + 2\Sigma t \frac{\partial \Sigma}{\partial t} \right), \\ \frac{d^2 \hat{c}}{dx^2} &= \frac{N'(d_1)}{x^2 \Sigma \sqrt{t}} \left[1 + 2xd_1 \sqrt{t} \frac{\partial \Sigma}{\partial x} + x^2 d_1 d_2 t \left(\frac{\partial \Sigma}{\partial x} \right)^2 + x^2 \Sigma t \frac{\partial^2 \Sigma}{\partial x^2} \right]. \end{aligned}$$

Hence the calculation formula using the implied volatilities is

$$\hat{\sigma}^2(t, x) = \frac{2 \frac{d\hat{c}}{dt}}{x^2 \frac{d^2\hat{c}}{dx^2}} = \frac{\Sigma^2 + 2\Sigma t \frac{\partial \Sigma}{\partial t}}{1 + 2x d_1 \sqrt{t} \frac{\partial \Sigma}{\partial x} + x^2 d_1 d_2 t \left(\frac{\partial \Sigma}{\partial x} \right)^2 + x^2 \Sigma t \frac{\partial^2 \Sigma}{\partial x^2}}.$$

Practical Implementation

The continuum of implied volatilities is needed to calculate the local volatilities. However, only a set of discrete values is quoted on the market. The following procedures may be considered for the implementation:

1. For each considered maturity, a smooth, twice differentiable function with few parameters (say, a quadratic function) is calibrated for representing the implied volatilities of all strikes. The volatility on both tails can be capped at reasonable levels
2. The local volatilities are then calculated according to Dupire's formula with the parametric function obtained from above.

Because of the asymmetric form of the volatility smile, the parametric function for the implied volatilities may be composed of two segments, like a quadratic function calibrated for the left side and another quadratic function calibrated for the right side, with eventually the forward as the pivoting point.

Remark 19.1 Although the local volatility model is the market standard model for equity options, it is not suitable for pricing products with forward skew sensitivities (e.g. the forward starting call-spread or the cliquet options). Suppose the maturity is T , the option's starting time is t , and the pricing time is 0. The local volatility surface is known to become flatter as time goes by. Therefore, the skew will be progressively lost for the time period $[t, T]$ with increasing t .

19.3 Stochastic Volatility Model

Stochastic volatility models take into account the skew/kurtosis of the underlying's distribution. They show better dynamics compared to the local volatility model, which makes them suitable for pricing products with forward skew sensitivities like cliquet options.

We outline here the *Heston model* which is one of the most popular stochastic volatility models [39]. The SDEs of Heston model are

$$\begin{cases} dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t, \\ dv_t = \kappa(\theta - v_t)dt + \sigma \sqrt{v_t} dX_t, \end{cases}$$

where W_t and X_t are correlated Brownians with $d\langle W, X \rangle_t = \rho dt$, κ is the mean reversion speed for the variance v_t , θ is the mean reversion level, and σ is the volatility of volatility (often read as vol of vol).

Denote $U(S, v, t)$ for the option value. The PDE for U can be obtained directly by applying the following risk premia to produce a “risk-neutral” measure

$$\begin{cases} d\tilde{W}_t = dW_t + \frac{\mu - r}{\sqrt{v_t}} dt, \\ d\tilde{X}_t = dX_t + \frac{\lambda}{\sigma \sqrt{v_t}} dt, \end{cases}$$

where $\lambda(S, v, t)$ is called market price of volatility risk. Applying Itô–Doebelin formula,

$$\begin{aligned} d(e^{-rt} U_t) = e^{-rt} & \left[\left(\frac{\partial U}{\partial t} + \frac{1}{2} v S^2 \frac{\partial^2 U}{\partial S^2} + \rho \sigma v S \frac{\partial^2 U}{\partial S \partial v} + \frac{1}{2} \sigma^2 v \frac{\partial^2 U}{\partial v^2} - rU \right) dt \right. \\ & \left. + \frac{\partial U}{\partial S} dS_t + \frac{\partial U}{\partial v} dv_t \right]. \end{aligned}$$

Replacing dS_t , dv_t and equating $[\dots] dt$ to 0, we obtain the **PDE for Heston model**:

$$\begin{aligned} & \frac{\partial U}{\partial t} + \frac{1}{2} v S^2 \frac{\partial^2 U}{\partial S^2} + \rho \sigma v S \frac{\partial^2 U}{\partial S \partial v} + \frac{1}{2} \sigma^2 v \frac{\partial^2 U}{\partial v^2} + r S \frac{\partial U}{\partial S} - rU \\ & = [-\kappa(\theta - v) + \lambda(S, v, t)] \frac{\partial U}{\partial v}. \end{aligned}$$

All the parameters in the model are calibrated by the market prices of (vanilla) options.

The meaning of market price of volatility risk $\lambda(S, v, t)$ can be shown in the below. Suppose that we only delta-hedge the option U with the underlying asset: $\Pi_t = U_t - \Delta_t S_t$, then

$$\begin{aligned} d\Pi_t = & \left(\frac{\partial U}{\partial t} + \frac{1}{2} v S^2 \frac{\partial^2 U}{\partial S^2} + \rho \sigma v S \frac{\partial^2 U}{\partial S \partial v} + \frac{1}{2} \sigma^2 v \frac{\partial^2 U}{\partial v^2} \right) dt \\ & + \left(\frac{\partial U}{\partial S} - \Delta \right) dS_t + \frac{\partial U}{\partial v} dv_t. \end{aligned}$$

Hence, by setting $\Delta_t = \frac{\partial U}{\partial S}$ and combining with Heston PDE, we get

$$d\Pi_t - r\Pi_t dt = \frac{\partial U}{\partial v} dv_t + [-\kappa(\theta - v) + \lambda] \frac{\partial U}{\partial v} dt = \frac{\partial U}{\partial v} [\lambda(S, v, t) dt + \sigma \sqrt{v_t} dX_t],$$

where we have replaced dv_t under the actual measure in the Heston model.

We observe that $\lambda(S, v, t)$ represents the extra return on variance risk through the term $\frac{\partial U}{\partial v}$.

Closed-Form Solution for European Options Under Heston Model

We outline Heston's method for the closed-form solution of European call option price without going into the details. It is based on Fourier transform of the transition probability. Scott, L. generalized the method to jump-diffusion model with stochastic volatility [77].

Denote $\tau = T - t$.

1. Transform the pricing problem into two probability calculations

$$\begin{aligned} U_t &= e^{-r\tau} \widetilde{\mathbb{E}}_t^{S,v} [(S_T - K)^+] \\ &= e^{-r\tau} \widetilde{\mathbb{E}}_t^{S,v} [S_T \mathbf{1}_{\{S_T > K\}}] - K e^{-r\tau} \widetilde{\mathbb{E}}_t^{S,v} [\mathbf{1}_{\{S_T > K\}}] \\ &= S \mathbb{Q}\{S_T > K\} - K e^{-r\tau} \widetilde{\mathbb{P}}\{S_T > K\} \quad (\text{already presented in BSM model}) \\ &= e^x P_1(x, v, \tau) - K e^{-r\tau} P_2(x, v, \tau) \end{aligned}$$

where $x = \ln S$.

2. P_1 and P_2 are the probabilities (conditional expectations) for the option to finish in the money when x_t and v_t follow the stochastic processes in $\mathbb{Q}$ and $\mathbb{P}$. As Feynman–Kac Proposition is applicable to conditional expectations that are Markovian, we can obtain the PDEs followed by P_1 and P_2 respectively.
3. Determine the characteristic functions of x at T for P_j ($j = 1, 2$):

$$f_j(x, v; \psi) = \widetilde{\mathbb{E}}_j[e^{i\psi x}].$$

As $f_j(x, v; \psi)$ ($j = 1, 2$) is a conditional expectation and Markovian, Feynman–Kac Proposition can be applied to obtain the PDE followed by $f_j(x, v; \psi)$. It can be proved that $f_j(x, v; \psi)$ takes the form of $e^{C_j(\tau; \psi) + D_j(\tau; \psi)v + i\psi x}$. For obtaining $C_j(\tau; \psi)$ and $D_j(\tau; \psi)$, we apply $f_j(x, v; \psi)$ to the PDE, which results in two ODEs (Riccati equations) that can be solved analytically.

4. Calculate P_j with the characteristic functions $f_j(x, v; \psi)$,

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left(\frac{e^{-i\psi \ln(K)} f_j(x, v; \psi)}{i\psi} \right) d\psi.$$

Numerical Approaches for Closed-Form Solution

The main task is to evaluate P_1 and P_2 . The integrand is calculated using f_j at each discretized ψ and the integral can be approximated by numerical integration approach like the trapezoidal rule. The two integrals in P_1 and P_2 can also be combined to reduce the computational cost.

An alternative approach is to apply the Carr–Madan formula [15], in which the fast Fourier transform is carried out for the numerical integration and a damping factor for the option price is used to handle the singularity near $\psi = 0$.

19.4 Static Hedging for European Style Options

Static hedging, which is opposite to dynamic hedging, does not adjust the hedging portfolio once it has been set up. In this section, an important result is first presented, i.e., any European style option can be hedged statically by calls and puts provided that there are calls and puts with any strike price in the market.

Proposition 19.1 *Any European style option can be (theoretically) hedged with a static portfolio of calls and puts:*

$$\begin{aligned} & \tilde{\mathbb{E}}_t^{S_t} [e^{-r(T-t)} h(S_T)] \\ &= e^{-r(T-t)} (h(\lambda) + h'(\lambda)(F - \lambda)) + \int_{\lambda}^{\infty} h''(K) c(K) dK + \int_0^{\lambda} h''(K) p(K) dK, \end{aligned} \quad (19.1)$$

where h is a twice differentiable payoff function, S_t is the asset price at time t , $F = \tilde{\mathbb{E}}_t^{S_t} [S_T]$ is the forward price of the asset, r is the constant interest rate, $c(K)$ and $p(K)$ are European call and put prices of strike K , and λ is any fixed level of the asset price. In particular when $\lambda = F$, the formula is reduced as

$$e^{-r(T-t)} h(F) + \int_F^{\infty} h''(K) c(K) dK + \int_0^F h''(K) p(K) dK.$$

Proof We first prove a general decomposition of a twice differentiable function h :

$$\begin{aligned} h(S) &= h(\lambda) + \mathbf{1}_{\{S > \lambda\}} \int_{\lambda}^S h'_u du - \mathbf{1}_{\{S < \lambda\}} \int_S^{\lambda} h'_u du \\ &= h(\lambda) + \mathbf{1}_{\{S > \lambda\}} \int_{\lambda}^S \left(h'(\lambda) + \int_{\lambda}^u h''(v) dv \right) du \\ &\quad - \mathbf{1}_{\{S < \lambda\}} \int_S^{\lambda} \left(h'(\lambda) - \int_u^{\lambda} h''(v) dv \right) du \end{aligned}$$

$$\begin{aligned}
&= h(\lambda) + h'(\lambda)(S - \lambda) + \mathbf{1}_{\{S > \lambda\}} \int_{\lambda}^S \int_{\lambda}^u h''(v) dv du + \mathbf{1}_{\{S < \lambda\}} \int_S^{\lambda} \int_u^{\lambda} h''(v) dv du \\
&= h(\lambda) + h'(\lambda)(S - \lambda) + \mathbf{1}_{\{S > \lambda\}} \int_{\lambda}^S \int_v^S h''(v) du dv + \mathbf{1}_{\{S < \lambda\}} \int_S^{\lambda} \int_S^v h''(v) du dv \\
&= h(\lambda) + h'(\lambda)(S - \lambda) + \mathbf{1}_{\{S > \lambda\}} \int_{\lambda}^S h''(v)(S - v) dv + \mathbf{1}_{\{S < \lambda\}} \int_S^{\lambda} h''(v)(v - S) dv \\
&= h(\lambda) + h'(\lambda)(S - \lambda) + \int_{\lambda}^{\infty} h''(v)(S - v)^+ dv + \int_0^{\lambda} h''(v)(v - S)^+ dv.
\end{aligned}$$

Now, setting $S = S_T$, denoting $v = K$, multiplying both sides by discount factor $e^{-r(T-t)}$, and taking conditional expectation, we obtain the static hedging for a generalized European option:

$$\begin{aligned}
&\tilde{\mathbb{E}}_t^{S_t} [e^{-r(T-t)} h(S_T)] \\
&= e^{-r(T-t)} [h(\lambda) + h'(\lambda)(F - \lambda)] + \int_{\lambda}^{\infty} h''(K) c(K) dK + \int_0^{\lambda} h''(K) p(K) dK. \quad \square
\end{aligned}$$

From the above formula, in order to replicate the contingent claim on the left hand side, we need a continuously changing value K to complete the two integrals on the right hand side. In practical implementation, approximations will be applied because the calls and puts in the market only have a discrete set of strike prices.

A useful special case is the payoff of a log contract, i.e., $h(S_T) = \ln(S_T)$. The static hedging representation is given by

$$\tilde{\mathbb{E}}_t^{S_t} [\ln(S_T)] = \ln(\lambda) + \frac{F - \lambda}{\lambda} - e^{r(T-t)} \left[\int_{\lambda}^{\infty} \frac{c(K)}{K^2} dK + \int_0^{\lambda} \frac{p(K)}{K^2} dK \right]. \quad (19.2)$$

The equation (19.2) will be used for pricing and hedging the variance swap.

19.5 Variance Swap

A variance swap is an agreement to exchange the realized variance V during a fixed maturity T with n business days for a pre-specified variance strike K_{var} in the contract. The payoff at maturity is given by

$$N_{\text{var}}(V - K_{\text{var}}),$$

where N_{var} is a notional amount which converts the units from variance to denominated currency on the contract. One way to calculate V is the estimator for sample variance

$$V = \frac{252}{n} \sum_{i=1}^n \left(\ln \frac{S_i}{S_{i-1}} \right)^2,$$

in which the sample mean is negligible and selected as 0, and the number 252 is the annualization factor representing the number of business days in a year.

The contract often contains volatility linked quantities that people are more familiar with (but do not confuse it with volatility swap). Suppose a volatility swap has volatility strike K_σ , notional amount N_{vega} , and payoff $N_{\text{vega}}(\sqrt{V} - K_\sigma)$. The term N_{vega} is also known as the vega notional because the volatility swap payoff would be N_{vega} if the $\sqrt{V} - K_\sigma$ is 1 volatility point. Then we have the following two relations between variance and volatility swaps:

$$K_\sigma = \sqrt{K_{\text{var}}} \quad \text{and} \quad N_{\text{vega}} = N_{\text{var}} \times 2K_\sigma.$$

Given N_{vega} , one can determine the notional number N_{var} according to the above relation. This is done in order to make volatility and variance swaps comparable. Note that variance swaps usually have a cap in the payoff formula (especially for single stocks) for controlling the risk to an acceptable level (stock price could become zero).

Valuation of Variance Swap

Proposition 19.2 *Assume that the underlying follows the model*

$$dS_t = rS_t dt + \sigma_t S_t d\tilde{W}_t,$$

where there is no dividend but σ_t can be stochastic itself. We have the valuation of variance swap which is independent of the modelling for the volatility σ_t :

$$\text{Price of Variance Swap} = N_{\text{var}}(\tilde{\mathbb{E}}[V] - K_{\text{var}})e^{-rT},$$

where $V = \frac{1}{T} \int_0^T \sigma_t^2 dt$,

$$\tilde{\mathbb{E}}[V] = \frac{2}{T} \left[\ln \left(\frac{F}{\lambda} \right) + \frac{\lambda - F}{\lambda} + e^{rT} \int_\lambda^\infty \frac{c(K)}{K^2} dK + e^{rT} \int_0^\lambda \frac{p(K)}{K^2} dK \right].$$

In other words, the variance swap can be theoretically priced and hedged by a series of calls and puts.

Proof Applying Itô–Doebelin formula to $\ln(S_t)$, we get

$$d \ln(S_t) = \frac{dS_t}{S_t} - \frac{1}{2} \frac{(dS_t)^2}{S_t^2} = \frac{dS_t}{S_t} - \frac{1}{2} \sigma_t^2 dt.$$

After integration it becomes

$$\ln(S_T) = \ln(S_0) + \int_0^T \frac{dS_t}{S_t} - \frac{1}{2} \int_0^T \sigma_t^2 dt.$$

Then the total variance is equal to

$$\begin{aligned} V &= \frac{1}{T} \int_0^T \sigma_t^2 dt = \frac{2}{T} \left[\ln(S_0) + \int_0^T \frac{dS_t}{S_t} - \ln(S_T) \right] \\ &= \frac{2}{T} \left[\ln(S_0) + rT + \int_0^T \sigma_t d\tilde{W}_t - \ln(S_T) \right] = \frac{2}{T} \left[\ln(F) + \int_0^T \sigma_t d\tilde{W}_t - \ln(S_T) \right] \end{aligned}$$

since the forward value is $F = S_0 e^{rT}$ with constant r . Taking the (conditional) risk-neutral expectation on both sides and making the substitution (19.2), we obtain

$$\begin{aligned} \tilde{\mathbb{E}}[V] &= \frac{2}{T} [\ln(F) - \tilde{\mathbb{E}}[\ln(S_T)]] \\ &= \frac{2}{T} \left[\ln\left(\frac{F}{\lambda}\right) + \frac{\lambda - F}{\lambda} + e^{rT} \int_{\lambda}^{\infty} \frac{c(K)}{K^2} dK + e^{rT} \int_0^{\lambda} \frac{p(K)}{K^2} dK \right]. \quad \square \end{aligned}$$

Usually the value λ is chosen to be $F = S_0 e^{rT}$ such that the formula is greatly simplified as

$$\tilde{\mathbb{E}}[V] = \frac{2e^{rT}}{T} \left[\int_F^{\infty} \frac{c(K)}{K^2} dK + \int_0^F \frac{p(K)}{K^2} dK \right].$$

Exercises

1. (i) Derive the Kolmogorov forward equation for the Brownian motion W_t .
(ii) Verify that the following transition probability function

$$p(t, T; x, y) = \frac{1}{\sqrt{2\pi(T-t)}} e^{-\frac{(y-x)^2}{2(T-t)}}$$

satisfies the equation in (i).

2. Use the Breeden–Litzenberger formula and perform $\frac{\partial^2}{\partial K^2}$ on the Black–Scholes call price c (assuming zero dividend) to recover the log-normal probability density function of the underlying at maturity T , i.e.,

$$f(T, K) = \frac{1}{K\sigma\sqrt{2\pi T}} e^{-\frac{[\ln K - \ln S_0 - (r - \frac{1}{2}\sigma^2)T]^2}{2\sigma^2 T}},$$

where K is also the dummy variable of the underlying.

3. Assume that the underlying follows the classic Black–Scholes–Merton model

$$dS_t = rS_t dt + \sigma S_t d\tilde{W}_t.$$

Let

$$f(S, \sigma) = \frac{2e^{rT}}{T} \left[\int_{Se^{rT}}^{\infty} \frac{c(S, K, \sigma)}{K^2} dK + \int_0^{Se^{rT}} \frac{p(S, K, \sigma)}{K^2} dK \right]$$

where S is the spot price and σ is the implied volatility of $c(S, K, \sigma)$ and $p(S, K, \sigma)$, the Black–Scholes call and put prices respectively with strike price K and time to maturity T .

- (i) Show that $\frac{\partial c}{\partial \sigma} = \frac{\partial p}{\partial \sigma}$ and then derive $\frac{\partial f}{\partial \sigma}$.
- (ii) Show that $\frac{\partial^2 f}{\partial S \partial \sigma} = 0$.

Chapter 20

Jump-Diffusion Models



Market prices of financial assets often show jumps caused by unpredictable events or news. The market closing-opening is also a source of price jumps. The pure Brownian motion based diffusion models do not admit large asset price move in a short period of time. Adding jumps to diffusion can show skewed distributions with fat tail which are difficult to produce from BSM model.

In this chapter, we give an introduction to jump processes and jump-diffusion models. For detailed mathematical analysis, we refer to [79].

20.1 Compound Poisson Process

The jump size for a Poisson process N_t is always 1. The *Compound Poisson process* G_t extends the Poisson process by taking *i.i.d.* random variables for the jump sizes:

$$G_t = \sum_{i=1}^{N_t} v_i, \quad t \geq 0,$$

where v_i represents the random variable for the i -th jump with $\mathbb{E}[v_i] = \eta, \forall i$. Let λ denote the intensity for the Poisson process. Here are some properties of compound Poisson process:

- (i) Mean: $\mathbb{E}[G_t] = \eta\lambda t$
- (ii) Martingale property: $G_t - \eta\lambda t = G_t - \mathbb{E}[G_t]$ is a martingale
- (iii) Independent and stationary increments: $G_t - G_s \stackrel{(d)}{\Longleftrightarrow} G_{t-s}$.

20.1.1 Simulating a Poisson Process and Compound Poisson Process

The key of simulating a Poisson process N_t lies in the arrival times T_i , where their differences are i.i.d. exponentially distributed random variables. Consider generating N_T , $T > 0$ starting from $N_0 = 0$. The procedure is as follows (assuming $T_0 = 0$):

1. generate a uniform variable $u_i \in \mathcal{U}[0, 1]$
2. generate an exponential variable $\tau_i = -\frac{1}{\lambda} \ln u_i$ and calculate an arrival time by $T_i = T_{i-1} + \tau_i$ (if $T_i > T$ then stop the procedure)
3. let $N_t = i$ for $[T_i, T_{i+1})$.

Here the inverse sampling is performed and two facts are used: firstly u_i and $1 - u_i$ are equivalent, and the inverse c.d.f. of exponential distribution is explicit.

Simulating G_t is similarly done once the distribution of jump sizes v_i is specified.

20.2 Stochastic Calculus for Jump-Diffusion Processes

Let $\mathcal{F}_t$ be the natural filtration generated by W_t and G_t . Next we introduce the stochastic calculus for jump processes. A *pure jump process*, denoted by J_t , has finite jumps in any time interval and is constant between jumps. Also it is assumed that no jump takes place at time 0 and $J_0 = 0$ is the initial value. The Poisson process N_t and compound Poisson process G_t are pure jump processes. For pure jump processes, we have

$$dJ_t = \delta J_t = J_t - J_{t-} = (J_t - J_{t-})dN_t.$$

We define a *jump-diffusion process* as a right continuous process as follows:

$$X_t = X_0 + \int_0^t \theta_u du + \int_0^t \phi_u dW_u + J_t,$$

where θ_u and ϕ_u are adapted processes. Note that $X_t^c \equiv X_0 + \int_0^t \theta_u du + \int_0^t \phi_u dW_u$ is the continuous Itô process in the earlier context. The addition of J_t introduces a finite number of jumps to X_t^c in finite time. Since $\delta X_t = \delta J_t$, the jump process can also be written as

$$X_t = X_t^c + J_t = X_t^c + \sum_{0 < u \leq t} \delta J_u = X_t^c + \sum_{0 < u \leq t} \delta X_u.$$

In differential form, we have $dX_t = dX_t^c + dJ_t$.

Previously it is shown that a stochastic integration with continuous martingale integrator and predictable integrand gives a martingale. This property can be extended when the integrator is a martingale with jumps. See [66] for more details. The calculus of jump-diffusion processes is given below. The proofs can be found in [79].

Proposition 20.1 *The cross variation for two jump-diffusion processes $X_t = X_t^c + J_t^X$ and $Y_t = Y_t^c + J_t^Y$ is given by*

$$\langle X, Y \rangle_t = \langle X^c, Y^c \rangle_t + \langle J^X, J^Y \rangle_t = \int_0^t \phi_u^X \phi_u^Y du + \sum_{0 < u \leq t} \delta J_u^X \delta J_u^Y,$$

or in the differential form $d\langle X, Y \rangle_t = d\langle X^c, Y^c \rangle_t + d\langle J^X, J^Y \rangle_t$. In particular, the cross variation of any continuous process X_t^c and pure jump process J_t is zero, i.e.,

$$\langle X^c, J \rangle_t = 0. \quad (20.1)$$

Theorem 20.1 (One-Dimensional Itô–Doebelin Formula for Jump-Diffusion Process) *Let $f(u, x)$ be a sufficiently smooth function of real variables u and x . Suppose X_u is a jump process. Then for $u < t$,*

$$\begin{aligned} f(t, X_t) &= f(0, X_0) + \int_0^t f_u(u, X_u) du + \int_0^t f_x(u, X_u) dX_u^c \\ &\quad + \frac{1}{2} \int_0^t f_{xx}(u, X_u) d\langle X^c, X^c \rangle_u + \sum_{0 < u \leq t} [f(u, X_u) - f(u, X_{u-})], \end{aligned}$$

where $f_u(u, X_u) = \frac{\partial f}{\partial u}(u, X_u)$, $f_x(u, X_u) = \frac{\partial f}{\partial x}(u, X_u)$ and $f_{xx}(u, X_u) = \frac{\partial^2 f}{\partial x^2}(u, X_u)$.

The difference from the ordinary Itô's formula lies in the last term.

Theorem 20.2 (Two-Dimensional Itô–Doebelin Formula for Jump-Diffusion Process) *Let X_u and Y_u be jump processes and $f(u, x, y)$ whose derivatives appearing in the below are defined and continuous. Then*

$$\begin{aligned} f(t, X_t, Y_t) &= f(0, X_0, Y_0) + \int_0^t f_u(u, X_u, Y_u) du + \int_0^t f_x(u, X_u, Y_u) dX_u^c \\ &\quad + \int_0^t f_y(u, X_u, Y_u) dY_u^c + \frac{1}{2} \int_0^t f_{xx}(u, X_u, Y_u) d\langle X^c, X^c \rangle_u \\ &\quad + \frac{1}{2} \int_0^t f_{yy}(u, X_u, Y_u) d\langle Y^c, Y^c \rangle_u + \int_0^t f_{xy}(u, X_u, Y_u) d\langle X^c, Y^c \rangle_u \end{aligned}$$

$$\begin{aligned}
& + \int_0^t f_{xy}(u, X_u, Y_u) d\langle X^c, Y^c \rangle_u + \frac{1}{2} \int_0^t f_{yy}(u, X_u, Y_u) d\langle Y^c, Y^c \rangle_u \\
& + \sum_{0 < u \leq t} (f(u, X_u, Y_u) - f(u, X_{u-}, Y_{u-})).
\end{aligned}$$

The two dimensional Itô's formula leads to the following Itô's product rule.

Corollary 20.1 (Itô's Product Rule for Jump Processes) *Let X_t and Y_t be two jump processes. Then,*

$$X_t Y_t = X_0 Y_0 + \int_0^t Y_u dX_u^c + \int_0^t X_u dY_u^c + \langle X^c, Y^c \rangle_t + \sum_{0 < u \leq t} (X_u Y_u - X_{u-} Y_{u-}),$$

which can be equivalently written as integration by parts:

$$X_t Y_t = X_0 Y_0 + \int_0^t Y_{u-} dX_u + \int_0^t X_{u-} dY_u + \langle X, Y \rangle_t.$$

20.3 Jump-Diffusion Asset Pricing Model

The jump-diffusion model assumes that, under the original probability measure $\mathbb{P}$, the SDE for the asset is

$$dS_t = \mu S_t dt + \sigma S_t dW_t + S_{t-} d(G_t - \eta \lambda t).$$

The diffusion part is the same as Black–Scholes model. The additional jump part is given by the *compensated* compound Poisson process $G_t - \eta \lambda t$, which is a $\mathbb{P}$ -martingale. The jump v_i 's are *i.i.d.* random variables and we assume that they have continuous distribution with density $f(y)$ satisfying $v_i \geq -1$ (as S_t becomes 0 for $v_i = -1$). For example, in Merton's model [57], $v_i = X_i - 1$ where X_i follows log-normal distribution. The term η is the expected jump size with $\eta = \mathbb{E}[v_i] = \int_{-1}^{\infty} y f(y) dy$.

Define $\tilde{\mathbb{P}}$ with a Radon–Nikodym derivative Z_t :

$$\left. \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} \right|_{\mathcal{F}_t} = Z_t = e^{-\theta W_t - \frac{1}{2} \theta^2 t}, \text{ with } \theta = \frac{\mu - r}{\sigma}.$$

Then, under $\tilde{\mathbb{P}}$, $\tilde{W}_t = W_t + \theta t$ is a Brownian and the discounted asset price is a martingale. The measure $\tilde{\mathbb{P}}$ implies

$$dS_t = rS_t dt + \sigma S_t d\tilde{W}_t + S_{t-} d(G_t - \eta \lambda t). \quad (20.2)$$

Note that $\tilde{\mathbb{P}}$ is not a unique risk-neutral measure. The case presented here states that the real-world and risk-neutral distribution for the jumps are identical, thus keeping the intensity and expected jump size, λ and η unchanged.

Proposition 20.2 *The solution to (20.2) under $\tilde{\mathbb{P}}$ is*

$$S_t = S_0 e^{\sigma \tilde{W}_t + (r - \eta \lambda - \frac{1}{2} \sigma^2)t} \prod_{i=1}^{N_t} (v_i + 1).$$

Proof Consider the pure jump process $Z_t = \prod_{i=1}^{N_t} (v_i + 1) = \prod_{0 < s \leq t} (1 + \delta G_s)$ which satisfies $dZ_t = Z_{t-} dG_t$. Define a continuous process $S_t^c = S_0 e^{\sigma \tilde{W}_t + (r - \eta \lambda - \frac{1}{2} \sigma^2)t}$, which is a geometric Brownian motion solving

$$dS_t^c = (r - \eta \lambda) S_t^c dt + \sigma S_t^c d\tilde{W}_t.$$

From (20.1), $\langle S^c, Z \rangle_t = 0$. Applying Itô's product rule to $S_t = S_t^c Z_t$,

$$\begin{aligned} S_t &= S_t^c Z_t = S_0 + \int_0^t S_u^c dZ_u + \int_0^t Z_{u-} dS_u^c \\ &= S_0 + \int_0^t S_u^c Z_{u-} dG_u + (r - \eta \lambda) \int_0^t Z_{u-} S_u^c du + \sigma \int_0^t Z_{u-} S_u^c d\tilde{W}_u \\ &= S_0 + r \int_0^t S_u du + \sigma \int_0^t S_u d\tilde{W}_u + \int_0^t S_{u-} d(G_u - \eta \lambda u). \end{aligned}$$

The differential form of the above is the SDE (20.2). □

The solution S_t has the jump size of $\delta S_t = S_t - S_{t-} = S_{t-} \delta G_t$, which leads to $(S_t - S_{t-}) dN_t = S_{t-} dG_t$. This will be used later for discussion of hedging.

20.4 European Call or Put Option Pricing

Consider a European call or put option with strike price K and maturity T .

Theorem 20.3 *Assume the stock price follows the jump-diffusion model (20.2). Then, the risk-neutral price of a European call or put*

$$V_t = \tilde{\mathbb{E}}[e^{-r(T-t)} h(S_T) | \mathcal{F}_t], \quad h(S_T) = (S_T - K)^+ \text{ or } (K - S_T)^+,$$

is given by

$$V_t = \sum_{n=0}^{\infty} e^{-\lambda\tau} \frac{(\lambda\tau)^n}{n!} \tilde{\mathbb{E}}[BS(\tau, \bar{S}(v_i, n))],$$

where the time to maturity $T - t$ is denoted as τ ,

$$\bar{S}(v_i, n) = S_t e^{-\eta\lambda\tau} \prod_{i=1}^n (v_i + 1),$$

and $BS(\tau, \bar{S})$ is the BSM call or put option formula with τ as tenor and $\bar{S}$ as spot.

Proof

$$\begin{aligned} V_t &= \tilde{\mathbb{E}}[e^{-r\tau} h(S_T) | \mathcal{F}_t] \\ &= \tilde{\mathbb{E}} \left[e^{-r\tau} h \left(S_t e^{\sigma \tilde{W}_\tau + (r - \eta\lambda - \frac{1}{2}\sigma^2)\tau} \prod_{i=1}^{N_\tau} (v_i + 1) \right) \right], \end{aligned}$$

where the properties of independent and stationary increments are used. Using the law of total expectation with the probability of n jumps,

$$\begin{aligned} V_t &= \tilde{\mathbb{E}} \left[\tilde{\mathbb{E}} \left[e^{-r\tau} h \left(S_t e^{\sigma \tilde{W}_\tau + (r - \eta\lambda - \frac{1}{2}\sigma^2)\tau} \prod_{i=1}^{N_\tau} (v_i + 1) \right) \middle| N_\tau = n \right] \right] \\ &= \sum_{n=0}^{\infty} e^{-\lambda\tau} \frac{(\lambda\tau)^n}{n!} \tilde{\mathbb{E}} \left[e^{-r\tau} h \left(S_t e^{\sigma \tilde{W}_\tau + (r - \eta\lambda - \frac{1}{2}\sigma^2)\tau} \prod_{i=1}^n (v_i + 1) \right) \right] \\ &= \sum_{n=0}^{\infty} e^{-\lambda\tau} \frac{(\lambda\tau)^n}{n!} \tilde{\mathbb{E}}[e^{-r\tau} h(\bar{S}(v_i, n)) e^{\sigma \tilde{W}_\tau + (r - \frac{1}{2}\sigma^2)\tau}] \\ &= \sum_{n=0}^{\infty} e^{-\lambda\tau} \frac{(\lambda\tau)^n}{n!} \tilde{\mathbb{E}}[\tilde{\mathbb{E}}[e^{-r\tau} h(\bar{S}(v_i, n)) e^{\sigma \tilde{W}_\tau + (r - \frac{1}{2}\sigma^2)\tau} | \mathcal{F}_t]] \\ &= \sum_{n=0}^{\infty} e^{-\lambda\tau} \frac{(\lambda\tau)^n}{n!} \tilde{\mathbb{E}}[BS(\tau, \bar{S}(v_i, n))]. \end{aligned}$$

□

One can see that the pricing formula is a summation of Black–Scholes prices multiplied by the corresponding probability of jumps, or in other words, an expectation over all jump occurrences. The pricing formula can be further simplified as a closed-form solution under Merton's lognormal jump-diffusion model; see [57] for reference.

20.5 PIDE for a European Style Option

Consider a T -maturity European option price $V(t, S_t)$ with underlying S_t at time t . We have the following theorem for the governing equation of V .

Theorem 20.4 *Assume the underlying stock follows the jump-diffusion model (20.2). The price $V(t, S_t)$ for a European style option paying out $V(T, S_T)$ at maturity T satisfies the PIDE (partial integro-differential equation):*

$$\frac{\partial V}{\partial t} + (r - \eta\lambda)S_t \frac{\partial V}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2} + \lambda\{\tilde{\mathbb{E}}[V(t, (\nu + 1)S_{t-})] - V(t, S_{t-})\} = rV,$$

where $\tilde{\mathbb{E}}[V(t, (\nu + 1)S_{t-})] = \int_{-1}^{\infty} V(t, (z + 1)S_{t-})\tilde{f}(z)dz$ if jump density $\tilde{f}$ is known.

Proof Denote the process $A_t = -rV + \frac{\partial V}{\partial t} + (r - \eta\lambda)S_t \frac{\partial V}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2}$. Applying Itô's formula and product rule, we have

$$e^{-rt}V_t = V_0 + \int_0^t e^{-ru}A_u du + \int_0^t e^{-ru}\sigma S_u \frac{\partial V}{\partial S_u} d\tilde{W}_u + \sum_{0 < u \leq t} e^{-ru}[V(u, S_u) - V(u, S_{u-})].$$

If the jump happens at time u , then $S_u = (\nu + 1)S_{u-}$. Therefore, the last term is

$$\sum_{0 < u \leq t} e^{-ru}[V(u, S_u) - V(u, S_{u-})] = \int_0^t e^{-ru}[V(u, (\nu + 1)S_{u-}) - V(u, S_{u-})]dN_u.$$

Splitting the integrator into $dN_u = d(N_u - \lambda u) + \lambda du$,

$$\begin{aligned} e^{-rt}V_t &= V_0 + \int_0^t e^{-ru}A_u du + \int_0^t e^{-ru}\sigma S_u \frac{\partial V}{\partial S_u} d\tilde{W}_u \\ &\quad + \int_0^t e^{-ru}[V(u, (\nu + 1)S_{u-}) - V(u, S_{u-})]d(N_u - \lambda u) \\ &\quad + \int_0^t e^{-ru}[V(u, (\nu + 1)S_{u-}) - V(u, S_{u-})]\lambda du. \end{aligned}$$

Taking expectation, terms with $d\tilde{W}_u$ and $d(N_u - \lambda u)$ become 0. Then,

$$\tilde{\mathbb{E}}[e^{-rt}V_t|\mathcal{F}_0] = V_0 + \int_0^t e^{-ru}A_u du + \int_0^t e^{-ru}\{\tilde{\mathbb{E}}[V(u, (\nu + 1)S_{u-})] - V(u, S_{u-})\}\lambda du.$$

Because $e^{-rt} V_t$ should be a martingale under $\tilde{\mathbb{P}}$, the du term should be 0, which proves the result. $\square$

Applying the PIDE, we obtain

$$\begin{aligned} e^{-rt} V_t = V_0 - \int_0^t e^{-ru} \lambda \{ \tilde{\mathbb{E}}[V(u, (v+1)S_{u-})] - V(u, S_{u-}) \} du \\ + \int_0^t e^{-ru} \sigma S_u \frac{\partial V}{\partial S_u} d\tilde{W}_u + \int_0^t e^{-ru} [V(u, S_u) - V(u, S_{u-})] dN_u, \end{aligned} \quad (20.3)$$

which will be later used in the context of hedging.

20.6 Discussion on Hedging Under Jump-Diffusion Model

We conduct the analysis on a short call position. The delta hedging portfolio for the call $c(t, S_t)$ is Π_t which satisfies

$$d\Pi_t = r\Pi_t dt + \Delta_t(dS_t - rS_t dt),$$

with $\Pi_0 = c(0, S_0)$. Then,

$$\begin{aligned} d(e^{-rt} \Pi_t) &= e^{-rt} (-r\Pi_t dt + d\Pi_t) = e^{-rt} (\Delta_t dS_t - r\Delta_t S_t dt) \\ &= e^{-rt} [\Delta_t \sigma S_t d\tilde{W}_t + \Delta_t S_{t-} d(G_t - \eta\lambda t)] \\ &= e^{-rt} [\Delta_t \sigma S_t d\tilde{W}_t + \Delta_t (S_t - S_{t-}) dN_t - \Delta_t S_{t-} \eta\lambda dt], \end{aligned}$$

where we use $dG_t = \frac{S_t - S_{t-}}{S_{t-}} dN_t$. From (20.3), we have

$$\begin{aligned} d(e^{-rt} c(t, S_t)) &= e^{-rt} \left\{ -\lambda \left[\int_{-1}^{\infty} [c(t, (z+1)S_{t-}) - c(t, S_{t-})] \tilde{f}(z) dz \right] dt \right. \\ &\quad \left. + \sigma S_t \frac{\partial c}{\partial S_t}(t, S_{t-}) d\tilde{W}_t + [c(t, S_t) - c(t, S_{t-})] dN_t \right\}. \end{aligned}$$

Applying $\Delta_t = \frac{\partial c}{\partial S_t}(t, S_{t-})$ and $\eta = \int_{-1}^{\infty} z \tilde{f}(z) dz$, the hedging result for time interval dt is

$$\begin{aligned}
& d(e^{-rt}\Pi_t - e^{-rt}c(t, S_t)) \\
&= e^{-rt} \left\{ \lambda \left[\int_{-1}^{\infty} [c(t, (z+1)S_{t-}) - c(t, S_{t-}) - zS_{t-} \frac{\partial c}{\partial S_t}(t, S_{t-})] \tilde{f}(z) dz \right] dt \right. \\
&\quad \left. - [c(t, S_t) - c(t, S_{t-}) - \frac{\partial c}{\partial S_t}(t, S_{t-})(S_t - S_{t-})] dN_t \right\}.
\end{aligned}$$

As the function $c(t, x)$ is strictly convex in x , we have

$$c(t, x_2) - c(t, x_1) > (x_2 - x_1) \frac{\partial c}{\partial x}(t, x_1), \quad \text{for all } x_2 > x_1 \geq 0.$$

Therefore,

- between jumps, the hedging portfolio outperforms the option
 (in this case, dN_t is 0 and $c(t, (z+1)S_{t-}) - c(t, S_{t-}) > zS_{t-} \frac{\partial c}{\partial S_t}(t, S_{t-})$ so that $d(e^{-rt}\Pi_t - e^{-rt}c(t, S_t)) > 0$)
- at jump times, the option outperforms the hedging portfolio
 (because $c(t, S_t) - c(t, S_{t-}) > \frac{\partial c}{\partial S_t}(t, S_{t-})(S_t - S_{t-})$)
- the expected difference is zero.

Exercises

1. Let X_t and Y_t be two jump processes. Show that

$$\begin{aligned}
\int_0^t Y_u - dX_u + \int_0^t X_u - dY_u + \langle X, Y \rangle_t &= \int_0^t Y_u dX_u^c + \int_0^t X_u dY_u^c + \langle X^c, Y^c \rangle_t \\
&\quad + \sum_{0 < u \leq t} (X_u Y_u - X_{u-} Y_{u-}).
\end{aligned}$$

2. Consider a stock modelled by Merton's default model [58], which is a special jump diffusion model with one jump to zero value, i.e., $\eta = -1$ in (20.2) and also $\nu = -1$.
 - (i) Derive the governing PIDE for a European call option with strike K and maturity T on this stock.
 - (ii) Obtain its time- t price.

3. Consider a stock modelled by Merton's jump-diffusion model [57] where $v_i = X_i - 1$ and X_i follows i.i.d. log-normal distribution, i.e., $\log(X_i) \sim \mathcal{N}(\mu_J, \sigma_J^2)$.

- (i) Show that $1 + \eta = e^{\mu_J + \frac{1}{2}\sigma_J^2}$.
(ii) Show that the European call/put option price is given as

$$\sum_{n=0}^{\infty} e^{-\bar{\lambda}\tau} \frac{(\bar{\lambda}\tau)^n}{n!} BS(\tau, S_t, \sigma_n, r_n)$$

where $\bar{\lambda} = \lambda(1 + \eta)$, BS is the classic Black–Scholes call/put prices with proxy volatility $\sigma_n = \sqrt{\sigma^2 + \frac{n\sigma_J^2}{\tau}}$ and proxy risk-free rate $r_n = r - \lambda\eta + n \frac{\log(1+\eta)}{\tau}$.

Chapter 21

Interest Rate Term Structure Modelling



In Part II Chap. 17, we have presented the pricing for Vanilla interest rate products which do not require the modelling of interest rate term structure. However, structures with the features such as path-dependency or callability usually require the modelling of term structure for pricing and risk management. In this chapter, we first present the concepts and the relationships between zero coupon bond, short rate and forward rate which are essential for interest rate term structure modelling. Then we introduce the Heath–Jarrow–Morton framework with the no-arbitrage condition, which will give birth to different short rate models. Lastly we study the Libor market model which is more and more used in practice.

As the interest rate term structure modelling is fairly complicated, we limit the presentation to the basic concepts and important results. For a comprehensive study and practical implementation, we refer to [2–4].

21.1 Continuous-Time Modelling of Interest Rate

There are different models for describing the dynamics of the term structure of interest rates. The definition of interest rates used in a model (known as “state-variables”) is often different from the interest rates implied from the contracts of financial instruments. Actually, the interest rates related measures can be constructed from the state variables. The principal component analysis (PCA) on historical data of interest rates confirmed that the dynamics of the term structure are mainly explained by 3 factors: the first factor for parallel shift, the second for yield curve pivoting and the third for curvature change. For describing the dynamics of the term structure, models with more than 2 factors would give satisfaction if efficient computation algorithms can be implemented.

21.1.1 Zero Coupon Bond

In continuous-time modelling, the zero coupon bonds (also known as discount bond), denoted by $P_t(T)$, are considered as the base tradable assets. A zero coupon bond $P_t(T)$ represents the time- t value of an instrument paying out \$1 value at maturity T . The probability $\tilde{\mathbb{P}}$ under which the discounted zero coupon bond prices are martingales is called the risk-neutral probability measure which allows dynamic hedging of derivatives instruments.

The zero coupon bonds for different maturities can be extracted from market traded instruments as explained in Part I, Chap. 2, Section “Yield Curve Construction”. Conversely, any interest rate instrument in the market can be represented by zero coupon bonds. For pricing interest rate derivatives, we usually need zero coupon bonds and stochastic discount factors (e.g. $e^{-\int_0^t r_u du}$) under $\tilde{\mathbb{P}}$ with term structure models.

Example 21.1 The forward USD 3M Libor rate at time T can be represented by

$$\left[\frac{P_t(T)}{P_t(T+3M)} - 1 \right] / \delta_{(T, T+3M)},$$

where the year fraction $\delta_{(T, T+3M)}$ is calculated with the relevant day count convention.

21.1.2 Short Rate r_t

The short rate r_t is the stochastic instantaneous interest rate previously used to define the money market account numéraire

$$M_t = e^{\int_0^t r_u du}$$

and M_t represents the accrued value of \$1 at time t .

Relationship between the market price of zero coupon bond $P_t(T)$ and the short rate r_t is as follows. From the risk-neutral pricing formula $P_t(T) = M_t \tilde{\mathbb{E}}[\frac{P_T(T)}{M_T} | \mathcal{F}_t]$ and $P_T(T) = 1$, we obtain

$$P_t(T) = \tilde{\mathbb{E}}[e^{-\int_t^T r_u du} | \mathcal{F}_t]. \quad (21.1)$$

21.1.3 Forward Rate $f_t(T)$

The *time- t discrete forward rate* for the time interval $(T, T + \delta T)$ is defined by the following $\mathcal{F}_t$ -measurable stochastic process $f_t(T, T + \delta T)$:

$$f_t(T, T + \delta T) = -\frac{\ln P_t(T + \delta T) - \ln P_t(T)}{\delta T},$$

which satisfies

$$\exp(-f_t(T, T + \delta T)\delta T) = \frac{P_t(T + \delta T)}{P_t(T)},$$

i.e., the interest rate at time t to borrow between T and $T + \delta T$. Then, the *forward rate* is defined as

$$f_t(T) = \lim_{\delta T \rightarrow 0} -\frac{\ln P_t(T + \delta T) - \ln P_t(T)}{\delta T} = -\frac{\partial \ln P_t(T)}{\partial T}.$$

Relationship between the market price of zero coupon bond $P_t(T)$ and the forward rate is

$$P_t(T) = e^{-\int_t^T f_t(u) du}.$$

Note that when $t = T$, the forward rate is reduced as the short rate $r_t = f_t(t)$.

21.2 Heath–Jarrow–Morton Framework

The Heath–Jarrow–Morton (HJM) framework analyzes the properties of no-arbitrage interest rate term structure models. It models the dynamics of forward rate as

$$f_t(T) = f_0(T) + \int_0^t \alpha_u(T) du + \int_0^t \sigma_u(T) \cdot d\mathbf{W}_u,$$

where $\sigma_t(T)$ and the Brownian motion $\mathbf{W}_t$ are n -dimensional processes and an inner product is carried out for the last term. The particular specification of the terms $\alpha_t(T)$ and $\sigma_t(T)$ will lead to different modelling choices for $f_t(T)$. An assumption of no-arbitrage will relate $\alpha_t(T)$ and $\sigma_t(T)$ and give the HJM no-arbitrage model for $f_t(T)$.

Proposition 21.1 (HJM Non-Arbitrage Model)

$$\begin{cases} f_t(T) = f_0(T) + \int_0^t \tilde{\alpha}_u(T) du + \int_0^t \sigma_u(T) \cdot d\tilde{\mathbf{W}}_u, \\ \tilde{\alpha}_t(T) = \sigma_t(T) \cdot \int_t^T \sigma_t(u) du \quad (\text{known as “HJM no-arbitrage condition”}), \end{cases}$$

where $\tilde{\mathbf{W}}_t$ is an n -dimensional vector of Brownians under $\tilde{\mathbb{P}}$, the money market account risk-neutral probability measure.

Proof Under interest rate modelling, the zero coupon bonds are tradable assets with strictly positive values. Hence, under $\tilde{\mathbb{P}}$, any zero coupon bond deflated by the numéraire $M_t = e^{\int_0^t r_u du}$ is a martingale, which implies

$$\frac{dP_t(T)}{P_t(T)} = r_t dt - \sigma_t^P(T) \cdot d\tilde{\mathbf{W}}_t.$$

Here the minus sign before σ^P is for notational convenience because it can be absorbed into the Brownian motion. Then by Itô's lemma,

$$d \ln P_t(T) = \left[r_t - \frac{1}{2} \sigma_t^P(T) \cdot \sigma_t^P(T) \right] dt - \sigma_t^P(T) \cdot d\tilde{\mathbf{W}}_t$$

or

$$\ln P_t(T) = \ln P_0(T) + \int_0^t \left[r_u - \frac{1}{2} \sigma_u^P(T) \cdot \sigma_u^P(T) \right] du - \int_0^t \sigma_u^P(T) \cdot d\tilde{\mathbf{W}}_u.$$

From $f_t(T) = -\frac{\partial \ln P_t(T)}{\partial T}$, we obtain

$$df_t(T) = \frac{\partial \sigma_t^P(T)}{\partial T} \cdot \sigma_t^P(T) dt + \frac{\partial \sigma_t^P(T)}{\partial T} \cdot d\tilde{\mathbf{W}}_t.$$

Defining $\sigma_t(T) = \frac{\partial \sigma_t^P(T)}{\partial T}$, we have

$$df_t(T) = \left(\sigma_t(T) \cdot \int_t^T \sigma_t(u) du \right) dt + \sigma_t(T) \cdot d\tilde{\mathbf{W}}_t$$

since $\int_t^T \sigma_t(u) du = \sigma_t^P(u)|_t^T$ and $\sigma_t^P(t) = 0$. □

We note that $f_0(T)$ can be obtained from the market quoted zero coupon prices $P_0(T)$ with $f_0(T) = -\frac{\partial \ln P_0(T)}{\partial T}$. It means that at time-0, HJM model is consistent with the market price for the interest rate instruments which can be composed of zero coupon bonds. Therefore, HJM model is arbitrage-free by construction.

From the no-arbitrage condition, we see that $\sigma_t(T)$ specifies the entire model. A particular implementation of HJM model consists of a specification of $\sigma_t(T)$. In particular, the risk-neutral SDE for $P_t(T)$ is also specified by $\sigma_t(T)$ as

$$\frac{dP_t(T)}{P_t(T)} = r_t dt - \int_t^T \sigma_t(u) du \cdot d\tilde{\mathbf{W}}_t. \quad (21.2)$$

The problem of the general form of HJM model is that, for obtaining the zero coupon bonds between (t_1, t_2) , a continuum of $f_t(s)$, $(t_1 \leq s \leq t_2)$ is required. In the practical implementation, the reduction of the dimensions is realized through the following approaches:

- applying particular specification of $\sigma_t(T)$ for obtaining lower dimensional Markovian models (cf. short rate model);
- modelling the discrete forward rate $f_t(t_1, t_2)$ directly (cf. LMM model in Sect. 21.4).

21.2.1 No-Arbitrage Short Rate Model

Under $\tilde{\mathbb{P}}$, the modelling of short rate r_t often requires a few factors for describing the dynamics of the whole term structure. Any no-arbitrage short rate model should satisfy the HJM model as

$$r_t = f_t(t) = f_0(t) + \int_0^t \tilde{\alpha}_u(t) du + \int_0^t \sigma_u(t) \cdot d\tilde{\mathbf{W}}_u. \quad (21.3)$$

It is straightforward to get the mean and variance of r_t , when σ is deterministic:

$$\tilde{\mathbb{E}}[r_t] = f_0(t) + \int_0^t \tilde{\alpha}_u(t) du \quad \text{and} \quad \tilde{\text{Var}}(r_t) = \int_0^t \sigma_u(t) \cdot \sigma_u(t) du.$$

21.2.2 Markovian Characterization for Short Rate Models

Performing differential¹ on (21.3), we obtain the SDE for r_t ,

$$dr_t = df_t(t) = f'_0(t)dt + \tilde{\alpha}_t(t)dt + \int_0^t \frac{\partial \tilde{\alpha}_u(t)}{\partial t} du dt + \sigma_t(t) \cdot d\tilde{\mathbf{W}}_t + \int_0^t \frac{\partial \sigma_u(t)}{\partial t} \cdot d\tilde{\mathbf{W}}_u dt$$

¹We know that $d\left(\int_0^t f(t, u) du\right) = f(t, t)dt + \int_0^t df(t, u)du$ for any smooth function $f(t, u)$.

Actually, $d\left(\int_0^t f(t, u) du\right) = \lim_{\Delta t \rightarrow 0} \left[\int_0^{t+\Delta t} f(t + \Delta t, u) du - \int_0^t f(t, u) du \right] = \lim_{\Delta t \rightarrow 0} \left[\int_t^{t+\Delta t} f(t + \Delta t, u) du + \int_0^t f(t + \Delta t, u) du - \int_0^t f(t, u) du \right] = f(t, t)dt + \int_0^t df(t, u)du.$

Similarly, we have $d\left(\int_0^t f(t, u) dW_u\right) = f(t, t)dW_t + \int_0^t df(t, u)dW_u.$

or

$$dr_t = \left[f'_0(t) + \int_0^t \frac{\partial \tilde{\alpha}_u(t)}{\partial t} du + \int_0^t \frac{\partial \sigma_u(t)}{\partial t} \cdot d\tilde{\mathbf{W}}_u \right] dt + \sigma_t(t) \cdot d\tilde{\mathbf{W}}_t \quad (21.4)$$

since $\tilde{\alpha}_t(t) = 0$. The drift contains a stochastic term $\int_0^t \frac{\partial \sigma_u(t)}{\partial t} \cdot d\tilde{\mathbf{W}}_u$ which is path dependent. Therefore, r_t is not Markovian in general.

The non-Markovian feature makes the implementation difficult. Taking the example of Tree method, a non-Markovian model may lead to non-recombining branches, and an explosive number of nodes. The finite difference method is only applicable under Markovian models. However, by choosing some special function forms for $\sigma_u(t)$, it is possible to obtain Markovian models for r_t .

Definition 21.1 (Volatility Separation Condition) The volatility $\sigma_t(T)$ is said to satisfy the separation condition if it takes the form

$$\sigma_t(T) = \mathbf{g}(t)h(T), \quad \text{where } \mathbf{g}(t) \text{ is a vector-valued function and } h(T) > 0.$$

Proposition 21.2 If $\sigma_t(T)$ satisfies the volatility separation condition, then

$$dr_t = \kappa(t)(\theta(t) - r_t)dt + \sigma_t(t) \cdot d\tilde{\mathbf{W}}_t \quad \text{i.e., } r_t \text{ is Markovian,}$$

where

$$\kappa(t) = -\frac{h'(t)}{h(t)}, \quad \theta(t) = \frac{f'_0(t)}{\kappa(t)} + \frac{h^2(t)}{\kappa(t)} \int_0^t \mathbf{g}^\top(u)\mathbf{g}(u)du + f_0(t).$$

This result can be extended to the case where h is an n -dimensional vector $\mathbf{h}$, and $\mathbf{g}$ an $n \times n$ matrix-valued function.

Proof It suffices to calculate the drift term of dr_t in (21.4), i.e.,

$$\int_0^t \frac{\partial \tilde{\alpha}_u(t)}{\partial t} du \quad \text{and} \quad \int_0^t \frac{\partial \sigma_u(t)}{\partial t} \cdot d\tilde{\mathbf{W}}_u.$$

Firstly from the definition of $\tilde{\alpha}$ in the no-arbitrage condition,

$$\tilde{\alpha}_u(t) = \sigma_u(t) \cdot \int_u^t \sigma_u(s)ds = h(t)\mathbf{g}^\top(u) \int_u^t h(s)\mathbf{g}(u)ds.$$

Direct operations give

$$\begin{aligned} \int_0^t \frac{\partial \tilde{\alpha}_u(t)}{\partial t} du &= \int_0^t \left[h'(t) \mathbf{g}^\top(u) \int_u^t h(s) \mathbf{g}(u) ds + h^2(t) \mathbf{g}^\top(u) \mathbf{g}(u) \right] du \\ &= -\kappa(t) \int_0^t \tilde{\alpha}_u(t) du + \int_0^t h^2(t) \mathbf{g}^\top(u) \mathbf{g}(u) du. \end{aligned}$$

Similarly,

$$\begin{aligned} \int_0^t \frac{\partial \sigma_u(t)}{\partial t} \cdot d\tilde{\mathbf{W}}_u &= h'(t) \int_0^t \mathbf{g}^\top(u) d\tilde{\mathbf{W}}_u = -\kappa(t) \left(\int_0^t \sigma_u(t) \cdot d\tilde{\mathbf{W}}_u \right) \\ &= -\kappa(t) \left(r_t - f_0(t) - \int_0^t \tilde{\alpha}_u(t) du \right). \end{aligned}$$

Hence,

$$\begin{aligned} dr_t &= \left(f'_0(t) + \int_0^t \frac{\partial \tilde{\alpha}_u(t)}{\partial t} du + \int_0^t \frac{\partial \sigma_u(t)}{\partial t} \cdot d\tilde{\mathbf{W}}_u \right) dt + \sigma_t(t) \cdot d\tilde{\mathbf{W}}_t \\ &= \kappa(t)(\theta(t) - r_t)dt + \sigma_t(t) \cdot d\tilde{\mathbf{W}}_t. \end{aligned} \quad \square$$

Proposition 21.3 *If $\sigma_t(T)$ satisfies the volatility separation condition and $\mathbf{g}, h$ are deterministic, then*

$$P_t(T) = e^{C_t(T) - D_t(T)r_t},$$

where

$$\begin{aligned} C_t(T) &= \ln \frac{P_0(T)}{P_0(t)} - \int_t^T \left[\int_0^t \left(h(z) \mathbf{g}^\top(u) \int_t^z h(s) \mathbf{g}(u) ds \right) du \right] dz + D_t(T) f_0(t), \\ D_t(T) &= \frac{\int_t^T h(u) du}{h(t)}. \end{aligned}$$

Proof Firstly using the volatility separation condition in (21.3) gives

$$r_t = f_t(t) = f_0(t) + \int_0^t \left(h(t) \mathbf{g}^\top(u) \int_u^t h(s) \mathbf{g}(u) ds \right) du + \int_0^t h(t) \mathbf{g}^\top(u) d\tilde{\mathbf{W}}_u.$$

Then,

$$\begin{aligned}
f_t(T) &= f_0(T) + \int_0^t \left(h(T) \mathbf{g}^\top(u) \int_u^T h(s) \mathbf{g}(u) ds \right) du + \int_0^t h(T) \mathbf{g}^\top(u) d\tilde{\mathbf{W}}_u \\
&= f_0(T) + \frac{h(T)}{h(t)} \left[\int_0^t \left(h(t) \mathbf{g}^\top(u) \int_u^t h(s) \mathbf{g}(u) ds \right) du \right. \\
&\quad \left. + \int_0^t h(t) \mathbf{g}^\top(u) d\tilde{\mathbf{W}}_u \right] + \int_0^t \left(h(T) \mathbf{g}^\top(u) \int_t^T h(s) \mathbf{g}(u) ds \right) du \\
&= f_0(T) + \frac{h(T)}{h(t)} (r_t - f_0(t)) + \int_0^t \left(h(T) \mathbf{g}^\top(u) \int_t^T h(s) \mathbf{g}(u) ds \right) du
\end{aligned}$$

and

$$\begin{aligned}
P_t(T) &= e^{-\int_t^T f_t(u) du} \\
&= \exp \left\{ - \int_t^T f_0(u) du \right\} \\
&\quad \exp \left\{ - \frac{\int_t^T h(u) du}{h(t)} (r_t - f_0(t)) \right. \\
&\quad \left. - \int_t^T \left[\int_0^t \left(h(z) \mathbf{g}^\top(u) \int_t^z h(s) \mathbf{g}(u) ds \right) du \right] dz \right\} \\
&= e^{C_t(T) - D_t(T)r_t}.
\end{aligned}$$

Note that

$$\exp \left\{ - \int_t^T f_0(u) du \right\} = \frac{\exp \{ - \int_0^T f_0(u) du \}}{\exp \{ - \int_0^t f_0(u) du \}} = \frac{P_0(T)}{P_0(t)}. \quad \square$$

21.3 Short Rate Models

One-factor short rate models are the easiest to handle. However, because there is only one Brownian for the whole term structure, the bond prices for all the maturities are perfectly correlated. So, the one-factor models are only suitable for structures which are mainly sensitive to parallel shifts. Two-factor models capture the yield curve pivoting, which gives satisfaction for the pricing of many interest rate products.

21.3.1 Hull–White One-Factor Model

In Hull–White (HW) model [41], r_t follows an OU process:

$$dr_t = \kappa(t)(\theta(t) - r_t)dt + \sigma d\tilde{W}_t.$$

We consider the widely-used case with deterministic $\theta(t)$, and constants κ and σ . It is a no-arbitrage model because it can be derived from HJM No-Arbitrage Model by setting $\sigma_t(T) = \sigma e^{-\kappa(T-t)}$.

r_t follows an OU process which is a Gaussian process. Its dynamics are known to be *mean reverting* which means that, with $\kappa > 0$, r_t tends to rise when $r_t < \theta(t)$ and tends to fall when $r_t > \theta(t)$. The parameter κ represents the reverting speed. As a Gaussian process, the model is fully specified by its mean and variance which can be easily obtained. The zero coupon bonds can be calculated with closed-form formula. HW model belongs to the so-called *arbitrage-free* models which can calibrate the initial term structure of the yield curve of the market thanks to the time-dependent parameters. HW model is the most applied model in practice for structures which are mainly sensitive to parallel shifts of the yield curve.

Applying $h(T) = \sigma e^{-\kappa T}$ and $g(t) = e^{\kappa t}$ to the results in Proposition 21.2, we obtain

Corollary 21.1 *For Hull–White model,*

$$\begin{aligned}\theta(t) &= f_0(t) + \frac{f'_0(t)}{\kappa} + \frac{\sigma^2}{2\kappa^2}(1 - e^{-2\kappa t}), \\ \mathbb{E}[r_t] &= f_0(t) + \frac{\sigma^2}{2\kappa^2}(1 - e^{-\kappa t})^2, \\ \widetilde{\text{Var}}(r_t) &= \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t}), \\ P_t(T) &= e^{C_t(T) + D_t(T)r_t},\end{aligned}$$

where

$$\begin{aligned}C_t(T) &= \ln \frac{P_0(T)}{P_0(t)} - D_t(T) \frac{\partial \ln P_0(t)}{\partial t} - \frac{\sigma^2}{4\kappa^3}(e^{-\kappa T} - e^{-\kappa t})^2(e^{2\kappa t} - 1), \\ D_t(T) &= \frac{1}{\kappa}[1 - e^{-\kappa(T-t)}].\end{aligned}$$

The yield for zero coupon bond is $C_t(T) + D_t(T)r_t$. This implies perfect correlation of yield levels over all maturities. As a consequence, the model is not suitable for structures having significant sensitivities to changes other than parallel shift of the yield curve.

HW model is equivalent to the one-factor case of LGM (Linear Gaussian Model) which takes the form $r_t = f_0(t) + x_t$ where x_t is the random state variable following an OU process. Working with $x_t = r_t - f_0(t)$ has the advantage of avoiding the non smoothness issues of $f_0(t)$ from bootstrapped prices of the market zero coupon bonds.

Remark 21.1 By setting $\theta(t)$ to a constant b , we get Vasicek model $dr_t = \kappa(b - r_t)dt + \sigma d\tilde{W}_t$. Unlike Hull–White model, Vasicek model is an equilibrium model which is not arbitrage free in the sense that the initial term structure of the market may not be satisfied.

21.3.2 Two-Factor LGM (Linear Gaussian Model)

In one-factor models, the yields of different maturities are perfectly correlated. For products that are sensitive to the yield curve changes other than parallel shift, more factors need to be introduced. For most of the interest rate structures, two factors give satisfaction for capturing the variability of the term structure.

Two-factor LGM is among the most popular ones used in practice. It is a no-arbitrage model and it allows analytical solution for the zero coupon bonds. A detailed analysis can be found in [3]. The standard form of two-factor LGM may be written as follows:

$$r_t = f_0(t) + x_t^1 + x_t^2,$$

where x_t^1, x_t^2 are the factors following the mean reverting process

$$d\mathbf{x}_t = [\mathbf{y}_t \mathbf{1} - \kappa(t)\mathbf{x}_t]dt + \sigma(t)^\top d\tilde{\mathbf{W}}_t,$$

with

$$\begin{aligned} \mathbf{x}_t &= [x_t^1, x_t^2]^\top, \quad \mathbf{1} = [1, 1]^\top, \\ \tilde{\mathbf{W}}_t &= [\tilde{W}_t^1, \tilde{W}_t^2]^\top, \text{ where } \tilde{W}_t^1, \tilde{W}_t^2 \text{ are independent Brownians,} \\ \sigma(t) &= \begin{bmatrix} \sigma_{11}(t) & 0 \\ \sigma_{21}(t) & \sigma_{22}(t) \end{bmatrix}. \end{aligned}$$

The other parameters are as follows which are derived from HJM model with separability condition on σ so that the model is Markovian:

$$\begin{aligned} \kappa(t) &= \text{diag}(\kappa_1(t), \kappa_2(t)), \\ \mathbf{h}(t) &= \begin{bmatrix} e^{-\int_0^t \kappa_1(u)du} \\ e^{-\int_0^t \kappa_2(u)du} \end{bmatrix}, \end{aligned}$$

$$\mathbf{g}(t) = \begin{bmatrix} \sigma_{11}(t)e^{\int_0^t \kappa_1(u)du} & 0 \\ \sigma_{21}(t)e^{\int_0^t \kappa_1(u)du} & \sigma_{22}(t)e^{\int_0^t \kappa_2(u)du} \end{bmatrix},$$

$$\mathbf{y}_t = \mathbf{H}(t) \left(\int_0^t \mathbf{g}(u)^\top \mathbf{g}(u) du \right) \mathbf{H}(t), \quad \mathbf{H}(t) = \text{diag}(\mathbf{h}(t)).$$

The model is fully specified by $\kappa_1, \kappa_2, \sigma_{11}, \sigma_{21}$ and σ_{22} .

Extension for Volatility Skew

LGM model (one and two factors) can be extended to take into account the volatility skew observed in the vanilla instruments for pricing certain structures that are sensitive to volatility dynamics. One extension applies local volatility to $g(t)$ in LGM model which assumes the volatility separability:

$$\sigma_t(T) = g(t)h(T) \quad \text{with} \quad g(t) = g(t, x_t, y_t).$$

A further extension consists of applying a stochastic term to $g(t, x_t, y_t)$:

$$\sqrt{z(t)}g(t, x_t, y_t)$$

where z follows a CIR process which is uncorrelated to x_t . We refer to [3] for detailed discussions.

21.3.3 CIR (Cox-Ingersoll-Ross) One-Factor Model

The original CIR model is given by

$$dr_t = \kappa(\theta - r_t)dt + \sigma\sqrt{r_t}d\tilde{W}_t,$$

where κ, θ, σ are all constants.

Similar to Vasicek model, it is an equilibrium model which is not arbitrage-free. The particular property of the model is that it guarantees the positivity of the short rate which used to be a much desired feature before the negative interest rates becoming a reality for several major currencies in the 2010s.

Similar to Hull-White model, CIR model can be derived from HJM No-Arbitrage Short Rate Model by setting $\sigma_t(T) = \sigma\sqrt{r_t}e^{-\kappa(T-t)}$. Under CIR model, the zero coupon bonds admit an analytical solution as follows [26]:

$$P_t(T) = e^{A_t(T) - B_t(T)r_t}.$$

21.3.4 Affine-Yield Models

A model satisfying the below features is called an affine-yield model:

- the short rate r_t is an affine function (linear plus translation) of the factors Y_t
- the zero coupon bond prices are of the form $P_t(T) = e^{C_t(T) - D_t(T) \cdot Y_t}$ where $C_t(T)$ and $D_t(T)$ are nonrandom functions ($D_t(T)$ in vector form).

It can be proved that an affine-yield model takes the below vector form [29]:

$$d\mathbf{Y}_t = (\lambda_0 + \phi \mathbf{Y}_t)dt + \Sigma \begin{bmatrix} \sqrt{\alpha_1 + \beta_1^\top \mathbf{Y}_t} \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sqrt{\alpha_n + \beta_n^\top \mathbf{Y}_t} \end{bmatrix} d\mathbf{W}_t,$$

where

$$\begin{array}{ll} \lambda_0, \beta_i : & \text{vectors} & \alpha_i : & \text{scalars} \\ \phi, \Sigma : & \text{matrices} & \mathbf{Y}_t : & \text{vector of factors (state variables)} \\ \mathbf{W}_t : & \text{Brownian motion vector} \end{array}$$

The Vasicek and CIR models are affine-yield models.

For a one-factor model, we have $r_t = Y_t$ (cf. the result of HW model for example) which implies that the yields of zero coupon bonds for different maturities are all linearly dependent on r_t , i.e. 100% correlated.

Interest Rate Derivatives Pricing with Short Rate Models

For pricing interest rate derivatives, short rate models can be applied. As usual, the model parameters are calibrated to match the market price of liquid instruments. Closed-form solutions for zero coupon bonds exist for the popular one-factor models. PDE and Tree methods are applicable for low dimension models. In Monte Carlo simulation, the increments of the short rate and the stochastic discount factor can be generated for each time-step. The zero coupon bond prices, if required, can also be simulated by using the solution of (21.2) (cf. the Monte Carlo method for BSM model). We recall that the price of zero coupon bond is the expected value of stochastic discount factor as shown by (21.1). For pricing a bond option with Monte Carlo, the random zero coupon bond prices need to be generated.

Here we take the example of the pricing of a caplet with Monte Carlo method. Assume that it is a call option on USD 3M Libor.

- generate one random path under $\tilde{\mathbb{P}}$ of the applied short rate model;
- calculate the 3M Libor rate with the stochastic discount factor values for this path;

- (iii) calculate the present value of the option payoff for this path;
- (iv) repeat the above steps and average the present value of the simulated payoffs to obtain the option price.

21.4 Libor Market Model (LMM)

LMM (also known as BGM, Brace–Gatarek–Musiela) models directly the forward Libor rates that are observable through FRA and Swap contracts which are replicable by zero coupon bonds. We have seen that each forward Libor rate $L_t(T_i)$ is a martingale under its natural risk-neutral probability measure $\tilde{\mathbb{P}}^{T_{i+1}}$. LMM consists of applying the same numéraire for all the Libor rates when pricing with the term structure. If we use $\tilde{\mathbb{P}}^{T_{n+1}}$ (the last one is a common choice known as the terminal measure, but we may choose any other $\tilde{\mathbb{P}}^{T_i}$ for all $L_t(T_k)$ ($k = 0, 1, \dots, n$), we will need to adjust the drifts accordingly. Assume the following SDE

$$dL_t(T_k) = \sigma_k \cdot d\tilde{\mathbf{W}}_t^{T_{k+1}}$$

where the volatility is deterministic and modelled either as

$$\sigma_k = \begin{cases} \sigma_k(t)L_t(T_k), & \text{for log normal model,} \\ \sigma_k(t), & \text{for normal model,} \end{cases}$$

and $\tilde{\mathbf{W}}_t^{T_{k+1}}$ is a multi-dimensional Brownian motion. By definition, the components of a multi-dimensional Brownian motion are independent one-dimensional Brownian motions.

Note that each $L_t(T_k)$ can be equivalently expressed with a one-dimensional Brownian motion

$$dL_t(T_k) = \|\sigma_k\| d\tilde{X}_t^{T_{k+1}}$$

where $d\tilde{X}_t^{T_{k+1}} = \frac{1}{\|\sigma_k\|} \sigma_k \cdot d\tilde{\mathbf{W}}_t^{T_{k+1}}$. However, one-dimensional Brownian terms $\tilde{X}_t^{T_{i+1}}$ and $\tilde{X}_t^{T_{k+1}}$ ($\forall i, k$) are correlated by

$$d\tilde{X}_t^{T_{i+1}} d\tilde{X}_t^{T_{k+1}} = \frac{\sigma_i \cdot \sigma_k}{\|\sigma_i\| \|\sigma_k\|} dt \equiv \rho_{i,k} dt.$$

According to (17.1) of Sect. 17.2, the Radon–Nikodym derivative process for the numéraire change from $\tilde{\mathbb{P}}^{T_{i+1}}$ to $\tilde{\mathbb{P}}^{T_i}$ is

$$\left. \frac{d\tilde{\mathbb{P}}^{T_i}}{d\tilde{\mathbb{P}}^{T_{i+1}}} \right|_{\mathcal{F}_t} = Z_t(T_i) = \frac{P_t(T_i)/P_0(T_i)}{P_t(T_{i+1})/P_0(T_{i+1})} = \frac{1 + \delta_i L_t(T_i)}{1 + \delta_i L_0(T_i)},$$

where we have used $L_t(T_i) = \frac{P_t(T_i) - P_t(T_{i+1})}{\delta_i P_t(T_{i+1})}$ and δ_i is the time fraction $\delta_i = T_{i+1} - T_i$. The Radon–Nikodym derivative admits to

$$dZ_t(T_i) = \frac{\delta_i dL_t(T_i)}{1 + \delta_i L_0(T_i)} = \frac{1 + \delta_i L_t(T_i)}{1 + \delta_i L_0(T_i)} \frac{\delta_i dL_t(T_i)}{1 + \delta_i L_t(T_i)} = Z_t(T_i) \frac{\delta_i dL_t(T_i)}{1 + \delta_i L_t(T_i)}.$$

Hence, applying Girsanov theorem in a vectorized fashion gives

$$\begin{aligned} d\tilde{\mathbf{W}}_t^{T_n} &= -\frac{d\langle Z_t(T_n), \tilde{\mathbf{W}}_t^{T_{n+1}} \rangle}{Z_t(T_n)} + d\tilde{\mathbf{W}}_t^{T_{n+1}} = -\frac{\delta_n \sigma_n}{1 + \delta_n L_t(T_n)} dt + d\tilde{\mathbf{W}}_t^{T_{n+1}}, \\ d\tilde{\mathbf{W}}_t^{T_{n-1}} &= -\frac{d\langle Z_t(T_{n-1}), \tilde{\mathbf{W}}_t^{T_n} \rangle}{Z_t(T_{n-1})} + d\tilde{\mathbf{W}}_t^{T_n} = -\sum_{i=n-1}^n \frac{\delta_i \sigma_i}{1 + \delta_i L_t(T_i)} dt + d\tilde{\mathbf{W}}_t^{T_{n+1}}, \\ &\vdots \\ d\tilde{\mathbf{W}}_t^{T_{k+1}} &= -\sum_{i=k+1}^n \frac{\delta_i \sigma_i}{1 + \delta_i L_t(T_i)} dt + d\tilde{\mathbf{W}}_t^{T_{n+1}}, \quad k < n. \end{aligned}$$

We obtain the SDE for $L_t(T_k)$ under the risk-neutral probability measure $\tilde{\mathbb{P}}^{T_{n+1}}$

$$dL_t(T_k) = \sigma_k \cdot \left[-\sum_{i=k+1}^n \frac{\delta_i \sigma_i}{1 + \delta_i L_t(T_i)} dt + d\tilde{\mathbf{W}}_t^{T_{n+1}} \right], \quad 0 \leq t \leq T_k,$$

or

$$dL_t(T_k) = \left[-\sum_{i=k+1}^n \frac{\delta_i \rho_{k,i} \|\sigma_k\| \|\sigma_i\|}{1 + \delta_i L_t(T_i)} dt + \sigma_k \cdot d\tilde{\mathbf{W}}_t^{T_{n+1}} \right], \quad 0 \leq t \leq T_k.$$

Factor Reduction

For avoiding a too large number of factors (Brownians) for long term products based on Libor rates, we can apply the factor reduction technique. Let $\tilde{B}_t^j$ ($j = 1, \dots, d$, typically $d < 10$) be independent Brownians and replace $d\tilde{X}_t^{T_{i+1}} = \frac{1}{\|\sigma_i\|} \sigma_i \cdot d\tilde{\mathbf{W}}_t^{T_{i+1}}$ by $\sum_{j=1}^d \theta_j^{T_{i+1}} d\tilde{B}_t^j$. Here, $\{\theta_j^{T_{i+1}}\}_{j=1, \dots, d}^{i=0, \dots, n}$ are calibrated to produce $d\tilde{X}_t^{T_{i+1}} d\tilde{X}_t^{T_{k+1}} = \rho_{i,k} dt$, $\forall i, k$.

Readers may refer to [3, 84] for further analysis on calibration and rank reduction for LMM.

Stochastic Volatility

The basic version of LMM does not handle volatility skew. It is possible to extend the model with a stochastic component in the volatility term like $\sqrt{v_t}\sigma_i$ where v_t is a positive process. The stochastic process on the volatility is often common to all maturities in the implementation.

21.5 Convexity Adjustment

For some of the financial products (e.g. in-arrears swap, CMS), there is inconsistency between the “natural” measure of the random variable process and the measure used for pricing. A so-called convexity adjustment needs to be applied for the valuation. We take the example of a Libor in-arrears cash flow where both the Libor fixing and the payment happen at the same time for the period:

$$PV_t = \delta \cdot P_t(T_i) \tilde{\mathbb{E}}^{T_i}[L_{T_i}(T_i)|\mathcal{F}_t],$$

where $\delta = T_{i+1} - T_i$, and as we know, $L_t(T_i)$ is martingale under $\tilde{\mathbb{P}}^{T_{i+1}}$ (not $\tilde{\mathbb{P}}^{T_i}$). The Radon–Nikodym derivative process for the numéraire change from $\tilde{\mathbb{P}}^{T_{i+1}}$ to $\tilde{\mathbb{P}}^{T_i}$ is

$$\left. \frac{d\tilde{\mathbb{P}}^{T_i}}{d\tilde{\mathbb{P}}^{T_{i+1}}} \right|_{\mathcal{F}_t} = Z_t(T_i) = \frac{P_t(T_i)/P_0(T_i)}{P_t(T_{i+1})/P_0(T_{i+1})} = \frac{1 + \delta L_t(T_i)}{1 + \delta L_0(T_i)} \quad (t < T_i).$$

Let σ denote the volatility of the $\mathbb{P}^{T_{i+1}}$ martingale $L_t(T_i)$. The convexity correction for L_t is defined as:

$$\begin{aligned} \Delta &= \tilde{\mathbb{E}}^{T_i}[L_{T_i}(T_i)|\mathcal{F}_t] - \tilde{\mathbb{E}}^{T_{i+1}}[L_{T_i}(T_i)|\mathcal{F}_t] = \frac{1}{Z_t(T_i)} \tilde{\mathbb{E}}^{T_{i+1}}[Z_{T_i}(T_i)L_{T_i}(T_i)|\mathcal{F}_t] \\ &\quad - \tilde{\mathbb{E}}^{T_{i+1}}[L_{T_i}(T_i)|\mathcal{F}_t] \\ &= \frac{1}{1 + \delta L_t(T_i)} \tilde{\mathbb{E}}^{T_{i+1}}[L_{T_i}(T_i)(1 + \delta L_{T_i}(T_i))|\mathcal{F}_t] - L_t(T_i) \\ &= \frac{1}{1 + \delta L_t(T_i)} \{L_t(T_i) + \delta \tilde{\mathbb{E}}^{T_{i+1}}[L_{T_i}^2(T_i)|\mathcal{F}_t]\} - L_t(T_i). \end{aligned}$$

In the above equation, $\tilde{\mathbb{E}}^{T_{i+1}}[L_{T_i}^2(T_i)|\mathcal{F}_t]$ can be priced and hedged using caplets as explained in Sect. 19.4 “Static Hedging for European Style Options” in Chap. 19. Here, we estimate the convexity adjustment value by assuming the distribution of $L_t(T_i)$ to be log normal:

$$\begin{aligned}
\Delta &= \frac{1}{1 + \delta L_t(T_i)} \left\{ L_t(T_i) + \delta \tilde{\mathbb{E}}^{T_i+1} \left[\left(L_t(T_i) e^{-\frac{1}{2}\sigma^2(T_i-t) + \sigma(\tilde{W}_{T_i}^{T_i+1} - \tilde{W}_t^{T_i+1})} \right)^2 \right] \right\} - L_t(T_i) \\
&= L_t(T_i) \left\{ \frac{1 + \delta L_t(T_i) e^{\sigma^2(T_i-t)} \tilde{\mathbb{E}}^{T_i+1} \left[e^{-\frac{1}{2}(2\sigma)^2(T_i-t) + 2\sigma(\tilde{W}_{T_i}^{T_i+1} - \tilde{W}_t^{T_i+1})} \right]}{1 + \delta L_t(T_i)} - 1 \right\} \\
&= L_t(T_i) \left[\frac{1 + \delta L_t(T_i) e^{\sigma^2(T_i-t)}}{1 + \delta L_t(T_i)} - 1 \right] \quad (\text{expectation of exponential martingale is 1}) \\
&= L_t(T_i) \theta [e^{\sigma^2(T_i-t)} - 1] \quad \left(\text{define } \theta = \frac{\delta L_t(T_i)}{1 + \delta L_t(T_i)} \right) \\
&\approx L_t(T_i) \theta \sigma^2 (T_i - t).
\end{aligned}$$

Exercises

1. Show that the forward rate is a martingale under the T -forward measure, i.e., for $t \leq T$,

$$\tilde{\mathbb{E}}^T[f_T(T)|\mathcal{F}_t] = f_t(T).$$

2. Assume a one-factor short rate model under the HJM framework with constant volatility, i.e., $\sigma_t(T) = \sigma$. Show that

$$f_t(T) = f_0(T) + \sigma^2 t \left(T - \frac{t}{2} \right) + \sigma \tilde{W}_t$$

and

$$dr_t = \left(\frac{\partial f_0(t)}{\partial t} + \sigma^2 t \right) dt + \sigma d\tilde{W}_t$$

which is a special case of the *Ho-Lee model* related to the slope of the initial yield curve.

3. Given the following two models under the risk-neutral measure for asset price and bond price respectively:

$$\begin{aligned}
dS_t &= r_t S_t dt + \sigma_S S_t d\tilde{W}_t, \\
dP_t(T) &= r_t P_t(T) dt - \sigma_P P_t(T) d\tilde{W}_t,
\end{aligned}$$

where r_t is a stochastic process of interest rate, σ_S and σ_P are positive constants.

- (i) Let $\text{For}_t(T) = S_t/P_t(T)$ be the forward price at time t and denote $\sigma = \sigma_S + \sigma_P$. Show that $d\text{For}_t(T) = \sigma \text{For}_t(T) d\tilde{W}_t^T$, where $\tilde{W}_t^T$ is the Brownian motion under the T -forward measure.
- (ii) Consider a European cash-or-nothing binary call on the asset S_t , expiring at time T with strike price K . Show that its time- t value is given by

$$P_t(T)\tilde{\mathbb{E}}^T[\mathbf{1}_{\{S_T > K\}}|\mathcal{F}_t] = P_t(T)\tilde{\mathbb{E}}^T[\mathbf{1}_{\{\text{For}_T(T) > K\}}|\mathcal{F}_t] = P_t(T)N(d_2),$$

where

$$d_2 = \frac{\log(\text{For}_t(T)/K) - \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}}.$$

Chapter 22

Credit Modelling



In this chapter, we study the case of default in credit modelling. Structural models and intensity models are first covered. We then study the pricing of some basket credit products, where correlated default is the main issue. Interested readers may refer to [40] and [12]. The counterparty risk measures are introduced at the end of the chapter.

22.1 Credit Modelling

22.1.1 Structural Models

Structural models link the default occurrence to the value of the firm. Merton [56] pioneered this approach in 1974. Denote A_t for the asset value at time t . Assume that the total debt of the firm is L for the considered time horizon T . At time T , the debt holder (creditor) will receive the full amount L or what is left of the firm, i.e., $D_T(T) = \min\{L, A_T\} = L - (L - A_T)^+$. The value of the debt at time t ($t < T$) is then

$$\begin{aligned} D_t(T) &= \widetilde{\mathbb{E}}[e^{-r(T-t)} D_T(T) | \mathcal{F}_t] = \widetilde{\mathbb{E}}[e^{-r(T-t)} (L - (L - A_T)^+) | \mathcal{F}_t] \\ &= Le^{-r(T-t)} - p_t(A_t, L, T). \end{aligned}$$

Here $\mathcal{F}_t$ is the natural filtration of A_t . The term $p_t(A_t, L, T)$ is a put option value with A_t as underlying, L as strike and T as maturity. So, the valuation only relies on a European put option pricing.

Let S_t be the equity value of the firm at time t . Since $A_t = S_t + D_t(T)$, we have

$$S_t = A_t - D_t(T) = A_t - Le^{-r(T-t)} + p_t(A_t, L, T) = c_t(A_t, L, T),$$

where the last equality is obtained by put-call parity. Therefore, the equity is valued as a European call option (with payoff $(A_T - L)^+$ since $A_T = S_T + D_T(T) = (A_T - L)^+ + \min\{L, A_T\}$).

The model can be further improved by letting the default time be the first time that $A_t < L$. As the asset value is not observable on daily basis, practitioners favor intensity models for credit derivatives modelling where the model parameters can be implied from the markets quotes of liquid credit instruments. As the structural model links the debt and equity value, it has found its application in the strategy of capital structure arbitrage. The model can be used for calculating the stock delta for hedging the position in the debt of the company.

In Moody's KMV¹ and CreditMetrics² models, the asset value of the i -th firm is assumed to follow the SDE

$$dA_t^i/A_t^i = \mu^i dt + \sigma_i dW_t^i.$$

The probability of default for the period $(0, T)$ is calculated from the *distance-to-default* which is defined as

$$d^i(T) = \frac{\mathbb{E}[\ln A_T^i] - \ln L^i}{\sigma_i \sqrt{T}} = \frac{\ln(A_0^i/L^i) + (\mu^i - \frac{1}{2}\sigma_i^2)T}{\sigma_i \sqrt{T}},$$

where L^i is the debt of the i -th firm.

22.1.2 Intensity Models

In intensity models (also known as reduced-form models), the *risk-neutral* distribution of the default time τ is

$$\tilde{\mathbb{P}}\{\tau > t\} = 1 - F(t) = e^{-\int_0^t \lambda_u du}, \quad t > 0,$$

where $\tilde{\mathbb{P}}\{\tau > t\}$ is the *survival probability* and $F(t)$ represents the default probability. When λ is a constant, τ follows the exponential distribution. It is the first jump time of a Poisson process. Note that λ can be calculated from the market quote of CDS and be used for pricing other credit derivatives. $\tilde{\mathbb{P}}\{\tau < t\}$ is the probability of default before t , and

¹Founded by Stephen Kealhofer, John McQuown and Oldřich Vašíček and later acquired by Moody's.

²RiskMetrics was established in 1989 by J.P. Morgan and acquired by MSCI in 2010.

$$\tilde{\mathbb{P}}\{t_1 < \tau < t_2\} = \tilde{\mathbb{P}}\{\tau > t_1\} - \tilde{\mathbb{P}}\{\tau > t_2\}$$

is the marginal probability of default between time t_1 and $t_2 (> t_1)$. From the result in Chap. 10 Sect. 10.9, $\lambda_t dt$ is the probability of default in $(t, t + dt)$, conditioned by no default before t .

In this chapter, $P_t(T)$ will denote the stochastic discount factor, $P_t(T) = e^{-\int_t^T r_u du}$. Let $\mathcal{F}_t$ be the filtration for the information on default free market until t which includes $\{\tau > t\}$. In intensity models, the default time τ is not a stopping time of $\mathcal{F}_t$. However, it is a stopping time of the enlarged filtration $\mathcal{G}_t = \mathcal{F}_t \cup \sigma(\{\tau \leq s\}, s < t)$.

Proposition 22.1 (Filtration Switching Formula) *Let X be an integrable random variable, then*

$$\mathbf{1}_{\{\tau > t\}} \mathbb{E}[X | \mathcal{G}_t] = \mathbf{1}_{\{\tau > t\}} \frac{\mathbb{E}[X \mathbf{1}_{\{\tau > t\}} | \mathcal{F}_t]}{\mathbb{E}[\mathbf{1}_{\{\tau > t\}} | \mathcal{F}_t]} \quad a.s.$$

Sketch of Proof Note that if $A \in \mathcal{G}_t$, then $A \cap \{\tau > t\} = B \cap \{\tau > t\}$ for some event $B \in \mathcal{F}_t$. So, there should exist $\mathcal{F}_t$ -measurable r.v. Y_t , s.t.

$$\mathbf{1}_{\{\tau > t\}} \mathbb{E}[X | \mathcal{G}_t] = \mathbf{1}_{\{\tau > t\}} Y_t.$$

Taking conditional expectation $\mathbb{E}[\mathbf{1}_{\{\tau > t\}} \mathbb{E}[X | \mathcal{G}_t] | \mathcal{F}_t] = \mathbb{E}[\mathbf{1}_{\{\tau > t\}} Y_t | \mathcal{F}_t]$, we obtain

$$Y_t = \frac{\mathbb{E}[X \mathbf{1}_{\{\tau > t\}} | \mathcal{F}_t]}{\mathbb{E}[\mathbf{1}_{\{\tau > t\}} | \mathcal{F}_t]}. \quad \square$$

Example 22.1 (Zero Coupon Bond Modelling) We take the zero coupon bonds for illustrating the modelling for bonds. Denote

R : Recovery Rate in case of default (usually, $R \approx 40\%$ for senior bonds)

LGD : Loss Given Default ($LGD = 1 - R$)

DP : Default Probability ($DP = \mathbb{E}[\mathbf{1}_{\{\tau \leq T\}} | \mathcal{G}_t]$).

The value of a zero coupon bond before default is

$$D_t(T) = \tilde{\mathbb{E}}[P_t(T)(\mathbf{1}_{\{\tau > T\}} + R \cdot \mathbf{1}_{\{\tau \leq T\}}) | \mathcal{G}_t] = \tilde{\mathbb{E}}[P_t(T)(1 - LGD \cdot \mathbf{1}_{\{\tau \leq T\}}) | \mathcal{G}_t].$$

If the short rate r_t is deterministic, we have

$$D_t(T) = P_t(T)(1 - LGD \times DP).$$

If $R = 0$, from filtration switching formula in Proposition 22.1, we obtain the value of a zero coupon bond before default

$$\begin{aligned}
 D_t(T) &= \tilde{\mathbb{E}}[P_t(T)\mathbf{1}_{\{\tau>T\}}|\mathcal{G}_t] = \frac{\tilde{\mathbb{E}}[P_t(T)\mathbf{1}_{\{\tau>T\}}\mathbf{1}_{\{\tau>t\}}|\mathcal{F}_t]}{\tilde{\mathbb{E}}[\mathbf{1}_{\{\tau>t\}}|\mathcal{F}_t]} = \frac{\tilde{\mathbb{E}}[P_t(T)\mathbf{1}_{\{\tau>T\}}|\mathcal{F}_t]}{\tilde{\mathbb{E}}[\mathbf{1}_{\{\tau>t\}}|\mathcal{F}_t]} \\
 &= \frac{\tilde{\mathbb{E}}[\tilde{\mathbb{E}}[P_t(T)\mathbf{1}_{\{\tau>T\}}|\mathcal{F}_T]|\mathcal{F}_t]}{\tilde{\mathbb{E}}[\mathbf{1}_{\{\tau>t\}}|\mathcal{F}_t]} = \frac{\tilde{\mathbb{E}}[P_t(T)e^{-\int_0^T \lambda_u du}|\mathcal{F}_t]}{e^{-\int_0^t \lambda_u du}} \\
 &= \tilde{\mathbb{E}}[e^{\int_t^T -(r_u + \lambda_u) du}|\mathcal{F}_t].
 \end{aligned}$$

22.2 CDS (Credit Default Swap)

As presented in Part I, Chap. 6, CDS (Credit Default Swap) remains the most widely used credit derivative instrument despite the issues in the financial crisis of 2008. Let N be the notional amount and R the recovery rate which is usually assumed to be around 40% for senior debts. Assume that

$T_i, i = 1, \dots, m$: dates on which the premium and the protection are paid

Premium Leg: $\sum_{i=1}^m sN(T_i - T_{i-1})\mathbf{1}_{\{\tau>T_i\}}$, s is the spread

Protection Leg: $N(1 - R)\mathbf{1}_{\{\tau \leq T_m\}}$ (usually, $R \approx 40\%$ for senior debt).

We have the following

$$\begin{aligned}
 LEG_{premium} &= s\tilde{\mathbb{E}}\left[\sum_{i=1}^m (T_i - T_{i-1})\mathbf{1}_{\{\tau>T_i\}}P(T_i)\right] \\
 LEG_{protection} &= \tilde{\mathbb{E}}[(1 - R)\mathbf{1}_{\{\tau \leq T_m\}}P(\tau)].
 \end{aligned}$$

Equating the two legs, we obtain the market pricing of CDS spread

$$s = \frac{\tilde{\mathbb{E}}[(1 - R)\mathbf{1}_{\{\tau \leq T_m\}}P(\tau)]}{\tilde{\mathbb{E}}\left[\sum_{i=1}^m (T_i - T_{i-1})\mathbf{1}_{\{\tau>T_i\}}P(T_i)\right]}.$$

In practice, the CDS spread is quoted in the market. By means of intensity model, we can imply asset's default probability for other purposes (e.g. convertible bond pricing).

22.2.1 Credit Triangle

Let us assume that the hazard rate λ and risk-free rate r are deterministic and flat. We have the so-called *credit triangle* which reveals the relationship between the CDS spread s , the default rate λ and the recovery rate R :

$$s = (1 - R)\lambda.$$

Proof The PV of the protection leg is

$$\begin{aligned} LEG_{protection} &= \widetilde{\mathbb{E}}[(1 - R)\mathbf{1}_{\{\tau \leq T_m\}}P(\tau)] = (1 - R) \int_0^{T_m} e^{-ru} \lambda e^{-\lambda u} du \\ &= (1 - R)\lambda \int_0^{T_m} e^{-(r+\lambda)u} du. \end{aligned}$$

The PV of the premium leg is

$$\begin{aligned} LEG_{premium} &= s \widetilde{\mathbb{E}} \left[\sum_{i=1}^m (T_i - T_{i-1}) \mathbf{1}_{\{\tau > T_i\}} P(T_i) \right] \\ &= s \widetilde{\mathbb{E}} \left[\sum_{i=1}^m \widetilde{\mathbb{E}} \left[(T_i - T_{i-1}) \mathbf{1}_{\{\tau > T_i\}} e^{-rT_i} | \mathcal{G}_{T_i} \right] \right] \\ &= s \widetilde{\mathbb{E}} \left[\sum_{i=1}^m (T_i - T_{i-1}) e^{-\lambda T_i} e^{-rT_i} \right] = s \sum_{i=1}^m (T_i - T_{i-1}) e^{-(r+\lambda)T_i}. \end{aligned}$$

Assuming further that CDS works in continuous time, we have

$$LEG_{premium} \rightarrow s \int_0^{T_m} e^{-(r+\lambda)t} dt.$$

Equating the protection and premium legs, we obtain the result to be proved. $\square$

22.3 Pricing and Hedging of Basket Credit Derivatives

22.3.1 Copula

Copula (*Latin: link*) describes the dependence structure of random variables in a general way. It combines univariate distribution functions to form multivariate distribution functions.

Definition 22.1 Let C be a function $[0, 1]^n \rightarrow [0, 1]$. It is called “Copula” if there exists a vector $(U_1, U_2, \dots, U_n)$ in which each $U_i, i = 1, \dots, n$ is uniformly distributed, s.t.,

$$C(u_1, \dots, u_n) = \mathbb{P}\{U_1 \leq u_1, \dots, U_n \leq u_n\}, \forall (u_1, \dots, u_n) \in [0, 1]^n.$$

Theorem 22.1 (Sklar Theorem) Let F be an n -dimensional cumulative distribution function with margins F_i . Then there exists a copula C such that

$$F(x_1, \dots, x_n) = C(F_1(x_1), \dots, F_n(x_n)), \quad \forall x_1, \dots, x_n \in \mathbb{R}.$$

One can extract a copula from any distribution with $F_i(x_i) = u_i$,

$$C(u_1, \dots, u_n) = F(F_1^{-1}(u_1), \dots, F_n^{-1}(u_n)).$$

If $F(\cdot)$ is a multivariate normal distribution, then $C(\cdot)$ is called Gaussian copula which is widely used in pricing portfolio credit structures (CDO, First-To-Default Note, etc.).

22.3.2 One-Factor Gaussian Copula Model

In the one-factor Gaussian Copula model, the latent variables for the issuers are assumed to be correlated to a common market factor, usually representing the systemic risk of the economy. Let x_i be the latent variable for the i -th issuer which is a standard normal random. Then,

$$x_i = \beta_i \bar{Z} + \sqrt{1 - \beta_i^2} Z_i,$$

where

$\bar{Z}$: common market factor

Z_i : idiosyncratic risk of the i -th issuer

β_i : correlation of the i -th issuer to $\bar{Z}$. It may be either estimated with econometric methods (e.g. correlation derived from the prices of the stock and the index representing $\bar{Z}$), or implied from the market quotes of liquid credit derivative products.

$Z_1, \dots, Z_n, \bar{Z}$: independent standard normal variables.

Note that

- correlation $\rho_{i,j}$ between the issuers i and j equals to $\beta_i \beta_j$
- conditioned by the common market factor $\bar{Z}$, all latent variables for issuers are independent.

In the structural approach, the default occurs if x_i falls below a certain level γ_i at the considered time t . This implies that the probability of default by t is given by $F_i(t) = \tilde{\mathbb{P}}\{\tau_i \leq t\} = \tilde{\mathbb{P}}\{x_i \leq \gamma_i\} = N(\gamma_i)$. The conditional probability of default by time t for issuer i is

$$\tilde{\mathbb{P}}\{\tau_i \leq t | \bar{Z}\} = \tilde{\mathbb{P}}\{x_i \leq \gamma_i | \bar{Z}\} = N\left(\frac{\gamma_i - \beta_i \bar{Z}}{\sqrt{1 - \beta_i^2}}\right).$$

If the default time τ_i follows an exponential distribution with intensity λ , then

$$\gamma_i = N^{-1}(F_i(t)) = N^{-1}(1 - e^{-\lambda t}).$$

Replacing γ_i , we have

$$\tilde{\mathbb{P}}\{\tau_i \leq t | \bar{Z}\} = N\left(\frac{N^{-1}(1 - e^{-\lambda t}) - \beta_i \bar{Z}}{\sqrt{1 - \beta_i^2}}\right).$$

Suppose default probabilities are identical for all issuers and let $p_t = \tilde{\mathbb{P}}\{\tau_i \leq t | \bar{Z}\}$. The conditional probability of k defaults by time t can be obtained from binomial distribution

$$\tilde{\mathbb{P}}(k, t | \bar{Z}) = \frac{n!}{(n-k)!k!} p_t^k (1 - p_t)^{n-k}.$$

22.3.3 Pricing of Basket Credit Derivatives

The one-factor copula model is usually employed for pricing portfolio based credit derivatives, for example, First-To-Default Note and CDO. We illustrate the pricing of basket credit derivatives through the example of a synthetic CDO tranche with “Attachment point” A and “Detachment point” D (e.g. $A = 3\%$, $D = 6\%$). Denote

L_t : portfolio loss in percentage of the total notional by time t

$L_t^{A,D}$: loss between A and D in percentage of the tranche notional by time t

$T_1, \dots, T_m$: payment times for the interests from the CDO tranche

s : interest rate applied for the CDO tranche.

We have

$$L_t^{A,D} = [(L_t - A)\mathbf{1}_{\{A < L_t \leq D\}} + (D - A)\mathbf{1}_{\{L_t > D\}}]/(D - A)$$

$$LEG_{default} = \tilde{\mathbb{E}}\left[\int_0^{T_m} P(t) dL_t^{A,D}\right]$$

$$LEG_{premium} = s \tilde{\mathbb{E}}\left[\sum_{i=1}^m P(T_i)(T_i - T_{i-1})(1 - L_{T_i}^{A,D})\right].$$

Assuming that the interest rate for discounting is flat r , the spread of the tranche will be

$$s = \frac{\int_0^{T_m} e^{-rt} d\tilde{\mathbb{E}}[L_t^{A,D}]}{\sum_{i=1}^m (T_i - T_{i-1}) e^{-rT_i} (1 - \tilde{\mathbb{E}}[L_{T_i}^{A,D}])}.$$

Assuming that the portfolio is homogeneous, i.e. all the assets have the same default probability and recovery rate R , then

$$L_t = \sum_{i=1}^n \frac{1}{n} (1 - R_i) \mathbf{1}_{\{\tau_i < t\}} = \frac{1 - R}{n} k_t,$$

where n is the size of the portfolio and k_t is number of defaults by time t . The expected loss conditioned by the market factor $\bar{Z}$ is

$$\tilde{\mathbb{E}}[L_t | \bar{Z}] = \frac{1 - R}{n} \tilde{\mathbb{E}}[k_t | \bar{Z}] = \frac{1 - R}{n} \sum_{k=1}^n \tilde{\mathbb{P}}(k, t | \bar{Z}).$$

The expression of $\tilde{\mathbb{P}}(k, t | \bar{Z})$ is given in one-factor copula model analysis. Assuming that $\bar{Z}$ follows normal distribution, then

$$\tilde{\mathbb{E}}[L_t] = \int_{-\infty}^{\infty} \tilde{\mathbb{E}}[L_t | z] \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz,$$

which can be calculated with Gaussian Quadrature, Monte Carlo or any other efficient numerical integration methods.

Unlike the derivatives under Complete Market modelled with Brownian motion, the hedging of basket credit derivatives is challenging mainly due to the jumps. For example, the issuer of a First-To-Default Note makes Delta-hedging with the CDS of the single names. However, the hedge will not be sufficient for a default event which implies a big negative gamma position at default. Before the default happens, the issuer usually has a position of positive Theta and positive Gamma of the credit spreads of the single names.

22.4 Counterparty Risk Measures

In this section, we will introduce the most applied concepts and measures which are used for the management of counterparty risks. Many measures and regulatory requirements related to counterparty risks are being developed and implemented after the financial crisis of 2008. We refer to [16] for specialized presentation on counterparty risk management.

In the following, V_t stands for the MtM of the considered trade or portfolio. Monte Carlo simulation based on the physical probability measure is often necessary for calculating the risk measures while option pricing should always be done under risk-neutral measure.

PFE (Potential Future Exposure)

PFE measures the worst potential loss of a transaction with a given percentile α (usually 97.5%) during its whole life. Mathematically, $PFE_\alpha = \inf\{X | \mathbb{P}(V_t \leq X) \geq \alpha, t \in [0, T]\}$. PFE is an important measure for counterparty risk management. A bank may define the credit limit for trades (other than the loans) with a client by a total PFE amount.³ When the aggregated PFE reaches the limit, the bank will stop making new trades with the client. Another example of PFE application is the determination of IA (Independent Amount) for swaps.

Close-Out Risk

It is the potential loss during the period between the time the counterparty is deemed to default and the time the transaction is unwound. Usually, the considered period is 10 business days. The Close-Out Risk is measured with a similar percentile for PFE (e.g. 97.5%).

EPE (Expected Positive Exposure)

$EPE_t = \mathbb{E}[V_t^+]$. It measures the exposure for the counterparty risk as the negative MtM does not constitute a risk.

CVA (Credit Value Adjustment)

CVA is market price of the counterparty credit risk for the considered transaction or portfolio. It is represented mathematically as $CVA = LGD \times \sum_{t=0}^T e^{-rt} \mathbb{E}[V_t^+] = (1 - R) \int_0^T e^{-rt} V_t^+ \lambda_c e^{-\lambda_c t} dt$, where λ_c is the default intensity of the counterparty

³For a portfolio with tradable transactions only, close-out risk or VaR may be used instead of PFE for the risk limit, for instance, 10 day 90% VaR.

and R the recovery rate. Actually, the CVA represents only the theoretical hedging cost due to the lack of liquidity for the CDS to be used. In practice, the counterparty risk is often proxy hedged with the CDS on liquid indices or reference entities correlated to the target one. The CVA needs to be accounted for the determination of the profit of a trade's P/L. For giving the background, CVA was largely ignored by most of the financial institutions before the financial of 2008.

DVA (Debt Value Adjustment or Debit Value Adjustment)

It is the CVA seen by the counterparty. $DVA = (1 - R) \int_0^T e^{-rt} V_t^- \lambda_d e^{-\lambda_d t} dt$, λ_d is dealer's own default intensity.

Wrong-Way Risk

Wrong-Way Risk occurs when the exposure to a counterparty is adversely correlated with the credit quality of that counterparty. For example, buying a put option on gold from a gold producer is a wrong-way risk. Under current practice, the market risk of a transaction and the counterparty risk are modelled independently. Hence, an exogenous consideration for wrong-way risk is necessary. While trying to avoid wrong-way risks, one should not forget to encourage the transactions with "Right-Way" exposure.

Example 22.2 Consider a long position of an equity call option (assuming BSM model with zero div, repo) with initial price V_0 . Recall that the 97.5-th percentile of a standard normal variate is 1.96. Then,

$$\begin{aligned} PFE_T &\approx S_0 e^{(r - \frac{1}{2}\sigma^2)T + 1.96\sigma\sqrt{T}} - K \\ CloseOutRisk_t &\approx \Delta_t (S_t e^{(r - \frac{1}{2}\sigma^2)\delta + 1.96\sigma\sqrt{\delta}} - S_t) e^{-r\delta} - V_t, \delta = 14/365 \\ EPE_t &= \mathbb{E}[V_t^+] = \mathbb{E}[\mathbb{E}[e^{-r(T-t)}(S_T - K)^+ | \mathcal{F}_t]] \approx e^{rt} V_0 \\ CVA &\approx V_0 s_c T, s_c \text{ is the credit spread of the counterparty.} \end{aligned}$$

Basel II Risk Weight Function

For cushioning the risks and preventing excessive leverage, banks are required to hold a certain amount of *regulatory capital*. In 2004, Basel Committee on Banking Supervision (BCBS) proposed to set the capital requirement (K) equal to the Unexpected Loss for the exposure of the bank excluding the extreme losses. It

is admitted that the Expected Loss is a business cost which is well managed by the appropriate pricing and provisioning. **VaR (Value at Risk)** is a measure for extreme losses. It is defined as the maximum loss over the considered period given a certain percentile.⁴ For banks adopting Internal Ratings-Based (IRB) Approach, it is allowed to use their own internal measures for key drivers of credit risk as primary inputs to the capital calculation, which normally leads to lower regulatory capital amount compared to the Standardized Approach. The general formula proposed by BCBS is as follows

$$K = LGD \times (\mathbb{P}|_{\text{VaR}} - PD) \times Adj(M, PD), \quad (22.1)$$

where

LGD: Loss Given Default estimated in a stressed situation, in percentage of the *EAD* (Exposure At Default)

$\mathbb{P}|_{\text{VaR}}$: probability that the loss is below a threshold α_{VaR} (e.g. 99.9%)

PD: Probability of Default for the period of 1 year; From structural approach, default happens when latent variable $x < \gamma$ which implies $PD = N(\gamma)$

Adj(*M*, *PD*): adjustment factor for giving more weight for transactions with longer maturities *M* and lower *PD*.

In (22.1), $LGD \times \mathbb{P}|_{\text{VaR}}$ represents the VaR and $LGD \times PD$ the expected loss. The detailed calculation formula for *K*, known as Basel II Risk Weight Function, is given by BCBS as follows:

$$K = LGD \times \left[N \left(\frac{N^{-1}(PD) + \beta N^{-1}(99.9\%)}{\sqrt{1 - \beta^2}} \right) - PD \right] \times Adj(M, PD),$$

where $Adj(M, PD) = \frac{1 + (M - 2.5)b(PD)}{1 - 1.5b(PD)}$, $b(PD) = [0.11852 - 0.05478 \times \ln(PD)]^2$.

Now, we show that this formula is related to the one-factor correlation model.⁵ Assume the asset return x to be

$$x = \beta \bar{Z} + \sqrt{1 - \beta^2} Z, \quad \text{where } Z \sim \mathcal{N}(0, 1).$$

Then, we have

$$\mathbb{P}\{x < \gamma | \bar{Z}\} = \mathbb{P}\left\{Z < \frac{\gamma - \beta \bar{Z}}{\sqrt{1 - \beta^2}} \middle| \bar{Z}\right\} = N\left(\frac{\gamma - \beta \bar{Z}}{\sqrt{1 - \beta^2}}\right) = N\left(\frac{N^{-1}(PD) - \beta \bar{Z}}{\sqrt{1 - \beta^2}}\right).$$

⁴For instance, the 10 day 90% VaR of USD100m means that there is 10% probability that the loss will be more than USD100m for the period of 10 (business) days.

⁵Our simplified analysis may not reflect the rationale of BCBS (cf. [6] and [34]).

For the point at percentile $\alpha_{VaR} = 99.9\%$, the market factor satisfies $N(\bar{Z}) = 0.1\%$ or equivalently $-\bar{Z} = N^{-1}(99.9\%)$. Hence,

$$\mathbb{P}|_{VaR} = N\left(\frac{N^{-1}(PD) + \beta N^{-1}(99.9\%) }{\sqrt{1 - \beta^2}}\right).$$

Plugging $\mathbb{P}|_{VaR}$ back in (22.1) for K , we obtain the desired result. In Basel II, the capital requirement is 8% of the RWA (Risk Weighted Asset), i.e. $EAD \times K = 8\% \times RWA$ which means that the capital requirement is 8% of the total risk weighted assets. Therefore, we have

$$\begin{aligned} RWA = EAD \times LGD \times & \left[N\left(\frac{N^{-1}(PD) + \beta N^{-1}(99.9\%) }{\sqrt{1 - \beta^2}}\right) - PD \right] \\ & \times \frac{1 + (M - 2.5)b(PD)}{1 - 1.5b(PD)} \times 12.5. \end{aligned}$$

The supervisory asset correlation function depends on the type of exposure and is given by BCBS:

- corporate, bank or sovereign with the size of S million Euros:

$$\beta^2 = 0.12\lambda_1 + 0.24(1 - \lambda_1) + SizeAdj(S).$$

$$\text{where } \lambda_1 = \frac{1 - e^{-50PD}}{1 - e^{-50}} \text{ and } SizeAdj(S) = \begin{cases} 0, & S > 50, \\ -0.04 \times \left(1 - \frac{S-5}{45}\right), & S \in [5, 50], \\ -0.04, & S < 5. \end{cases}$$

- residential mortgage: $\beta^2 = 0.15$
- qualifying revolving retail: $\beta^2 = 0.04$
- other retail: $\beta^2 = 0.03\lambda_2 + 0.16(1 - \lambda_2)$, $\lambda_2 = \frac{1 - e^{-35PD}}{1 - e^{-35}}$.

Exercises

1. Consider Merton's structural model given by

$$dA_t = rA_t dt + \sigma_A A_t d\tilde{W}_t$$

and let $S_t = A_t - D_t(T)$ where S_t represents the equity value and $D_t(T)$ the debt. Show that the asset volatility σ_A and the equity volatility σ_S are related by

$$\sigma_S = \sigma_A N(d_1) \frac{A_t}{S_t}, \quad d_1 = \frac{\ln \frac{A_t}{L} + (r + \frac{1}{2}\sigma_A^2)(T - t)}{\sigma_A \sqrt{T - t}}.$$

2. Derive the survival probability by time T , i.e. $\tilde{\mathbb{P}}\{\tau > T\}$, when the intensity follows a stochastic CIR process as follows:

$$d\lambda_t = (b - a\lambda_t)dt + \sigma\sqrt{\lambda_t}d\tilde{W}_t,$$

in which it is assumed that the intensity is positive.

3. Calculate the PFE with $\alpha = 97.5\%$ for a long equity put option in which the asset follows the SDE as

$$dS_t = (r - q)S_t dt + \sigma S_t d\tilde{W}_t.$$

Chapter 23

Commodity Modelling



From the supply-and-demand relation, the commodity futures price $F_t(T)$ with maturity T at time t is

$$S_t e^{(r+y)(T-t)},$$

where y is a number reflecting *both* convenience yield and storage costs so it can be positive or negative. For simplicity, we still call it convenience yield in modelling.

One direct way for commodity modelling is to assume that the spot price follows the geometric Brownian motion under the risk-neutral measure

$$dS_t = (r + y)S_t dt + \sigma S_t d\tilde{W}_t$$

and direct calculation leads to

$$dF_t(T) = \sigma F_t(T) d\tilde{W}_t,$$

which satisfies the fact that the futures price $F_t(T)$ is a martingale. It is used for European options which depend mainly on final distributions. However, for path-dependent or some other exotic options, more advanced models may be required for capturing the mean reverting property of some of the commodities as well as the volatility skew.

Gabillon Spot Model

Gabillon model [33], originally proposed for oil futures, is one of the most popular models for commodity derivatives. Define the long-term price of a commodity as the price for delivery at infinite time. Under the risk-neutral probability measure, the spot price S_t is modelled along with the long term price process L_t as

$$\begin{cases} \frac{dS_t}{S_t} = (r_t + y_t)dt + \sigma_t^S d\tilde{W}_t^S, \\ \frac{dL_t}{L_t} = \sigma_t^L d\tilde{W}_t^L, \end{cases}$$

where r_t is the short rate, σ_t^S and σ_t^L are corresponding volatilities, and $d\tilde{W}_t^S d\tilde{W}_t^L = \rho dt$. More importantly, the convenience yield y_t is totally determined by S_t and L_t as

$$y_t = \kappa_t \ln(L_t/S_t) + \delta_t,$$

where κ_t is a positive process representing the speed of mean reversion and δ_t is an extra term.

Gabillon Futures Model

Now, we show how to derive the model for futures $F_t(T)$ from the Gabillon Spot Model. Applying the Itô's formula with the functions $\ln S_t$ and $\ln L_t$ on the Gabillon spot model, we obtain the following two SDEs:

$$\begin{cases} d \ln S_t = \left[r_t + y_t - \frac{1}{2}(\sigma_t^S)^2 \right] dt + \sigma_t^S d\tilde{W}_t^S \\ d \ln L_t = -\frac{1}{2}(\sigma_t^L)^2 dt + \sigma_t^L d\tilde{W}_t^L. \end{cases}$$

Denoting the quotient $X_t = S_t/L_t$ (such that $\ln X_t = \ln S_t - \ln L_t$) and plugging in the convenience term y_t , we obtain

$$d(\ln X_t) = (a_t - \kappa_t \ln X_t)dt + \sigma_t^X d\tilde{W}_t^X,$$

where $a_t = r_t + \delta_t + \frac{1}{2}[(\sigma_t^L)^2 - (\sigma_t^S)^2]$ and $\sigma_t^X d\tilde{W}_t^X = \sigma_t^S d\tilde{W}_t^S - \sigma_t^L d\tilde{W}_t^L$.

We see that $\ln X_t$ is an OU process that can be solved with the usual method. Let $K_t(T) = e^{-\int_t^T \kappa_s ds}$. Then,

$$d(K_t(T) \ln X_t) = K_t(T)(a_t dt + \sigma_t^X d\tilde{W}_t^X)$$

$$\ln X_T = K_T(T) \ln X_t + \int_t^T K_u(T)(a_u du + \sigma_u^X d\tilde{W}_u^X).$$

Putting $\ln L_T = \ln L_t - \frac{1}{2} \int_t^T (\sigma_u^L)^2 du + \int_t^T \sigma_u^L d\tilde{W}_u^L$ and the expression of $\tilde{W}_u^X$,

$$\begin{aligned} \ln S_T = & K_t(T) \ln S_t + (1 - K_t(T)) \ln L_t + \int_t^T [K_u(T) a_u - \frac{1}{2} (\sigma_u^L)^2] du \\ & + \int_t^T K_u(T) \sigma_u^S d\tilde{W}_u^S + \int_t^T (1 - K_u(T)) \sigma_u^L d\tilde{W}_u^L. \end{aligned}$$

In the following it is assumed that κ_t , σ_t^S and σ_t^L are deterministic. Then, S_T is log-normally distributed, and so is $S_T|\mathcal{F}_t$. From the property of log-normal distribution, we know that $\tilde{\mathbb{E}}[S_T|\mathcal{F}_t] = \exp\{\tilde{\mathbb{E}}[\ln S_T|\mathcal{F}_t] + \frac{1}{2} \tilde{\text{Var}}(\ln S_T|\mathcal{F}_t)\}$.

We have $\tilde{\mathbb{E}}[\ln S_T|\mathcal{F}_t] = A + \int_0^t K_u(T) \sigma_u^S d\tilde{W}_u^S + \int_0^t (1 - K_u(T)) \sigma_u^L d\tilde{W}_u^L$ because the expectations of the Brownian integrals from t to T are zero. A is a term that does not contain Brownians and needs not to be detailed here. The variance $\tilde{\text{Var}}(\ln S_T|\mathcal{F}_t) = B$ does not contain any Brownian terms either.

The futures price under the risk-neutral measure is given by $F_t(T) = \tilde{\mathbb{E}}[S_T|\mathcal{F}_t]$. Hence,

$$\begin{aligned} F_t(T) = & \tilde{\mathbb{E}}[S_T|\mathcal{F}_t] = \exp\left(A + \frac{1}{2} B\right) \\ & \exp\left(\int_0^t K_u(T) \sigma_u^S d\tilde{W}_u^S + \int_0^t (1 - K_u(T)) \sigma_u^L d\tilde{W}_u^L\right). \end{aligned}$$

Recall that futures price under the risk-neutral measure is a martingale (i.e. zero drift). Therefore, we obtain

$$\frac{dF_t(T)}{F_t(T)} = K_t(T) \sigma_t^S d\tilde{W}_t^S + (1 - K_t(T)) \sigma_t^L d\tilde{W}_t^L,$$

which is the Gabillon model for futures.

For modelling the skewed distribution for near expiries for different maturities T_i , we can add extra terms $\alpha_t^S(F)$ and $\alpha_t^L(F)$ which enhance and generalize the original Gabillon model:

$$\frac{dF_t(T_i)}{F_t(T_i)} = \alpha_t^S(F) K_t(T_i) \sigma_t^S d\tilde{W}_t^S + \alpha_t^L(F) (1 - K_t(T_i)) \sigma_t^L d\tilde{W}_t^L.$$

Here $\alpha_t^S(F)$ and $\alpha_t^L(F)$ can be calibrated from option prices.

Exercises

1. Consider Schwartz one-factor model [76] for commodity spot price

$$dS_t = (b - a \ln S_t) S_t dt + \sigma S_t d\tilde{W}_t,$$

where $a, b > 0$.

- (i) Show that $\ln S_t$ follows an OU process.
- (ii) Show that the futures price satisfies $dF_t(T) = \sigma e^{-a(T-t)} F_t(T) d\tilde{W}_t$, i.e., volatility decreases for longer maturity.

2. Consider the following Gabillon futures model

$$\frac{dF_t(T)}{F_t(T)} = K_t(T)\sigma^S d\tilde{W}_t^S + (1 - K_t(T))\sigma^L d\tilde{W}_t^L,$$

where $K_t(T) = e^{-\kappa(T-t)}$, $\sigma^S > 0$ and $\sigma^L > 0$.

- (i) Derive the volatility of $F_t(T)$.
- (ii) Given that $\kappa = 0.5$, $\sigma^S = 0.3$, $\sigma^L = 0.1$ and the short and long term prices are correlated by -0.4 . Plot the volatility term structure.

Part IV
Structured Products and Solutions

Chapter 24

Structured Products



24.1 Introduction

The direct investment in an asset offers linear and symmetric exposure to the performance of the asset. However, an investor's view on the market may be more complex than the directional perspective, e.g., performance cycles, relative performance of several assets, view on volatilities and correlations, and etc. The risk appetite and tolerance are different according to investors' profiles.

A structured product is a pre-packaged financial product for facilitating customized risk-return objectives based on the returns from certain investable assets. It is used as an alternative to direct investment. Structured products can offer the exposure for specific market views and desired risk profile under the constraints of financial budget and legal framework for investment. The derivatives hedging (or replication) technology benchmarked by Black–Scholes–Merton model has made it possible for banks to offer various structured products without warehousing the market risks. The activity of structured products has experienced a fantastic development since the late 1990s when the French banks pioneered the innovative equity-linked products to retail as well as institutional investors.

The major end investors of structured solutions are retail, high-net-worth (HNW) individuals and non-bank financial institutions (e.g. insurance or pension funds). The investment targets may be classified with the following categories:

- *Participation*: exposure to the performance of the underlying implemented with tracking/market access structures
- *Income*: enhanced yield with fixed, variable coupons or capped performance of the underlying
- *Protection*: prudent investment with no or limited potential loss
- *Leverage*: leveraged exposure to the performance of the underlying.

A structured product is composed of three building blocks, as described in the summary table below.

Asset: any investable asset with high liquidity and low transaction cost may be used as underlying asset

Payoff: it determines the return/risk of the product based on the performance of the underlying assets

Wrapper or Instrument: it is the legal form for the product.

Asset Class	- Equities - Interest Rate - Foreign Exchange - Commodities - Credit - Fund - Customized Index/Strategy -		
Payoff	Access Product - Direct Exposure - Leveraged Exposure	Option Based Structure - Principal Protected - Non Principal Protected	Dynamic Portfolio - Portfolio Insurance - Dynamic Asset Allocation
Wrapper	- Structured Note - Certificate - Structured Deposit - Structured Fund - Insurance Policy - Over-The-Counter (OTC) Option - Warrant - Swap - Islamic Shariah Compliant Product -		

24.2 Assets

The return of a structured product is linked to the performance of the underlying assets pre-defined in the contract. We give an overview of the investable assets used as underlyings in structured products.

Equity - Single Stock - Basket of Stocks - Equity Index -	Interest Rates - Libor Rate - Libor Swap Rate - OIS Rate - Futures -	Foreign Exchange - FX Spot Rate - Basket of Currencies -	Commodities - Futures - Index - Spot Price -
Credit - Corporate Bond - Corporate Bonds Index - Credit Index -	Fund - Mutual Fund - ETF - Hedge Fund - Managed Account -	Customized Index/Strategy - Smart Beta - Alternative Beta - Factors - Multi-Asset Strategy -	

Asset Return

The return rate of a financial asset for the period (s, t) , $s < t$ is defined as $r = P_t/P_s - 1$, where s, t are the indices for the time line and P_t is the price of the asset at time t . We recall that the usual measures for the rate of return include *price return* (dividend payments between (s, t) ignored), *total return* (all dividends re-invested back in the asset), and *excess return* (total return rate minus the effective financing cost, e.g. USD 3M Libor+0.30%). We refer to Chap. 1 for detailed explanation of the returns.

It is worth highlighting that, in general, the performance for structured products is measured from the start date of the product. Only in special cases the performance is not measured from the beginning but the explicit definition for the performance will be given for such products.

24.3 Payoff

There is a wide variety of payoffs and underlying assets for structured products. The categorization of structured products can be made in different ways. One categorization criteria is the principal protection level.

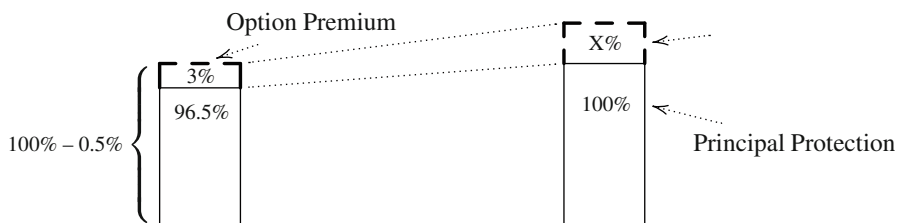
24.3.1 Principal Protected Products

Principal protection is an important feature for risk averse investors. It may be achieved by two types of techniques:

- Option based structures, which is also known as OBPI (Option Based Portfolio Insurance as opposed to CPPI)
- Dynamic asset allocation based structures, i.e. CPPI (Constant Proportion Portfolio Insurance) and TIPP (Time Invariant Portfolio Insurance) which are risk management techniques for adjusting investment exposure to mitigate gap risk and will be explained in Chap. 26.

The simplest option-based principal protected structure consists of spending the available premium (which is from the interest of the investment of the principal in fixed-income instrument) to buy a call option on the underlying asset. For example, a 1 year, 100% principal protected Note with notional amount of N may be structured by the issuer as follows:

- investing in a 1 year discount bond which will be redeemed at N . Assume that the upfront discount is $3.5\% \times N$
- assuming that $0.5\% \times N$ will be used for covering the issuance cost and the distribution cost, the available premium to spend will be $(3.5\% - 0.5\%) \times N = 3\% \times N$
- spending the available premium of $3\% \times N$ to buy a call option on the underlying with $PR = 37\%$ (*Participation Rate*, also known as *Gearing*)
- at maturity, the investor will receive $100\% \times N + PR \times N \times \max\{0, \frac{S_T}{S_0} - 1\}$.



24.3.2 Principal Non Protected Products

In a principal non protected product, the client takes certain risk on the performance of the underlying for a higher investment return. The most common risk taken by bullish investors consists of selling a put option on the underlying (cf. the analysis for Equity Linked Note in the next chapter as a typical example). The premium of the put option will be used to enhance the return. Other types of risk taking include range view, bearish view, combination of views on different assets, view on yield curve shape etc. which will be presented through the popular structures in the following chapter.

The maximum loss for the investor will depend on the structure: e.g. 100% of the invested amount for Note/Certificate/Warrant/long option, limited loss for a short put option but unlimited loss for a short call option or a Swap.

24.4 Wrapper or Instrument

The choice for the legal form of a structured product depends on the distribution channel, regulatory framework, the credit agreement between the counterparties, etc. The mainly used wrappers are as follows:

- *Structured Note*: a debt security, in Bearer or Registered form, which pays the interests and principal according to the performance of valuation assets (underlying assets) of the product
- *Certificate*: a registered security which entitles holder to payments linked to the performance of the underlying assets
- *Warrant*: it is a securitized option (could be vanilla or exotic option with European or American style); Warrants can be listed at the exchange (for retail investors) or issued under private placement (known as OTC Warrant). We note that the warrant as a structured product does not create dilution to the underlying stock because it involves only the secondary market shares.
- *Swap*: it helps to achieve leveraged return because the fund involved is the initial margin and variation margin. The initial margin is only return at maturity of the trade. The variation margins are exchanged during the life of the trade according to MtM of the trade over time. For example, under the ISDA/CSA agreement, the initial margin is called IA (Independent Amount) which is determined according to the volatility, liquidity and riskiness of the underlying assets (e.g. 15%). Swaps with regular settlements (or reset) have lower counterparty risk and require lower IA which implies higher leverage.
- *OTC Option*: it is a flexible instrument for options with customized features. The counterparties need to have a credit agreement in place before transacting OTC options, similarly to Swaps.
- *Structured Deposit*: offered by deposit taking institutions. Deposits are not transferable.
- *Structured Fund*: issued under the form of fund units which are not transferable
- *Insurance Policy*: structured product can be included in an insurance product. There exist transferable insurance policies.

Notes/Certificates/Warrants are transferable securities that are unsubordinated and unsecured obligations of the issuer and are ranked *pari passu* among themselves. The securities may be settled through a clearing house (e.g. Euroclear, Clearstream). DVP (Delivery Versus Payment) settlement helps to eliminate counterparty risk. For the investor, a structured product wrapped as a Note or a Certificate from the same issuer has no real difference.

Institutions normally transact Swaps and OTC options under an ISDA/CSA agreement (cf. Chap. 1, “OTC derivatives documentation and counterparty credit risk management”).

24.5 Risks of Structured Products

For investors, a structured product has the following risks in general:

- **Issuer's Credit Risk:** similar to any debt instrument, the credit-worthiness of the issuer is an important criteria;
- **Liquidity Risk:** the risk of not being able to early redeem the product at a reasonable price; For terminating a structured product before its maturity, the investor needs to “unwind” the product at the bid price (also known as secondary market bid price) from the product issuer/counterparty. The unwind price will include the transaction costs (e.g. 0.50% for equity structured products with short tenor);
- **Market Risk:** it is the risk from all the market factors (e.g. underlying's price or volatility) determining the price of the structured product.

Counterparty Credit Risk Management

For the securitized instruments such as note/certificate and warrants, the investor's worst loss is the total amount spent to buy the instrument. So, the issuer does not take any counterparty credit risk from the investors. However, the investors are bearing the credit risk of the issuer.

For OTC transactions between counterparties with an ISDA/CSA agreement, typically between the institutions, the collateral management is governed by CSA which contains, in particular, the type of acceptable collateral, the threshold and the MTA (Minimum Transfer Amount). Margin call will be triggered for one party if $(\text{Positive Net Exposure} - \text{Threshold} - \text{Collateral Held}) > \text{MTA}$. Let us take an example of a two-way ISDA/CSA agreement between party A and party B in which $\text{Threshold} = \text{USD}1\text{m}$, $\text{MTA} = \text{USD}500\text{k}$. In the situation where $\text{Net Exposure} = \text{USD}1.6\text{m}$ and $\text{Collateral Held} = 0$ for party A, a margin call will be triggered and the Margin amount will be $\text{USD}600\text{k}$ to be paid by party B.

Financial institutions, e.g. a private bank, may offer credit facilities to their clients. The credit risk management is often based on deed of charge instead of ISDA/CSA, in which the client's assets are used as collateral. The credit risk monitoring is based on the below quantities in general:

Add-On: specific to each trade for cushioning the forced unwind, it is the estimated loss for the institution from the trade under stressed scenario (e.g. two standard deviations of the price change) during a certain period of time (e.g. two weeks) that is necessary for unwinding the trade.

Negative MtM: it is the absolute value of the total negative mark-to-market value of all the trades of the client. The positive mark-to-market values are often neglected in the calculation which is a common conservative approach in practice.

Whenever $(\text{Add-On} + \text{Negative MtM}) > \text{Held Collateral}$, a margin call will be raised to the client for settling the difference.

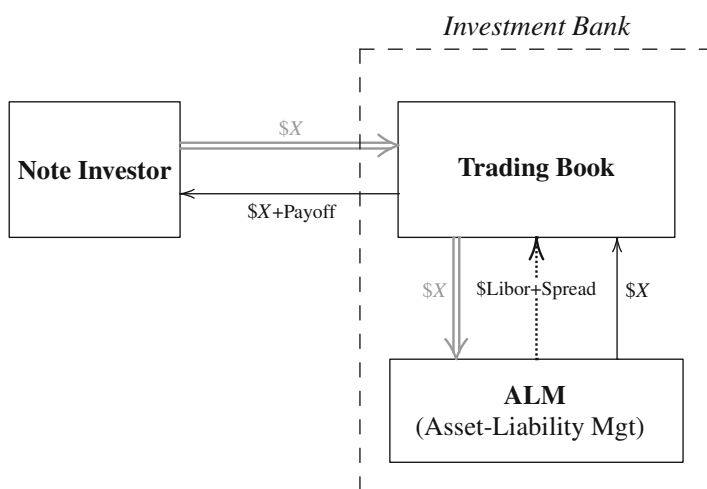
Netting of the exposure of the trades with the same counterparty/client will be dependent on the mutual agreement and applicable regulation.

Investment Financing

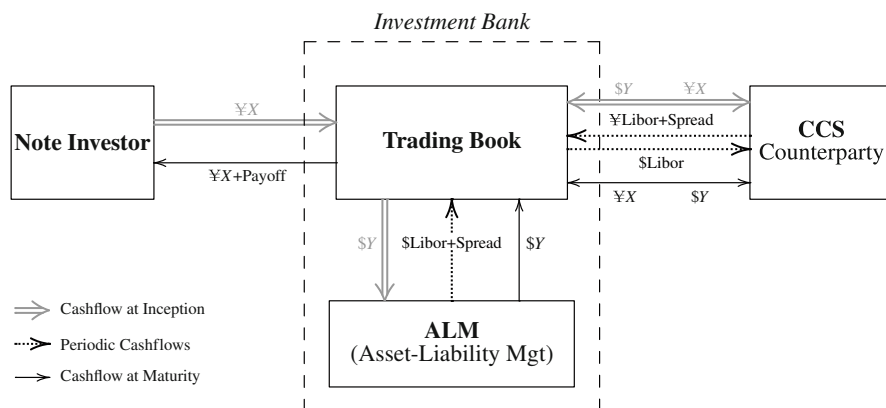
It is common for the private banks to allow their clients to make investment with leverage (e.g. investment in bonds, stocks or structured products). The leverage for a certain investment depends on the volatility of its MtM. The financing is applied by means of a *credit facility* on the whole portfolio of the client, in which, each investment is given three ratios: *Financing Ratio (FR)*, *Margin Call Ratio (MR)* and *Liquidation Ratio (LR)* (e.g. 70%/75%/80%). The bank may apply zero *FR* to the investments that the bank is not willing to finance. The cash in portfolio's currency is considered to have 100% *FR* for generalizing the calculation. The limit of the credit facility is calculated as $\sum_i MtM_i \times FR_i$. The client has the right to borrow from the bank up to the limit of the credit facility. Similarly, margin call level and liquidation level can be calculated from *MR* and *LR*. When the financing from the bank is greater than the margin call level, the bank will issue a margin call. If the financing from the bank is greater than the liquidation level, the bank may start to unwind/liquidate some of the investments of the client.

24.6 Funding Management for Structured Notes

The proceeds of structured notes are typically managed by the ALM (Asset Liability Management)/Corporate Treasury of the issuer. Trading book receives the interests of the proceeds from ALM. The following diagram illustrates the typical funding management for structured notes.



ALM manages only the domestic currency and eventually some other major currencies (e.g. USD). If the Note denomination currency is not managed by ALM, trading needs to make a CCS (Cross Currency Swap) for converting the currency into a major currency acceptable by ALM.



24.7 Back-Testing

Back-testing (also known as historical simulation) consists of simulating the return of the structured product by applying the real historical prices of the underlying. People try to use the time series data from the historical beginning (or from a meaningful start date if the data history is too long to be relevant). The backtesting process is described as follows:

- For the first date of the considered period, a back-test is realized by assuming that the structured product has started at that date. One simulated return of the product is obtained.
- Then, the following date will be chosen as the start date for the structured product and another simulated return will be obtained.
- This process is repeated until the expiry date of the product becomes the current date (in which case, the process cannot continue any further).

A distribution is hence obtained with all the simulated returns. It is customary to show the following back-testing results:

- Average IRR (Internal Rate of Return), Worst IRR, Best IRR
- Return distribution chart illustrating frequencies for return intervals.

In the product marketing materials, it is often highlighted that “past performance is not a good indication for future performances”. Unfortunately, this statement proved to be true with real life cases.

24.8 Business Organization

The structured product issuers are mainly investment banks which have the necessary resources: technical skills, systems, risk management capabilities and client base. There are a number of teams involved in the structured product activities.

The Front-Office teams normally refer to

Sales: advising and servicing clients with the suitable products

Structuring (or *Financial Engineering*): product solution and pricing

Trading: managing the risk of book of all clients' trades.

Other functions may include

Quantitative modelling: pricing and hedging systems

Risk: market risk and credit risk monitoring

Back-office: confirmation, settlement, reporting, and etc.

IT: systems for front-office, back-office, and etc.

24.9 Payoff Structuring Techniques

The payoff determines how the return is calculated from the performance of the assets. A rich set of payoffs is being offered in the market for meeting the targets of investors with specific market view and risk profile. The goal of payoff design consists of using the market view to offer the target return/risk profile with the price constraint of investors. For minimizing the cost, the payoff needs to offer attractive return when the market view is realized but gives little return or even suffers loss for other market conditions. Therefore, a straddle (long call plus long put) may not be a tempting payoff because the cost is too high. A structured product called "Twin-Win" that will be introduced in the next chapter is a more common form of straddle with an embedded risk-taking component which cheapens the price.

For a bullish structure in general, lower forward of the underlying leads to a lower price. So, the underlying selection will favor the ones with high dividend, high repo rate and low correlation for worst-of basket (if applicable). The denomination currency of the underlying may matter if it is different to the currency of the product. The spread of cross currency swap market and/or the correlation FX/underlying contribute to lower forward as well.

According to the investor's view and the budget for the product, the following technical features are often employed in the payoff design:

1. Basket Performance Variations

- Worst-of Basket Performance: the most popular multi-underlying structures are based on the worst performance of the underlyings due to the low cost. The features of a product based on worst-of basket often look attractive which may provide a comfortable protection cushion and/or high potential return.

On the other hand, the cost of a “Best-of” based bullish product is normally too high to get investors interested.

- Rainbow Basket Performance: for multi-underlying structures, the *Rainbow* performance is measured as the weighted average of the performances according to the performance ranking, e.g. 70% of the worst, 30% of the second worst. Worst-of performance is a special case of Rainbow performance.

2. Cap on the Return

The binary option is the basic version of products with capped return. It has the clear return target for the investor. This feature helps to reduce the cost of the structure. For the multi-underlying structures, the Cap may be applied to the averaged performance, to the best performance or to the performance of each underlying (known as *Individual Cap*, which is an effective mechanism for reducing cost).

3. Barrier

The barrier feature allows to achieve different purposes:

- providing protection: For example, short position of Down-and-In Put is often seen in the structured products where the investor takes downside risk to fund the return on the upside. The barrier protects the investor until the barrier level.
- price cheapening: If the barrier event reduces the potential gain or even incurring loss to the investor, the price of the structure will be cheapened consequently. In the product design, the barrier is set at the level which is unlikely to achieve according to the market view of the investor. One simple example is an Up-and-Out call option.
- return lock-in: The barrier applied over a period of time, either continuously or on discrete points, allows the investor to lock in the return over time. Binary options or one touch options are simple examples.

The observation for the barrier event can be of American style (sometimes noted as AKI – American Knock-In) or European style (sometimes noted as EKI – European Knock-In, or AXKI – At Expiry Knock-In). For basket based structure, the barrier is often observed on the worst-of performance. The best-of performance may be applied for range-bound products or bearish products.

Apart from the barrier event observed on specific underlying’s price, there are other types of barrier events:

- Target Redemption: the barrier event is considered to occur when the accumulated gain for the investor has achieved a certain predefined target. A variation consists of counting the accumulated number of times that the intrinsic value has been positive instead of the accumulated gain.
- Memory Touch: for the considered period, the barrier event is deemed to occur if the barrier has been hit by every underlying in a basket but not necessarily at the same time. This feature is often used as an additional condition for locking in a benefit for the investor.

4. Averaging

It consists of using the averaged price as reference for the payoff calculation. The volatility of the averaged price is lower than the volatility of price itself (e.g. daily closing price). Averaging makes the price cheaper for products selling volatilities. For the investors, the averaging mitigates the risk of abnormal price moves.

5. American Option Feature

The American option gives the termination right to the option holder. When the right is given to the product issuer, it is known as *Issuer Callable* structure. If the right is given to the investor, the structure is then known as *Puttable*. In the case of issuer callable structure, the premium from the call option is used for enhancing the product return.

24.10 Pricing and Hedging

The standard model for equity-linked products is the local volatility model which is used for the majority of the products. For interest rate products requiring term structure modelling, LGM model with one or two factors and Libor Market Model are the most popular ones. But models with more factors are needed for the products with more complex risks. For foreign exchange rate linked structures, stochastic volatility and local volatility models are frequently used in the market. The parameters for each model should be calibrated with the liquid market traded instruments. The hedging costs are considered in pricing by shifting the parameters on the conservative side.

For multi-asset products, Monte Carlo is the usual numerical method which offers the flexibility for payoff variations. For single underlying products, PDE and tree methods are more efficient.

The nature of structured products often introduces gap risk and correlation risk that are difficult or impossible to hedge. The techniques for mitigating these risks are explained in Chap. 16 covering exotic options in Part II.

Exercises

1. Consider the simplest option-based principal protected structure with a long call option. How will a low interest rate environment affect such a structure?
2. Suppose you are structuring a near-zero-cost risk reversal (short OTM put with strike K_1 and long OTM call with strike K_2 with $K_1 < K_2$) type of product for a potential client. The client has a view on the underlying and thinks that the put strike K_1 can be moved further away from spot, say, to be reduced by half. How would you modify the risk reversal payoff?
3. Describe the relationship between the price of a worst-of call option and its component asset volatilities.

Chapter 25

Popular Structured Products



With the demand from investors and the increasing hedging capability of the investment banks, new structured products keep being introduced to the market. Some of the most popular structures in practice are presented in the sequel for illustrating the limitless payoff variations. The product categorization is difficult to make because the same structure may be used for different asset classes and may offer yield enhancement, participation or leverage depending on the parameter specifications.

We choose to present the popular structures by the underlying asset class instead of technical feature, return profile or market view. The categorization by asset class follows the organization of business lines in practice. The naming of the products is not standardized across the issuers. There are product names suggesting technical features (e.g. Fixed Coupon Callable Note) and product names used for marketing campaigns (e.g. Himalaya, Athena, etc). It is common that the same structure is named differently by the different issuers.

We highlight that the structures presented in one asset class may be applicable to the other asset classes. For example, Target Redemption mechanism based products presented for FX underlyings can be applied to equities as well.

25.1 Equity Structured Products

Equity structured products offer the most diversified product types and solutions. Equity is a preferred asset class for retail investors who often invest in equity linked products with their views on particular stocks or stock market. On the other hand, it is relatively easier to structure product variations with equities due to the vast universe for the underlying as well as the implementation flexibility for hedging strategies.

25.1.1 Equity Linked Note (ELN) or Reverse Convertible Note

The Equity Linked Note (ELN, also known as Reverse Convertible Note) is one of the most popular wealth management products, especially for private investors. It offers yield enhancement by taking downside risk on equities. The ELN is often a short term product (typically, 1 month – 6 months), issued at a discount to the face value which implies the return of the investment. At maturity,

- if the underlying stock price has not dropped beyond a certain level (called Strike) which is set at a lower level than its initial price at inception, the investor will receive 100% of the face value of the Note;
- Otherwise, the investor will receive a number of stocks determined by the Strike price of the stock (physical settlement). Cash settlement with the equivalent value may also be applied to the trade.

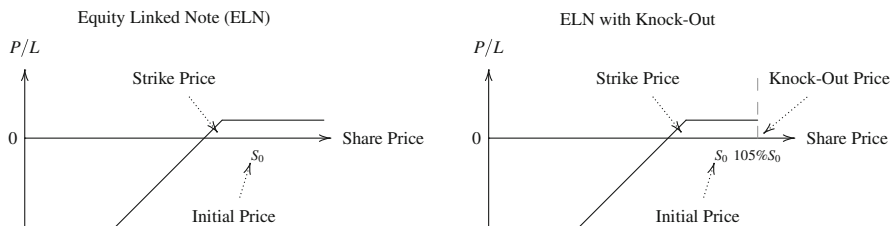
ELN is suitable for stable or bullish view of the underlying stock for the tenor of the trade. The investor enjoys an attractive return if the stock does not underperform significantly. In case that the investor will be delivered with stocks, the purchase price will be the Strike price which is lower than the initial stock price, implying a better return compared to the direct investment for the same period. On the risk hand, the investor will be obliged to buy the stocks at the Strike level even though the stock price dropped below this level. See Table 25.1 for an illustration.

Table 25.1 Illustration example of ELN (Equity Linked Note)

Issuer	XYZ bank
Tenor	1 Month
Denomination (N)	HKD 1 m
Issue price	$N \times (100\% - \text{Discount})$
Discount	0.85%
Underlying stock	[●]
Initial stock price (S_0)	HKD 102.10
Strike (K)	93%
Maturity payoff	If final stock price (S_T) $> K \times S_0$, Investor will receive $100\% \times N$; Otherwise, Investor will receive $N / (K \times S_0)$ stocks

If the currency of the stock (Ccy^S) is not the same currency for the Note (Ccy^N), the final payoff wording will be the following which involves the foreign exchange rate X_T (the number of Ccy^S for 1 unit of Ccy^N at time T) for calculating the number of stocks to be received by the investor:

Maturity payoff	If final stock price (S_T) $> K \times S_0$, Investor will receive $100\% \times N$; Otherwise, Investor will receive $X_T \times N / (K \times S_0)$ stocks
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A simple variation of the ELN, called KO ELN, consists of introducing an Up-and-Out Barrier (e.g. $105\% S_0$) which is typically observed on a daily basis. Upon Knock-Out event, the structure will be early terminated at par (i.e. 100% of Denomination).

We show that, in the basic version of ELN, the investor is selling a geared Put. Firstly, we note that

$$S_T \times \frac{N}{K \times S_0} = N \left(1 + \frac{1}{K} \times \frac{S_T}{S_0} - 1 \right) = N \left[1 - \frac{1}{K} \times \left(K - \frac{S_T}{S_0} \right) \right].$$

Now, denote the investor's final redemption value as $N \left[1 - \frac{1}{K} F(S_T, K) \right]$. According to the terms of the ELN contract, we have

$$F(S_T, K) = \begin{cases} 0, & \text{if } \frac{S_T}{S_0} \geq K, \\ K - \frac{S_T}{S_0}, & \text{if } \frac{S_T}{S_0} < K. \end{cases}$$

Hence, $F(S_T, K) = \max\{0, K - \frac{S_T}{S_0}\}$ which is the Put Option payoff with Strike K , and $\frac{1}{K}$ is the “gearing” term in front of the Put Option in the final redemption formula.

The same analysis can be made when the currency of the share (Ccy^S) is different than the currency for the Note (Ccy^N). The value of the stocks received by the investor will be $S_T \times [X_T \times N / (K \times S_0)]$ in Ccy^S , which is equivalent to $(S_T \times X_T \times N / (K \times S_0)) / X_T = N \left[1 - \frac{1}{K} \left(K - \frac{S_T}{S_0} \right) \right]$ in Ccy^N . We note that the Put Option is a Quanto Option in this case.

The Put Option can be priced under the Black–Scholes–Merton model with the implied volatility for the Strike. The local volatility model is normally used for pricing the variations. For the ELN structure, the investor is short volatility, long forward (long forward implies long interest rate, short dividend and repo rate). So, higher the volatility and lower the forward, lower the issue price of the product.

25.1.2 Fixed Coupon Callable Note (FCN) and Daily Accrual Callable (DAC)

Fixed Coupon Callable Note (FCN) is gaining increasing popularity as a yield enhancement product, particularly in low volatility market environment where it is difficult for ELNs to show attractive yield levels. Compared to ELN, the tenor is longer for FCN (typically 6M – 2Y). It offers fixed coupons that are paid periodically to investors. The underlying is typically the worst performer of a basket of two or three equity underlyings. There is an early termination feature: if the underlying's price is above the Knock-Out barrier at any of the observation dates, the note will be early redeemed at par (i.e. Denomination or its face value). At maturity (if not early terminated),

- if the underlying has not underperformed beyond a certain level (called Strike), the investor will redeem the note at par.
- Otherwise, the investor will receive the worst performing stock. The quantity will be determined by the Strike price.

See Table 25.2 for an illustration example of FCN.

Table 25.2 Illustration example of FCN (Fixed Coupon Callable Note)

Issuer	XYZ bank
Tenor	12 Months
Denomination (N)	HKD 1 m
Issue price	$100\% \times N$
Coupon	12% p.a. payable quarterly
Observations	Quarterly
Underlying stocks	S^1 and S^2
Underlying performance at time t (P_t)	$P_t = \min \left\{ \frac{S_t^1}{S_0^1}, \frac{S_t^2}{S_0^2} \right\}$ which is the worst of basket
Initial stock price	To be fixed at the inception of the trade
Strike level (K)	70%
Knock-Out barrier	100%
Early termination	If the stock price has breached Knock-Out barrier at any observation date, the Note will be early redeemed at par
Maturity payoff (if not early terminated)	If $P_T > K$, Investor will receive $100\% \times N$; Otherwise, Investor will receive $N/(K \times S_0^*)$ stocks of S^* (S^* is the worst performing stock at T) In case the currency of S^* is different than the currency of the note, the number of stocks will be $X_T \times N/(K \times S_0^*)$

Technically speaking, the perfect underlying for achieving high Coupon with low Strike is a basket of stocks with high volatility, low correlation and high dividend yield and repo rate. Evidently, increasing the number of stocks in the underlying

basket leads to a higher coupon level. The correlations between the foreign exchange rate and underlying stocks impact also the conditions of the structure. Readers can refer to Chap. 18 “Foreign Exchange Modelling”, Proposition 18.2 for the correlation impact on the drift, or “Quanto forward”.

Further variations may be introduced for satisfying the investor’s needs:

- applying ATM Strike for the Put but combined with a Knock-In barrier which provides a cushion for the downside risk
- Knock-Out barrier stepping down over time for making the early termination easier (e.g. 100% for the first month, 99% for the second month. . .)
- Coupon based on floating rate, e.g. 3M Libor + 10% p.a., and etc.

For investors who need to pay regular interests on a financing/borrow, FCN may be used to generate the regular cashflows. For instance, clients of private banks may use credit facilities (collateralized by clients’ assets) for making certain investment. The interests are charged on the loan obtained by the clients.

Daily (Range) Accrual Callable Note (DAC) is a variation of FCN in which, instead of paying a predefined coupon, the coupon payable at the end of each period (e.g. i -th period) is determined according to the below formula

$$Coupon_i = R\% \times \frac{n_i}{N_i} \times t_i$$

where

$R\%$: nominal annual interest rate (e.g. 15% p.a.)

n_i : number of days in the period that the underlying performance $P_{t_i} \geq Coupon Barrier$; For simplification, *Coupon Barrier* is often set at *Strike level*

N_i : total number of days in the period

t_i : year fraction for the period

25.1.3 Phoenix Callable Note

Phoenix Callable Note takes the same downside risk as FCN or DAC. On the earning side, the investor will receive either a higher coupon or a lower coupon (often set at zero) at the end of each period according to the performance of the underlying. If the performance is at or above the *Coupon Barrier* (e.g. 80%), the investor will receive the higher coupon. Otherwise, the investor will receive the lower coupon for that period. Compared to FCN, the higher coupon level is higher because of the condition to get the coupon. The pricing and hedging for Phoenix Callable Note is similar to FCN, except that, due to the digital feature of coupon, payoff smoothing needs to be applied on the coupon.

Table 25.3 contains an illustration example of Phoenix Callable Note.

Table 25.3 Illustration example of Phoenix Callable Note

Issuer	XYZ bank
Tenor	12 Months
Denomination (N)	HKD 1 m
Issue price	$100\% \times N$
Observations	Quarterly
Underlying stocks	S^1 and S^2
Underlying performance at time t (P_t)	$P_t = \min \left\{ \frac{S_t^1}{S_0^1}, \frac{S_t^2}{S_0^2} \right\}$ which is the worst of basket
Initial stock price	To be fixed at the inception of the trade
Strike level (K)	70%
Coupon barrier	80%
Coupon (for the i -th period)	18% (Higher coupon), if $P_{t_i} \geq \text{Coupon Barrier}$; 2% (Lower coupon), otherwise
Knock-Out barrier	100%
Early termination	If the stock price has breached Knock-Out barrier at any observation date, the Note will be early redeemed at par
Maturity payoff (if not early terminated)	If $P_T > K$, Investor will receive $100\% \times N$; Otherwise, Investor will receive $N/(K \times S_0^*)$ stocks of S^* (S^* is the worst performing stock at T) In case the currency of S^* is different from the currency of the note, the number of stocks will be $X_T \times N/(K \times S_0^*)$

25.1.4 Autocallable Structure

Autocallable products are very popular to retail investors, especially in South Korea (usually with a basket of two underlyings), in Japan (mostly with one underlying and is known as Uridashi for the product name) and in Europe (called Athena). It can be structured as a Note, Certificate or a Swap. The Swap format is used for the two institutions where one is the hedging party and the other one the distributor who sells Notes/Certificates to the end investors.

The Autocallable structure is an income product which may be early terminated if the underlying price satisfies a certain condition predefined in the contract, e.g. when an Up-and-Out barrier is hit. Upon the termination event, the investor will receive the principal plus a coupon equal to $n \times \text{Coupon Rate}$ where n is the number of observation periods until the termination event. This type of coupon is known as a *Snowball Coupon*. On the other side, the investor takes the downside risk with a short Put (often with a Knock-In barrier).

We note that the word autocallable implies that the early termination is triggered by the performance of the underlying. It marks the difference with the other types of early termination determined at the discretion of the product issuer (known as Issuer Callable) or the investor (known as Puttable). Actually, FCN and DAC are also autocallable products. The autocallable feature helps to lock in the gain and eliminate the risk for the investor. At the same time, it cheapens the cost of the structure because it avoids the potential bigger gain which grows over time with Snowball effect if the underlying performs well. See Table 25.4 for an example.

Table 25.4 Illustration example of an Autocallable Swap

Structure	Swap
Notional	KRW20bn
Tenor	3 years
Underlying	Worst performer of KOSPI200 Index and HSCEI Index
Funding	KRW CD Rate ^a + 0.15% p.a. quarterly
Observation	Every 6 months
Knock-Out barrier	90/90/85/85/80/80, in % of initial index levels (stepping down every 6 months)
KO coupon	6.5% p.a. if Knocked-Out
Put strike	100% (ATM)
Knock-In barrier	60%, daily observation on closing prices
Bonus coupon^b	$3 \times 6.5\%$, if not Knocked-In and not Knocked-Out until maturity

^a It is a floating interest rate in the money market of South Korea

^b This feature is optional. Careful readers may notice that the Autocallable product with Bonus Coupon feature offers a similar return to investors as an FCN under most of the scenarios of underlying performance

The hedging of Autocallable structures is tricky because of the *volatility convexity* behaviour which may make the product hedger buy volatility when its level rises and sell volatility when it falls. The local volatility model is widely used for pricing and hedging autocallable structures. But stochastic volatility models better capture the risks from the volatility variability. The price discontinuity of the structure due to the barriers may be smoothed out by the techniques of barrier shift and call/put spread presented in Chap. 16, “Exotic Options Pricing and Hedging”. As to the numerical method, PDE or tree method are the efficient ones for the structures with single underlying. Monte Carlo is universal for pricing all autocallable structures, especially those with a basket underlying.

25.1.5 Issuer Callable Structure

Issuer Callable Structure means that the issuer of the product has the right to early terminate the trade at one of the predefined dates or within the predefined time windows. Technically speaking, the return for the investor is enhanced by

the “option value” of the early termination right given (or, more precisely, sold) to the product issuer. The callable structures are usually designed in a way such that the early termination event provides an attractive benefit to the investors. The early termination feature is generally perceived by the investors as a lock-in mechanism for the gain. The drawback of the issuer callable product to the investors is the uncertainty as to when the product will be called (i.e. early terminated). See Table 25.5 for an illustration.

The structure where the call right is given to the investor instead of the issuer is known as a *Puttable* structure. This type of products is less common in the market because the call right given to the investor makes the product “more expensive” leading to reduced returns. As the investors normally determine the early termination/continuation according to their views on the market instead of the optimal option exercise time calculated by a pricing tool, the termination/continuation decision is often not optimal as seen by the hedging party, i.e. the product issuer. So, the hedging party is likely to realize hedging profit for such products.

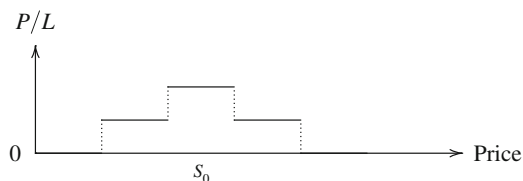
The pricing of multi-asset callable structures requires American Monte Carlo algorithms. The Least Square American Monte Carlo method proposed by Longstaff and Schwartz is widely used in practice (cf. Chap. 15 “American Options”). We highlight that this method gives only the lower bound of the optionality value which is conservative for the issuer callable products because it implies that the issuer “buys” the call option at a lower price. However, for Puttable structures, this method may lead to a too aggressive pricing for the product issuer due to the potential underestimation of the option value that the issuer “sells” to the investor.

Table 25.5 Illustration example of Issuer Callable Note

Structure	Note
Notional	TWD800m
Tenor	2 Years
Underlying performance	Worst performance of 5 stocks
High coupon	10% p.a.
Low coupon	2% p.a.
Coupon payment	Half yearly
<i>First coupon date</i>	High coupon is guaranteed
<i>From second coupon date onward</i>	→ High Coupon: if no stock has ever traded below Barrier in the period → Low Coupon: otherwise
Barrier	65%
Final redemption	→ 100%: if Barrier has never been hit (daily monitoring) → $\text{Min}\{100\%, \text{Final Price/Initial Price}\}$: otherwise
Issuer call	Every 6 months, Issuer has the right to early terminate the Note at par

25.1.6 *Wedding Cake*

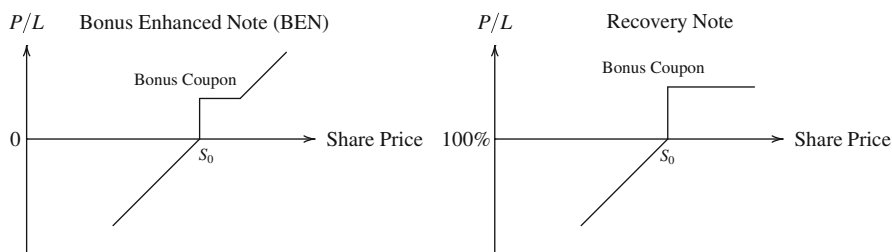
This payoff is suitable for investors with a range view. The investor will receive a payoff contingent to the scale of the price change for the observation period. The observation may be at the end of the period (known as European style) or monitored over the whole period (known as American style). The underlying for the product may be a single underlying or a basket of underlyings with observations on the worst and the best performances of the basket. The performances are measured against the initial prices at inception. This product is not based on a directional view. The investor enjoys the absolute return if the price change is moderate. The risk is the potential low return if the price has moved with a larger scale than expected.



For pricing, we may use the same pricing model for standard European or American Binary options. The discontinuity smoothing needs to be applied. Obviously, the structure is sensitive to the volatility of the underlying.

25.1.7 *Bonus Enhanced Note (BEN)*

Bonus Enhanced Note offers yield enhancement or participation according to the performance of the underlying. It is a bullish product offering unlimited upside on the *price return* of the underlying as shown in the following graph:



Actually, the BEN needs to be structured with stocks showing low forward (high dividend and repo rate). Technically, the investor forgoes the dividend and the security lending or repo fees for the benefit shown by the (diagonal) triangle in the graphic. The tenor should include the next ex-dividend date. See Table 25.6 for an example.

Table 25.6 Illustration example of Bonus Enhanced Note

Structure	Note
Notional (N)	USD1m
Tenor	3 months
Underlying	A dividend paying stock
Coupon	6.5% flat (i.e. 26% p.a.)
Strike (K)	100%
Final redemption	If $\frac{S_T}{S_0} \geq K$, Investor will receive $N \times \max \left\{ 1 + \text{Coupon}, \frac{S_T}{S_0} \right\}$; Otherwise, Investor will receive a number of $\frac{N}{K \times S_0}$ stocks

If a worst-of basket is used as underlying, it may be possible to have the below variations:

- a floor for limiting the downside risk, or
- a Knock-In barrier for the Put

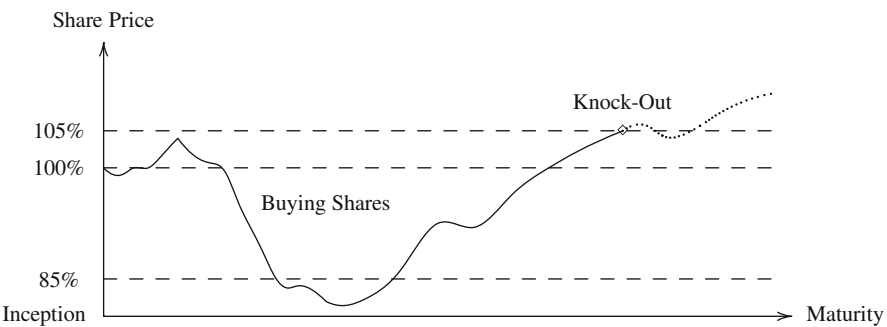
Another variation, known as *Recovery Note*, consists of offering a digital payoff as shown in the graphic. As suggested by its name, it is often proposed to the investors who have suffered loss from the holding of the underlying stock and wish to recover the loss if the price of the underlying stock will finish above the Strike at maturity. In such case, the investor should sell the stocks and buy the Recovery Note using the sale proceeds.

25.1.8 Accumulator

Accumulator is one of the most popular short term products for private investors in Asia. It also carries other names such as Knock-Out Forward, BBM (Buy Below the Market), etc. It is an OTC contract between the investor and the product issuer (e.g. a private bank) in which the investor buys, on a daily basis a certain number of shares at a fixed price, called Strike Price, which is at discount to the initial price (e.g. 80%) for the whole tenor of the contract. The shares are delivered to the investor periodically, e.g. every week or every month. If share price trades above a pre-defined Knock-Out barrier level (e.g. 105% of initial stock price) on any day, the investor will stop buying the shares and the trade is early terminated. The contract starts at the value of zero for both parties, similar to a Swap contract. However, the investors are normally required to post collateral for the transaction (e.g. 35% of total exposure) which is a general practice for the OTC transactions where the product issuer has credit risk exposure of the client. See Table 25.7 for an illustration.

Table 25.7 Illustration example of (non-leveraged) Accumulator

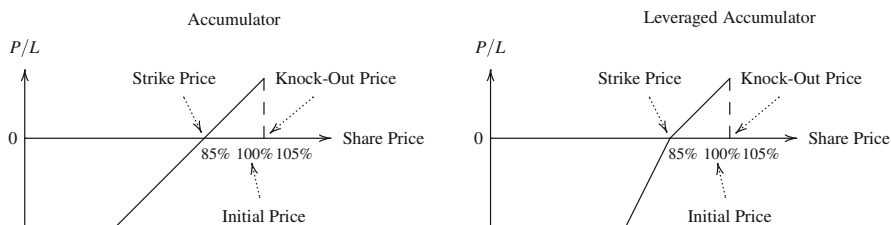
Maturity	52 weeks
Share name	[●]
Number of shares per day (<i>n</i>)	1,000
Knock-Out	105% of Initial Share Price, observed intraday If the share trades at or above 105% of the Initial Share Price at anytime, the trade will be terminated
Strike price	85% of Initial Share Price Client purchases <i>n</i> shares every day at Strike price
Settlement	Settlement at the end of every week
Termination	If no Knock-Out, the structure will terminate at the end of the Maturity



For achieving a lower Strike Price, investors often prefer to invest in the leveraged version of Accumulator in which the investor will buy a leveraged number of shares (e.g. two times) if the stock price is below the Strike Price on the considered date. For the above example, the terms may be modified as follows:

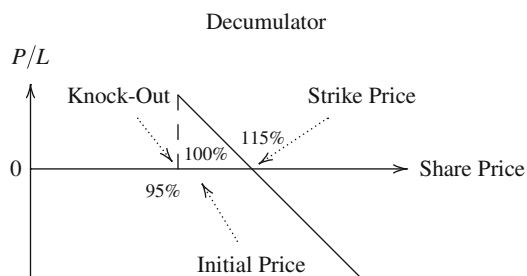
Strike price	85% of Initial Share Price
	For each business day, if the closing price of the share > Strike Price, Client purchases <i>n</i> shares at Strike Price; Otherwise, Client purchases $2 \times n$ shares at Strike price

Technically, Accumulator is a strip of call and put options with a common Up-and-Out barrier as illustrated in the graphic. In particular, the non-leveraged version is a strip of forward contracts with an Up-and-Out barrier. For the two times leveraged version, it is a strip of option combinations of long 1 call, short 2 puts with the same Up-and-Out barrier. The local volatility model is normally used for pricing the Accumulator. For the numerical methods, PDE or tree method are preferred to Monte Carlo which is less accurate and slower.



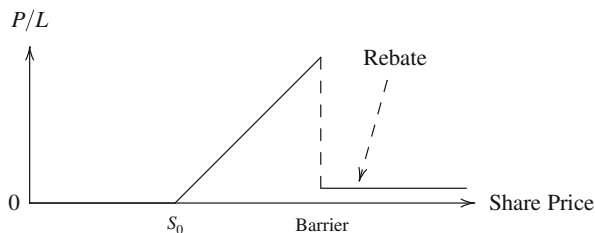
25.1.9 Decumulator

The opposite trade to Accumulator is the so-called Decumulator which consists of selling shares at premium to the initial share price. It is also known as Reverse Knock-Out Forward or SAM (Sell Above Market). Similar to the Accumulator, the investor will stop selling shares if the share price trades below a pre-defined Knock-Out barrier level on the downside.



25.1.10 Shark-Fin

Shark-Fin offers limited participation to the upside. Technically, Shark-Fin structure is just an Up-and-Out Call Option. The barrier observation is usually on continuous or daily basis. Because of its cheap price, Up-and-Out Call is often used to structure principal protected products (e.g. offered as a structured deposit in South Korean retail banks).

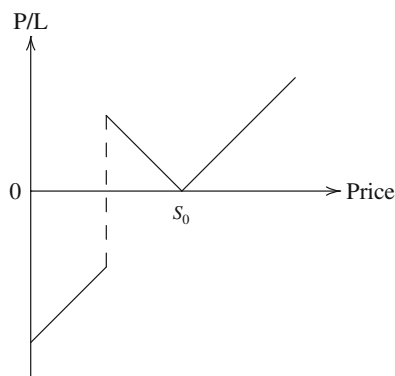


Shark-Fin is in fact a standard Up-and-Out barrier option. For equity underlying, the local volatility model is a good choice for pricing barrier options.

25.1.11 Twin-Win

The basic payoff of Twin-Win consists of a long position of Down-and-Out Straddle and a short position of Down-and-In Put with the same barrier level. The structure may be sold either as an option combination at zero cost (illustrated by the P/L graphic), or as a structured Note issued at par.

The product is suitable for a direction-less but limited downside market view. The investor enjoys the absolute return if the barrier on the downside is not hit. However, the investor may suffer substantial loss if the underlying price dropped beyond the barrier level. The structure is cheap when the volatility is high. Hence, it is often proposed to investors when the market has gone through a significant correction. In such situation, the volatilities are high but people believe that the market may not suffer another big downturn beyond the level of the protection (knock-in barrier).



A Twin-Win Note is a further tweak which is presented through the below example with a single stock as underlying. Worst-of basket may also be used as underlying. Table 25.8 contains an illustration of a Twin-Win Note.

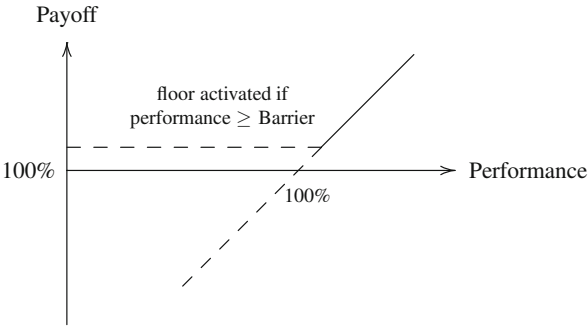
Table 25.8 Illustration of Twin-Win Note

Structure	Note
Notional (N)	USD1mio
Tenor	6 months
Underlying	Stock A
Performance (at t)	$P_t = \frac{S_t}{S_0}$, S_t is the closing price for date t
Observation dates	Monthly dates except the last one, i.e. 5 dates in total
KO barrier (H)	100%
KO event	Deemed to have occurred if $P_{t_i} \geq H$ on any Observation Date t_i
KO coupon	12.5% p.a., i.e. $\frac{i \times 12.5\%}{12}$ if knocked out at the i -th Observation Date
Strike (K)	100%
KI barrier (L)	75%
KI event	Deemed to have occurred if on any date t , $P_t < L$
Early termination	If KO Event occurred, the Note will be early redeemed at $N \times (1 + \text{KO Coupon})$
Maturity payoff	At maturity, if not early terminated, Investor will receive: → $N \times (1 + P_T - 1)$, if KI Event has never occurred; → $N \times (1 + \max\{0, P_T - 1\})$, if KI Event has occurred and $P_T \geq K$; → $N / (K \times S_0)$ units of stocks, if KI Event has occurred and $P_T < K$

The same pricing model for standard barrier options may be used for pricing Twin-Win. The price discontinuity of the structure due to the barrier should be smoothed out by the techniques of barrier shift (for American barrier) or call spread (for European barrier). The structure is quite sensitive to the volatility and the skew parameters.

25.1.12 Tracker⁺ Note

This structure offers the underlying’s performance with a conditional protection. The protection is activated if the underlying’s performance achieved a pre-defined barrier level during the tenor of the product. The underlying is usually a worst-of basket.



A variation consists of offering an additional condition to trigger the protection with the “memory-touch” mechanism: e.g. the barrier event is also deemed to happen if each underlying in the basket has breached the barrier at some time, independently or concurrently, during the observation period.

25.1.13 Leveraged Structures

A Leveraged Note offers a leveraged return of an investment with a fixed maturity. For example, a Note offering 2 times the performance of a stock, either positive or negative, net of financing cost for the leveraged investment. By the legal nature of Note, the maximum loss for the Note investor is limited to the purchase price of the Note (e.g. zero redemption value). So, taking a 2 times leveraged Note as example, the issuer will start to suffer loss when the stock price drops by more than 50% if the issuer simply hedged with 2 times investment in the underlying. For addressing this issue, the issuer usually implements one of the two mechanisms:

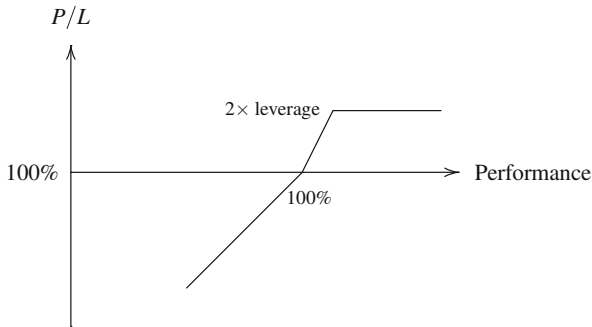
- **Optional payoff:** the Note redemption value is defined as $Notional \times \max \left\{ 1 + PR \times \left(\frac{S_T}{S_0} - 1 \right), 0 \right\}$ which is equivalent to $PR \times Notional \times \max \left\{ \frac{S_T}{S_0} - K, 0 \right\}$ where $K = 1 - \frac{1}{PR}$. Clearly, it is an ITM call option when $PR > 1$. The investment in the underlying will follow the option's delta.
- **Stop-loss:** this approach consists of introducing an early termination feature linked to a pre-agreed *stop loss* level. As soon as the Note value (or equivalently, the performance of the investment with leverage minus financing cost) drops below the *stop loss* level, the Note will be early redeemed at loss for the Note investor. For the risk analysis for the issuer, we refer to the gap-risk based asset allocation techniques in the next chapter.

The ITM (in-the-money) Call option is another structure for getting leveraged exposure to the performance of the underlying without paying too much time-value for the option. The popular CBBC warrant is an example of such product. For active investors, there may be a need to early terminate the trade for locking in the underlying's good performance. The American option may be a good choice due to its transparency of early termination price which is the intrinsic value.

Total Return Swap (TRS) is a preferred trade for institutional investors for leveraged exposure because the money engaged by the investor is only the initial margin (called IA – Independent Amount, e.g. 15% of the notional amount) and the variation margin during the life of the transaction. The ISDA/CSA agreement is required in general. The IA depends on the PFE (Potential Future Exposure – introduced in the Chap. 22, Sect. 22.4) of the TRS. An efficient way for reducing the IA consists of resetting the P/L regularly, e.g. monthly. The Swap is then called a resettable TRS.

25.1.14 *Booster Note*

Booster Note is a bullish product offering leveraged upside until the cap level. This is achieved by giving up the further upside, as shown in the graphic. A floor can be applied by lowering the cap level. If the underlying is a worst-of basket, it is possible to have a floor (e.g. 95%) and a high cap or even no cap.



25.1.15 *Stellar Note*

This is an income structure based on individually capped basket of underlyings. It pays regular coupons which are the positive basket performances at the observation dates. For investors who are bullish for the long term but concerned by the near term market volatility, this structure can be an alternative to fixed-income products.

See Table 25.9 for an illustration example of Stellar Note.

Table 25.9 Illustration example of Stellar Note

Structure	Note
Notional (N)	EUR10mio
Tenor	10Y
Underlying	$S^1 = \text{EURSTOXX50}$, $S^2 = \text{HSCEI}$, $S^3 = \text{SPX}$
Coupon payments	Yearly
Coupon(1)	3.5%
Coupon(i), $i = 2, \dots, 10$	$N \times \max\{0, P(t) - K\}$
Performance $P(t)$	$\frac{1}{3} \sum_{i=1}^3 \min \left\{ \frac{S_t^i}{S_0^i}, \text{Cap} \right\}$
Cap	106%
Strike (K)	100%
Final redemption	At par

25.1.16 *Cliquet Option*

Cliquet Option is an option using the future market prices of the underlying as references (e.g. option Strike) in the payoff, for example, the call option on the sum of the intrinsic values of forward starting options. The cap and floor are often applied on the intrinsic values in a Cliquet option as illustrated by the following example:

$$\max \left\{ 0, \sum_{i=1}^{12} \max \left\{ -4\%, \min \left\{ 2\%, \frac{S_i}{S_{i-1}} - 1 \right\} \right\} \right\},$$

where S_i is the price of the underlying at the end of the i -th month.

Cliquet option's premium can be very low if the cap level is low. It is sometimes used for structuring products with tiny premium amount to spend. However, for obtaining a good return with Cliquet structure, the underlying's price needs to continually perform in the right direction for most of the times during the tenor of the product.

The price of a Cliquet option is sensitive to forward skew due to the cap and floor levels applied on forward starting options. The local volatility model is not suitable for pricing Cliquet options because it does not preserve the forward skew of volatilities. Actually, the local volatility surface tends to flatten for the long term part. More advanced models with volatility dynamics, e.g. stochastic volatility models or jump-diffusion models are often used in practice for pricing Cliquet options. The Cliquet option often gives negative volatility convexity exposure to the issuer which makes the hedging difficult. The investment banks are cautious for offering this type of products.

25.1.17 *Himalaya*

Initially created by Société Générale at the time around year 2000, Himalaya structure is a typical example for demonstrating the flexibility and openness to imagination in financial engineering. The principal of the structure may be described by the following fictive example of the basic version:

1. Assume 5 yearly observation periods and initially 5 stocks as underlying.
2. At the end of each yearly observation period except the last one:
 - record the performance of the best performing stock in the underlying basket, and then,
 - remove this stock from the underlying basket for the following observation periods

3. For the last observation period which is the maturity date of the product, there will be only one stock remaining in the underlying basket. Its performance is then recorded for the last period.
4. The payoff of the structure for the investor is the arithmetic average of the recorded performances.

In a Himalaya structure, the best performances are locked-in at each period. However, the performance in the last period could be very low which may drag down the final payoff. As the performances at the observations are always measured with the initial stock prices, the performance for the last period might be very negative.

For equity linked products, local volatility model with Monte Carlo implementation can be used. The stock removing may create a discontinuity but it is normally small and may be neglected in pricing and hedging.

25.2 Fixed-Income Structured Products

Fixed-Income derivatives and structured products offer investment and hedging solutions linked to rates and credit for both asset and liability management. Fixed-Income derivatives such as Interest Rate Swap, Swaption, Cap/Floor, CDS are actively used by the financial market professionals and money managers. Various structured products have been introduced for financial institutions, corporates as well as retail investors. The tenors of Fixed-Income products are longer than those for equity and FX structured products in general.

25.2.1 Range Accrual

A (Daily) Range Accrual (also called Corridor) structure has a number of observation periods (e.g. every 6 months) until maturity of the trade. At the end of each period, the investor will receive a coupon calculated with the following formula:

$$Coupon_i = R\% \times \frac{n_i}{N_i} \times T_i$$

where

$R\%$: nominal annual interest rate (e.g. 8% p.a.)

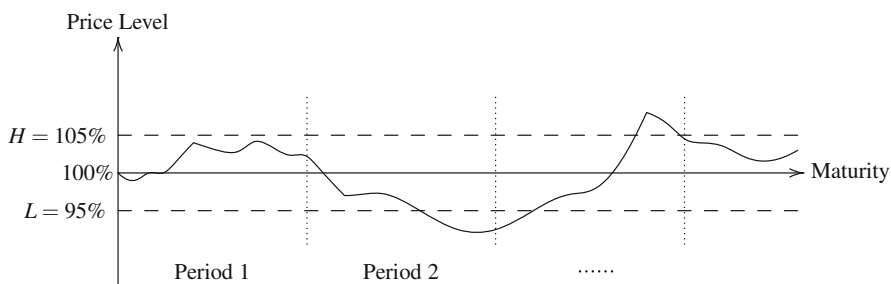
n_i : number of days during the period that the closing price of the underlying has been within the predefined range $[L, H]$

N_i : total number of days in the considered period

T_i : time fraction in years for the considered period.

Although, the structure is introduced in this section, it is however applicable to all asset classes: rates, equities, FX and commodities. For equity (and some commodities) underlying, it is normally structured with one sided barrier (i.e. the upper barrier is set to $+\infty$). The reason is that people often have bullish views on these underlyings due to the (natural) growth perspective. Daily Range Accrual is also a popular structure for hybrid products where investors have views on assets from different asset classes (e.g. equity, rate and FX).

For basket underlying, the Range Accrual products can be structured with the worst performance for the lower barrier observation and the best performance for the upper barrier observation. It means that the performances of all underlyings in the basket should stay within the range for the accrual. With the same option price, the basket version has wider range levels than the single underlying case.



Technically, a Range Accrual option can be decomposed into a sum of binary options. The short party of a Range Accrual option is buying volatility (and selling correlation for the basket version) in general. The pricing model for the binary options is applicable for Range Accrual options.

There are a few other interesting variations for Range Accrual structures:

- full coupon lock-in: as soon as the number of accrued days breaches a certain threshold, the full coupon will be paid to the investor;
- “accrual/decrual” structure: it adds a “decrual” feature which consists of reducing the number of accrued days if the underlying’s price is outside the range. For instance, we may design a Swap paying a coupon of $\max \left\{ -2\%, R\% \times \frac{n_i - 3 \times (N_i - n_i)}{N_i} \right\} \times T_i$. This decrual feature helps to enhance the level of $R\%$ when the normal range accrual structure is not attractive enough. The payoff is less smooth than a normal daily range accrual, but it is smoother than a binary coupon observed at the period end only.

25.2.2 Inverse Floater

The basic payoff of Inverse Floater is

$$N \times \max\{0, K - leverage \times r_T\},$$

where K is a constant reference level, *leverage* is the leverage factor and r_T is the floating rate (e.g. USD 3M Libor) level at maturity date T . It is clearly a European Put option on the floating rate.

25.2.3 Zero Coupon Callable Note

As indicated by its name, the Zero Coupon Callable Note is a zero coupon note which pays the coupon at the maturity or the early termination time. We illustrate the structure with a fictive example; see Table 25.10.

Table 25.10 Illustration example of a 10Y (1Y Non Call) Zero Coupon Callable Note

Issuer	XYZ
Currency	USD
Tenor	10Y
Denomination (N)	USD 100,000
Issue price	100% of denomination
IRR	4% p.a.
Early termination	At the discretion of the issuer
Callable date	Quarterly, starting from Year 2
Early redemption	$N \times (1 + IRR)^t$, where t is the number of years from inception to Early Redemption date
Final redemption	$N \times (1 + IRR)^{10}$

In this structure, the Note investor is short a call option (Bermudan option) for earning a higher return than the fixed interest rate from the same issuer for the same maturity.

25.2.4 Reverse Convertible Note on Rate

The Reverse Convertible Note for equities (also known as ELN) has been extended to rate underlyings recently. We show it through an example in Table 25.11.

With a similar analysis for the ELN, we can find that the final redemption is $N \times (1 + Coupon) - N \times \min \left\{ 1, \max \left\{ 0, \frac{1}{K} \left(K - \frac{R_T}{R_0} \right) \right\} \right\}$. Hence, the investor

Table 25.11 Illustration example of Reverse Convertible Note on USD CMS (Constant Maturity Swap) 10Y

Issuer	XYZ
Currency	USD
Tenor	1 month
Denomination (N)	USD 100,000
Issue price	100% of denomination
Underlying	USD CMS 10Y
Initial level of CMS 10Y (R_0)	1.95%
Coupon	5% p.a. payable at maturity
Strike (K)	75.5% (or CMS 10Y level 1.4723%)
Final USD CMS 10Y (R_T)	USD CMS 10Y level observed at maturity
Final redemption	if $R_T \geq K \times R_0$, Investor will receive N otherwise, Investor will receive $R_T \times \frac{N}{K \times R_0}$ floored at zero

is short a Put option on rate. As the underlying interest rate may become negative, the option has a cap for limiting the maximum loss to N .

25.2.5 Yield Spread Structure

Yield spread (e.g. 20Y CMS rate minus 2Y CMS rate) linked trades are also known as “Steepeners”. They provide opportunities to generate returns or to effectively hedge portfolios by taking the exposure on yield difference between the maturities. In the market, it is frequent to see the spread trades having long tenors and issuer callable features. See Table 25.12 for an example.

Table 25.12 Illustration example of Callable Steepener Note

Issuer	XYZ Bank
Tenor	5 Years, subject to early termination
Denomination (N)	USD 20mio
Issue price	$N \times 100\%$
Final redemption	$N \times 100\%$ (if not called by the issuer before the maturity)
Underlying	USD 10-year CMS rate – 2-year CMS rate
Quarterly coupons	$N \times \min\{6\%, \max\{0, 5.6 \times (10Y \text{ CMS rate} - 2Y \text{ CMS rate})\}/4\}$, if the note is not early terminated
Early termination	At the end of each quarter, Note issuer has the right to call the Note at par

For European spread options, there exist active broker markets for some pairs. The modelling can be realized on the two underlying CMS rates separately. Copula may be used for calibrating the Swaption volatilities of the two CMS rates and other parameters. For the structure with callable feature, interest rate term structure

modelling is required. At least two factors for the rates are necessary for capturing the yield spread dynamics and stochastic volatilities.

25.3 Foreign-Exchange Structured Products

Foreign exchange instruments and structured products are actively used by financial institutions and corporates for hedging and investment. Retail investors often use FX products for yield enhancement.

25.3.1 Dual Currency Investment (DCI)

It is a short term (1 week – 12 months) product for yield enhancement which involves a base currency and an alternative currency. The investor of a DCI with initial investment in base currency believes that the alternative currency will not depreciate against the base currency and wants to monetize this view for an enhanced yield. See Table 25.13 for an illustration example.

The initial investment of the investor is in base currency. At maturity of the DCI,

- if the FX Spot fixing at maturity is equal or higher than *Strike* (expressed as units of alternative currency per base currency): the investor will receive the total amount of principal plus interest in base currency;
- otherwise, investor will receive the total amount of principal plus interest in alternative currency converted with *Strike*.

Table 25.13 Illustration example Dual Currency Investment (DCI)

Base currency	USD
Alternative currency	JPY
Tenor	2 months
Notional (N)	USD 100,000
Spot reference (i.e. initial spot)	111.00
Strike (K)	1/113.01 (USD units per JPY)
Coupon (C)	1% (6% p.a.)
Final spot X_T	Expressed as USD units per JPY
Final redemption	If $X_T \geq K$, $N \times (1 + C) = 101,000$ in USD otherwise, $N \times (1 + C)/K = 11,414,010$ in JPY

In DCI, the investor's exposure is short Put option on alternative currency at *Strike* as shown by the final value measured in the same currency (which is USD in our example):

$$\begin{aligned}
 & N(1 + C)\mathbf{1}_{X_T \geq K} + (X_T N(1 + C)/K)\mathbf{1}_{X_T < K} \\
 &= N(1 + C) \left(1 - \frac{1}{K}(K - X_T)\mathbf{1}_{X_T < K} \right) \\
 &= N(1 + C) \left(1 - \frac{1}{K} \max\{0, K - X_T\} \right).
 \end{aligned}$$

25.3.2 *FX Accumulator*

Accumulators are actively traded on FX rates similarly to equity Accumulators. The tenors of 3M, 6M and 1 year are actively traded. See Table 25.14 for a Boosted/Leveraged product example.

Table 25.14 Illustration example of Boosted EUR/USD Accumulator – Daily Accrual, Monthly Settlement

Currency pair	EUR/USD
Tenor	12M
Daily notional	N euros
Spot ref	1.1780
Strike (K)	1.1490
Accrual	On each business day, Investor buys EUR/USD at Strike. The quantity is N if the fixing price is above Strike; Otherwise, the quantity will be $2 \times N$
Settlement	Monthly
Knock-Out rate	1.2080

25.3.3 *Target Redemption Forward (TRF)*

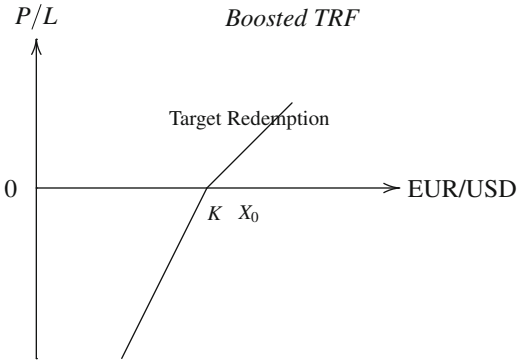
TRF (Target Redemption Forward) is a very popular structure for FX investors. The special feature of a Target Redemption structure is that when the accumulated coupons or intrinsic value for the investor achieved a certain level, the whole structure will be early terminated. See Table 25.15 for an illustration of boosted TRF. A variation consists of counting the accumulated number of times that the intrinsic value has been positive instead of the accumulated intrinsic value.

Table 25.15 Illustration example of a Boosted TRF

Currency pair	EUR/USD
Notional per fixing (<i>N</i>)	EUR200k
Tenor	12M, monthly settlement
Spot ref	1.1780
Strike (<i>K</i>)	1.1410
Intrinsic value	$\max\{ERR - K, 0\}$
Frequency	Monthly fixing and monthly settlement
Target	10 Big Figures (i.e. 0.1000), Exact (meaning that the total payout at Knock-Out is exactly 10 Big Figures)
Knock-Out event	If accumulated Intrinsic Value $\geq$ Target
Payoff for each fixing (before Knock-Out)	If $ERR \geq K$, investor buys <i>N</i> at <i>K</i> ; Otherwise, investor buys $2 \times N$ at <i>K</i>

Where *ERR* (Expiry Reference Rate) is the fixing at the monthly observation date

The TRF allows the investor to buy a certain currency at discount (or to sell at premium) until the early termination event which is triggered by the Target Redemption mechanism. A structured Note with Target Redemption is known as a *TARN*.



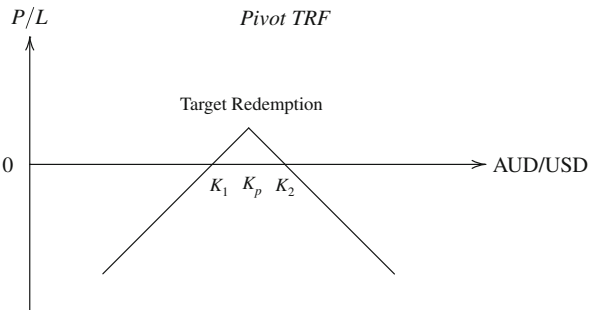
Variations of TRF

One popular variation is the Pivot structure where the investor will be buying the foreign currency at Lower Strike and selling at Higher Strike. It is suitable for clients believing that the FX rate will be trading in a range for the considered tenor. Target Redemption is a natural Knock-Out mechanism for Pivot structure. See Table 25.16 for an example.

Table 25.16 Illustration example of Pivot TRF

Currency pair	AUD/USD
Notional	AUD100k per Month
Tenor	12M, monthly fixing and settlement
Spot ref	$X_0, K_1 \leq X_0 \leq K_2$
Buy strike	K_1
Sell strike	K_2
Pivot strike	K_p
Target	0.2000 Exact
Intrinsic value	if $ERR < K_p$: $\max\{ERR - K_1, 0\}$ if $ERR \geq K_p$: $\max\{K_2 - ERR, 0\}$
Knock Out event	if accumulated Intrinsic Value $\geq$ Target
Settlements	if $ERR \leq K_p$: investor buys at K_1 if $ERR > K_p$: Investor sells at K_2

Where ERR (Expiry Reference Rate) is the fixing at the monthly observation date



We note that in the examples of Pivot, we have chosen $K_p - K_1 = K_2 - K_p$ (a butterfly structure). But in general, $K_p - K_1$ may differ from $K_2 - K_p$.

Similar to the Boosted TRF, the boosted version of Pivot TRF allows the investor to have wider Strike levels. We illustrate with graphics and fictive examples the two popular variations: Boosted Pivot TRF and One Sided Pivot TRF (Table 25.17).

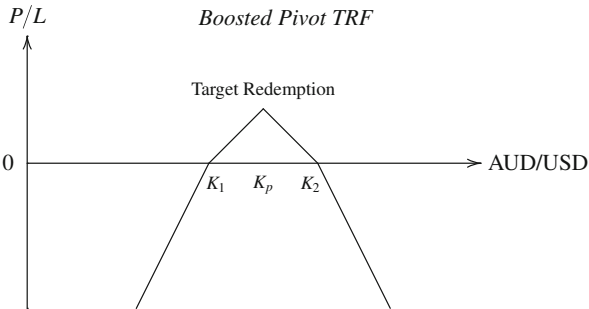
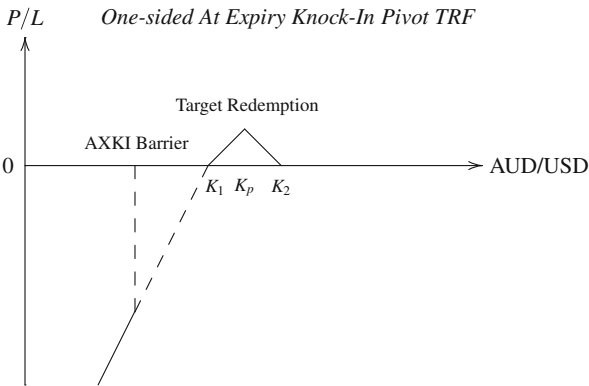


Table 25.17 Illustration example of one sided AXKI Pivot TRF

Currency pair	AUD/USD		
Notional	AUD1mio × AUD2mio per Month		
Tenor	12M, monthly settlement		
Spot ref	1.0359		
Lower strike (K_1)	0.9900	Upper strike (K_2)	1.0700
Pivot strike (K_p)	1.0300	AXKI barrier	0.9100
Target	0.1500 exact		
Intrinsic value	if $ERR < K_p$: $\max\{ERR - K_1, 0\}$ if $ERR \geq K_p$: $\max\{K_2 - ERR, 0\}$		
Knock Out event	if accumulated Intrinsic Value $\geq$ Target		
Settlements	if $ERR \geq K_1$ and $ERR \leq K_p$: investor buys AUD1mio at K_1 if $ERR > K_p$ and $ERR \leq K_2$: investor sells AUD1mio at K_2 if $ERR < AXKI$: investor buys AUD2mio at K_1 Otherwise, no settlement		

Where ERR (Expiry Reference Rate) is the fixing at the monthly observation date



25.4 Commodities Structured Products

In general, commodities linked structured products are less traded compared to the ones that are linked to other asset classes. Precious metals, Oil (Brent or WTI) are more often used as underlyings than other commodities such as the industrial metals and soft commodities.

We may list a few examples of structured products linked to commodities:

- Reverse Convertible linked to the price of Oil front-month futures: 6M USD Note, Put Strike 100%, American Knock-In Barrier at 72% of Initial Spot, 10% p.a. Coupon;
- DAC (Daily Range Accrual Callable): 2Y USD Note, Worst-of basket Gold/Silver/Platinum, Nominal Coupon 8%, Accrual Barrier = Put Strike = 87%, Autocall Barrier 100%.

25.5 Hybrid Structured Products

When the underlyings belong to different asset classes, the structured product is called a hybrid product. It allows the investor to take advantage of their views on more than one asset class. The structures for basket underlying presented so far are generally applicable to hybrid underlyings. In the following, we will show a few hybrid structures for illustration purpose.

25.5.1 Callable Daily Range Accrual

Table 25.18 illustrates an example with interest rate and equity as underlying. The interesting feature to observe is that the range for equity and the range for interest rate are set to their respective levels.

Table 25.18 Illustration example of Callable Daily Range Accrual

Structure	Note
Notional (N)	USD20mio
Tenor	10Y, NC1Y (No Call for the first year)
Underlying	SPX Index, USD Libor 3M
Equity condition	SPX Index closes at or above 50% of Initial level
Rate condition	USD Libor 3M fixing is below 5% (Spot Rate Ref 1.34%)
Quarterly coupon	Notional $\times 3.35\% \times m/M$, where m is the number of days both Equity Condition and Rate Condition are satisfied, and M is the total number of days in the period
Issuer call	Quarterly after Year 1, issuer has the right to early redeem the note at par
Final redemption (if not early redeemed)	100% of Notional

25.5.2 Best-of Profile

Best-of-Profile delivers the best performance of different asset portfolios. Each portfolio represents a specific investment profile. The profile of a portfolio can be achieved by

- choosing the assets from different asset classes
- weighting the assets differently according to the riskiness or geographic exposure.

Table 25.19 Illustration example of Best-of-Profile

Structure	Option
Notional	USD20mio
Tenor	3 Years
Underlying	3 Portfolios: Aggressive Portfolio (P_1): 50% Equity + 30% Commodities + 20% Bond Balanced Portfolio (P_2): 33% Equity + 33% Commodities + 34% Bond Conservative Portfolio (P_3): 20% Equity + 30% Commodities + 50% Bond where Equity = 33% S&P Index + 33% EuroSTOXX + 34% HSCEI Commodities = GSCI Commodities Index Bond = iBoxx Investment Grade Bond Index
Maturity payoff	$\max\{0, \max\{Perf(P_1), Perf(P_2), Perf(P_3)\} - 1\}$ where $Perf(P_i) = P_i(Final)/P_i(Initial), i = 1, 2, 3$

Best-Of Profile can be sold either as an option (cf. Table 25.19) or as a structured note. It is suitable for uncertain market view by offering the best performance among the investment portfolios. However, the expected gain is often limited because the portfolios tend to have low volatility.

In general, the product issuer is buying correlation between the portfolios but selling correlation between the assets within a portfolio. So, the correlation shifts should be applied accordingly in pricing.

25.5.3 ASCOT: Asset Swapped Convertible Option Transaction

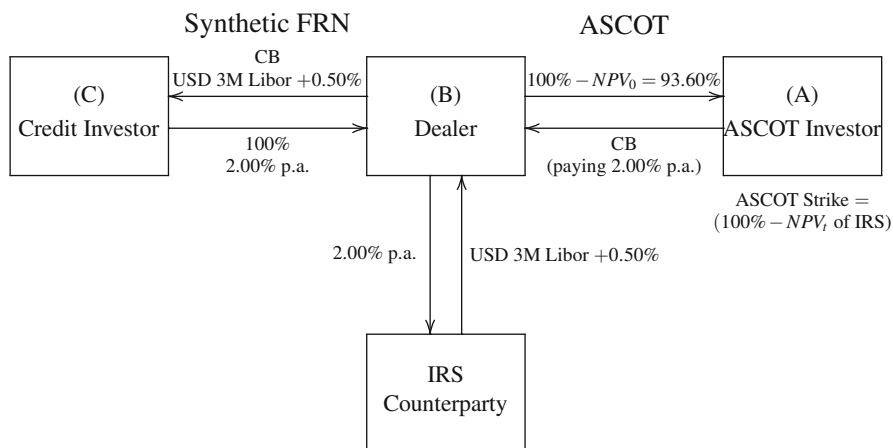
Convertible bonds (CBs) often present opportunities to credit and equity investors (cf. Sect. 25.8 Equity Derivatives for Corporates). But some credit investors may not be willing to handle the equity risk and vice versa for the equity investors.

ASCOT consists of separating a convertible bond into two related products for two types of investors:

- credit investor: a note paying attractive floating rate based coupons
- equity investor: an option providing return on the equity upside.

We illustrate the structure by a fictive example. Let us assume that

- a hedge fund (A), holding a CB, wants to free up the capital but keeps the upside exposure to the equity;
- an institutional investor (C) is interested by an investment bearing the credit risk of the CB issuer;
- a Dealer (B), usually an investment bank, arranges the trades with ASCOT.



Investor (C) buys from the Dealer (B) a synthetic FRN (Floating Rate Note) at 100% paying floating coupon (Libor + spread) which is callable by (B) at 100% plus accrued interest. The FRN is in fact an Asset Swap composed of the CB and an IRS (Interest Rate Swap). In the IRS, (C) pays to (B) the fixed coupon equal to the coupon of the CB and receives the floating coupon (Libor + spread). The “spread” (0.50% in the example) is agreed at inception which is related to the credit risk of the CB issuer. The initial NPV (Net Present Value) of (Floating Leg – Fixed Leg) of the IRS is NPV_0 .

Hedge fund (A) sells to Dealer (B) the CB at $(100\% - NPV_0)$. In return, (A) holds the ASCOT which is an American option giving the right to buy back the CB at $(100\% - NPV_t)$ where NPV_t is the unwind price of the IRS when the option is exercised.

25.5.4 Gap Note

Gap risk refers to the potential loss due to a sudden large move of the underlying’s price. In some structures such as CPPI or margin loans, the product issuers are exposed to gap risks. In the broker market, it is sometimes possible to buy protections on gap risk of equity indices through the product called *Stability Note* or *Gap Note*. See Table 25.20 for an illustration.

Table 25.20 Illustration example of Gap Note

Issuer	XYZ Bank
Tenor	6M
Denomination (N)	USD2mio
Issue price	$100\% \times N$
Underlying index (I)	S&P 500 (Bloomberg Code: SPX Index)
Coupon	USD 3M Libor + 350bps p.a., quarterly payable until maturity or Trigger Event
Strike (K)	90%
Final redemption	If no Trigger Event has occurred: $N \times 100\%$ If Trigger Event has occurred: $N \times (100\% - 10 \times PutSpread_{\tilde{t}})$ where $\tilde{t}$: the first date where Trigger Event occurred ($\tilde{t} - 1$ represents the previous business day) $PutSpread_{\tilde{t}} = \min\{10\%, \max\{0, K - I_{\tilde{t}}/I_{\tilde{t}-1}\}\}$, I_t is the Index closing price at date t
Trigger event	Trigger event is deemed to have occurred if $I_t/I_{t-1} < K$

This structure cannot be delta-hedged as a normal option. The investor of the note takes the Gap risk for income. The pricing is mostly driven by demand/offer. Jump diffusion models may be used for the consistency of the prices across different strikes/tenors.

25.6 Fund Linked Structured Products

Structured products are mostly linked to mutual funds rather than hedge funds. Mutual funds provide daily liquidity for subscription and redemption and can be sold to retail investors. They are highly regulated with transparent investment objectives and risk management methodologies. For being able to offer fund linked derivatives, the derivatives issuer needs to have an agreement with the fund manager for transacting the fund units at NAV without the subscription/redemption fees. It is customary for the fund manager to rebate part of the management fees on the transacted fund units for delta hedging to the derivatives issuer as an incentive for helping to increase the fund AUM (Assets Under Management). The rebate will be fairly priced in the structured products which helps to improve the features.

Hedge funds usually do not provide daily liquidity, which makes the hedging of options difficult. It is however possible to hedge the option on a managed account set-up by the derivatives issuer for replicating the hedge fund. The hedge fund manager will be the investment advisor so that the managed account will contain the same assets as the hedge fund itself. As the managed account is transparent for the invested assets, the derivatives issuer can make delta hedging on daily basis by adjusting the assets in the managed account.

In general, the fund linked structured products favor the ones with smooth gamma, non-negative delta and no discontinuity. Sometimes, for cheapening the fund linked derivative, a *synthetic dividend*¹ is introduced to accumulation share class of the fund. The payoff of the product will be linked to the newly created fund underlying with synthetic dividend which is cheaper in price due to the lower forward. The volatility market for funds is limited. Hence, the structured products often require Volatility Target on the underlying funds (cf. Sect. 26.1 on asset allocation). The derivatives issuers are reluctant to sell structures with volatility target that may require the issuer to buy volatility (e.g. in a call spread, the short option at the upper Strike makes the issuer buy volatility for this Strike). CPPI/TIPP which do not require volatility hedging are often used for funds, especially those with low volatility.

The maximum issue size for a fund linked structured product is limited to a certain percentage of the AUM of the fund. If the delta becomes important compared to the AUM of the fund, the product issuer (the hedging party) may have the majority control of the fund and may need to consolidate the fund investment into its balance sheet. In such case, the issuer may choose to give up the majority control by an official notice/request.

In the following, we will present a few popular structured products linked to funds through fictive examples. The most common purpose includes the principal protection, leveraged return and financing. Total Return Swap is an efficient derivative for having the exposure to funds' performance with financing.

25.6.1 Bullish Note

This structure offers a geared upside at maturity with a participation rate (PR) which can be lower or higher than 100%. The underlying can be a single fund or a basket of funds. The volatility target may be applied in general. The product example in Table 25.21 can offer leveraged upside due to its very low volatility. The volatility target is not applied because of the availability for volatility hedging in the market for this particular fund. The embedded payoff of the structure is a leveraged call option.

For principal non protected products, the below alternative payoff may be considered. It provides a geared upside. The downside exposure is not geared, making it comparable to the direct fund exposure. A floor, represented by "maximum loss" in the below formula, is often introduced in such type of structures:

$$PR \times \max \left\{ 0, \frac{S_T}{S_0} - 1 \right\} - \max \left\{ 0, \min \left\{ \text{maximum loss}, 1 - \frac{S_T}{S_0} \right\} \right\}.$$

¹The concept of synthetic dividend is also applied to total return indices or systematic strategies.

Table 25.21 Illustration example of Bullish Note

Structure	Note
Denomination (N)	USD100,000
Currency	USD
Tenor	3 Years
Issue price	100%
Underlying (S)	A fixed-income fund with low volatility
PR	150%
Redemption	$N \times \left(1 + PR \times \max \left\{ 0, \frac{S_T}{S_0} - 1 \right\} \right)$

25.6.2 Bullish Coupon Note

For the investors requiring periodic payoffs, we show an example of bullish coupon note linked to the performance of a fund; see Table 25.22. In this structure, the yearly payoff is the performance of the fund, measured from the beginning and divided by the number of years.

Table 25.22 Illustration example of Bullish Coupon Note

Structure	Note
Denomination (N)	USD100,000
Currency	USD
Tenor	3 Years
Issue price	100%
Underlying (S)	A fund with volatility target
Coupon for year i	$PR \times \max \left\{ 0, \frac{S_{t_i}}{S_0} - 1 \right\} / i$
PR	100%
Redemption	At par

A popular variation to this structure consists of offering the yearly coupon which is the fund's performance at the year end reduced by the coupons received previously: $Coupon(i) = \max \left\{ 0, \frac{S_{t_i}}{S_0} - 1 - \sum_{k=1}^{i-1} Coupon(k) \right\}$ for $i \geq 2$. Actually, this payoff is equivalent to paying the highest yearly performance to the investor.

25.7 Credit Linked Structured Products

25.7.1 Credit Linked Note (CLN)

CLN linked to a single reference entity is the most popular credit structured product sold to investors. Readers can refer to Part I, Chap. 6 for the product structuring. The CLN issuers often offer structures that are tailor made to satisfy the client's

need. It is common to see zero recovery in the credit linked notes sold to individual investors where the implied recovery rate is priced in the structure for yield enhancement.

In the years shortly before the financial crisis of 2008, CLNs linked to FTD (first-to-default) basket of reference entities were actively traded. The default event for FTD basket is the first credit event among all the reference entities in the basket. Hence, the default correlation is an important factor for pricing such CLNs. After the financial crisis of 2008, the implied default correlations have heightened so much that the product features with first-to-default basket often fail to be attractive enough for investors.

25.7.2 CLN on Credit Index

The CLN linked to a Credit Index is less risky compared to a single name CLN. The credit index helps to diversify the risk across the constituent reference entities. The liquidity on credit index is in general better than the liquidity for a single reference entity. We give one simple credit index linked note as illustration example; see Table 25.23. For sophisticated investors, index single tranche linked products (similar to CDO) can be structured.

Table 25.23 Illustration example of Credit Index Linked Note

Structure	Note
Tenor	5 Years
Outstanding principal amount	Initial value is USD10,000,000
Underlying	iTraxx Asia ex Japan IG Index: 40 constituents, each with 2.5% weighting
Coupon rate	USD Libor 3M + 1.5%
Quarterly coupon	Outstanding Principal Amount × Coupon Rate × Year Fraction
Credit event	Outstanding Principal Amount will be written off by 2.5% for each credit event (i.e. zero recovery). The Coupon will continue to accrue based on the reduced Outstanding Principal Amount
Redemption	Outstanding principal amount

25.8 Equity Derivatives for Corporates

There are several popular strategic solutions using equity derivatives for the purpose of

Financing equity linked financing structures can be used by corporates or their major shareholders. The margin loan and collar financing to be presented later belong to the type of financing known as *non-recourse* loan, meaning that the

lender can not go after borrower's other assets except those posted as collateral for the loan transaction.

Share Buy-Back there are different reasons for a corporate to buy back its own shares from the market. For example, corporates may need shares to pay the executive employees as part of their incentive compensation. New issuance of shares will dilute the value of existing shares. The corporates often choose to buy back shares from the market.

Share Disposal Corporates may need to sell large quantity of shares that they hold for strategic reasons.

25.8.1 *Financing*

Convertible Bonds (CB)

A company may raise financing through different channels: direct capital increase, right issues, preferred shares, bonds, and etc. Convertible Bond is an alternative to bond. It gives bondholders the right to convert the CB into a fixed number of common shares of a certain stock. It is a hybrid structure of (Bond + Call) with early termination feature triggerable by either a Put or a (Soft) Call event.

For investors of convertible bonds, there are the below attractive features

- upside potential of the underlying stock
- downside protection when the underlying stock is performing poorly
- investment or arbitrage opportunity for equity option investors or credit investors

Hedge funds hold an important part of issued convertible bonds because this hybrid credit/equity instrument allows them to implement various trading strategies. For derivatives professionals, convertible bonds offer the exposure to the implied volatility of the underlying stocks (which are often under priced in the CBs) that are not blue-chips in general.

For fast-growing companies, convertible bond is a preferred instrument for financing. The advantages for the issuer of the convertible bond include

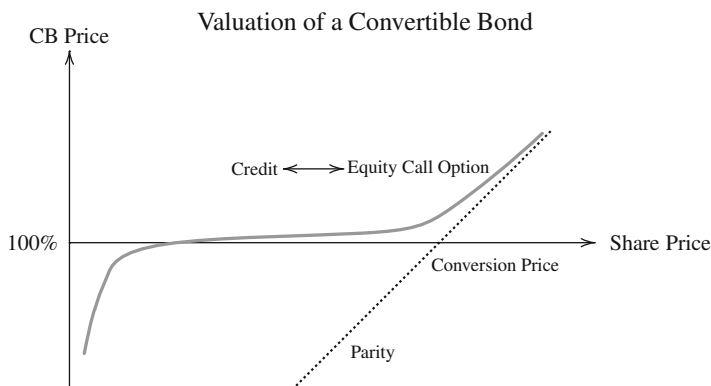
- lower financing cost because the coupon offered by a convertible bond is lower compared to a normal bond
- delayed dilution for common stock and the earnings-per-share (EPS)
- issuer is selling at premium to the prevailing stock price

Convertible bond is a hybrid structure involving equity and credit modelling. The local volatility model for equity part combined with a jump component for credit modelling is a natural approach. For interested readers, we refer to [1, 14].

In Table 25.24, we summarize the common features of convertible bonds together with an example. The graphics give further illustration for the schedule and the valuation of the convertible bonds.

Issue Date Put Date Soft Call Period Maturity Date

Conversion Period



Typical features	Explanation	Example
Issuer		XYZ company
Underlying share		XYZ common share
Maturity	The tenor of the CB	5Y
Denomination	Face value of the CB	USD10,000
Coupon rate	Interest rate offered by the CB	2.75% p.a.
Conversion price	Price at which the convertible bond is converted into common stocks	10.76
Conversion ratio	Number of shares of a common stock received by the CB holder when the bond is converted; It is equal to Denomination/Conversion Price	929.368
Put	Investor's right to redeem the bond at a pre-determined price on a pre-determined date prior to maturity	On April 24, 2011, the bondholder will have the right to redeem the bond at 111.54% of the principal amount
Soft call	The convertible bond may be called if the underlying share price breaches a pre-specified upper limit for at least a pre-defined number of days during the soft call period	After April 24, 2011, the Issuer has the right to early redeem the bond at 130% of Denomination, if the share has been traded above 130% of Conversion Price for at least 20 during 30 consecutive trading days
Early redemption		100% of denomination
Final redemption		100% of denomination if no early redemption occurred

Typical features	Explanation	Example
Issuer		XYZ company
Underlying share		XYZ common share
Maturity	The tenor of the CB	5Y
Denomination	Face value of the CB	USD10,000
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Early redemption		100% of denomination
Final redemption		100% of denomination if no early redemption occurred

Margin Loan

Stock Margin Loan is a simple solution for obtaining financing with the shares of the corporate or any other tradable shares as collateral. The ratio of the borrowed amount to the collateral value is called Loan-To-Value (LTV). The LTV is higher when the collateral has low volatility and high liquidity. For stocks, the LTV often ranges from 30% to 60%. The size of the loan is conditioned by both the averaged daily trading turn-over volume (ADTV) and the free-float of the stock.

In a stock margin loan transaction, the margin call mechanism is defined in the contract. As soon as the collateral value drops by a certain threshold (e.g. 15%), the margin call (mostly in cash) will be triggered. In case of non payment for margin call, forced termination is applicable.

The shares used as collateral are kept in a special account (e.g. escrow account). Hence, the legal ownership of the shares still belongs to the borrower. This leads to the following advantages of Stock Margin Loan to the borrowers:

- the voting right is retained for the borrower for the shares used as collateral
- there is no regulatory disclosure requirement for the substantial share holders.

The drawbacks of stock margin loan include:

- LTV is often quite low (e.g. below 50%).
- Margin Call which needs to be handled over the whole tenor for the loan.

Collar Financing

Collar Financing is an attractive trade which is applicable if the underlying share has sufficient liquidity for allowing the hedging of the embedded derivatives. The word Collar is used due to the long Put and short Call options employed in the typical structure as explained in the below. Actually, the Call option is introduced for reducing the financing cost and it is not mandatory. However, the Put option is mandatory as it “transforms” the Credit Risk into Market Risk for the lender. In a typical Collar financing trade, the borrower

- purchases a Put Option with Lower Strike (e.g. 80%) from the lender
- optionally, sells a Call Option with Higher Strike (e.g. 115%) to the lender for lowering the total cost
- delivers shares to the lender as collateral
- obtains a loan (e.g. at a rate of $\text{Libor} + 3.5\%$).

The lender can lend an amount up to the PV of the Put Strike. At maturity, the borrower pays back the loan. The lender returns the collateral which may be adjusted by the difference between the strike price and the prevailing share price (Collar payoff).

For the borrower, the advantages of Collar financing include

- No Margin Call: this is because the value decrease of the shares in collateral is compensated by the valuation increase of the Put option.
- High LTV (Loan-To-Value): close to the Put Strike
- Low Financing Cost: the Call Option lowers the financing cost and enables financing at a reasonable interest rate level;
- Downside Protection: the Put Option hedges the downside risk of the shares for the borrower.

Compared to the stock margin loan structure, the drawbacks of Collar financing are as follows:

- Upside share performance is capped at the Call Strike
- Voting right is lost during the tenor of the trade
- Potential disclosure requirement for substantial shareholders (according to the applicable regulation).

Actually, Collar financing may be structured and transacted as an in-the-money Call Spread (e.g. 80–115%). In this case, the “borrower” needs to sell the stocks and buy the in-the-money call spread with a fraction of the selling proceeds. The choice of the structure will depend on the context of the borrower and the lender.

25.8.2 *Shares Buy-Back*

Structures for Fixed Price but Uncertain Quantity

The corporate may use derivatives for buying back shares at discount to initial price. In the below structures, the price is fixed at discount to the initial price at inception of the trade. However, the quantity that the corporate will purchase depends on the future market price of the underlying stock.

- Put Writing: the corporate sells an OTM Put Option, i.e. the obligation to buy shares at strike price, set at discount to the initial price, if the share price drops below the strike. The corporate gains the option premium. If the stock price has not increased more than the option premium in percentage, the corporate effectively lowered the buy-back cost compared to the straight purchase at initial price.
- Accumulator: the corporate buys a number of shares at a fixed price at discount to the initial stock price every day. The trade will be early terminated if share price trades above a pre-defined Knock-Out barrier on the upside.

Structures for Fixed Quantity but Uncertain Price

Accelerated Share Buy-Back (ASB) is a popular trade used by corporates, especially in North America, for the buy-back of a fixed quantity of shares within a time window. The purchase price is uncertain but at a discount to the averaged value of the share for the purchase period. The trade can be described as follows:

- The corporate enters into a contract, at zero cost, with an investment bank to buy a fixed number of shares based on the average price over the accumulation period (equally weighted) with a discount (e.g. 1%). The reference price is usually VWAP (volume weighted average price).
- The investment bank buys shares gradually from the market but has the discretion for the effective daily quantities and termination date. Actually, the investment bank tends to buy more when the price is low, and buy less (or even sell shares) when the share price is high. As a result, the investment bank guarantees a final purchase price at a discount to the average price over the accumulation period.

This structure is a kind of American option. The quantities that the bank buys over the period are determined by the delta of the option. The vega hedging will also be necessary because otherwise, the bank would not be able to hedge the trade should the share price keep rising (implying very low realized volatility). PDE pricer under the local volatility model may be used in practice.

Apart from the ASB trade, there is a systematic execution strategy which tends to outperform the averaged price during the buy-back period with equal weight. It consists of buying back the quantity that is determined by the running average of the share price:

- if the current price is higher than the running average, a smaller quantity will be bought;
- if the current price is lower than the running average, a greater quantity will be bought.

25.8.3 *Shares Disposal*

The solutions for Shares Disposal are the opposite ones compared to Share Buy-Back, i.e.

- Writing Call instead of writing Put
- Decumulator instead of Accumulator
- Accelerated Share Disposal which allows to sell a fixed number of shares at premium of the average price over the disposal period (equally weighted).

Exercises

1. Implement a Monte Carlo pricer for a standard domestic ELN with an Up-and-Out barrier and tweak the underlying volatility parameter to study the short volatility effect. Assume the underlying follows the BSM with no dividend.
2. Consider an accumulator with the following modified structure. Every month, the client can buy from IB 1/12-th of the total number of shares at a discounted price of the initial spot price of the share. But if the KO level is achieved at the settlement day, the structure will be terminated (the client will stop buying share going forward).

Total number of shares	1 million
Tenor	12 Months
Settlement frequency	Monthly
Strike	?% of Initial Spot
Knock out level	105% of Initial Spot
Initial spot	HKD 8.8
Interest rate	0.35% per annum
Volatility	27.5%

Solve the Strike (in %, accurate to 2 decimal places) for 0 premium assuming the Black–Scholes model and continuous compounding.

3. A Chinese bank offers a 2 year, 100% principal protected note to their clients. The product is described in the below:
 - at the inception, the client purchases the note at par (100% notional);
 - at the end of i -th quarter,
 - if both underlying stocks trade at or above their initial spot levels, the client will receive a coupon of $\text{notional} \times (100\% + i \times \text{Snowball Coupon Rate}/4)$ and the product is early terminated;
 - otherwise, the client receives nothing and the product continues;
 - at maturity date, if no early termination happened, the client will receive 100% notional.

On the bank side,

- the bank receives the proceeds of 100% notional from the client;
- Treasuries division of the bank pays a funding of 3.5% per annum (quarterly payable) until the product's termination date;
- Treasuries division pays back 100% notional to the client at the termination date;
- The bank makes a back-to-back swap hedge with an IB.

The terms of the swap are as follows: Bank pays funding to IB which is 3.5% p.a. every quarter.
IB pays the return of the below to the Bank:

Notional	xxx
Tenor	2 years
Denomination	CNY
Underlying stocks	1 Japanese stock/1 US stock
Initial spot	8975/443
Observation	Quarterly, on the worst performance
Knock out level	100% of initial spot levels
Snowball coupon rate	? % per annum
KO coupon if KO at i -th quarter	$(\text{Snowball coupon rate}/4) \times i$

Market data – US stock		Market data – US stock	
Volatility of the stock	30%	Volatility of the stock	28%
Correlation stock – USD/CNY	10%	Correlation stock – JPY/CNY	–40%
Volatility of USD/CNY	2%	Volatility of JPY/CNY	11%
USD rate	0.38%	JPY rate	0.13%
Correlation stock US – stock JP 60%			

- (i) Implement a Monte Carlo pricer with the above market data;
- (ii) Solve for the “Snowball Coupon Rate”.

4.

- (i) What is a correlation skew?
- (ii) What is the difference between “Issuer Callable” and “Autocallable” structure?
- (iii) Is the local volatility model suitable for pricing a Forward Starting Call Spread? Why?
- (iv) What type of American option that Longstaff–Schwartz (Least-Square) method is less suitable to apply? Why?
- (v) An investment bank (IB) wants to apply Black–Scholes–Merton model to price a Call option on the excess return of a mutual fund. The IB will receive a rebate (retrocession) of 0.30% from the management fees charged by fund manager. Please write down the SDE (Stochastic Differential Equation) to be used for the pricing.
- (vi) What is the risk for the investor in a tri-party repo based product?

5. For a structured Note, there are time lags between the trade and issue date, final valuation date and maturity date. In pricing, the actual dates used for the final valuation and the dates used for discounting are different. In this exercise, the description as well as the parameters are close to a real FCN product.

Tenor	6 Months + 2 Weeks
Trade date	03 Oct 2017 (Tue)
Issue date	17 Oct 2017 (Tue)
Final valuation date (T)	17 Apr 2018 (Tue)
Maturity date	24 Apr 2018 (Tue)
Denomination (N)	USD 100k
Currency	USD
Issue price	100% of N
Coupon	1% flat paid at each monthly observation (i.e. $\sim 12\%$ p.a.)
Put Strike (K)	To be calculated
Underlying basket	Stock A (US stock) and Stock B (Japanese stock)
Performance (P)	Worst-of underlying basket
Observation	17 Nov 2017, 19 Dec 2017, 17 Jan 2018, 17 Feb 2018, 17 Mar 2018, 17 Apr 2018
Knock-Out barrier	100% at each monthly observation date
Spot reference – S_0	USD 50.00 for Stock A, JPY 20000.00 for stock B
Final redemption at maturity	If $P_T \geq K$ then 100% of N Otherwise, $X_T \times N / (K \times S_0^*)$ shares of S^* , where S_0^* is the initial price of the worst performing stock at maturity; $X_T = 1$ if S^* is US stock, otherwise, X_T is the prevailing USD/JPY exchange rate at Final Valuation Date
Issuer/Distributor margin	0.60%

Assume BSM model for pricing with the market parameters as follows:
USD interest rate is 1.6% p.a. flat (30/360). JPY interest rate is 0%.

	Stock A	Stock B
Volatility	38%	40%
Dividend rate	0%	0%
Repo rate (continuous, ACT/365)	2%	4%

The volatility USD/JPY is 9.50%.

Use Excel-VBA coding to do the below:

- Estimate Strike K with Monte Carlo Simulation (at least 10,000 paths)
- Calculate the 95% confidence Interval for the issue price with the Strike fixed at 90%.

Correlation	Stock A	Stock B	USD/JPY
Stock A	1		
Stock B	0.50	1	
USD/JPY	0.25	0.20	1

Note: It is often useful to use the same seed for the random number generation in VBA so that the simulation result can be reproduced. In Excel-VBA it can be done by generating the first random number with `Rnd(-2)` and the following ones with `Rnd()`.

Chapter 26

Dynamic Asset Allocation



Dynamic asset allocation is the mechanism for controlling the timing and quantity of tradable assets such as equities, bonds, currencies and commodities. It can be used to achieve one or more of the following targets:

- reducing the risk
- providing protection
- creating relatively stable returns
- dynamically changing the beta (market) exposure across different assets according to the growth/performance cycles.

In this chapter, we will introduce a few popular dynamic asset allocation strategies. Firstly there are methods that mainly involve allocation between risky assets and nearly risk-free ones, including the volatility target strategy for controlling risk, and the two important gap risk-based approaches known as CPPI and TIPP. Then, we will present the allocation methods on a basket of risky assets, such as the classic mean-variance framework and its extension, the Black–Litterman model. In the following, the concept of *exposure* or *weight* is represented by the *percentage* of the total capital invested in a certain asset.

Assume we have n assets, risky or not. Let $P_{i,t}$ be the price of the i -th asset at time t , and NAV_t be the total net asset value at the same time. If the exposure at time $t - 1$ for the i -th asset is $w_{i,t-1}$ and all exposures add up to 1, then

$$\frac{NAV_t}{NAV_{t-1}} = \sum_{i=1}^n w_{i,t-1} \frac{P_{i,t}}{P_{i,t-1}}, \quad \text{or} \quad \frac{NAV_t - NAV_{t-1}}{NAV_{t-1}} = \sum_{i=1}^n w_{i,t-1} \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}},$$

where the latter format puts emphasis on the return rate.

Consider $n = 2$ and the second asset M_t is an investment in money market. Then we have the following relation which is used in the subsequent sections

$$\frac{NAV_t}{NAV_{t-1}} = w_{t-1} \frac{P_t}{P_{t-1}} + (1 - w_{t-1}) \frac{M_t}{M_{t-1}}. \quad (26.1)$$

Example 26.1 (constant mix strategy) If the exposure is a time-invariant constant $w_t = w$, then the rebalancing strategy is called *constant mix*. For instance, $w = 80\%$ implies 80% of capital invested into risky assets and 20% in money market. Note that it is the exposure that stays constant. Keeping risky assets exposed at 80% level may require frequent rebalancing. When $w = 100\%$, it is also called the *buy-and-hold* strategy.

26.1 Volatility Target for Risk Control

Volatility target mechanism aims at controlling the volatility of the exposure to the underlying asset (or portfolio). It is realized through dynamic adjustment of the exposure w_t according to the following rule:

$$w_t = \min \left\{ \text{Cap}, \frac{\sigma_{\text{target}}}{\sigma_{\text{realized}}} \right\},$$

where

σ_{target} : required target level for the volatility of the strategy (e.g. 5%);

σ_{realized} : realized volatility of the asset at time t over an observation window (e.g. calculated from the daily returns in the past 6 months);

Cap: maximum exposure (e.g. 150%) for limiting the leverage, hence the leveraging cost.

The global NAV_t of the volatility target strategy at time t is based on the asset portfolio return and the return on the investment in money market as shown in (26.1). A threshold for rebalancing can be introduced to avoid frequent transactions of small amount for the exposure changes, i.e., there is no adjustment if the adjustment amount is less than the threshold.

Volatility target has several advantages. It is commonly applied to investment strategies for

1. controlling the risk, as it tends to
 - deleverage when the realized volatility of the asset σ_{realized} increases, leading to decreased exposure to the asset and increased investment in money market.
 - leverage when the realized volatility of the asset σ_{realized} decreases, leading to increased exposure to the asset and decreased investment in money market. If $w_t > 100\%$, funding cost will be charged for the leverage.
2. offering options on the assets on which we cannot normally trade volatilities in the market (e.g. mutual funds which have no volatility quote in the market). Pricing and hedging become possible because of volatility target. However, the

option writers are only comfortable to offer the payoffs with positive Vega (i.e. option writer sells volatility). This is because the risk is limited for the hedging party to sell the volatility at a level slightly above the volatility target (e.g. 1.5% above) but it is difficult to determine the volatility level to buy when the realized volatility of the asset is very low as the cap prevents the leverage from increasing for achieving the target volatility.

26.2 Gap Risk Based Asset Allocation

A gap risk in portfolio management is the risk caused by a significant asset's price change from one level to another with no trading possibility in between, for instance, the change between closing and open prices or a sudden large market move due to an unpredictable event/news.

A gap risk-based asset allocation strategy allows an investor to maintain some exposure to the risky asset while providing protection against gap risk on the downside. Denote

$Floor_t$: guaranteed minimum level

G : estimated gap risk for the underlying in % (e.g. 25% sudden drop for an equity index)

$Multiplier$: it is defined as $1/G$, e.g. $Multiplier = \frac{1}{25\%} = 4$

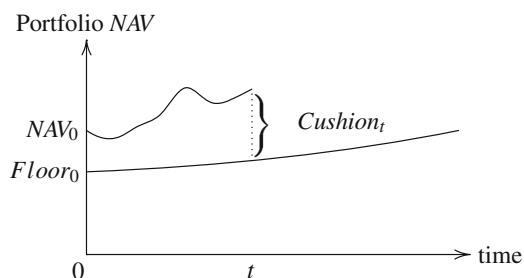
$Cushion_t$: equals to $NAV_t - Floor_t$ – all costs and fees which is the maximum loss under the constraint of $Floor$.

$Floor_t$ is the level that the product issuer (i.e. hedging party) does not want NAV_t to fall below. It can be a zero coupon bond price at t which allows to get the full principal back at maturity T , or a percentage of NAV_t or even constant levels for any t (e.g. straight line over time). The choice depends on the product offered to investors (Fig. 26.1).

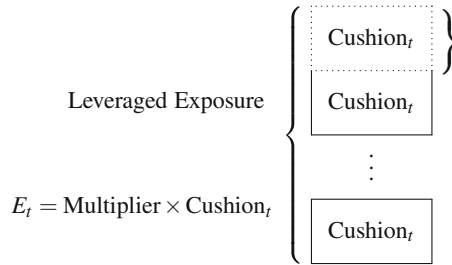
For protecting the investment to $Floor_t$ against the gap risk G , the *absolute* exposure to risky asset at t can be set at

$$E_t = Multiplier \times Cushion_t$$

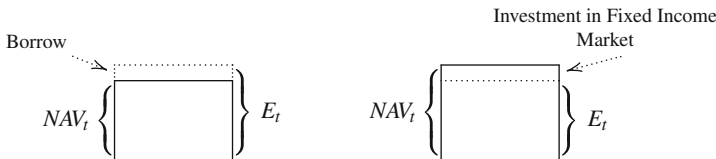
Fig. 26.1 An increasing floor with time t



and the relative exposure is $w_t = E_t / NAV_t$. Should the gap risk occur and the asset price drop exactly by G , the loss for the investor will be just $Cushion_t$ (as the loss is $E_t \times G = Multiplier \times Cushion_t \times G = \frac{1}{G} \times Cushion_t \times G = Cushion_t$). In case that the price drop is more than G , the product issuer will take the loss beyond $Cushion_t$ but not the investor. For this reason, people say that the issuer is shorting a Gap Put option.



The gap risk-based asset allocation strategy consists of dynamically adjusting the exposure according to the above formula over time. If $E_t > NAV_t$ at some time t , the investor needs to borrow for financing the leveraged investment. Otherwise, the difference is invested in the fixed income market.



When NAV_t drops to $Floor_t$, there will be no more investment in the risky asset until product maturity T . The strategy is then said to be “cash-locked”, and the investor has to forfeit any further upside in the risky asset. Note that the gap risk-based method is different from the option-based principal-protecting structure where the floor is invested in zero coupon bond for getting back 100% principal at maturity while the remaining money is invested in an option offering the exposure to the asset which is not impacted by NAV_t .

Example 26.2 Consider $G = 25\%$ and $Multiplier = 1/G = 4$, such that the exposure is $E_t = 4 \times Cushion_t$. Assume there is no cost or fee. Under the exposure E_t , the worst loss caused by $G \geq 25\%$ equals to $Cushion_t$ for the investor. Suppose at inception $NAV_0 = \$100$ and $Floor_0 = \$85$. Then,

$$E_0 = Multiplier \times Cushion_0 = 4 \times (\$100 - \$85) = \$60.$$

It means $w_0 = E_0/NAV_0 = 60\%$ of the portfolio is invested in the risky asset. Furthermore, the remaining 40% is invested in risk-free fixed-income assets.

There are two popular structures based on gap risk approach in the market: CPPI and TIPP which will be presented in the following sections. Assets with low volatility are more suitable for these structures than the assets with high volatility. CPPI linked to fixed-income mutual funds is a typical example.

The gap risk-based allocation can be used to provide capital protection without using options. It is particularly useful when the underlying has no volatility market. The drawback of this allocation mechanism is that, if the underlying is volatile, it risks to make the investor “buy high/sell low”, similar to a short Gamma position.

Similar to the volatility target strategy, a *threshold* is often set for the exposure adjustment in the gap risk-based approach for reducing the number of small transactions. One must also pay attention to the timing of rebalancing, as the gap risk might bring the portfolio below the floor level more easily if the rebalancing dates are not frequent (e.g. weekly).

There are numerous variations including the aforementioned fixed floor level for avoiding interest rate impact on the cushion and floor, guaranteed minimum exposure level to the risky asset for avoiding to be cash-locked, time-dependent multiplier, or even a volatility target for controlling the exposure. The above variations may be hybrid options with risks linked to interest rate and the risky asset for the hedging party.

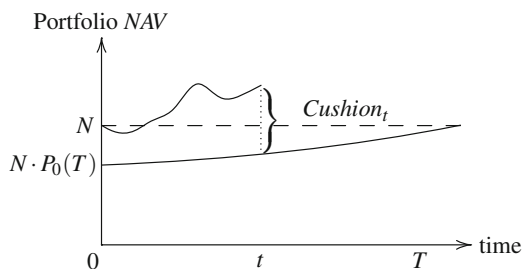
26.2.1 CPPI (Constant Proportion Portfolio Insurance)

In its basic version, CPPI [63] offers 100% principal protection of notional N ($= NAV_0$) at product maturity T . A zero coupon bond maturing at T with time- t price $P_t(T)$ is used for the principal guarantee. See Fig. 26.2 for reference.

The implementation is described as follows:

1. Investing $E_t = \text{Multiplier} \times \text{Cushion}_t$ in the risky asset, where $\text{Cushion}_t = NAV_t - N \times P_t(T)$ – cost and fees.
2. Investing the amount $NAV_t - E_t$ into some “risk-free” fixed-income assets (bills, bonds, etc.). If $NAV_t - E_t$ is negative, borrowing will be needed.

Fig. 26.2 An illustration for CPPI. Note that the bond floor may fluctuate because of interest rate



3. If NAV_t touches $N \times P_t(T)$, the investment will be fully in the T -maturity zero coupon bond, which costs $N \times P_t(T)$ for receiving N at maturity. The CPPI is said to be cash-locked.

For CPPI modelling, readers are referred to [24] which uses a jump-diffusion model in continuous-time framework.

26.2.2 TIPP (Time Invariant Portfolio Protection)

TIPP [30] is a structure without the need of a fixed maturity. Based on the Gap risk based leveraging/deleveraging mechanism, it guarantees a floor percentage p (e.g. 80%) to the running highest NAV (denoted as $\widehat{NAV}_t$), which is the highest net asset value since inception until t . The floor level is thus

$$\text{Floor}_t = p \times \widehat{NAV}_{t-1} = p \times \max_{0 \leq s < t-1} NAV_s$$

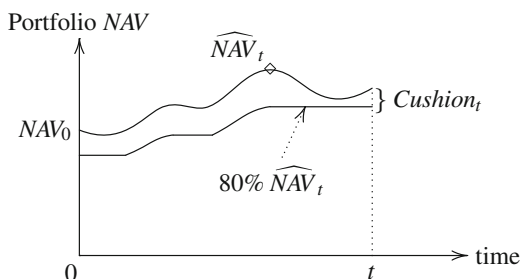
which is non-decreasing in time. See Fig. 26.3 for reference. The idea resembles the trailing stop method for securing profit in trading strategies. The implementation of TIPP is described as follows:

1. Investing $E_t = \text{Multiplier} \times \text{Cushion}_t$ in the risky asset, where $\text{Cushion}_t = NAV_{t-1} - p \times \widehat{NAV}_{t-1}$ – running cost and fees
2. Invest $NAV_t - E_t$ in money market at overnight rate (borrowing is needed if this amount is negative).
3. Theoretically speaking, if the interest rate is positive, there will never be a “cash-locked” situation because the investment of the guaranteed amount will accrue a small cushion every day. But it will take a long time to obtain any meaningful cushion.

26.3 Portfolio Asset Allocation

In the following, we will turn our focus to how to allocate exposure to multiple risky assets (classes).

Fig. 26.3 An illustration of TIPP with $p = 80\%$



We first list below the general notations used throughout this section:

n	number of assets in the portfolio
w_i	weighting/exposure of the i -th asset
$\mathbf{w} = [w_1, \dots, w_n]^\top$	column vector of weights for the asset allocation
μ_i	return of the i -th asset
r_i	expected return of the i -th asset ($r_i = \mathbb{E}[\mu_i]$)
σ_i	standard deviation (volatility) of the i -th asset
ρ_{ij}	correlation between the i -th and the j -th assets
$\Sigma = [\rho_{ij}\sigma_i\sigma_j]_{n \times n}$	covariance matrix for the asset returns
$\mathbf{1} = [1, \dots, 1]^\top$	n -dimensional column vector of value 1

The returns μ_i are random variables of the i -th asset. And the quantities $r_i = \mathbb{E}[\mu_i]$, $\sigma_i^2 = \text{Var}(\mu_i)$, and $\rho_{ij}\sigma_i\sigma_j = \text{Cov}(\mu_i, \mu_j)$ are assumed to be known.¹

To quantify a portfolio, we consider the portfolio return $\mu = \sum_{i=1}^n w_i \mu_i$, a random variable with weights $\mathbf{w}$ as parameters. Let the *expected* portfolio return be $\hat{r} = \mathbb{E}[\mu] = \sum_{i=1}^n w_i r_i$. Furthermore, for

σ^2 portfolio return variance

$\text{Cov}(\mu, \hat{\mu})$ Covariance of two portfolio returns μ and $\hat{\mu}$ with weights $\mathbf{w}$ and $\hat{\mathbf{w}}$

The two numbers σ^2 and $\text{Cov}(\mu, \hat{\mu})$ can be represented by a real quadratic form.

Proposition 26.1 *The portfolio return variance and the covariance of two portfolio returns can be given in matrix format as*

$$\begin{aligned}\text{Var}(\mu) &= \sigma^2 = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \rho_{ij} \sigma_i \sigma_j = \mathbf{w}^\top \Sigma \mathbf{w}, \\ \text{Cov}(\mu, \hat{\mu}) &= \sum_{i=1}^n \sum_{j=1}^n w_i \hat{w}_j \rho_{ij} \sigma_i \sigma_j = \mathbf{w}^\top \Sigma \hat{\mathbf{w}}.\end{aligned}$$

Proof From the definition of covariance, we have

$$\begin{aligned}\text{Cov}(\mu, \hat{\mu}) &= \mathbb{E} \left[\left(\sum_{i=1}^n w_i \mu_i - \mathbb{E} \left[\sum_{i=1}^n w_i \mu_i \right] \right) \left(\sum_{j=1}^n \hat{w}_j \mu_j - \mathbb{E} \left[\sum_{j=1}^n \hat{w}_j \mu_j \right] \right) \right] \\ &= \mathbb{E} \left[\sum_{i=1}^n w_i (\mu_i - r_i) \sum_{j=1}^n \hat{w}_j (\mu_j - r_j) \right] = \sum_{i=1}^n \sum_{j=1}^n w_i \hat{w}_j \rho_{ij} \sigma_i \sigma_j.\end{aligned}$$

Letting $\hat{\mu} = \mu$ gives the result of the portfolio return variance σ^2 . $\square$

¹For instance, historical data from a window are taken to sample μ_i , and sample statistics can be used for r_i , σ_i , and ρ_{ij} .

In the following sections, we first introduce a class of allocation methods which only make use of asset volatilities σ_i 's and correlations ρ_{ij} 's and do not involve estimating the asset returns r_i 's. Then, we will present the portfolio approaches aiming to determine the asset weighting vector $\mathbf{w} = [w_1, \dots, w_n]^\top$ such that the portfolio expected return $\hat{r}$ and the portfolio risk σ are optimized.

26.3.1 Risk-Based Methods

The risk of an asset is often measured by its volatility, which can be estimated and used for asset allocation.

Note that the trivial allocation known as *equally weighted portfolio* with all weights equal to $w_i = \frac{1}{n}$ is a special case in this class.

1. Risk parity

The risk parity strategy consists of allocating the same risk quantity (i.e. volatility) across all the component assets of the portfolio. For instance, it can be achieved by periodically resetting the weights so that

$$w_i \sigma_i = w_j \sigma_j, \quad \forall i, j = 1, 2, \dots, n.$$

To determine the weighting w_i of asset i for achieving risk parity given the volatilities of each asset, we note that

$$1 = \sum_{j=1}^n w_j = \sum_{j=1}^n \frac{w_j \sigma_j}{\sigma_j}.$$

Pulling w_i out of the summation gives

$$w_i = \frac{1/\sigma_i}{\sum_{j=1}^n (1/\sigma_j)}.$$

Risk parity is easy to implement and particularly convenient for long-only scenario. When assets go bad, they usually are highly volatile, and risk parity places less weight on such assets.

2. Risk budgeting

Risk budgeting consists of allocating a risk budget for each asset in the portfolio.

We know that $\sigma^2 = \mathbf{w}^T \Sigma \mathbf{w} = \sum_{i=1}^n w_i (\Sigma \mathbf{w})_i$, or, equivalently, $1 = \sum_{i=1}^n w_i \frac{(\Sigma \mathbf{w})_i}{\sigma^2}$.

It is intuitive to set the risk budgets p_i as

$$p_i = w_i \frac{(\Sigma \mathbf{w})_i}{\sigma^2},$$

which satisfies $\sum_i p_i = 1$. Then, one can solve w_i 's from the above relation with the weight constraint $\sum_i w_i = 1$.

The risk budgeting method has the advantage of simplicity. The investor may achieve the desired return profile through his risk appetite/tolerance. For example, one may apply

- 30% on equities,
- 20% on emerging market bonds,
- 40% on investment grade bonds,
- 10% on commodities.

Actually, the weights obtained from Risk Budgeting method can be reduced to the risk parity weights when all asset pairs are correlated with the same correlation level ρ and the risk budgets are equal [70].

3. Most diversified portfolio

The method is proposed by Choueifaty and Coignard [20]. The idea is to determine the weights w_i by solving the following maximization problem, in which the target function is known as the *diversification ratio*

$$\max_{\mathbf{w}} \frac{\sum_{i=1}^n w_i \sigma_i}{\sigma} = \max_{\mathbf{w}} \frac{\sum_{i=1}^n w_i \sigma_i}{\sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}}.$$

Once again the expected returns r_i 's are not required in the derivation of w_i 's, and the problem can be solved by numerical methods.

26.3.2 Modern Portfolio Theory

The risk-based allocation methods only utilize volatility and, sometimes, the correlation. On the other hand, the portfolio theory pioneered by Nobel laureate Harry Markowitz [53] makes use of both portfolio return and portfolio variance. This problem is named as mean-variance optimization (MVO), or the modern portfolio theory (MPT). It consists of solving the optimal asset allocation $\mathbf{w}$ with either of the following equivalent problems:

- given the portfolio variance, solving for the highest expected return
- given the portfolio expected return, solving for lowest portfolio variance.

For instance, given an *expected* portfolio return $\hat{r}$, the optimal asset allocation is the solution to the following quadratic programming problem:

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w},$$

subject to the constraints

$$\begin{cases} \mathbf{r}^\top \mathbf{w} = \hat{r}, \\ \mathbf{1}^\top \mathbf{w} = 1, \end{cases}$$

where $\mathbf{r} = [r_1, \dots, r_n]^\top$ are the *expected* returns of the assets.

Proposition 26.2 *The solution to the MVO problem is $\mathbf{w} = \mathbf{h} + \hat{r}\mathbf{g}$ (assuming Σ is invertible), where $\mathbf{h}, \mathbf{g}$ are vectors and are independent of $\hat{r}$ (i.e. common to all solution $\mathbf{w}$) given by*

$$\mathbf{h} = \Sigma^{-1} \frac{\mathbf{b}\mathbf{1} - \mathbf{a}\mathbf{r}}{\mathbf{b}\mathbf{c} - \mathbf{a}^2} \quad \text{and} \quad \mathbf{g} = \Sigma^{-1} \frac{\mathbf{c}\mathbf{r} - \mathbf{a}\mathbf{1}}{\mathbf{b}\mathbf{c} - \mathbf{a}^2},$$

where $\mathbf{a}, \mathbf{b}$, and $\mathbf{c}$ are scalars

$$\mathbf{a} = \mathbf{1}^\top \Sigma^{-1} \mathbf{r}, \quad \mathbf{b} = \mathbf{r}^\top \Sigma^{-1} \mathbf{r}, \quad \text{and} \quad \mathbf{c} = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}.$$

Proof The following proof is extracted from [54].

First order condition on the Lagrangian

$$\mathcal{L}(\mathbf{w}, \lambda_1, \lambda_2) = \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w} + \lambda_1 (\hat{r} - \mathbf{r}^\top \mathbf{w}) + \lambda_2 (1 - \mathbf{1}^\top \mathbf{w})$$

implies

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \Sigma \mathbf{w} - \lambda_1 \mathbf{r} - \lambda_2 \mathbf{1} = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda_1} = \hat{r} - \mathbf{r}^\top \mathbf{w} = 0 \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial \lambda_2} = 1 - \mathbf{1}^\top \mathbf{w} = 0.$$

From the first equation, we obtain $\mathbf{w} = \Sigma^{-1}(\lambda_1 \mathbf{r} + \lambda_2 \mathbf{1})$. Replacing $\mathbf{w}$ in the two constraint equations, we have

$$\mathbf{r}^\top \mathbf{w} = \mathbf{r}^\top \Sigma^{-1} (\lambda_1 \mathbf{r} + \lambda_2 \mathbf{1}) = \hat{r},$$

$$\mathbf{1}^\top \mathbf{w} = \mathbf{1}^\top \Sigma^{-1} (\lambda_1 \mathbf{r} + \lambda_2 \mathbf{1}) = 1.$$

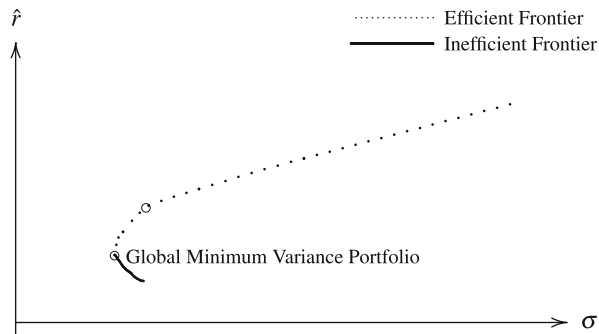
Using the notations of $\mathbf{a}, \mathbf{b}$, and $\mathbf{c}$, we get $\begin{bmatrix} \mathbf{b} & \mathbf{a} \\ \mathbf{a} & \mathbf{c} \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} \hat{r} \\ 1 \end{bmatrix}$. Hence,

$$\lambda_1 = \frac{\mathbf{c}\hat{r} - \mathbf{a}}{\mathbf{b}\mathbf{c} - \mathbf{a}^2}, \quad \lambda_2 = \frac{\mathbf{b} - \mathbf{a}\hat{r}}{\mathbf{b}\mathbf{c} - \mathbf{a}^2},$$

and

$$\mathbf{w} = \Sigma^{-1} \frac{\mathbf{r}(\mathbf{c}\hat{r} - \mathbf{a}) + \mathbf{1}(\mathbf{b} - \mathbf{a}\hat{r})}{\mathbf{b}\mathbf{c} - \mathbf{a}^2} = \mathbf{h} + \hat{r}\mathbf{g}. \quad \square$$

Fig. 26.4 Illustration of efficient frontier on the risk-return plane



The proposition implies that, once Σ and $\mathbf{r}$ are estimated and $\hat{r}$ is specified, the scalars $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ can be calculated accordingly and used to determine $\mathbf{w}$.

Note that *any* portfolio can be simply characterized by a pair of numbers $(\hat{r}, \sigma)$ and therefore it is convenient to visualize portfolios as a point on the $\hat{r}$ - σ plane. See Fig. 26.4 for reference. In the MVO framework, for each given $\hat{r}$, the solution $\mathbf{h} + \hat{r}\mathbf{g}$ is the set of weights that makes the resulting portfolio have minimum risk σ and

$$\sigma^2 = \mathbf{w}^T \Sigma \mathbf{w} = (\mathbf{h} + \hat{r}\mathbf{g})^T \Sigma (\mathbf{h} + \hat{r}\mathbf{g}) = \frac{\mathbf{b} - 2\mathbf{a}\hat{r} + \mathbf{c}\hat{r}^2}{\mathbf{bc} - \mathbf{a}^2}.$$

Therefore, the set of MVO solutions for different $\hat{r}$ constitutes a hyperbola on the $\hat{r}$ - σ plane, and the upward sloping part is called the *efficient frontier*. To gain higher return, risk-averse investors have to move up the efficient frontier and endure more risk.

If a risk-free asset (with “risk-free” rate r_f) is part of the portfolio [80], the mean-variance minimization problem becomes

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w},$$

subject to a new target return constraint

$$\mathbf{r}^T \mathbf{w} + (1 - \mathbf{w}^T \mathbf{1})r_f = \hat{r}.$$

The weights for risky assets together with the risk-free asset sum up to 1 and this condition is naturally embedded in the above constraint. Note that $\mathbf{w}^T \mathbf{1} > 1$ means borrowing from the money market. The method of Lagrange multiplier gives

$$\mathbf{w} = (\hat{r} - r_f) \Sigma^{-1} (\mathbf{r} - r_f \mathbf{1}) / K,$$

where $K = (\mathbf{r} - r_f \mathbf{1})^T \Sigma^{-1} (\mathbf{r} - r_f \mathbf{1}) = \mathbf{b} - 2\mathbf{a}r_f + \mathbf{c}r_f^2$ is positive because Σ is positive definite. Using $\sigma^2 = \mathbf{w}^T \Sigma \mathbf{w}$, we obtain

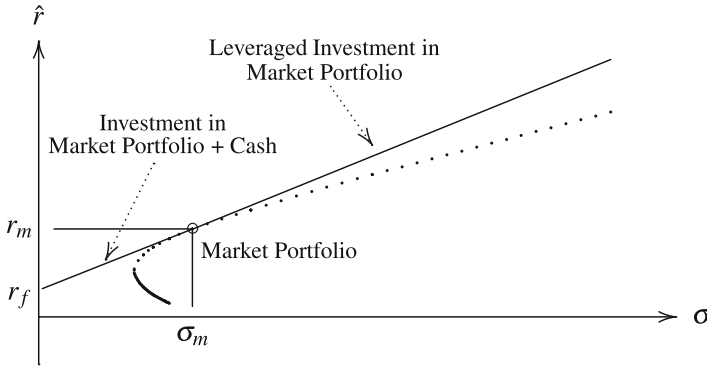


Fig. 26.5 Illustration of efficient frontier. The terms r_m and σ_m represent respectively the expected return and volatility of the market portfolio. Sharpe ratio of market portfolio is then $\frac{r_m - r_f}{\sigma_m}$

$$\hat{r} = r_f + \sqrt{K}\sigma.$$

Hence, the efficient frontier of the portfolio forms a straight line on the risk-return plane which is known as the Capital Allocation Line (CAL). The CAL has an intercept at r_f and the slope of the CAL is the Sharpe ratio, i.e., excess return over volatility

$$\frac{\hat{r} - r_f}{\sigma}.$$

The CAL happens to be tangential to the efficient frontier of the portfolio without the risk-free asset (i.e. $\mathbf{1}^\top \mathbf{w} = 1$). In a perfect universe, all investors will like to hold the tangency portfolio (hence the name Market Portfolio). The part of the line above Market Portfolio involves borrowing at r_f and leveraging the investment in the Market Portfolio, while the segment of CAL below Market Portfolio implies investment in both Market Portfolio and the risk-free asset. See Fig. 26.5.

In general, an MVO-based strategy is to minimize the portfolio risk

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}, \quad \text{subject to } \mathbf{w} \in \Omega,$$

where Ω gives the constraints of the weight $\mathbf{w}$, such as the previously used target portfolio return $\mathbf{r}^\top \mathbf{w} = \hat{r}$ and total weight sum $\mathbf{1}^\top \mathbf{w} = 1$. It is often necessary to set other (linear) constraints on $\mathbf{w}$, such as a long only scenario (i.e. $w_i \geq 0$), or certain weight limits on some of the assets (e.g. $w_2 + w_3 \leq 30\%$, $0 \leq w_1 + w_4 + w_5 \leq 100\%$, etc). The constrained optimization problem can be resolved with numerical methods.

MVO is an important tool for asset managers who are mandated to make discretionary investment decisions. The business model of investment banks consists of offering financial services but not discretionary investment management. With the

development of structured business, banks also started to employ MVO approach to offer products with rule-based strategy investing in the portfolios on the efficient frontier. The implementation is described as follows:

- The strategy typically invests into an asset universe across different asset classes (through indices, ETFs or large mutual funds).
- It often applies a fixed target portfolio volatility.
- The portfolio is rebalanced periodically, e.g., every month, quarter or semester.
- Between the rebalancing dates, the volatility target mechanism may be applied for controlling the portfolio volatility.
- It is possible to write options on the strategy if the volatility target is applied.

26.3.3 Black–Litterman Model

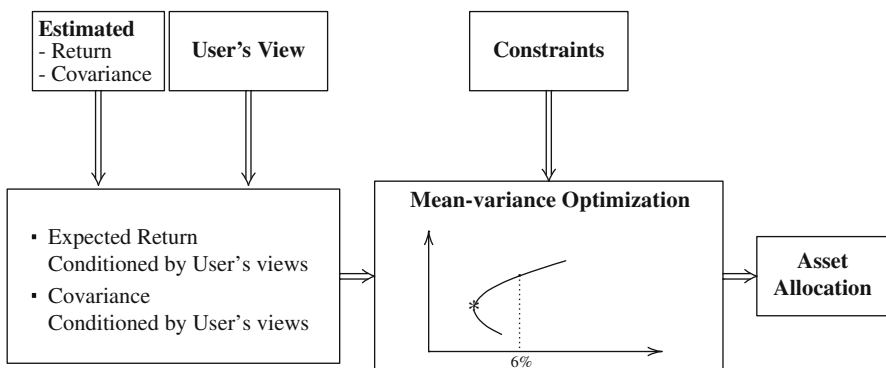
The application of modern portfolio theory faces a well-known problem: although the covariances of the assets can be approximately estimated, it is difficult to come up with reasonable estimates of the returns. Moreover, it is known that the MVO weights are very sensitive to the expected returns, and assets with higher return will be awarded with extremely larger weights.

Black–Litterman (BL) model, which was first proposed by Fischer Black and Robert Litterman at Goldman Sachs [11], extends the mean-variance analysis by combining the estimated market returns with user’s view on the estimation uncertainty as well as the relationships between the returns of the assets. The model is described as follows. From

- the estimated returns and covariances (“market prior”), and
- the user’s views expressed as linear combinations of the estimated returns,

the BL model determines the input parameters for the MVO problem (“posterior”), i.e.,

- the expected returns conditioned by user’s views;
- the covariance conditioned by user’s view.



The user of BL model can express

- their view on the uncertainty of the estimated returns $\mathbf{r}$ by applying a distribution instead of fixed values: $\mathbf{r} \sim \mathcal{N}(\boldsymbol{\pi}, \tau \boldsymbol{\Sigma})$, where τ is a scalar
- their market view by means of a linear combination of the estimated returns $\mathbf{Pr}$ where $\mathbf{P}$ is the pick matrix. For example, with $\mathbf{P} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$ and $\mathbf{Pr} = \begin{bmatrix} 5\% \\ 10\% \end{bmatrix}$, the user gives the following relative view:
 1. Asset 1 outperforms Asset 3 by 5%
 2. return of Asset 2 is 10%.
- his uncertainty about the market view, denoted by $\boldsymbol{\Omega}$. The matrix $\boldsymbol{\Omega}$ is often assumed to be diagonal so as to avoid covariance between views. If $\boldsymbol{\Omega}$ is the zero matrix, all the views have 100% certainty.

The expected return and its covariance conditional on user's views are given below using a Bayesian approach.

Proposition 26.3 (Black–Litterman) *Denote*

“market prior” expected return $\mathbf{r}$ follows a normal distribution $\mathbf{r} \sim \mathcal{N}(\boldsymbol{\pi}, \tau \boldsymbol{\Sigma})$, in which τ allows the user to express the estimation uncertainty;
“market views” of the user $\mathbf{q}|\mathbf{r} \sim \mathcal{N}(\mathbf{Pr}, \boldsymbol{\Omega})$.

Then, $\mathbf{r}|\mathbf{q}$ (“posterior”) follows a multivariate normal distribution with

$$\begin{aligned} \mathbb{E}[\mathbf{r}|\mathbf{q}] &= \boldsymbol{\pi} - \tau \boldsymbol{\Sigma} \mathbf{P}^T (\boldsymbol{\Omega} + \tau \mathbf{P} \boldsymbol{\Sigma} \mathbf{P}^T)^{-1} (\mathbf{P} \boldsymbol{\pi} - \mathbf{q}), \\ \text{Cov}(\mathbf{r}, \mathbf{r}|\mathbf{q}) &= \tau \boldsymbol{\Sigma} - \tau^2 \boldsymbol{\Sigma} \mathbf{P}^T (\boldsymbol{\Omega} + \tau \mathbf{P} \boldsymbol{\Sigma} \mathbf{P}^T)^{-1} \mathbf{P} \boldsymbol{\Sigma}. \end{aligned}$$

Proof The proof follows the approach in [46]. Firstly the unbiased estimator gives

$$\mathbb{E}[\mathbf{q}] = \mathbb{E}[\mathbb{E}[\mathbf{q}|\mathbf{r}]] = \mathbb{E}[\mathbf{Pr}] = \mathbf{P} \boldsymbol{\pi}.$$

Applying the law of total covariance (see Proposition A.1 in Appendix A), we have

$$\begin{aligned} \text{Cov}(\mathbf{r}, \mathbf{q}) &= \mathbb{E}[\text{Cov}(\mathbf{r}, \mathbf{q}|\mathbf{r})] + \text{Cov}(\mathbb{E}[\mathbf{r}|\mathbf{r}], \mathbb{E}[\mathbf{q}|\mathbf{r}]) \\ &= \mathbb{E}[\text{Cov}(\mathbf{r}, \mathbf{q}|\mathbf{r})] + \text{Cov}(\mathbf{r}, \mathbb{E}[\mathbf{q}|\mathbf{r}]). \end{aligned}$$

From the definition of conditional covariance (in the multivariate sense), we have

$$\text{Cov}(\mathbf{r}, \mathbf{q}|\mathbf{r}) = \mathbb{E}[(\mathbf{r} - \mathbb{E}[\mathbf{r}|\mathbf{r}]) (\mathbf{q} - \mathbb{E}[\mathbf{q}|\mathbf{r}])^T | \mathbf{r}] = \mathbb{E}[(\mathbf{r} - \mathbf{r}) (\mathbf{q} - \mathbb{E}[\mathbf{q}|\mathbf{r}])^T | \mathbf{r}] = 0.$$

Hence, we obtain

$$\text{Cov}(\mathbf{r}, \mathbf{q}) = \text{Cov}(\mathbf{r}, \mathbb{E}[\mathbf{q}|\mathbf{r}]) = \text{Cov}(\mathbf{r}, \mathbf{Pr}) = \text{Var}(\mathbf{r}) \mathbf{P}^T = \tau \boldsymbol{\Sigma} \mathbf{P}^T,$$

where $\mathbf{P}$ is taken out of the covariance operator by Proposition A.2 in Appendix A. And similarly

$$\begin{aligned}\text{Cov}(\mathbf{q}, \mathbf{q}) &= \mathbb{E}[\text{Cov}(\mathbf{q}, \mathbf{q}|\mathbf{r})] + \text{Cov}(\mathbb{E}[\mathbf{q}|\mathbf{r}], \mathbb{E}[\mathbf{q}|\mathbf{r}]) \\ &= \text{Var}(\mathbf{q}|\mathbf{r}) + \text{Cov}(\mathbf{P}\mathbf{r}, \mathbf{P}\mathbf{r}) = \mathbf{\Omega} + \tau \mathbf{P}\mathbf{\Sigma}\mathbf{P}^\top.\end{aligned}$$

Therefore $\mathbf{q} \sim \mathcal{N}(\mathbf{P}\boldsymbol{\pi}, \mathbf{\Omega} + \tau \mathbf{P}\mathbf{\Sigma}\mathbf{P}^\top)$. Coupling the normally distributed $\mathbf{r}$ and $\mathbf{q}$, and applying the conditional normal distribution (see Proposition A.4 in Appendix A) with

$$\begin{bmatrix} \mathbf{r} \\ \mathbf{q} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix}\right),$$

where

$$\begin{cases} \boldsymbol{\mu}_1 = \boldsymbol{\pi}, & \boldsymbol{\mu}_2 = \mathbf{P}\boldsymbol{\pi}, \\ \boldsymbol{\Sigma}_{11} = \text{Var}(\mathbf{r}) = \tau \boldsymbol{\Sigma}, \\ \boldsymbol{\Sigma}_{22} = \text{Cov}(\mathbf{q}, \mathbf{q}) = \mathbf{\Omega} + \tau \mathbf{P}\mathbf{\Sigma}\mathbf{P}^\top, \\ \boldsymbol{\Sigma}_{12} = \text{Cov}(\mathbf{r}, \mathbf{q}) = \tau \boldsymbol{\Sigma}\mathbf{P}^\top, \end{cases}$$

we obtain that the posterior $\mathbf{r}|\mathbf{q}$ follows the multivariate normal distribution with

$$\begin{aligned}\mathbb{E}[\mathbf{r}|\mathbf{q}] &= \boldsymbol{\pi} - \tau \boldsymbol{\Sigma}\mathbf{P}^\top (\mathbf{\Omega} + \tau \mathbf{P}\mathbf{\Sigma}\mathbf{P}^\top)^{-1} (\mathbf{P}\boldsymbol{\pi} - \mathbf{q}), \\ \text{Cov}(\mathbf{r}, \mathbf{r}|\mathbf{q}) &= \tau \boldsymbol{\Sigma} - \tau^2 \boldsymbol{\Sigma}\mathbf{P}^\top (\mathbf{\Omega} + \tau \mathbf{P}\mathbf{\Sigma}\mathbf{P}^\top)^{-1} \mathbf{P}\boldsymbol{\Sigma}.\end{aligned}\quad \square$$

Remark 26.1 The posterior estimated return $\mathbb{E}[\mathbf{r}|\mathbf{q}]$ also admits another form known as the “master formula” given by [38]

$$\mathbb{E}[\mathbf{r}|\mathbf{q}] = [(\tau \boldsymbol{\Sigma})^{-1} + \mathbf{P}^\top \boldsymbol{\Omega}^{-1} \mathbf{P}]^{-1} [(\tau \boldsymbol{\Sigma})^{-1} \boldsymbol{\pi} + \mathbf{P}^\top \boldsymbol{\Omega}^{-1} \mathbf{q}].$$

It can also be checked that

$$\text{Cov}(\mathbf{r}, \mathbf{r}|\mathbf{q}) = ((\tau \boldsymbol{\Sigma})^{-1} + \mathbf{P}^\top \boldsymbol{\Omega}^{-1} \mathbf{P})^{-1}.$$

See Proposition A.5 in Appendix A.

Exercises

1. Show that the equal weighting asset allocation scheme is essentially a “buy low/sell high” strategy.

2. Consider a CPPI strategy with a risky asset $dS_t = \mu S_t dt + \sigma S_t dW_t$ and a risk-free asset $dM_t = rM_t dt$ as floor. Let $C_t = NAV_t - M_t$ denote the cushion process and m is the multiplier. Hence, the total return of the portfolio consisting S_t and M_t is

$$\frac{dNAV_t}{NAV_t} = \frac{mC_t}{NAV_t} \frac{dS_t}{S_t} + \left(1 - \frac{mC_t}{NAV_t}\right) \frac{dM_t}{M_t}.$$

- (i) Show that

$$\frac{dC_t}{C_t} = [m\mu + (1-m)r]dt + m\sigma dW_t.$$

- (ii) Show that for $t > 0$,

$$C_t = C_0 e^{(1-m)(r + \frac{m}{2}\sigma^2)t} \left(\frac{S_t}{S_0}\right)^m.$$

3. (Global Minimum Variance) When solving the MVO problem, one can remove the constraint $\mathbf{r}^\top \mathbf{w} = \hat{r}$ and obtain the *Global Minimum Variance* portfolio. Using the method of Lagrange multiplier, prove that the weights take the form of

$$\mathbf{w} = \frac{\boldsymbol{\Sigma}^{-1} \mathbf{1}}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \mathbf{1}}.$$

4. Assume two risky assets with estimated annualized volatility 30% and 10% respectively. They have a correlation factor of 0.7.
- Calculate the weights for a risk parity portfolio
 - Calculate the annualized volatility of an equally weighted portfolio
 - Calculate the weights for a global minimum variance portfolio.

Chapter 27

Systematic Strategy



27.1 Introduction

Structured business has experienced a fantastic development since the late 1990s. At the early stage, much of the effort has been spent on the payoff engineering. In a few years, the payoff engineering has achieved its maturity stage: some “exotic” structures are now considered as flow products and the pricing is more and more efficient.

Guided by the research works in finance and the experience of asset managers, financial engineers in investment banks start to focus on rule-based investment strategies known as *systematic strategies*. The investment allocation in such strategies is rebalanced periodically, e.g. monthly, with an algorithm following a set of rules. The input data for the algorithm should be publicly available. Also the rules are disclosed to investors in general. Systematic strategies can be categorized as follows

Beta strategy: a strategy with its returns correlated to the performance of market traded assets (e.g., bonds, stocks) or benchmark (indices);

Smart beta strategy: a strategy aiming to provide higher Sharpe ratio than the market traded assets; Its return is related to the performance of “factors” such as value or momentum other than just the market traded assets through, typically, long-only exposure.

Alternative beta strategy: a strategy designed to achieve uncorrelated returns to the market traded assets with full exposure to “factors”; The strategy is generally implemented with diversified and/or long-short positions.

The advantage of a systematic strategy includes

- low cost: the management cost is low compared to an actively managed fund;
- transparency: the strategy rules are made transparent to investors;

- consistency: the investment decisions are algorithm-driven, which avoids the impact of human behaviours; hence in general they can be backtested and automated;
- protection: with the techniques of volatility target or CPPI, principal protected products may be offered with systematic strategies. The traditional actively managed funds do not offer principal protection.

On the other hand, there are several well-known drawbacks for systematic strategies:

- non-flexibility: The rules are never perfect. Amending the rules is difficult if there are existing products linked to the strategy. For changing the rules, a new strategy needs to be set up and the investor should unwind the existing trade for investing into the new index. In comparison, the fund managers have the flexibility for making the investment decisions.
- time lag: Most of the strategies use historical data for obtaining signals. The regime change may be detectable only after a time lag.
- backtesting optimization: If the rebalancing is not on daily basis but on fixed schedules (e.g. every month end), the strategy parameters are sometimes optimized to these schedules for achieving a nice backtesting result. Using the same parameters, the results might be quite different if the rebalancing schedules are shifted to other days. Actually, a systematic strategy may show excellent backtested performance but deliver poor real performance after its launch.
- pricing risk: Any implemented strategy is proprietary to an investment bank. Very often, only that bank can offer products linked to the strategy. Investors may not be able to use competition to achieve pricing efficiency.

27.2 Asset Pricing Models

Asset pricing models relate asset returns to their driving factors and are useful in building systematic strategies. Factors may be identified with the techniques of econometrics. The factor models normally take a linear form.

27.2.1 Capital Asset Pricing Model (CAPM)

CAPM, which is proposed by Treynor [81], Sharpe [78], Lintner [49], and Mossin [59] independently, is based on the modern portfolio theory with the presence of a risk-free asset and assumes that the market factor is the only source of price risk. In practice, it is described by a simple linear regression model:

$$\mu_i - r_f = \beta_i(\mu_m - r_f) + \epsilon_i,$$

where

μ_i : return of asset i

μ_m : return of the market factor common to all assets

r_f : “risk-free” rate in fixed-income market

β_i : represents the correlation of asset i to the market factor, i.e.,

$$\beta_i = \frac{\text{Cov}(\mu_i, \mu_m)}{\text{Var}(\mu_m)}$$

ϵ_i : idiosyncratic component which has zero expected value (white noise).

The regression is carried out between the excess returns $\mu_i - r_f$ and $\mu_m - r_f$. The correlation to the market factor β_i is also the slope of the regression line. The market factor brings about the unavoidable systematic risk and ϵ_i carries the asset-specific idiosyncratic risk which can be diversified away.

Index Model

It is an extension of CAPM by introducing a non systematic return called *Alpha*, or Jensen’s Alpha. Consider a fund manager and their return variable R_{fund} . Regression against the market factor gives

$$R_{\text{fund}} - r_f = \alpha_{\text{fund}} + \beta_{\text{fund}}(\mu_m - r_f) + \epsilon_{\text{fund}},$$

where α_{fund} is the unexplained return which is uncorrelated to μ_m . The index model incorporates the impact of the investment skills of a fund manager, which is quantified by the value α_{fund} .

The theoretical framework of CAPM is based on market equilibrium and assumes market conditions such as zero transaction cost, arbitrary divisibility of underlying asset, and etc. It states that the expected return of the non-systematic component is zero and

$$\mathbb{E}[\mu_i] = r_f + \beta_i(\mathbb{E}[\mu_m] - r_f),$$

which gives the theoretical expected return of the i -th asset in an efficient market. When $\beta_i = 0$, the i -th asset is uncorrelated to the market and expects to earn the risk-free rate. Under CAPM, investors share the same beliefs about return and risk and will be rewarded for the risk taken. They are equally exposed to the risky assets and hold the same *tangency portfolio* for the highest Sharpe ratio, which turns out to be the market portfolio in MVO analysis.

27.2.2 Arbitrage Pricing Theory (APT)

The arbitrage pricing theory, proposed by Stephen Ross [71], extends CAPM by allowing multiple risk factors, resulting in a multiple linear regression formulation,

$$\mu_i - r_f = \sum_{k=1}^K \beta_{ik}(F_k - r_f) + \epsilon_i, \quad (27.1)$$

where μ_i , r_f , ϵ_i are defined as before, and F_k is the return of the k -th systematic risk factor which is common to all assets (hence no index i). The i -th asset's return is now described by a set of betas β_{ik} (also known as *factor loadings*), generalizing the single market factor in CAPM.

APT uses fewer assumptions than CAPM. Statistical methods, such as principal component analysis (PCA), may help to identify the factors in (27.1) from the given asset returns μ_i , $i = 1, 2, \dots, n$. However, the generated factors may sometimes bear no economic meaning. Alternative methods are as follows.

Time series regression: Consider only the i -th asset and its returns μ_i 's are realized in a time series. Let F_k be *asset-independent* and *observable* factors (say, macroeconomic variables such as inflation, GDP or fundamental variables such as exchange rates) also given as time series. Then, a multiple regression is performed between μ_i and the K factors, and the factor loadings β_{ik} for asset i are determined. See the Chen–Roll–Ross [19] macroeconomic model for reference.

Cross-sectional regression: All assets are considered but only at the same time point t . Let the asset returns be μ_1 , μ_2 , and etc. at t . The factor loadings β_{ik} are assumed to be known in this case, e.g., asset-specific variables such as asset momentum or industry classification. Then, a multiple regression is employed to determine the factor realizations F_k at time t . See BARRA approach [35] for reference.

Others: There are also other approaches, say the Fama–MacBeth regression [31].

Recall that in the mean-variance analysis, the estimation of the input variables may be heavy when the number of assets n is large. The APT describes the asset returns by systematic factors, which are usually much smaller in size (known as *dimensionality reduction* in machine learning and statistics). The reduced estimation complexity for covariance Σ is achieved by

$$\Sigma = \mathbf{B}\Sigma_f\mathbf{B}^\top + \mathbf{D},$$

where $\mathbf{B}$ is the n -by- K matrix of factor loadings, Σ_f is the K -by- K factor covariance matrix, and $\mathbf{D}$ is the specific risk covariance matrix.

27.3 Factors and Factor-Based Strategies

In addition to the market factor in CAPM, significant advances have been made over the past two decades in understanding the factors impacting the *risk premium*, i.e., the return in excess of the risk-free rate through the exposure to the (risk) factors. As more data are becoming available, the number of such factors is increasing significantly and this phenomenon is described as a “zoo of new factors” by John Cochrane [23]. Nowadays, factors are made into indices by index providers such as MSCI. They are also used as investment strategies in factor-tracking ETFs or mutual funds.

In the following, we introduce some risk factors which are well studied in academic research and believed to offer risk premia from empirical tests. Evidently, the list of factors has no limit.

Momentum

Momentum strategy is based on the belief that increases in the price of a security will be likely followed by additional gains and vice versa for declining values. This strategy consists of taking long position in the assets with upward trends (and eventually short sell securities with downward trends).

For the implementation, practitioners need to make the choices regarding:

- observation window: for calculating price momentum, the common choices are 3 months, 6 months, or 1 year.
- return calculation: the return type may be total return, z -score¹ or some other return types. The historical data may be weighted equally or exponentially for giving higher importance to recent data.
- other signals: e.g., the crossing of short-term moving average over long-term moving average.

Value

Value factor investing has long existed and is designed to long the under-valued assets and eventually short over-valued assets. Prices are not directly used to represent the factor but instead normalized financial ratios are used, e.g.

- price-to-book (P/B) ratio
- forward price/earnings (P/E) ratio

¹ z -score is defined as $\frac{\text{return} - \text{averaged return}}{\text{standard deviation of the return}}$.

- dividend yield
- cash-flow to price ratio.

Equity valuation, bond rating and other financial analysis may be found on public or private sources:

- Bloomberg (“ANR” command, showing buy/sell/hold ratings from analysts)
- Conviction list from research teams of investment banks
- Financial analytics systems, e.g., STARMINE.

Size

Size factor is generally defined by the market capitalization (abbreviated as “cap”) and aims at investing into either of the following categories:

- small caps
- mid caps (medium size corporates)
- big caps.

A common size strategy is set up by buying small caps and short-selling big caps. Practical implementation might take into account the liquidity and transaction cost when trading small caps. The borrow availability and cost also need to be considered in building the short position.

Carry

Carry trade aims at creating value by carrying over time a certain position and earning on the accrual (amongst other things). For example, FX carry trade consists of

- borrowing in a low interest rate currency, e.g., JPY
- investing in a high interest rate currency, e.g., AUD.

This strategy will be a winning trade which earns approximately the interest rate difference if the FX rate has not changed significantly at the end of the trade tenor.

Volatility

Volatility strategies capture the opportunities from volatility exposures. For example:

- Receiving implied volatility and paying realized volatility: it is based on the observation that the implied volatility is often higher than the realized volatility.

- *Tail-risk hedging* strategy based on implied volatility (through trades on VIX or forward starting variance swaps): it jumps up at extreme market event.
- *Call overwriting* (or *Buy/Write*) strategy: it consists of holding a long position of certain assets and selling (either regularly or opportunistically) short term call options on (some of) these assets.

Geography

This factor targets to take advantage of either

- the high growth potential of certain geographical regions, or
- the growth differential between different geographical regions.

An example is to long emerging market and short developed market. When used in a BARRA-type factor model, the “Geography” factor can be represented by a dummy variable in the regression, i.e., taking the value 1 if belonging to a certain region and 0 otherwise.

Multi-factor Models

Factors are often combined to alleviate the possible underperformance of a specific factor during a certain period. The following well-known model is a typical example of implementable multi-factor models.

Example 27.1 The famous investable “Fama–French three-factor model” [32] included “Size” and “Value” factors in addition to the “Market” factor of CAPM. It is later extended by Carhart with an additional “Momentum” factor, leading to the “Fama–French–Carhart four-factor model” [13]:

$$\mu_i - r_f = \alpha_i + \beta_i \times (\mu_m - r_f) + s_i \times SMB + h_i \times HML + w_i \times WML + \epsilon_i,$$

where the size factor *SMB* (Small Minus Big), the value factor *HML* (High Minus Low), the momentum factor *WML* (Winner Minus Loser) are not obtained from direct observations but from a long-short type of factor mimicking portfolio:

SMB: difference between the returns on diversified portfolios of small stocks and big stocks;

HML: difference between the returns on diversified portfolios of high book-to-market stocks and low book-to-market stocks;

WML: difference between the returns on diversified portfolios of the winners and losers of the past year.

27.4 Other Systematic Strategies

Systematic strategies are not limited to factor-based ones. Any rule-based investment constitutes a systematic strategy.

In the market, we have seen many strategies based on dynamic asset allocation mechanism applied on a basket of assets from different asset classes. For example, several investment banks have implemented MVO-based strategies making dynamic investment on a basket of indices or ETFs covering equity, bond and commodities market (e.g. a basket of 10 underlyings). Risk parity is also a popular systematic strategy due to its simplicity.

27.5 Performance Measuring

Building systematic strategies is based on the *observed* daily or monthly returns of various assets, which are used to estimate the expected returns and covariance matrix. Once the strategy is set up, we can evaluate its backtested performance for k days/months. Take the example of global minimum variance strategy for n assets which determines the asset weights $\mathbf{w} = [w_1, w_2, \dots, w_n]^\top$ via the covariance matrix such that the expected variance of the portfolio is minimized.

Suppose at time $i = 1, 2, \dots, k$, the return of the j -th asset is r_{ij} , then the strategy (portfolio) return in the backtesting period is

$$\mathbf{r}_s = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_k \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{k1} & r_{k2} & \cdots & r_{kn} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix},$$

which is the realized version of the random variable $\mathbf{r}^\top \mathbf{w}$ in Chap. 26.

Basic descriptive statistics of $\mathbf{r}_s$ describe the systematic strategy such as

- mean: measuring the overall profitability
- standard deviation: a commonly accepted measure for risk
- skewness: whether winning small constantly but losing big once in a while, or the other way around
- kurtosis: tail events, the scenario of having large gain or loss.

The performance of a systematic strategy is generally measured with the same indicators used for funds. See Chap. 1 for the *ex-ante* definitions of Sharpe ratio, information ratio, Treynor ratio, Sortino ratio and maximum drawdown.

For instance, the *ex-post* calculation of Sharpe ratio is as follows. The mean of $\mathbf{r}_s$ minus the corresponding risk-free rate (or, the real financing cost) of the same period

gives the excess return, which is then divided by the volatility or sample standard deviation of r_s .

27.6 Customized Index

A *customized index* is an investment strategy with its return values published on a market data system, e.g. Bloomberg. Any systematic strategy can be presented and traded as a customized index.

One important type of customized index is the *rolling futures* index which consists of investing in futures continually. It is an efficient way to gain beta exposure to the underlying asset. The return of rolling futures is close to excess return due to the low capital cost for futures transactions. The simplest implementation consists of investing in the nearest (front-month) futures because of its high liquidity, and changing (rolling) to the next nearest futures a few days before the expiry date of the nearest futures. The downward sloping futures price curve (backwardation) makes “profit” on rolling. Conversely, in the upward sloping (contango) situation, the index suffers “loss” on rolling.

The investment banks often publish their systematic strategies as customized indices. The implementation of a customized index includes two documentations.

1. *Index Rules*: providing all the details including the investment strategy, costs and the exact actions to take upon eventual events. The below elements are often seen in the index rules.
 - Index fees (management fees)
 - Corporate actions treatment: dividend, spin-off, merger, etc.
 - Tax: dividend tax, capital gain tax, etc.
 - Investment restrictions: market access, securities borrow, conflicts of interests, etc.
 - FX exposure: composite return or hedging with FX forward
 - Liquidity: caps on each name according to its liquidity
 - Transaction cost, funding cost, securities lending benefits
 - and etc.
2. *Index Risk Disclosure*: explaining the risks as well as the investor suitability.

Exercises

1. Find out the definition of turnover rate for a portfolio. Does an actively managed portfolio or a passively managed one have a larger turnover rate and why is that? What kind of risk is related to a high turnover rate?

2. Consider CAPM and let $\mathbf{w}_{\text{mkt}} = [w_1, w_2, \dots, w_n]^\top$ denote the market capitalization weight such that the market return is

$$\mu_m = \sum_{j=1}^n w_j \mu_j$$

and Σ denotes the covariance matrix of asset returns. Show that the expected excess returns are

$$[\mathbb{E}[\mu_1] - r_f, \mathbb{E}[\mu_2] - r_f, \dots, \mathbb{E}[\mu_n] - r_f]^\top = \frac{\mathbb{E}[\mu_m] - r_f}{\text{Var}(\mu_m)} \Sigma \mathbf{w}_{\text{mkt}}.$$

Note that this formulation can be used for constructing the prior return in the Black–Litterman model.

3. Suppose the daily returns of a systematic strategy are given by

$$\mathbf{r}_s = [2.5986\%, -0.4661\%, -1.1829\%, -1.3517\%, 0.7311\%]^\top$$

for a certain week. Calculate the maximum drawdown for this period.

Appendix A

Theorems in Probability Theory

In this appendix, we have included some theorems in probability theory, especially ones in multivariate analysis.

A.1 Law of Total Covariance

Let X , Y , and Z be random variables in the same probability space. The conditional covariance of X and Y given Z is defined as

$$\text{Cov}(X, Y|Z) = \mathbb{E}[(X - \mathbb{E}[X|Z])(Y - \mathbb{E}[Y|Z])|Z] = \mathbb{E}[XY|Z] - \mathbb{E}[X|Z]\mathbb{E}[Y|Z].$$

Proposition A.1 (Law of Total Covariance)

$$\text{Cov}(X, Y) = \mathbb{E}[\text{Cov}(X, Y|Z)] + \text{Cov}(\mathbb{E}[X|Z], \mathbb{E}[Y|Z])$$

Proof Taking $\mathbb{E}[\cdot]$ on both sides of $\text{Cov}(X, Y|Z)$ gives

$$\mathbb{E}[\text{Cov}(X, Y|Z)] = \mathbb{E}[XY] - \mathbb{E}[\mathbb{E}[X|Z]\mathbb{E}[Y|Z]].$$

Then,

$$\begin{aligned} \text{Cov}(X, Y) &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \\ &= \mathbb{E}[\text{Cov}(X, Y|Z)] + \mathbb{E}[\mathbb{E}[X|Z]\mathbb{E}[Y|Z]] - \mathbb{E}[X]\mathbb{E}[Y] \\ &= \mathbb{E}[\text{Cov}(X, Y|Z)] + \mathbb{E}[\mathbb{E}[X|Z]\mathbb{E}[Y|Z]] - \mathbb{E}[\mathbb{E}[X|Z]]\mathbb{E}[\mathbb{E}[Y|Z]] \\ &= \mathbb{E}[\text{Cov}(X, Y|Z)] + \text{Cov}(\mathbb{E}[X|Z], \mathbb{E}[Y|Z]). \end{aligned} \quad \square$$

A.2 Multivariate Random Variable

Let $\mathbf{X}$ be a column vector of random variables $[X_1, X_2, \dots, X_p]^\top$ on some probability space. The expectation $\mathbb{E}[\mathbf{X}]$ is also a column vector, containing the expectations in each dimension. The variance $\text{Var}(\mathbf{X})$ is a $p \times p$ matrix defined as

$$\text{Var}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^\top],$$

which is also known as the variance-covariance matrix. Let $\mathbf{Y}$ be another q -dimensional multivariate random variable. The $p \times q$ *cross-covariance* matrix between $\mathbf{X}$ and $\mathbf{Y}$ is defined as

$$\text{Cov}(\mathbf{X}, \mathbf{Y}) = \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{Y} - \mathbb{E}[\mathbf{Y}])^\top] = \mathbb{E}[\mathbf{XY}^\top] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{Y}]^\top.$$

Let $\mathbf{A}$ and $\mathbf{B}$ be non-random $n \times p$ and $m \times q$ matrices respectively. We have the following proposition.

Proposition A.2 *For the linear transformations $\mathbf{AX}$ and $\mathbf{BY}$, we have*

$$\begin{aligned}\mathbb{E}[\mathbf{AX}] &= \mathbf{A}\mathbb{E}[\mathbf{X}], \\ \text{Var}(\mathbf{AX}) &= \mathbf{A}\text{Var}(\mathbf{X})\mathbf{A}^\top, \\ \text{Cov}(\mathbf{AX}, \mathbf{BY}) &= \mathbf{A}\text{Cov}(\mathbf{X}, \mathbf{Y})\mathbf{B}^\top.\end{aligned}$$

Proof The expectation follows directly from linearity. Then, we only need to show the covariance case

$$\begin{aligned}\text{Cov}(\mathbf{AX}, \mathbf{BY}) &= \mathbb{E}[\mathbf{AX}(\mathbf{BY})^\top] - \mathbb{E}[\mathbf{AX}]\mathbb{E}[\mathbf{BY}]^\top \\ &= \mathbf{A}\mathbb{E}[\mathbf{XY}^\top]\mathbf{B}^\top - \mathbf{A}\mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{Y}]^\top\mathbf{B}^\top \\ &= \mathbf{A}(\mathbb{E}[\mathbf{XY}^\top] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{Y}]^\top)\mathbf{B}^\top = \mathbf{A}\text{Cov}(\mathbf{X}, \mathbf{Y})\mathbf{B}^\top. \quad \square\end{aligned}$$

A.3 Multivariate Normal Distribution

Assume that the multivariate random variable $\mathbf{X}$ follows the multivariate normal distribution with $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Proposition A.3 (Linear Transformation Property of Multivariate Normal Vector) *Let $\mathbf{A}$ be an $n \times n$ non-random matrix. The linear transformation $\mathbf{AX}$ also follows the multivariate normal distribution, with*

$$\mathbf{AX} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top).$$

Proof Denote $\mathbf{X}$ as $[X_1, X_2, \dots, X_n]^\top$. By definition of multivariate normal distribution, any linear combination of $X_i, i = 1, \dots, n$ is normally distributed. As each entry of the vector $\mathbf{AX}$ is a linear combination of $X_i, i = 1, \dots, n$, any linear combination of the entries must also be normally distributed. Hence, $\mathbf{AX}$ follows the multivariate normal distribution which is fully characterized by the expectation and covariance. From Proposition A.2, we obtain

$$\mathbb{E}[\mathbf{AX}] = \mathbf{A}\mathbb{E}[\mathbf{X}] = \mathbf{A}\boldsymbol{\mu} \quad \text{and} \quad \text{Cov}(\mathbf{AX}, \mathbf{AX}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top. \quad \square$$

A.4 Conditional Normal Distribution

Consider an n -dimensional normally distributed random vector partitioned into two parts as

$$\begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

where

$$\boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix} \text{ is an } n \times 1 \text{ vector,}$$

$$\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix} \text{ is an } n \times n \text{ symmetric positive-semidefinite matrix,}$$

$\mathbf{X}_1, \boldsymbol{\mu}_1$ are $p \times 1$ vectors, $\mathbf{X}_2, \boldsymbol{\mu}_2$ are $q \times 1$ vectors, $p + q = n$, $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^\top$, and $\boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{12}^\top$.

Proposition A.4 (Conditional Normal Distribution) *The conditional distribution of $\mathbf{X}_1$ given $\mathbf{X}_2$ is normal:*

$$\mathbf{X}_1 | \mathbf{X}_2 = \mathbf{x}_2 \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{X}_1 | \mathbf{X}_2 = \mathbf{x}_2}, \boldsymbol{\Sigma}_{\mathbf{X}_1 | \mathbf{X}_2 = \mathbf{x}_2}),$$

where

$$\boldsymbol{\mu}_{\mathbf{X}_1 | \mathbf{X}_2 = \mathbf{x}_2} = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} (\mathbf{x}_2 - \boldsymbol{\mu}_2),$$

$$\boldsymbol{\Sigma}_{\mathbf{X}_1 | \mathbf{X}_2 = \mathbf{x}_2} = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21}.$$

Evidently, we have similar result for $\mathbf{X}_2 | \mathbf{X}_1 = \mathbf{x}_1$.

Proof Consider the centered random vectors $\mathbf{Y} = \mathbf{X}_1 - \boldsymbol{\mu}_1$ and $\mathbf{X} = \mathbf{X}_2 - \boldsymbol{\mu}_2$. Define $\mathbf{Z} = \mathbf{Y} - \mathbf{A}\mathbf{X}$ where $\mathbf{A} = \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}$. From

$$\begin{bmatrix} \mathbf{Z} \\ \mathbf{X} \end{bmatrix} = \begin{bmatrix} \mathbf{I} - \mathbf{A} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{Y} \\ \mathbf{X} \end{bmatrix} = \begin{bmatrix} \mathbf{I} - \mathbf{A} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{X}_1 - \boldsymbol{\mu}_1 \\ \mathbf{X}_2 - \boldsymbol{\mu}_2 \end{bmatrix},$$

we see that $\mathbf{Z}$ and $\mathbf{X}$ are jointly normally distributed vectors, and so are $\mathbf{Z}$ and $\mathbf{X}_2$. The distribution of $\mathbf{Z}$ is

$$\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21})$$

because $\mathbb{E}[\mathbf{Z}] = \mathbb{E}[\mathbf{Y} - \mathbf{A}\mathbf{X}] = \mathbb{E}[\mathbf{Y}] - \mathbf{A}\mathbb{E}[\mathbf{X}] = \mathbf{0}$ and

$$\begin{aligned} \text{Var}(\mathbf{Z}) &= \text{Var}(\mathbf{Y} - \mathbf{A}\mathbf{X}) \\ &= \text{Var}(\mathbf{Y}) + \mathbf{A}\text{Var}(\mathbf{X})\mathbf{A}^\top - \mathbf{A}\text{Cov}(\mathbf{X}, \mathbf{Y}) - \text{Cov}(\mathbf{Y}, \mathbf{X})\mathbf{A}^\top \\ &= \boldsymbol{\Sigma}_{11} + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{22}(\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1})^\top - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} - \boldsymbol{\Sigma}_{12}(\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1})^\top \\ &= \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}. \end{aligned}$$

Now, we show that $\mathbf{Z}$ and $\mathbf{X}_2$ are uncorrelated:

$$\begin{aligned} \text{Cov}(\mathbf{Z}, \mathbf{X}_2) &= \text{Cov}(\mathbf{Y}, \mathbf{X}_2) - \text{Cov}(\mathbf{A}\mathbf{X}, \mathbf{X}_2) \\ &= \text{Cov}(\mathbf{X}_1 - \boldsymbol{\mu}_1, \mathbf{X}_2) - \mathbf{A}\text{Cov}(\mathbf{X}_2 - \boldsymbol{\mu}_2, \mathbf{X}_2) = \boldsymbol{\Sigma}_{12} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{22} = \mathbf{0}. \end{aligned}$$

It is a known property that uncorrelated joint normal vectors are independent. Hence, $\mathbf{Z}$ and $\mathbf{X}_2$ are independent normal vectors. Now, we can determine the distribution of $\mathbf{X}_1|\mathbf{X}_2 = \mathbf{x}_2$ by

$$\begin{aligned} \mathbb{E}[\mathbf{X}_1|\mathbf{X}_2 = \mathbf{x}_2] &= \mathbb{E}[\boldsymbol{\mu}_1 + \mathbf{Y}|\mathbf{X}_2 = \mathbf{x}_2] = \mathbb{E}[\boldsymbol{\mu}_1 + \mathbf{Z} + \mathbf{A}\mathbf{X}|\mathbf{X}_2 = \mathbf{x}_2] \\ &= \boldsymbol{\mu}_1 + \mathbb{E}[\mathbf{Z}|\mathbf{X}_2 = \mathbf{x}_2] + \mathbf{A}(\mathbf{x}_2 - \boldsymbol{\mu}_2) \\ &= \boldsymbol{\mu}_1 + \mathbb{E}[\mathbf{Z}] + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2) \\ &= \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2) \end{aligned}$$

and

$$\begin{aligned} \text{Var}(\mathbf{X}_1|\mathbf{X}_2 = \mathbf{x}_2) &= \text{Var}(\mathbf{Y}|\mathbf{X}_2 = \mathbf{x}_2) = \text{Var}(\mathbf{Z} + \mathbf{A}\mathbf{X}|\mathbf{X}_2 = \mathbf{x}_2) \\ &= \text{Var}(\mathbf{Z} + \mathbf{A}(\mathbf{x}_2 - \boldsymbol{\mu}_2)|\mathbf{X}_2 = \mathbf{x}_2) \\ &= \text{Var}(\mathbf{Z}) = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}. \end{aligned}$$

□

Proposition A.5 (Woodbury matrix identity) *Let $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ be matrices with compatible dimensions. Then,*

$$(\mathbf{A} + \mathbf{CBD})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{C}(\mathbf{B}^{-1} + \mathbf{DA}^{-1}\mathbf{C})^{-1}\mathbf{DA}^{-1}.$$

Proof Let $\mathbf{I}$ be an identity matrix with suitable size. Then,

$$\begin{aligned} & (\mathbf{A} + \mathbf{CBD})(\mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{C}(\mathbf{B}^{-1} + \mathbf{DA}^{-1}\mathbf{C})^{-1}\mathbf{DA}^{-1}) \\ &= (\mathbf{A} + \mathbf{CBD})\mathbf{A}^{-1} - (\mathbf{A} + \mathbf{CBD})\mathbf{A}^{-1}\mathbf{C}(\mathbf{B}^{-1} + \mathbf{DA}^{-1}\mathbf{C})^{-1}\mathbf{DA}^{-1} \\ &= \mathbf{I} + \mathbf{CBDA}^{-1} - (\mathbf{C} + \mathbf{CBDA}^{-1}\mathbf{C})(\mathbf{B}^{-1} + \mathbf{DA}^{-1}\mathbf{C})^{-1}\mathbf{DA}^{-1} \\ &= \mathbf{I} + \mathbf{CBDA}^{-1} - \mathbf{CB}(\mathbf{B}^{-1} + \mathbf{DA}^{-1}\mathbf{C})(\mathbf{B}^{-1} + \mathbf{DA}^{-1}\mathbf{C})^{-1}\mathbf{DA}^{-1} \\ &= \mathbf{I} + \mathbf{CBDA}^{-1} - \mathbf{CBDA}^{-1} \\ &= \mathbf{I}. \end{aligned}$$

□

Appendix B

Representation of CIR Process as a Squared Bessel Process

Some of the important stochastic processes used in financial modelling such as CEV process (used in SABR model) and CIR process (used in CIR short rate model and Heston model) can be represented by a time-changed Squared Bessel process. This helps to understand the properties of these processes because Squared Bessel process is well studied and many properties are well known. Here, we will only illustrate the representation of CIR process. We refer to [43] for rigorous and detailed presentation in general.

B.1 Squared Bessel Process

Squared Bessel Process is one of the most important one-dimensional stochastic processes. It is defined as a generalization of the process followed by the squared norm of a multi-dimensional Brownian motion. Denote $Z_t^2 = \sum_{i=1}^k (B_t^i)^2$ where $\mathbf{B}_t = (B_t^1, \dots, B_t^k)$ is a k -dimensional Brownian motion and $Z_t = \|\mathbf{B}_t\|$. By Ito–Doebelin formula, we obtain

$$dZ_t^2 = \sum_{i=1}^k 2B_t^i dB_t^i + kdt = 2Z_t \sum_{i=1}^k \frac{B_t^i}{\|\mathbf{B}_t\|} dB_t^i + kdt = 2Z_t dW_t + kdt,$$

where we use the fact that $W_t = \sum_{i=1}^k \int_0^t \frac{B_s^i}{\|\mathbf{B}_s\|} dB_s^i$ is a one-dimensional Brownian motion which can be verified with Levy's Theorem.

Squared Bessel Process is defined as the generalization of the process followed by Z_t^2 with k extended to non-integer values, i.e. $k \in \mathbb{R}$:

$$dX_t = 2\sqrt{X_t}dW_t + kdt.$$

In our context, we only consider $k \geq 0$ and strictly positive initial value $X_0 > 0$.

X_t follows *Chi-squared* distribution extended to non-integer degrees of freedom (cf. [43]). An important focus is the condition for reaching zero. The following features are useful for the applications in financial modelling

- (i) $k > 2$: X_t will never reach 0 and $X_t \rightarrow +\infty$ as $t \rightarrow +\infty$.
- (ii) $k = 2$: X_t will never reach 0 and X_t is *recurrent*, meaning that the probability for X_t to get infinitely close to any non negative value in any finite time is strictly positive.
- (iii) $0 < k < 2$: X_t reaches 0 in finite time. However, it leaves 0 instantaneously.
- (iv) $k = 0$: X_t reaches 0 in finite time and will stay at 0 afterwards.

For the proofs, we refer to [44] for $k = 2$. For a comprehensive analysis, we refer to [43, 68] and [51]. For helping to understand the properties, we mention some intuitions. In the case of one dimension (i.e. $k = 1$), $\sqrt{X_t}$ is a standard Brownian motion before hitting 0. So, it reaches 0 in finite time as it is the case for standard Brownian motion. For $0 \leq k < 1$, the process can be written as the sum of a Squared Bessel process with dimension 1 and a strictly decreasing term: $dX_t = (2\sqrt{X_t}dW_t + 1dt) + (k - 1)dt$. So, X_t reaches 0 for $0 \leq k < 1$. Similarly, for $k > 2$ with $dX_t = (2\sqrt{X_t}dW_t + 2dt) + (k - 2)dt$, X_t will never reach 0 if it never reaches zero for $k = 2$. Whenever the point 0 is reached, the staying time should be zero (i.e. it leaves 0 instantaneously). Otherwise, the SDE can not hold because in such case, the kdt term is strictly positive but all the other terms are zero for the time period that X_t stayed at zero.

B.2 Representation of CIR Process

Consider the CIR process which follows the below SDE under the assumption that $b > 0$, $r_0 > 0$, $\sigma > 0$ and $a \in \mathbb{R}$,

$$dr_t = (a - br_t)dt + \sigma\sqrt{r_t}dW_t.$$

First, we try to remove r_t in the drift. Applying Ito's Product Rule to $Y_t = e^{bt}r_t$, we have

$$dY_t = e^{bt}dr_t + br_te^{bt}dt = e^{bt}(adt + \sigma\sqrt{r_t}dW_t) = ae^{bt}dt + \sigma\sqrt{e^{bt}}\sqrt{Y_t}dW_t.$$

Denote $\tau(t)$ such that $t = \frac{\sigma^2}{4} \int_0^{\tau(t)} e^{bs}ds$ for any given t . $\tau(t)$ exists because $f(x) = \frac{\sigma^2}{4} \int_0^x e^{bs}ds$ is an increasing continuous function with $f(0) = 0$ and $f(+\infty) = +\infty$. Hence, integration from 0 to $\tau(t)$ gives

$$Y_{\tau(t)} = r_0 + a \int_0^{\tau(t)} e^{bs} ds + 2 \int_0^{\tau(t)} \frac{\sigma}{2} \sqrt{e^{bs}} \sqrt{Y_s} dW_s.$$

The first integral is equivalent to the following which is straightforward from the definition of $\tau(t)$,

$$a \int_0^{\tau(t)} e^{bs} ds = \frac{4a}{\sigma^2} t.$$

For the second integral, we will apply the so-called *time-change*. Actually, for any continuous martingale M_t , it is known that $M_t = W_{\langle M_t, M_t \rangle}$. This helps to understand time-change which implies:

$$\int_0^T \beta_t dM_t = \int_0^{\langle M_T, M_T \rangle} \beta_{u(t)} dM_{u(t)} = \int_0^{\langle M_T, M_T \rangle} \beta_{u(t)} dW_t,$$

where $u(t)$ is the value such that $\langle M_{u(t)}, M_{u(t)} \rangle = t$.

Now, for our case, denote $M_t = \int_0^t \frac{\sigma}{2} \sqrt{e^{bs}} dW_s$. M_t is a continuous martingale and $\langle M_{\tau(t)}, M_{\tau(t)} \rangle = \frac{\sigma^2}{4} \int_0^{\tau(t)} e^{bs} ds = t$. Applying time-change, we have for the second integral,

$$\int_0^{\tau(t)} \frac{\sigma}{2} \sqrt{e^{bs}} \sqrt{Y_s} dW_s = \int_0^{\tau(t)} \sqrt{Y_s} dM_s = \int_0^t \sqrt{Y_{\tau(s)}} dW_s.$$

And all together,

$$Y_{\tau(t)} = r_0 + \frac{4a}{\sigma^2} t + 2 \int_0^t \sqrt{Y_{\tau(s)}} dW_s.$$

Setting $X_t = Y_{\tau(t)}$, $X_0 = r_0$ and $k = \frac{4a}{\sigma^2}$, we obtain the representation of CIR process as a Squared Bessel process

$$X_t = r_0 + kt + 2 \int_0^t \sqrt{X_s} dW_s.$$

From the properties of Squared Bessel process, we obtain the following:

- if $a \geq \frac{\sigma^2}{2}$, then r_t will never reach zero.
- if $0 < a < \frac{\sigma^2}{2}$, then r_t has a non zero probability to touch 0 in any time interval but r_t will leave 0 instantaneously if it does.

References

1. Andersen, L.B.G., Buffum, D.: Calibration and implementation of convertible bond models. *J. Comput. Finance* **7**(2), 1–34 (2004)
2. Andersen, L.B.G., Piterbarg, V.V.: Interest Rate Modeling. Volume 1: Foundations and Vanilla Models, 1st edn. Atlantic Financial Press, London (2010)
3. Andersen, L.B.G., Piterbarg, V.V.: Interest Rate Modeling. Volume 2: Term Structure Models, 1st edn. Atlantic Financial Press, London (2010)
4. Andersen, L.B.G., Piterbarg, V.V.: Interest Rate Modeling. Volume 3: Products and Risk Management, 1st edn. Atlantic Financial Press, London (2010)
5. Bachelier, L.: Théorie de la spéculation. *Annales Scientifiques de l'École Normale Supérieure* **3**(17), 21–86 (1900)
6. Basel Committee on Banking Supervision: An Explanatory Note on the Basel II IRB Risk Weight Functions. BIS (2005) <https://www.bis.org/bcbs/irbriskweight.pdf>
7. Becker, M.: Exact simulation of final, minimal and maximal values of Brownian motion and jump-diffusions with applications to option pricing. *Comput. Manage. Sci.* **7**(1), 1–17 (2010)
8. Bergomi, L.: Stochastic Volatility Modeling, 1st edn. Chapman and Hall/CRC Financial Mathematics. CRC Press, Boca Raton (2016)
9. Bisesti, L., Castagna, A., Mercurio, F.: Consistent pricing and hedging of an FX options book. *Kyoto Econ. Rev.* **74**(1), 65–83 (2005)
10. Black, F., Scholes, M.: The pricing of options and corporate liabilities. *J. Polit. Econ.* **81**(3), 637–654 (1973)
11. Black, F., Litterman, R.: Asset allocation: combining investor views with market equilibrium. Goldman Sachs Fixed Income Research Note (1990)
12. Brigo D., Pallavicini A., Torresetti, R.: Credit Models and the Crisis: A Journey into CDOs, Copulas, Correlations and Dynamic Models. Wiley, Hoboken (2010)
13. Carhart, M.M.: On persistence in mutual fund performance. *J. Finance* **52**, 57–82 (1997)
14. Carr, P., Javaheri, A.: The forward PDE for European options on stocks with fixed fractional jumps. *Int. J. Theor. Appl. Finance* **8**(2), 239–253 (2005)
15. Carr, P., Madan, D.B.: Option valuation using the fast Fourier transform. *J. Comput. Finance* **2**(4), 61–73 (1999)
16. Cesari, G., Aquilina, J., Charpillon, N., Filipovic, Z., Lee, G., Manda, I.: Modelling, Pricing, and Hedging Counterparty Credit Exposure. Springer Finance, Berlin/Heidelberg (2009)
17. Chan, R.H., Ng, M.K.: Conjugate gradient methods for Toeplitz systems. *SIAM Rev.* **38**(3), 427–482 (1996)
18. Chan, R.H., Jin, X.-Q.: An Introduction to Iterative Toeplitz Solvers. SIAM, Philadelphia (2007)

19. Chen, N.-F., Roll, R., Ross, S.A.: Economic forces and the stock market. *J. Bus.* **59**(3), 383–403 (1986)
20. Choueifaty, Y., Coignard, Y.: Toward maximum diversification. *J. Portf. Manage.* **35**(1), 40–51 (2008)
21. Clark, I.J.: *Foreign Exchange Option Pricing – A Practitioner’s Guide*, 1st edn. Wiley Finance, Chichester (2011)
22. Clark, I.J.: *Commodity Option Pricing – A Practitioner’s Guide*, 1st edn. Wiley Finance, Chichester (2014)
23. Cochrane, J.H.: Presidential address: discount rates. *J. Finance* **66**(4), 1047–1108 (2011)
24. Cont, R., Tankov, P.: Constant proportion portfolio insurance in presence of jumps in asset prices. *Math Finance* **19**(3), 379–401 (2009)
25. Cox, J.C., Ross, S.A., Rubinstein, M.: Option pricing: a simplified approach. *J. Financ. Econ.* **7**(3), 229–263 (1979)
26. Cox, J.C., Ingersoll, J.E., Ross, S.A.: A theory of the term structure of interest rates. *Econometrica* **53**, 385–407 (1985)
27. Craig, J.D., Sneyd, A.D.: An alternating-direction implicit scheme for parabolic equations with mixed derivatives. *Comput. Math. Appl.* **16**(4), 341–350 (1988)
28. Crank, J., Nicolson, P.: A practical method for numerical evaluation of solutions of partial differential equations of the heat conduction type. *Math. Proc. Camb. Philos. Soc.* **43**(1), 50–67 (1947)
29. Duffie, D., Kan, R.: A yield-factor model of interest rates. *Math. Finance* **6**(4), 379–406 (1996)
30. Estep, T., Kritzman, M.: TIPP: insurance without complexity. *J. Portf. Manage.* **14**(4), 38–42 (1988)
31. Fama, E.F., MacBeth, J.D.: Risk, return, and equilibrium: empirical tests. *J. Polit. Econ.* **81**(3), 607–636 (1973)
32. Fama, E.F., French, K.R.: Common risk factors in the returns on stocks and bonds. *J. Financ. Econ.* **33**, 3–56 (1993)
33. Gabillon, J.: The term structures of oil futures prices. Oxford Institute for Energy Studies, Working paper (1991)
34. Gordy, M.B.: A risk-factor model foundation for ratings-based bank capital rules. *J. Financ. Intermed.* **12**, 199–232 (2003)
35. Grinold, R., Kahn, R.: *Active Portfolio Management: A Quantitative Approach for Producing Superior Returns and Controlling Risk*, 2nd edn. McGraw-Hill, New York (1999)
36. Gatheral, J.: *The Volatility Surface – A Practitioner’s Guide*, 1st edn. Wiley Finance, Hoboken (2006)
37. Glasserman, P.: *Monte Carlo Methods in Financial Engineering*, 1st edn. Springer, New York (2003)
38. He, G., Litterman, R.: The intuition behind Black–Litterman model portfolios. Goldman Sachs Quantitative Resources Group Working Paper (1999)
39. Heston, S.L.: A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Rev. Financ. Stud.* **6**(2), 327–343 (1993)
40. Hull, J.: *Options, Futures, and Other Derivatives*, 10th edn. Pearson, New York (2017)
41. Hull, J., White, A.: Pricing interest-rate derivative securities. *Rev. Financ. Stud.* **3**(4), 573–592 (1990)
42. Jacod, J., Protter, P.: *Probability Essentials*, 2nd edn. Springer, Berlin/Heidelberg (2004)
43. Jeanblanc, M., Yor, M., Chesney, M.: *Mathematical Methods for Financial Markets*, 1st edn. Springer, London (2009)
44. Karatzas, I., Shreve, S.E.: *Brownian Motion and Stochastic Calculus*, 2nd edn. Springer, New York (1988)
45. Kwok, Y.-K.: *Mathematical Models of Financial Derivatives*, 2nd edn. Springer, New York (2008)
46. Kolanovic, M., Wei, Z.: *Equity Risk Premia Strategies – Risk Factor Approach to Portfolio Management*. J.P. Morgan (2014)

47. Lamberton, D.: Optimal stopping and American options. Ljubljana Summer School on Financial Mathematics (2009)
48. Leland, H.E.: Option pricing and replication with transactions costs. *J. Finance* **40**(5), 1283–1301 (1985)
49. Lintner, J.: The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *Rev. Econ. Stat.* **47**(1), 13–37 (1965)
50. Longstaff, F.A., Schwartz, E.S.: Valuing American options by simulation: a simple least-squares approach. *Rev. Financ. Stud.* **14**(1), 113–147 (2001)
51. Lowther, G.: Bessel Processes. Almost Sure (2010). <https://almostsure.wordpress.com/2010/07/28/bessel-processes/>
52. Markit Group Limited: Markit Credit Indices – A Primer (2013)
53. Markowitz, H.M.: Portfolio selection. *J. Finance* **7**(1), 77–91 (1952)
54. Merton, R.C.: An analytic derivation of the efficient portfolio frontier. *J. Financ. Quant. Anal.* **7**(4), 1851–1872 (1972)
55. Merton, R.C.: Theory of rational option pricing. *Bell J. Econ. Manage. Sci.* **4**(1), 141–183 (1973). The RAND Corporation
56. Merton, R.C.: On the pricing of corporate debt: The risk structure of interest rates. *J. Finance* **29**(2), 449–470 (1974)
57. Merton, R.C.: Option pricing when underlying stock returns are discontinuous. *J. Financ. Econ.* **3**, 125–144 (1976)
58. Merton, R.C.: Continuous-Time Finance. Wiley-Blackwell, Cambridge (1992)
59. Mossin, J.: Equilibrium in a capital asset market. *Econometrica* **34**(4), 768–783 (1966)
60. Øksendal, B.: Stochastic Differential Equations: An Introduction with Applications, 6th edn. Springer, New York (2003)
61. Overhaus, M., Bermudez, A., Buehler, H., Ferraris, A., Jorindson, C., Lamnouar, A.: Equity Hybrid Derivatives, 1st edn. Wiley Finance, Hoboken (2007)
62. Papoulis, A.: Probability, Random Variables, and Stochastic Processes, 2nd edn. McGraw-Hill, New York (1984)
63. Perold, A.F.: Constant proportion portfolio insurance. Harvard Business School, Working paper (1986)
64. Pinto, J.E., Henry, E., Robinson, T.R., Stowe, J.D.: Equity Asset Valuation, 3rd edn. Wiley, New York (2015)
65. Piterbarg, V.: Funding beyond discounting: collateral agreements and derivatives pricing, Risk.net (2010)
66. Protter, P.E.: Stochastic Integration and Differential Equations, 2nd edn. Springer, New York (2005)
67. Rannacher, R.: Finite element solution of diffusion problems with irregular data. *Numer. Math.* **43**, 309–327 (1984)
68. Revuz, D., Yor, M.: Continuous Martingales and Brownian Motion, 3rd edn. Springer, New York (1999)
69. Rogers, L.C.G.: Monte Carlo valuation of American options. *Math. Finance* **12**(3), 271–286 (2002)
70. Roncalli, T.: Introduction to Risk Parity and Budgeting, 1st edn. Chapman and Hall/CRC, Boca Raton (2013)
71. Ross, S.: The arbitrage theory of capital asset pricing. *J. Econ. Theory* **13**(3), 341–360 (1976)
72. Ross, S., Westerfield, R., Jaffe, J.: Corporate Finance, 10th edn. McGraw-Hill, New York (1976)
73. Rudin, W.: Real and Complex Analysis, 3rd edn. McGraw-Hill Education, Singapore (1986)
74. Samuelson, P.: Rational theory of warrant pricing. *Ind. Manage. Rev.* **6**, 13–31 (1965)
75. Schofield, N.C.: Commodity Derivatives: Markets and Applications, 1st edn. Wiley Finance, Chichester (2007)
76. Schwartz, E.S.: The stochastic behavior of commodity prices: implications for valuation and hedging. *J. Finance* **52**(3), 923–973 (1997)

77. Scott, L.O.: Pricing stock options in a jump-diffusion model with stochastic volatility and interest rates: applications of Fourier inversion methods. *Math. Finance* **7**, 345–358 (1997)
78. Sharpe, W.F.: Capital asset prices: a theory of market equilibrium under conditions of risk. *J. Finance* **19**(3), 425–442 (1964)
79. Shreve, S.E.: *Stochastic Calculus for Finance II: Continuous-Time Models*, 1st edn. Springer, New York (2004)
80. Tobin, J.: Liquidity preference as behavior towards risk. *Rev. Econ. Stud.* **25**(2), 65–86 (1958)
81. Treynor, J.L.: Market value, time, and risk, Unpublished manuscript dated August 8, 1961, no. 95–209 (1961)
82. White, R.: *Local Volatility*. OpenGamma Quantitative Research (2013)
83. Wilmott, P.: *Derivatives: The Theory and Practice of Financial Engineering*, 1st edn. Wiley, New York (1998)
84. Wu, L.: *Interest Rate Modeling: Theory and Practice*, 1st edn. Chapman & Hall/CRC Financial Mathematics. CRC Press, Boca Raton (2009)

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